

Practical 5

Aim: Install and Deploy Data Protection Manager

Writeup:

This image shows a single page of white paper with horizontal blue lines. The lines are evenly spaced and run across the width of the page, typical of notebook paper or a document template. There are no margins, text, or other markings on the page.

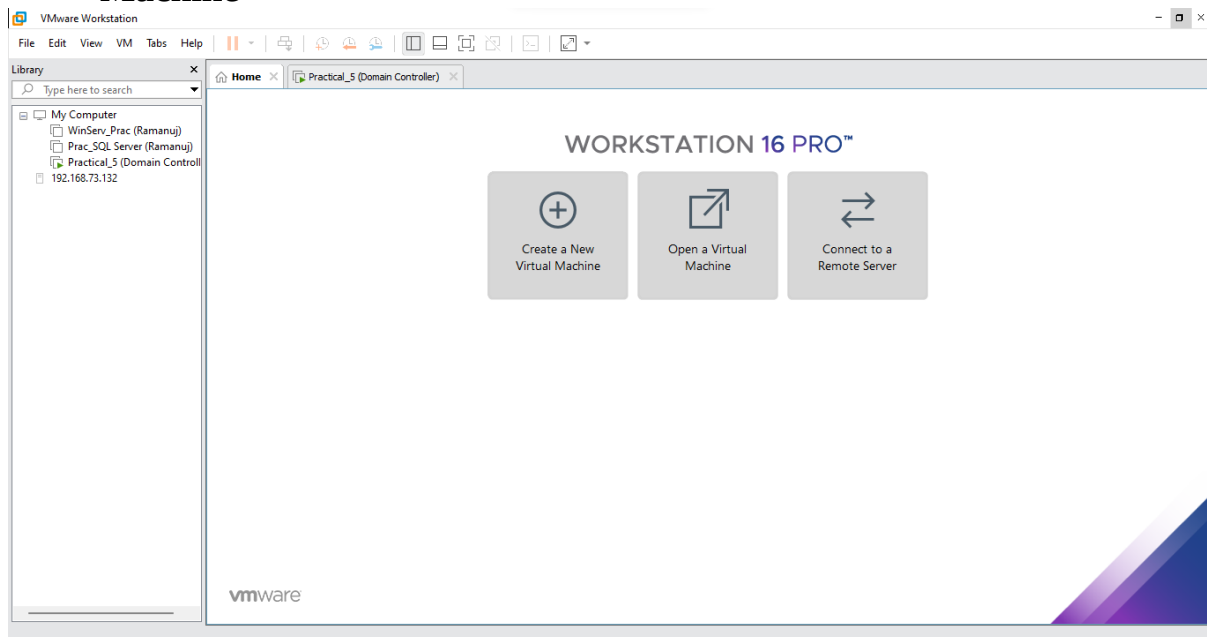
5-A: Installing DPM

Pre-requisites:

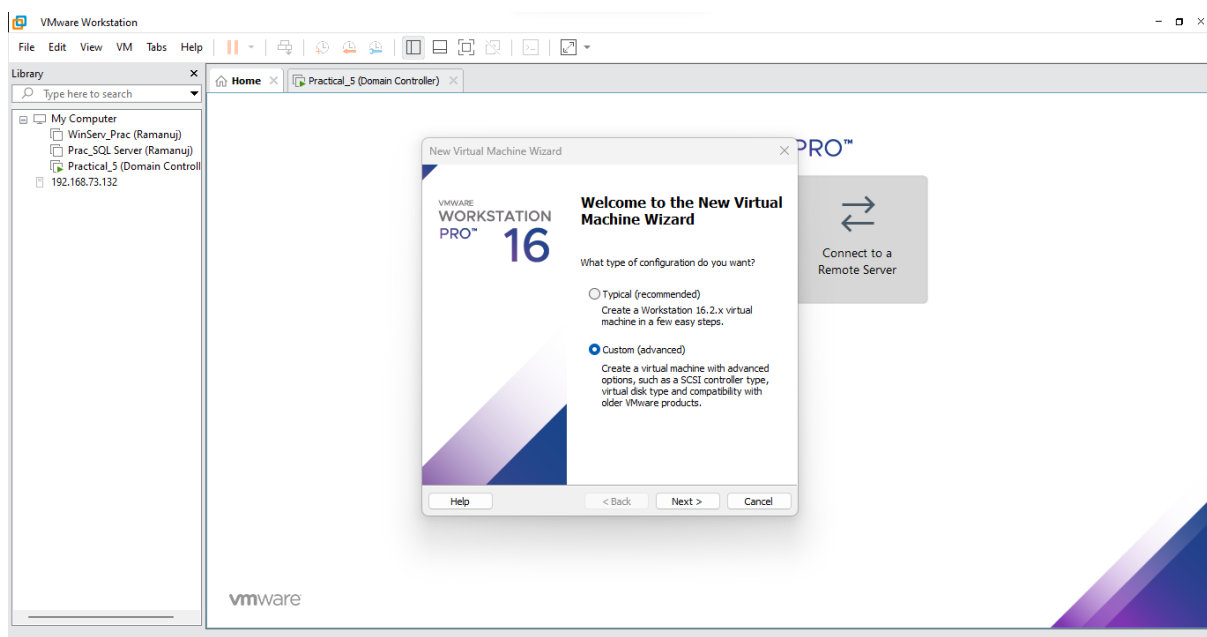
- Domain Controller
- User containing SQL Server

Step 1: We will be using the same Domain Controller created in Practical 2 and only have to create a new User VM.

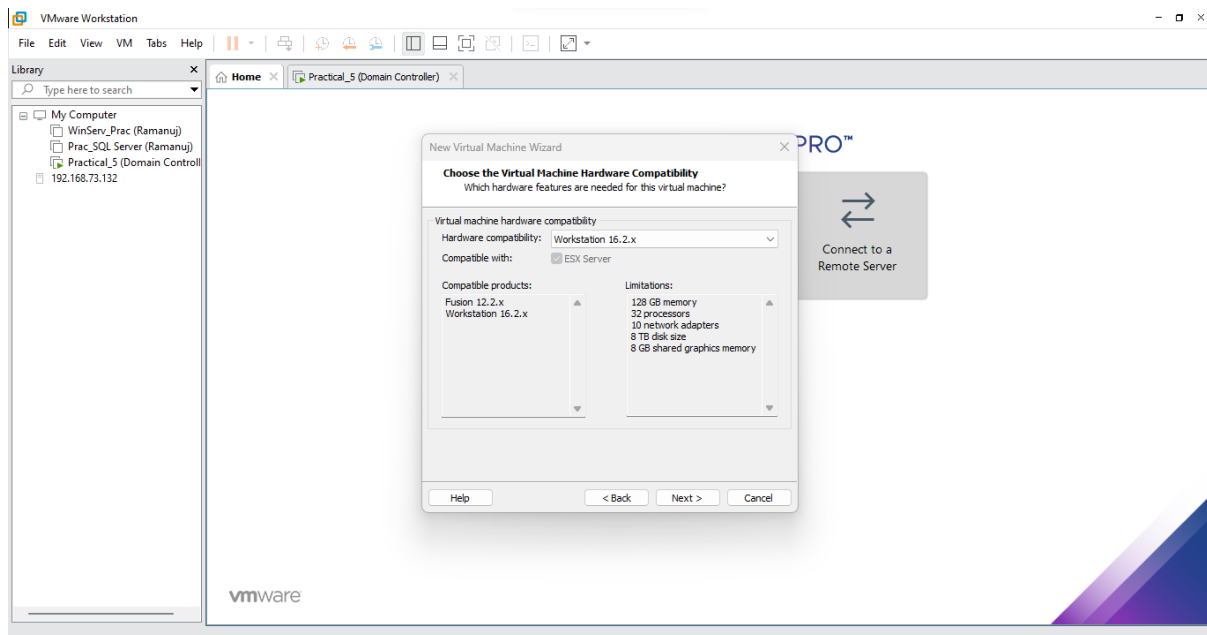
- Open **VMware Workstation** and Click on **Create a New Virtual Machine**



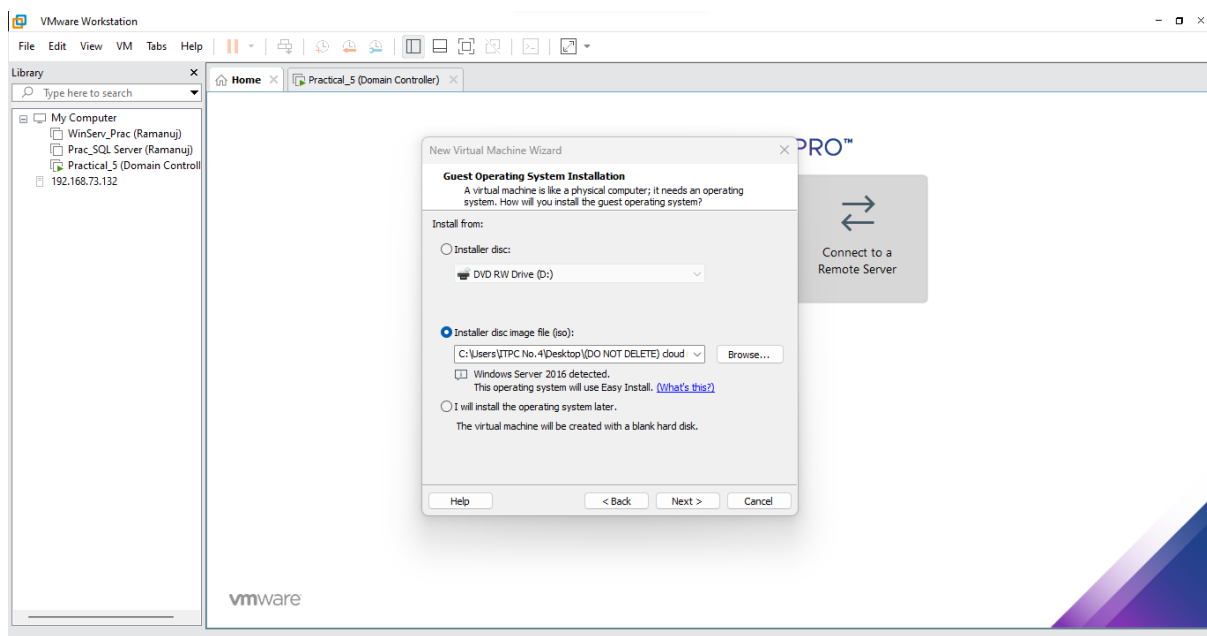
- Select **Custom** and Click **Next**



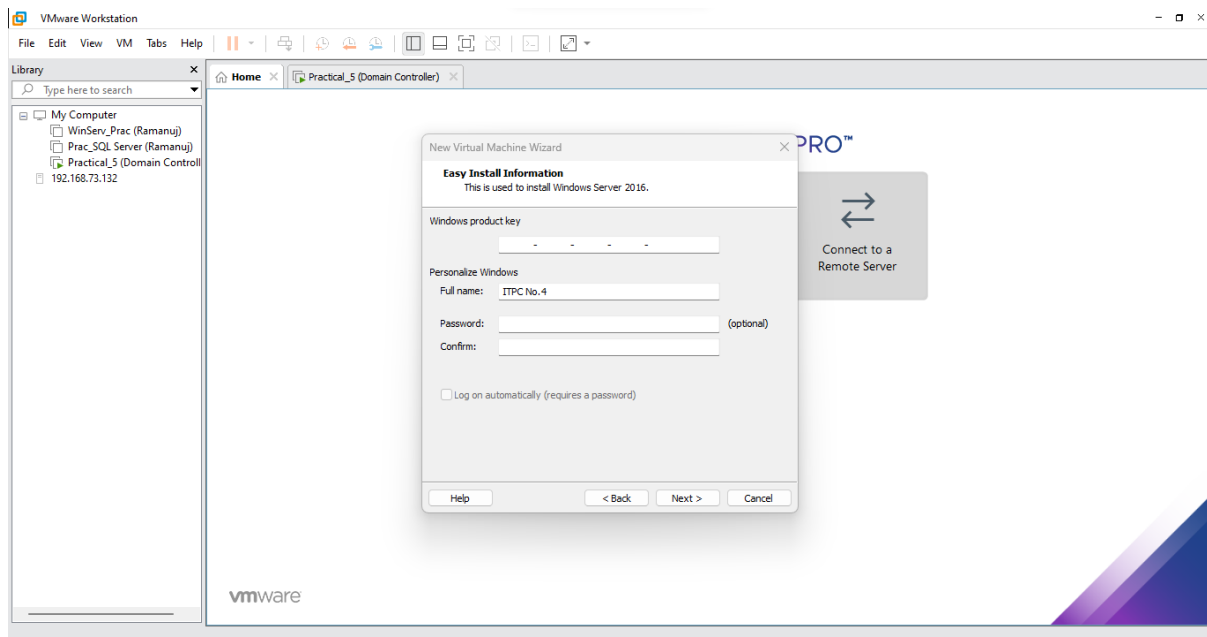
- Keep default values and Click **Next**



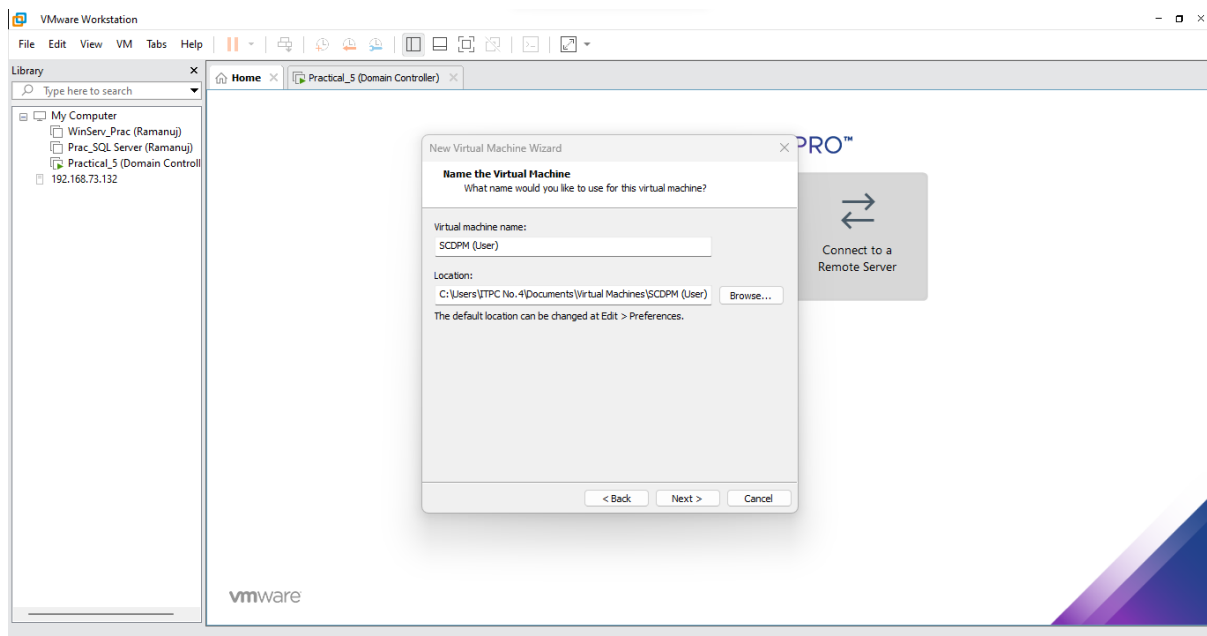
- Select **Installer disc image file (iso)** and browse for **the 2016 Windows Server iso** on your host PC



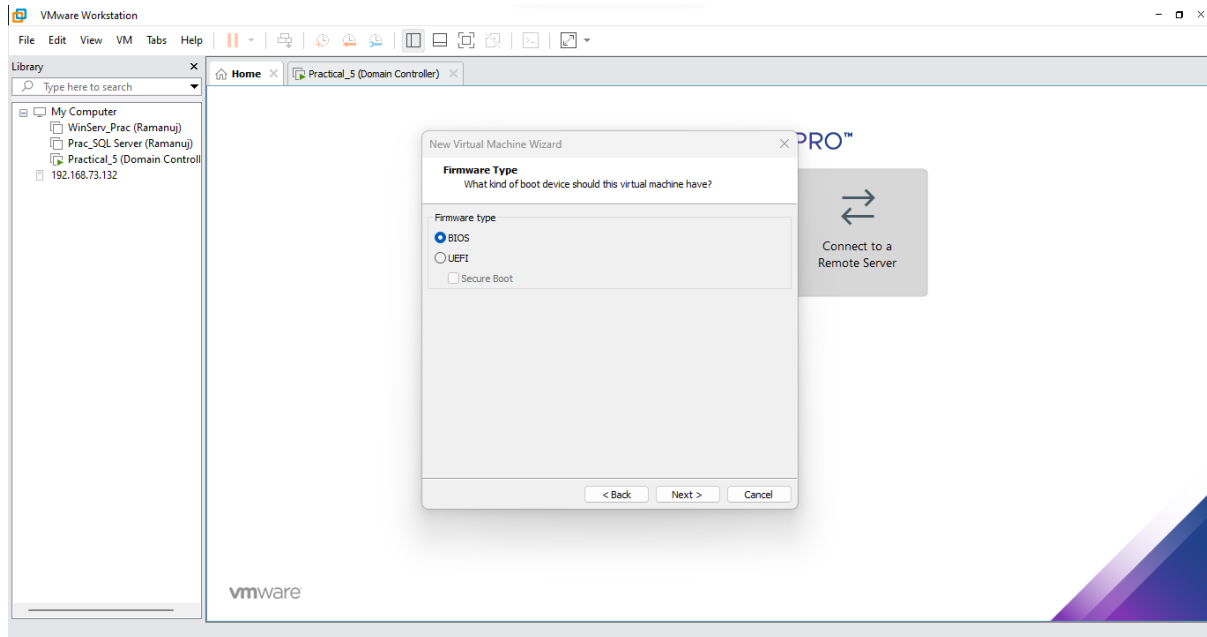
- Keep default values and Click **Next**



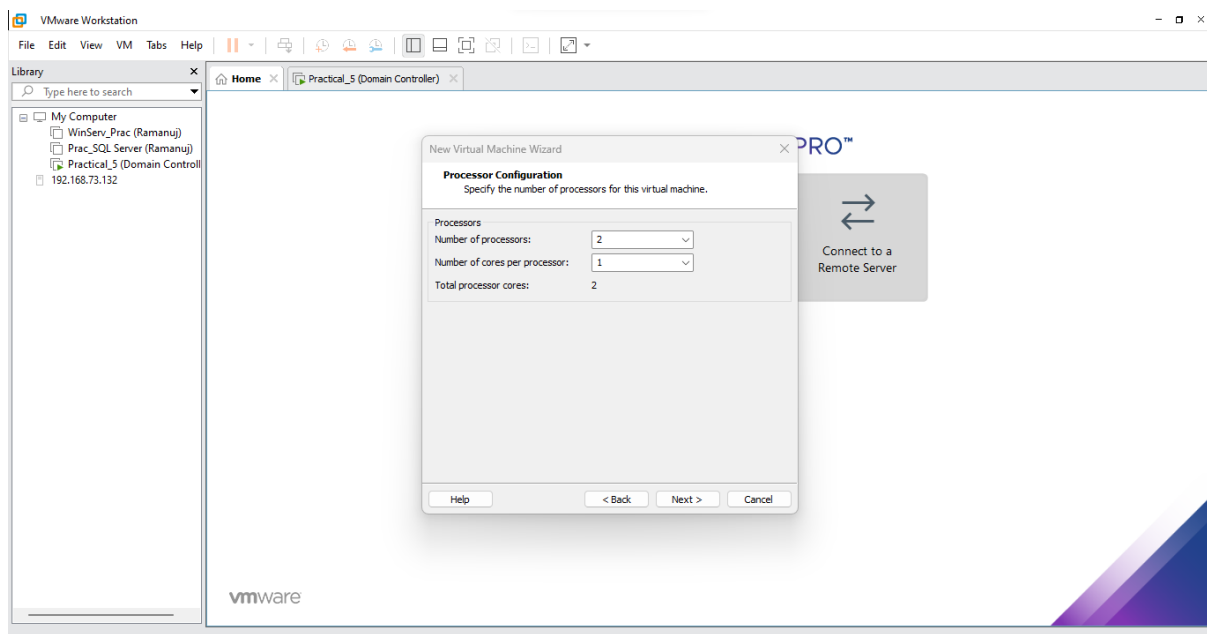
- Name your virtual machine (Here it is called **SCDPM (User)**) and Click **Next**



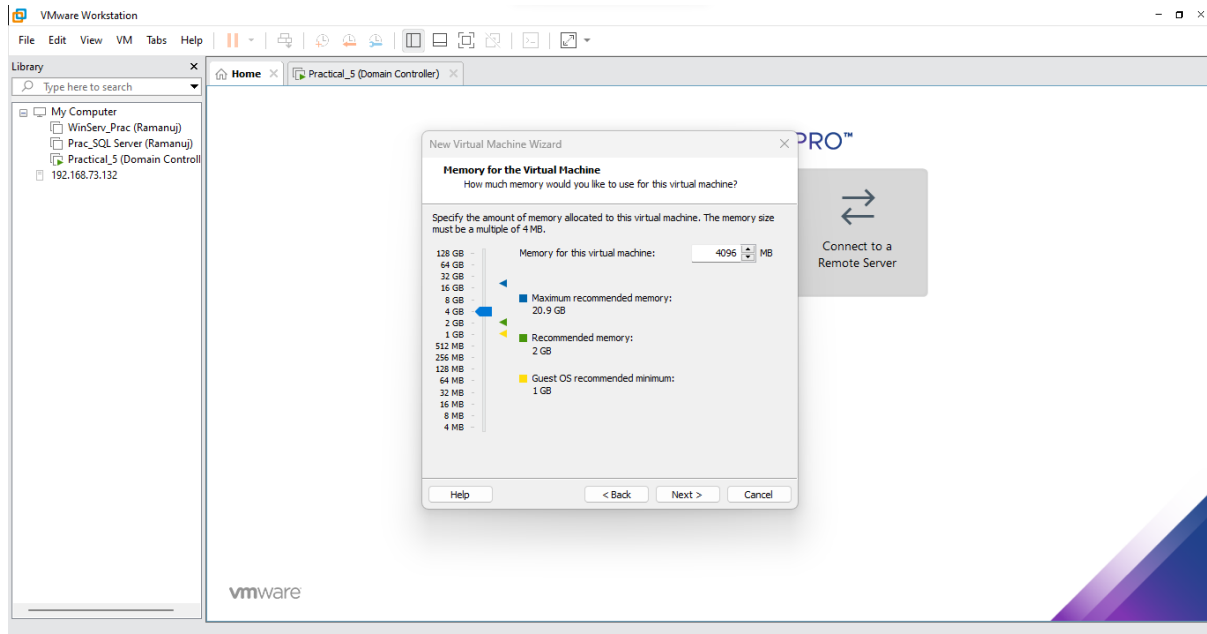
- In Firmware type Select **BIOS** and Click **Next**



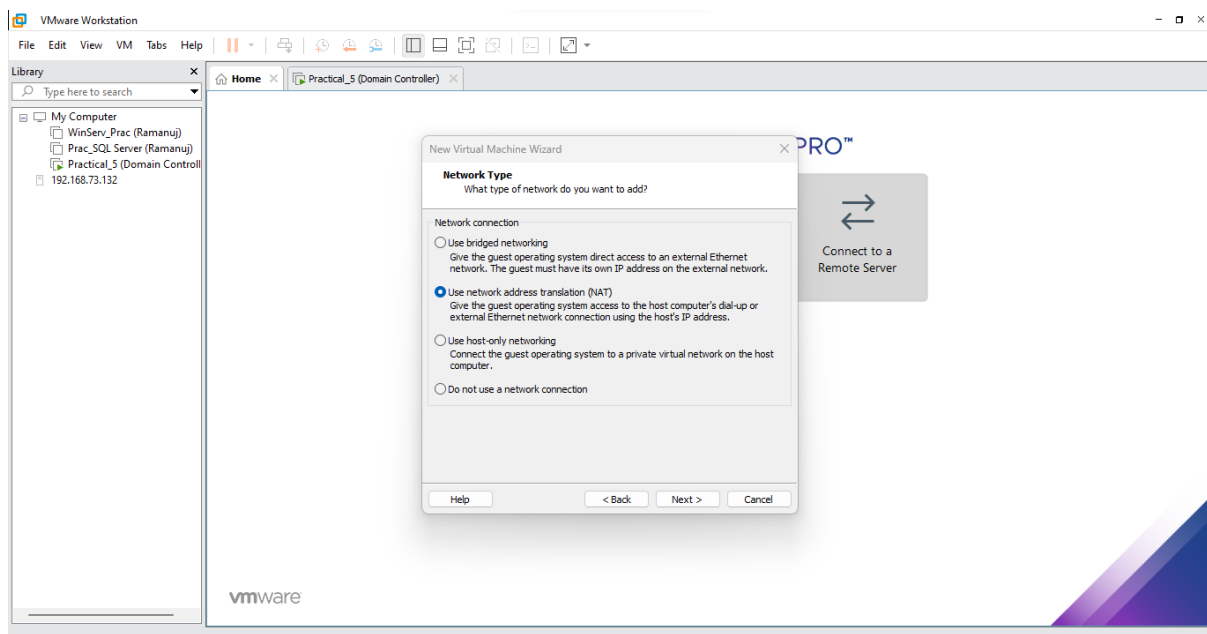
- Keep default values and Click **Next**



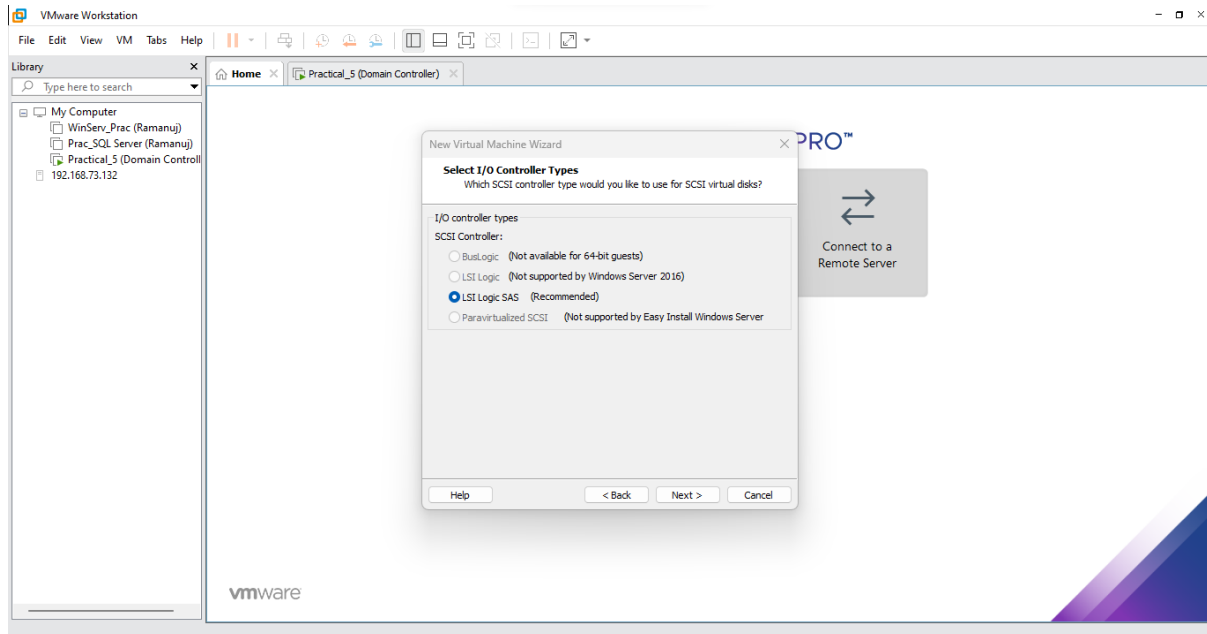
- Change the RAM value to **4096 MB** and Click **Next**



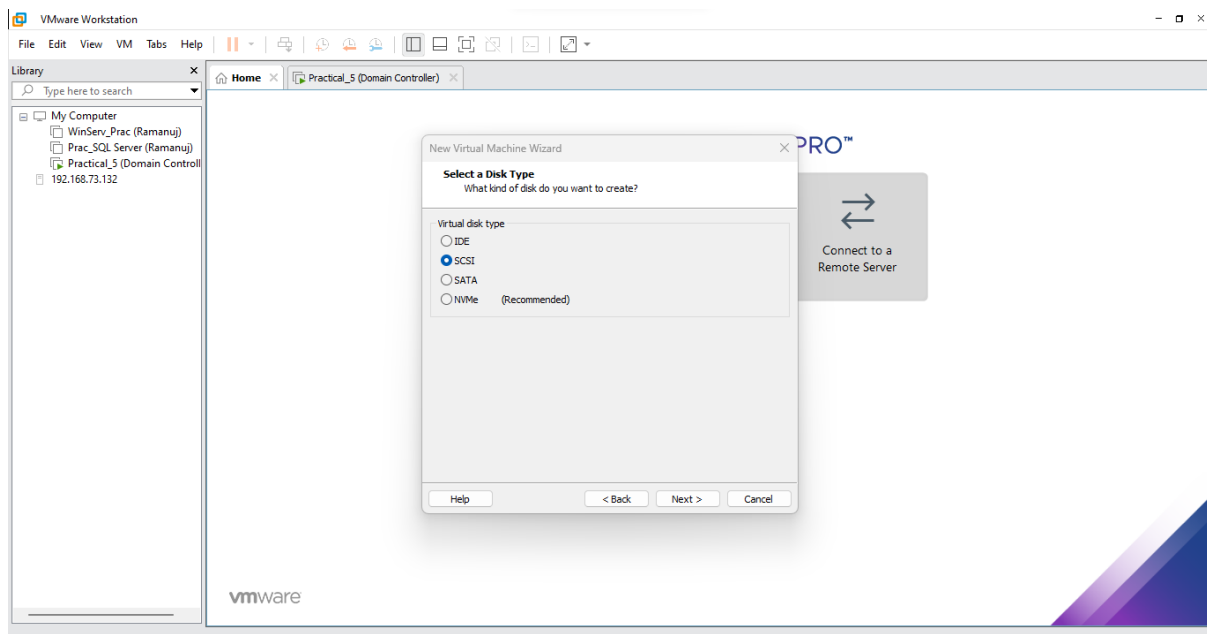
- Select **Use network address translation (NAT)** and Click **Next**



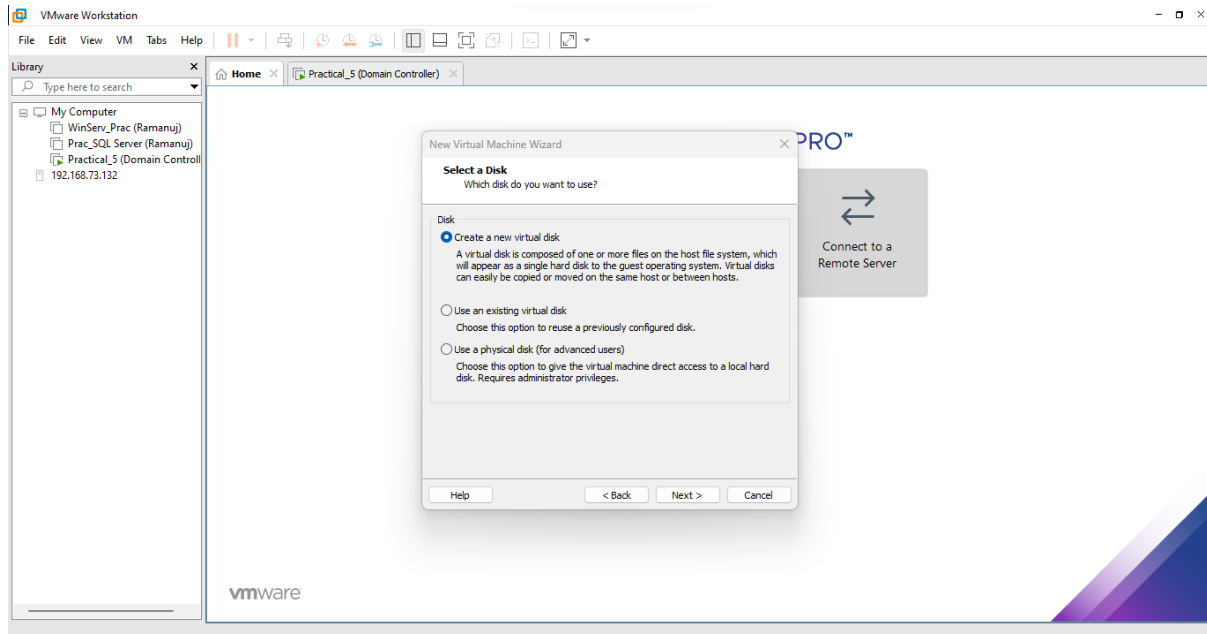
- Keep default values and Click **Next**



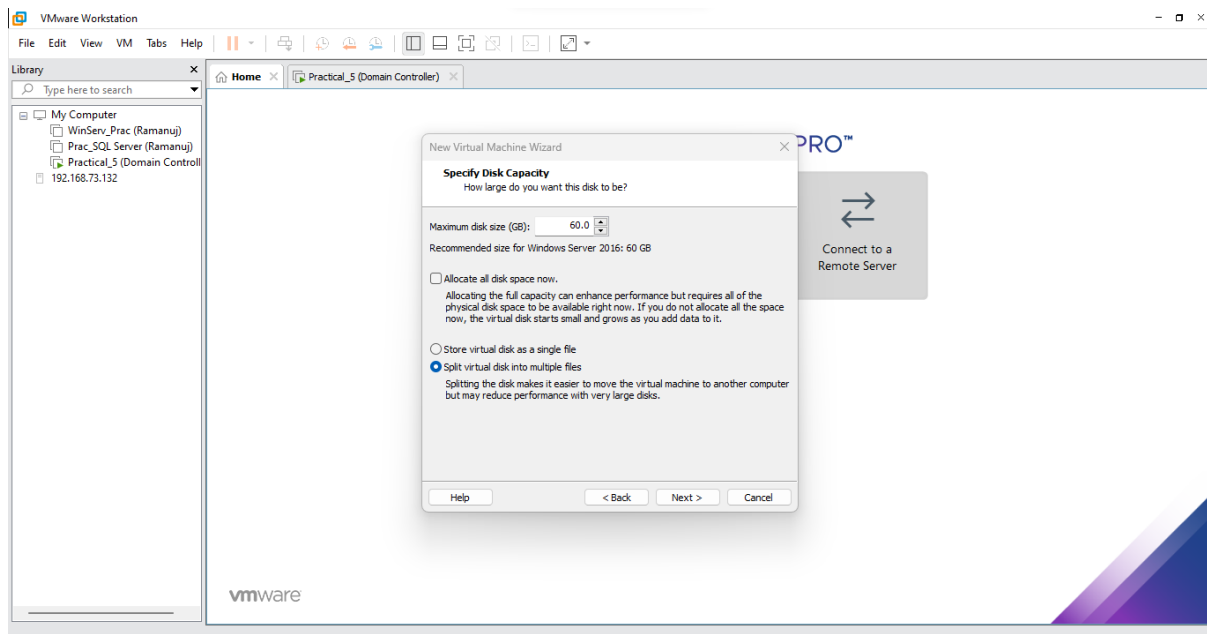
- Select Virtual disk type as **SCSI** and Click **Next**



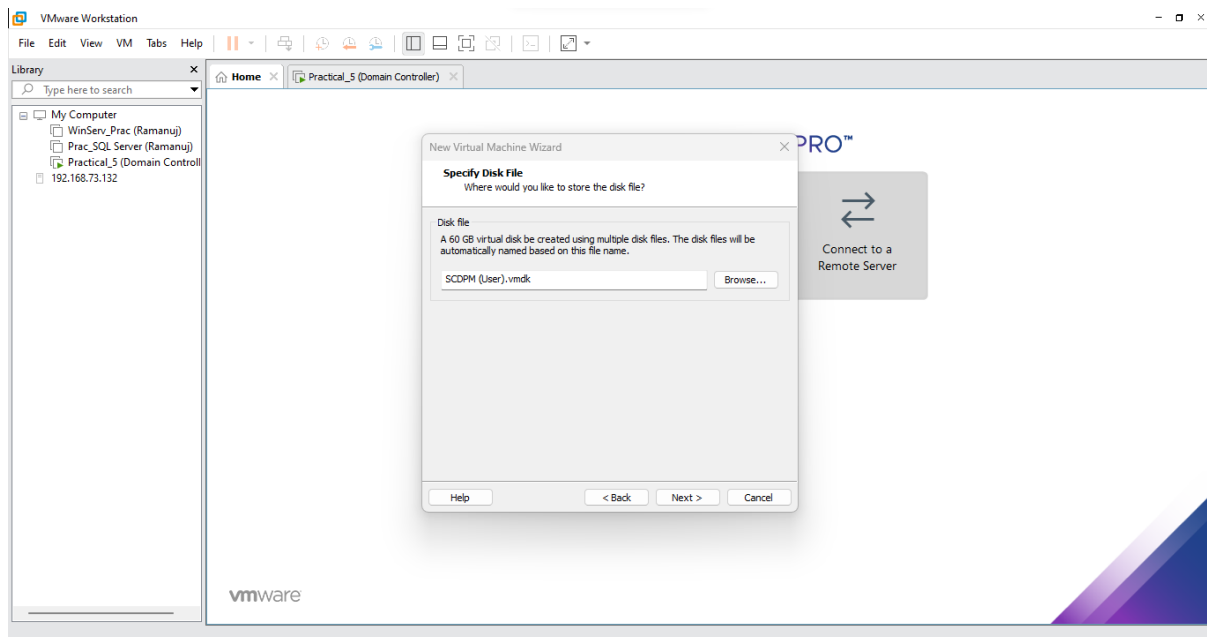
- Keep default values and Click **Next**



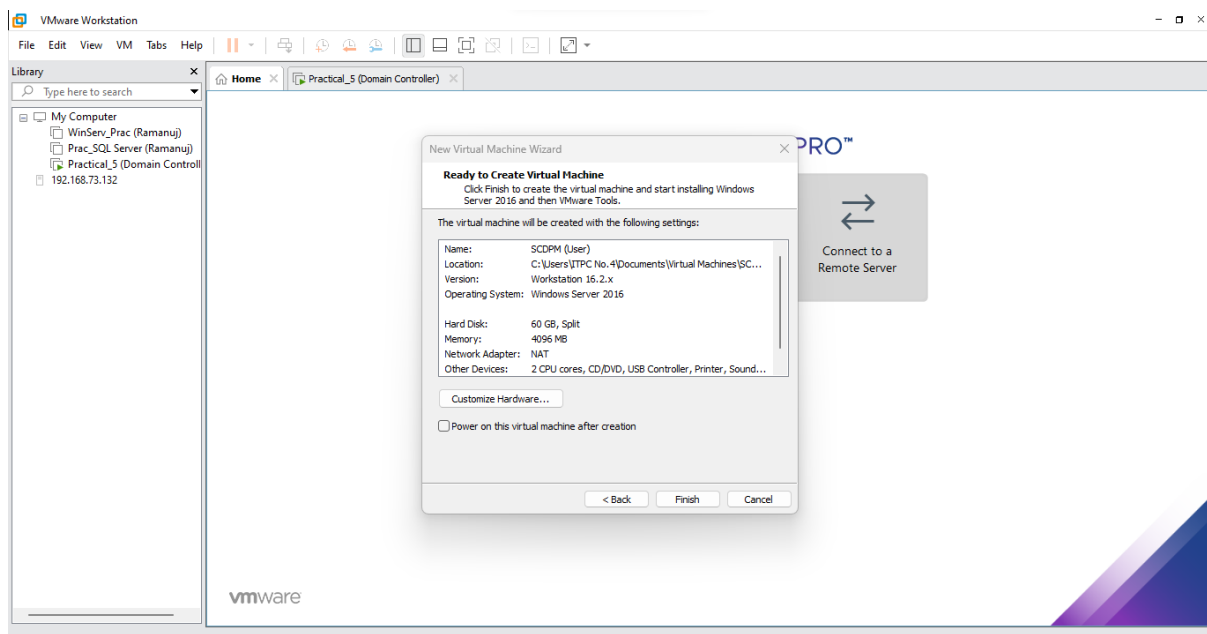
- Keep default value of **60 GB** and Click **Next**



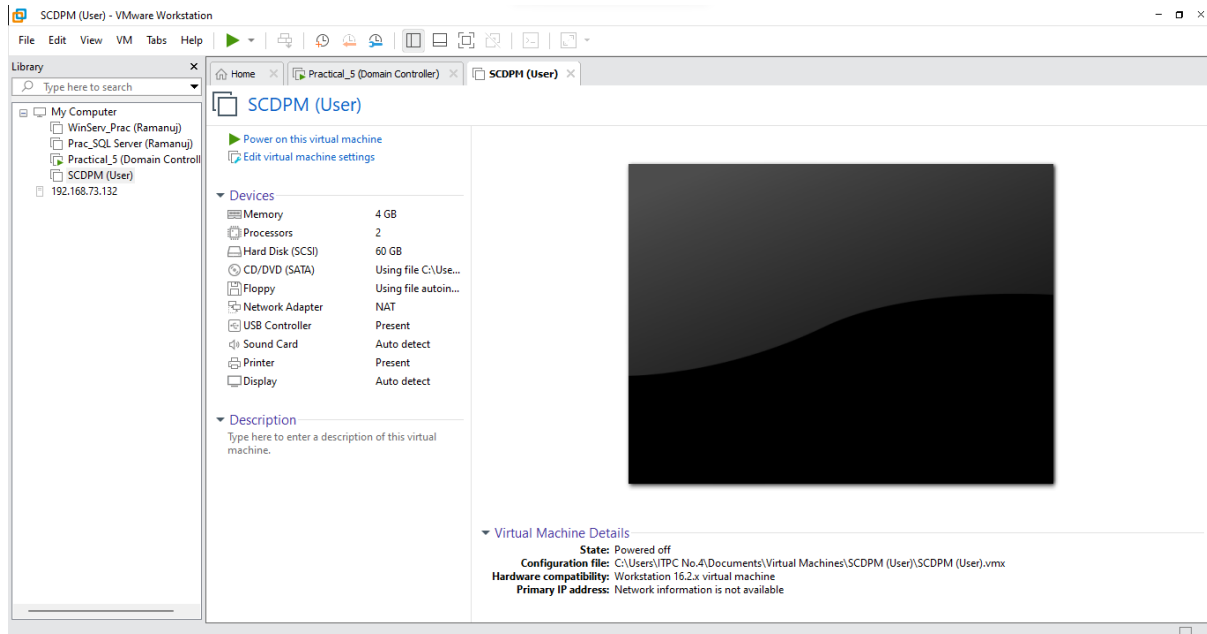
- Keep default values and Click **Next**



- Review the summary of your virtual machine and Click **Finish**

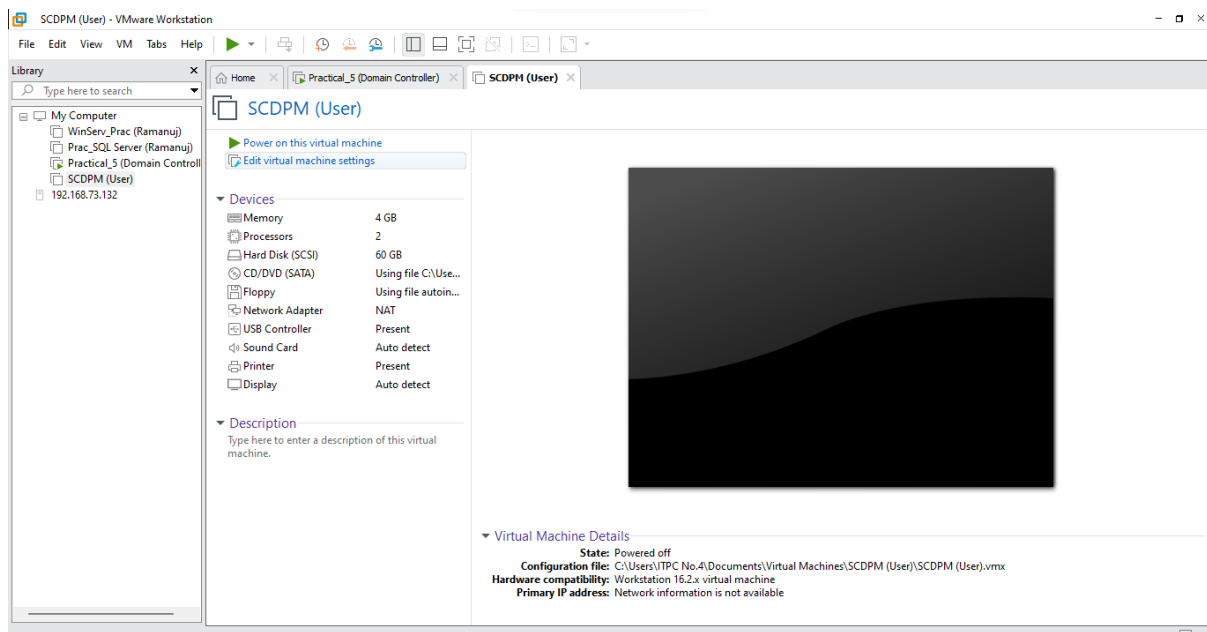


- Your virtual machine has been created

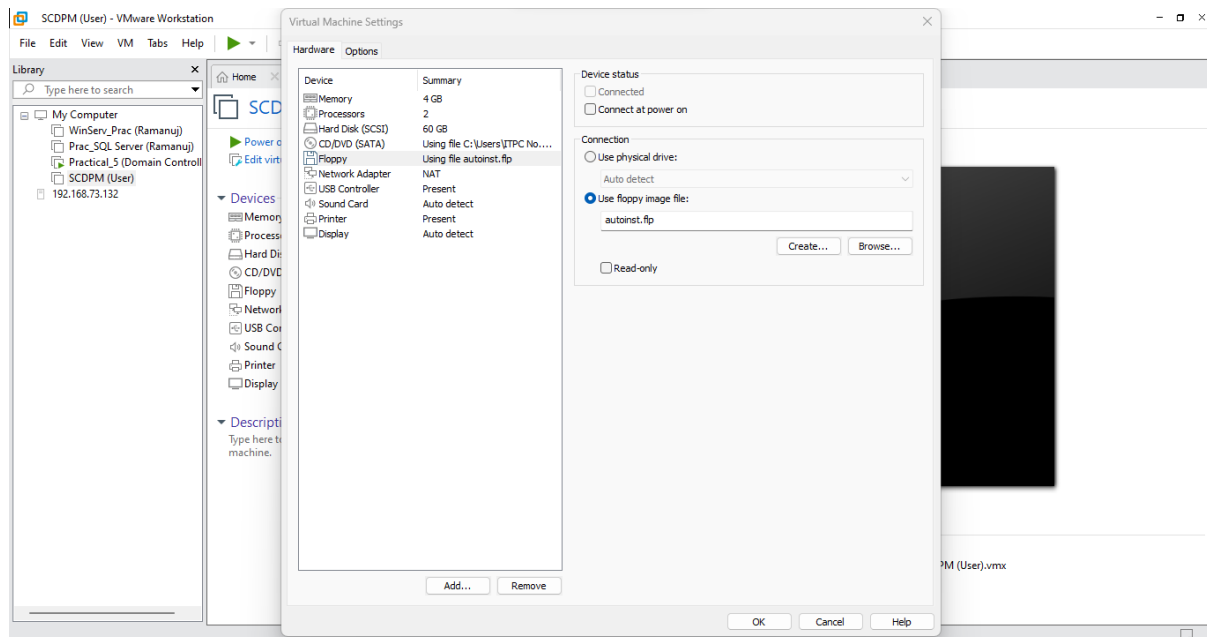


Step 2: Before Powering On the virtual machine a few change need to be made in the **virtual machine settings**

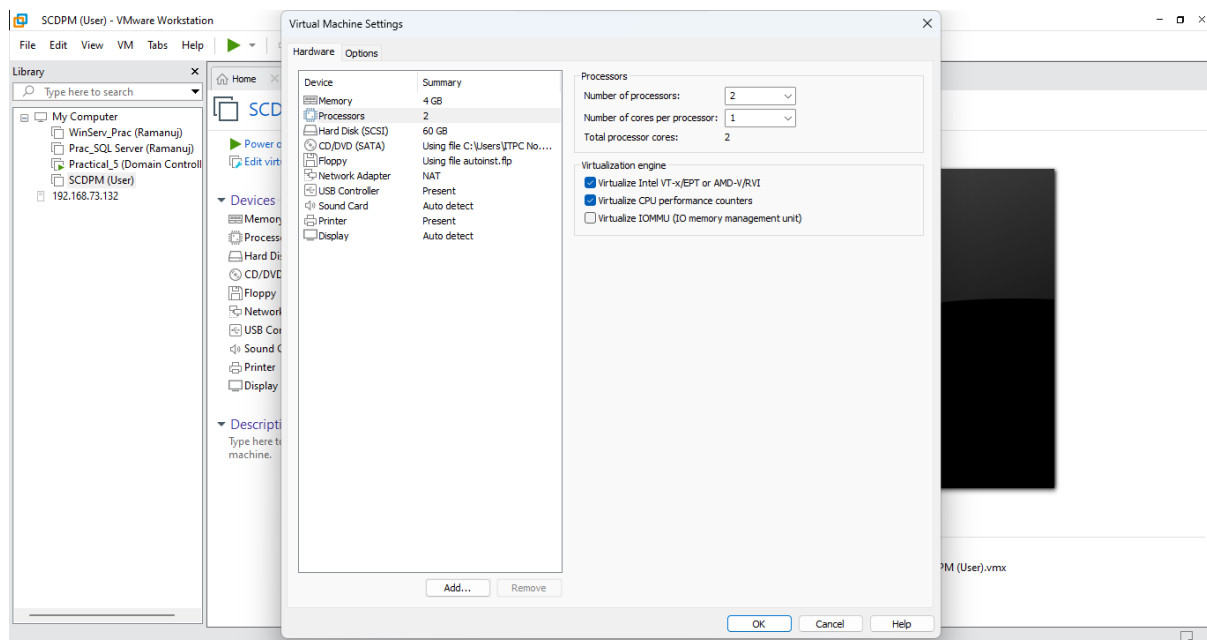
- Click on **Edit virtual machine settings**



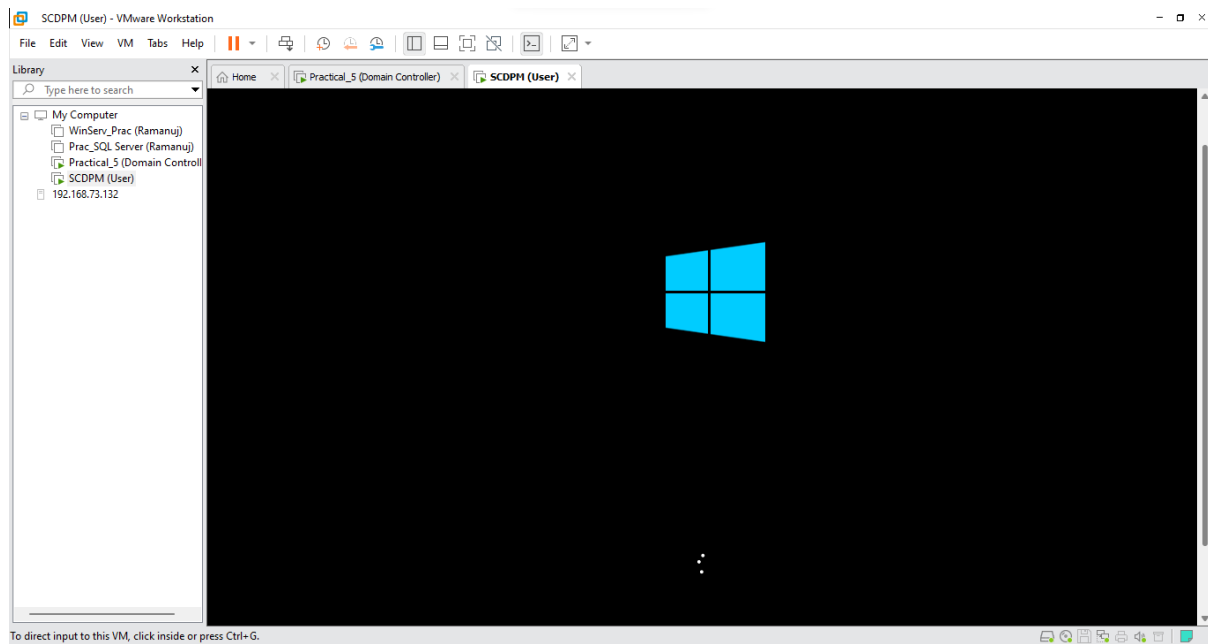
- Within virtual machine settings Select **Floppy** and **Un-Click Connect on Power On**



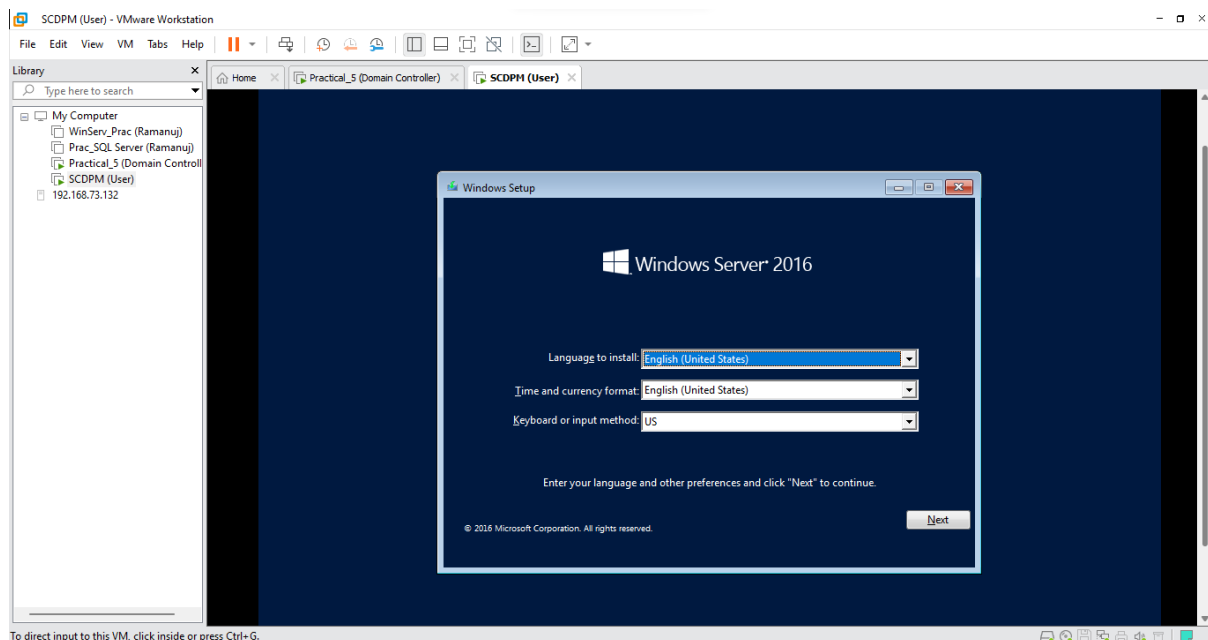
- Select **Processors** and Click on **Virtualized INTEL** and **Virtualized CPU performance counters** and Click **Ok**



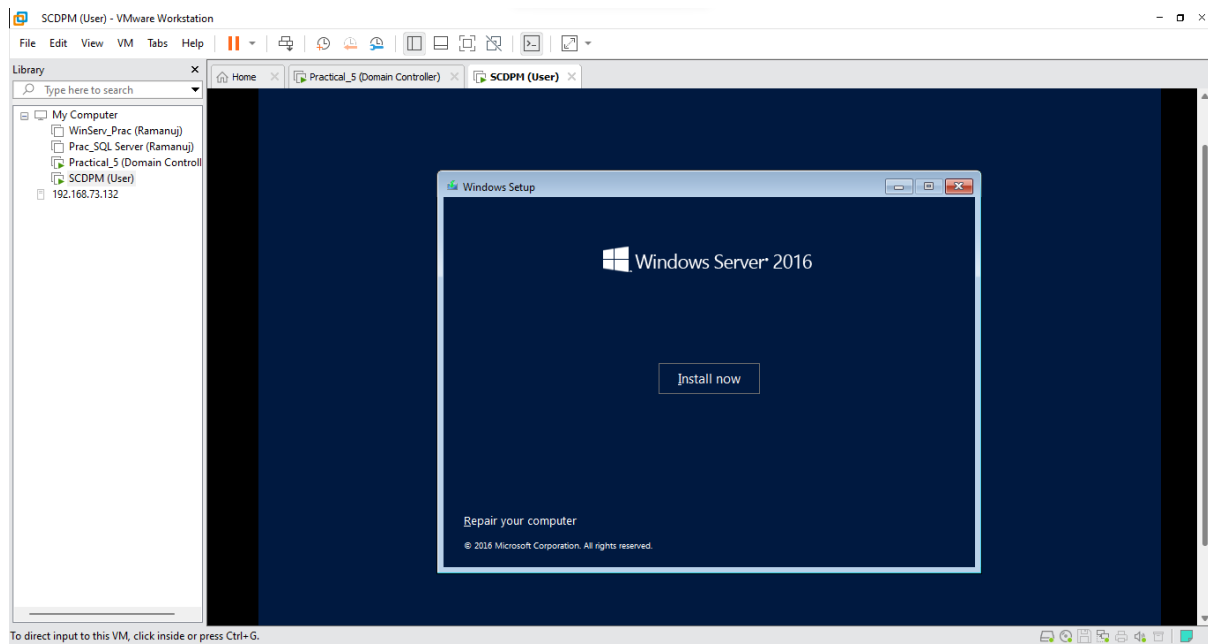
- Now Power On your virtual machine



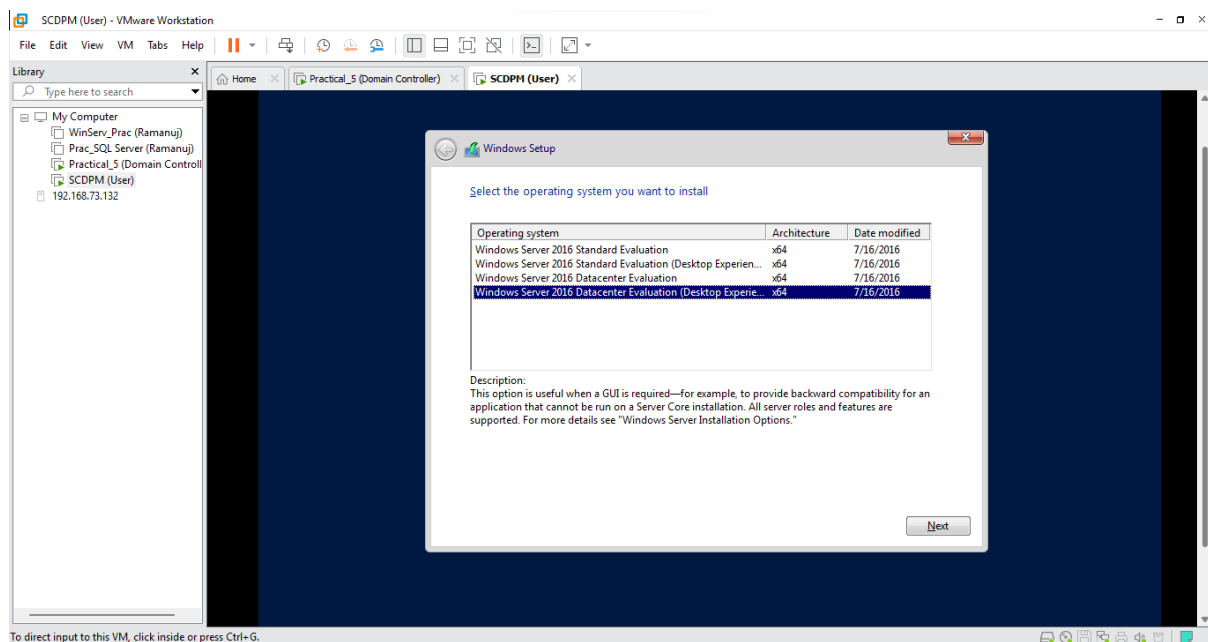
- The Windows Server Installer initialize, Keep default values and Click **Next**



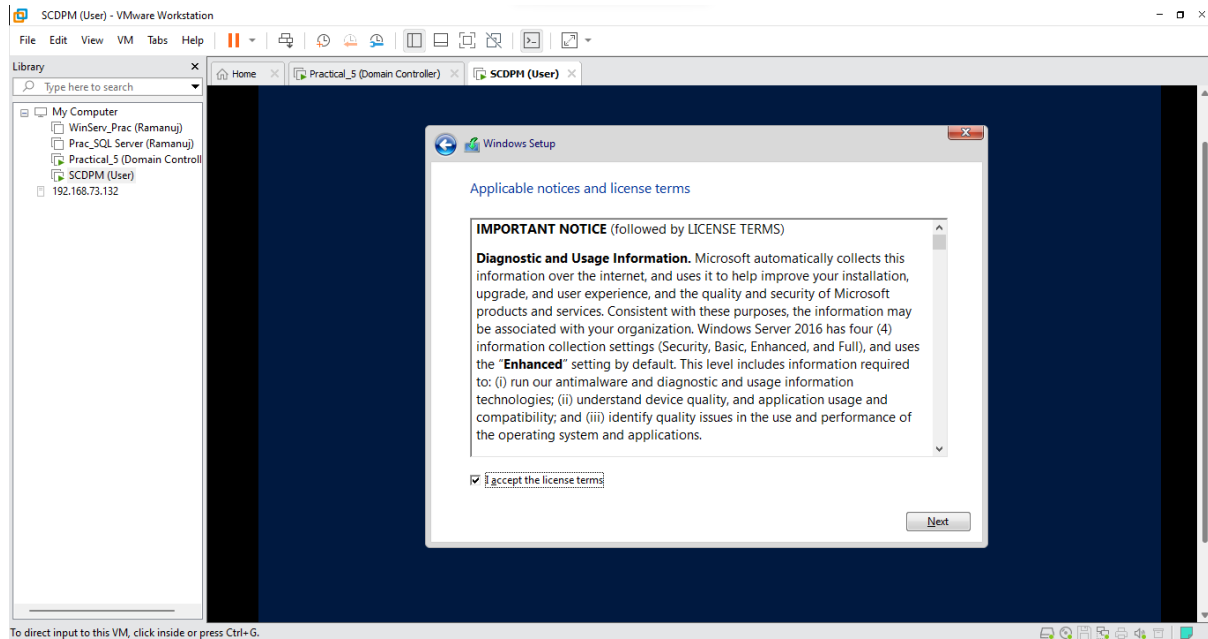
- Click on **Install Now**



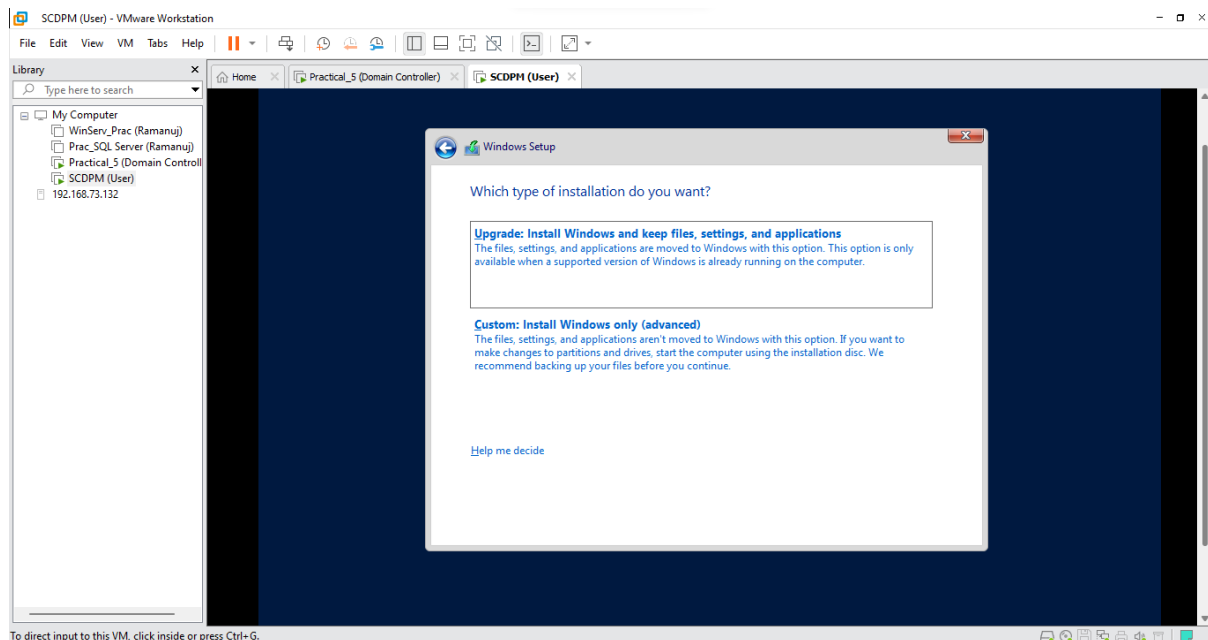
- In Operating Systems Select **Windows Server 2016 Datacenter Evaluation (Desktop Experience)** and Click Next



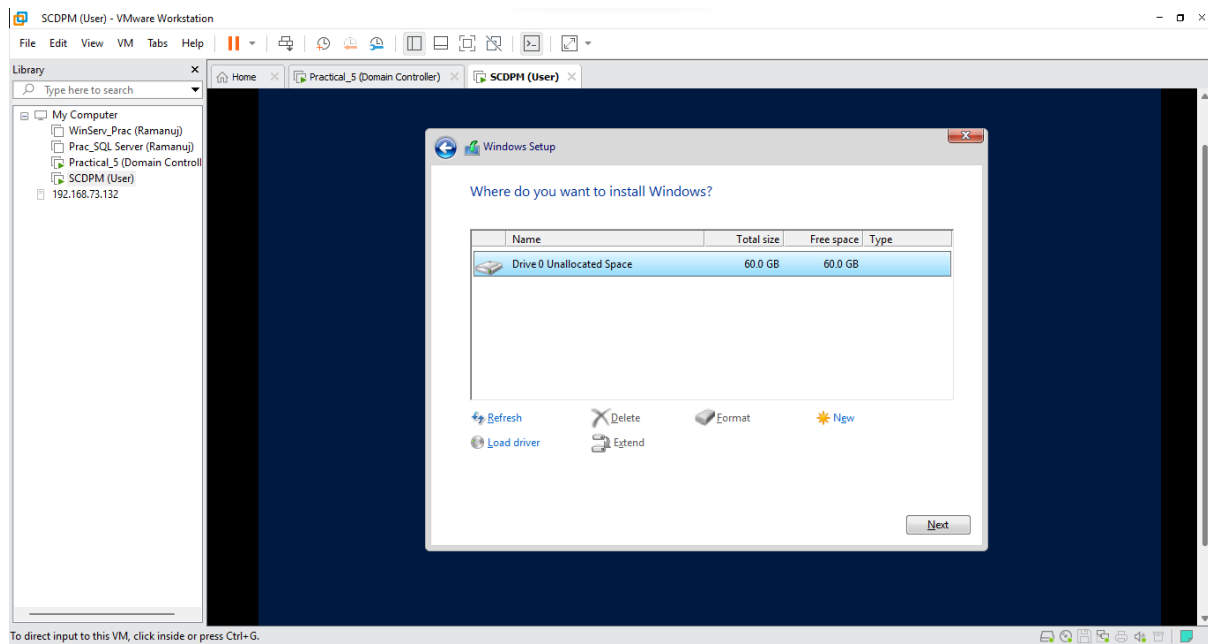
- Select **I accept these license terms** and Click **Next**



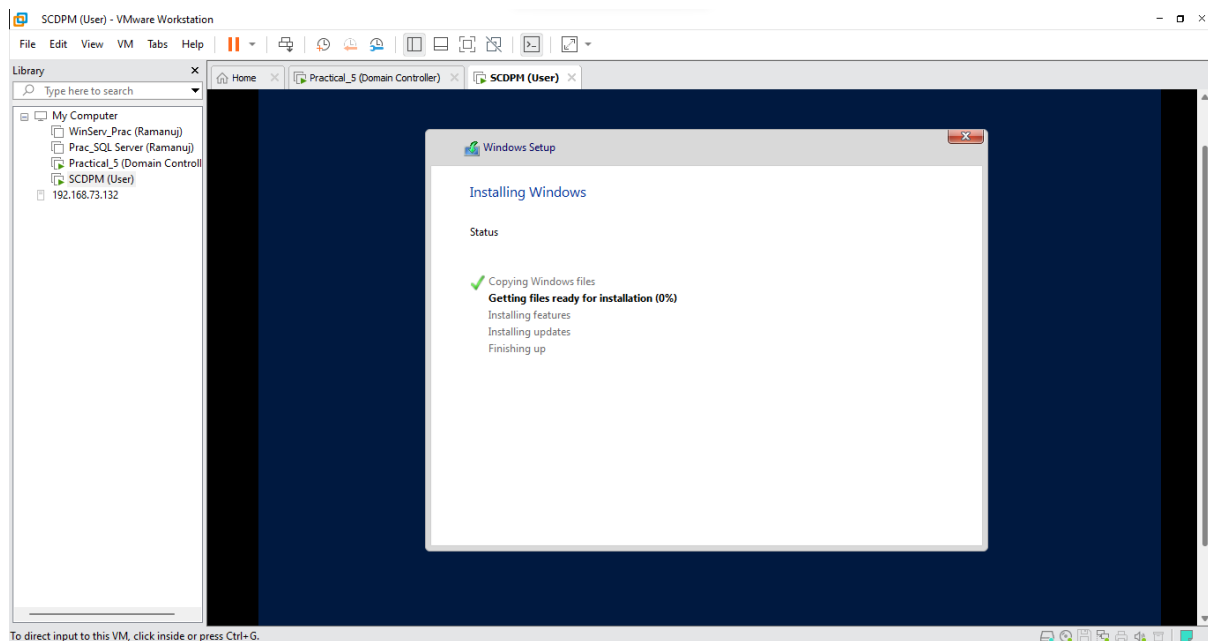
- Select **Custom**



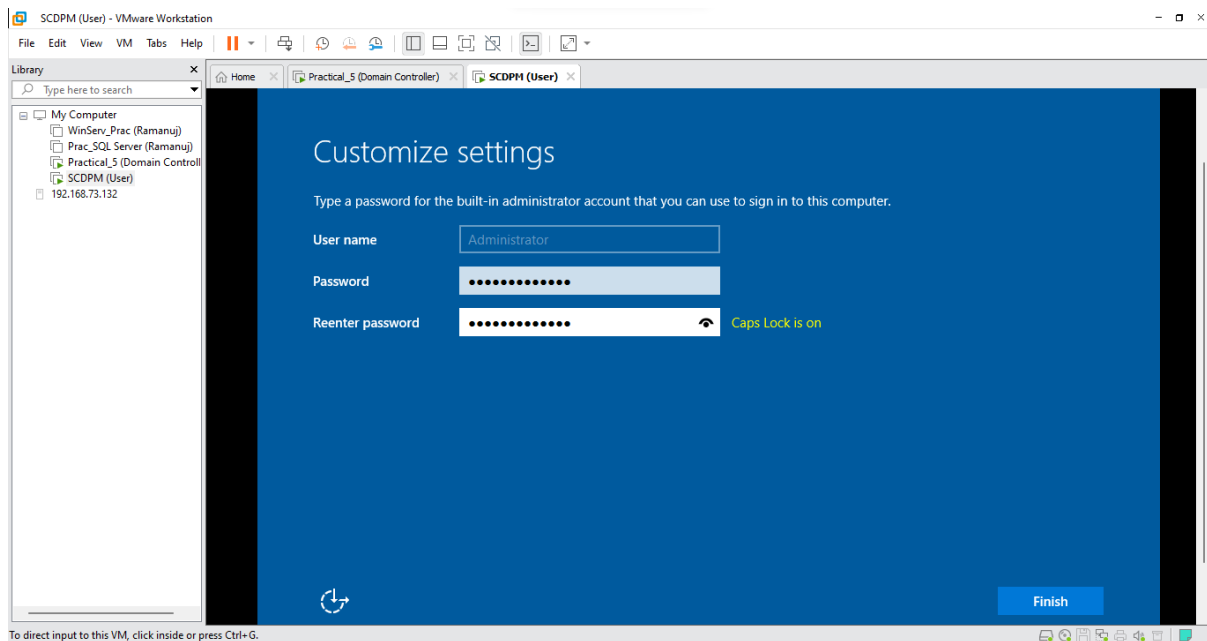
- Keep default values and Click **Next**



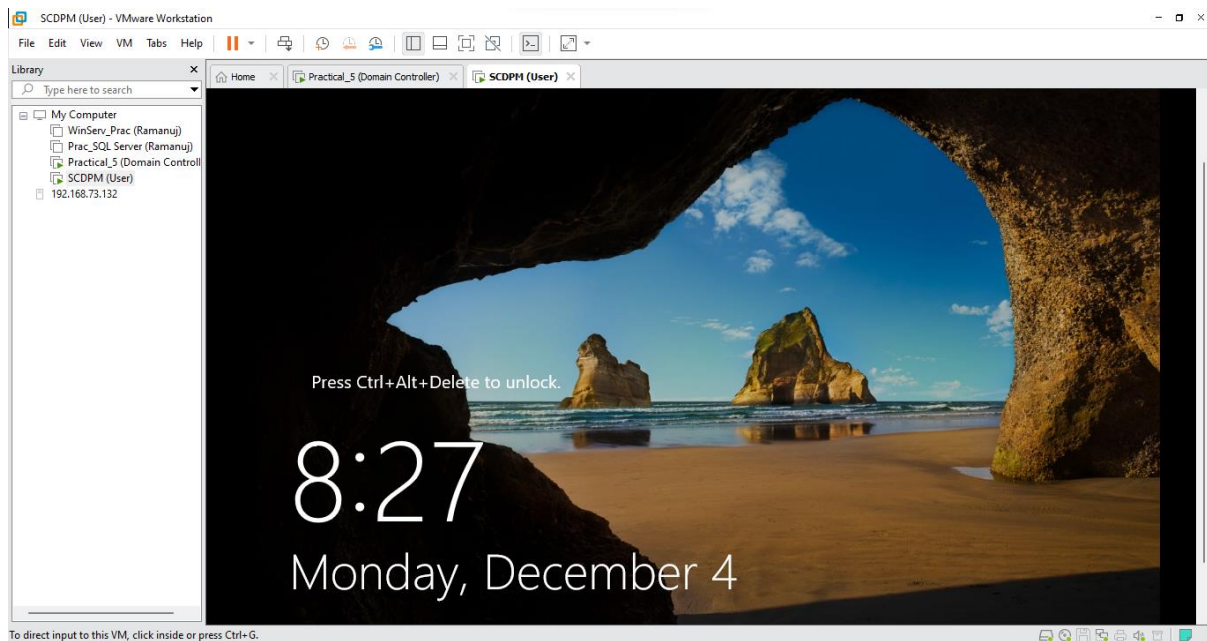
- Windows Server will begin installing



- After installation type in a password for the Windows Server and Click **Finish**

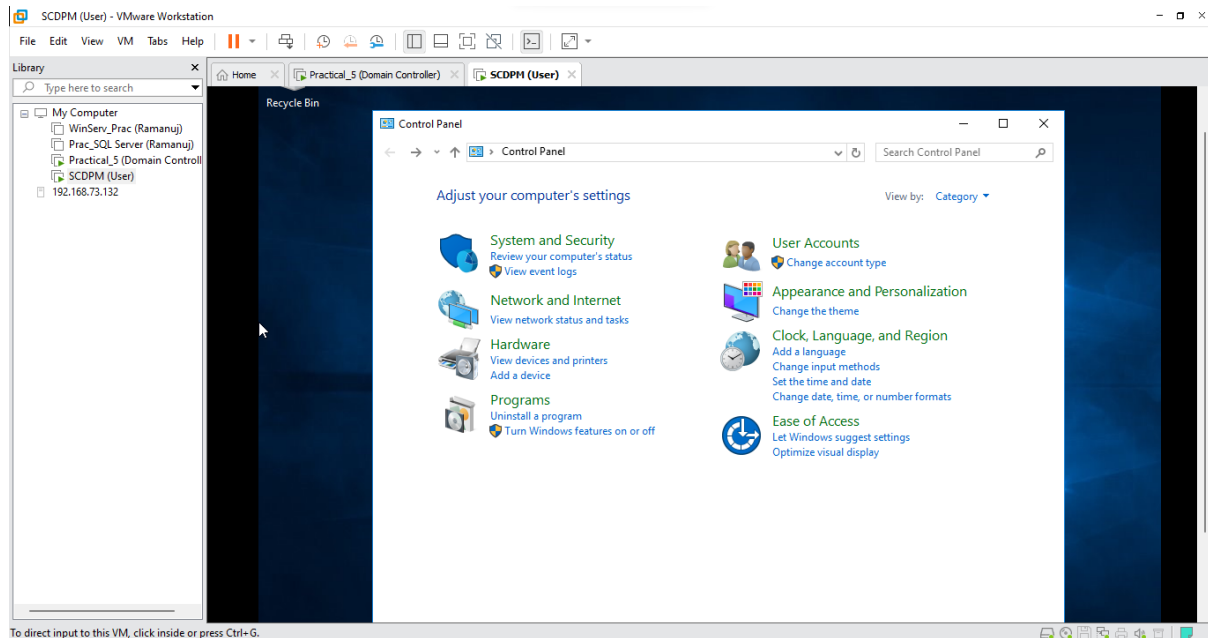


- Your Windows Server has now been successfully created

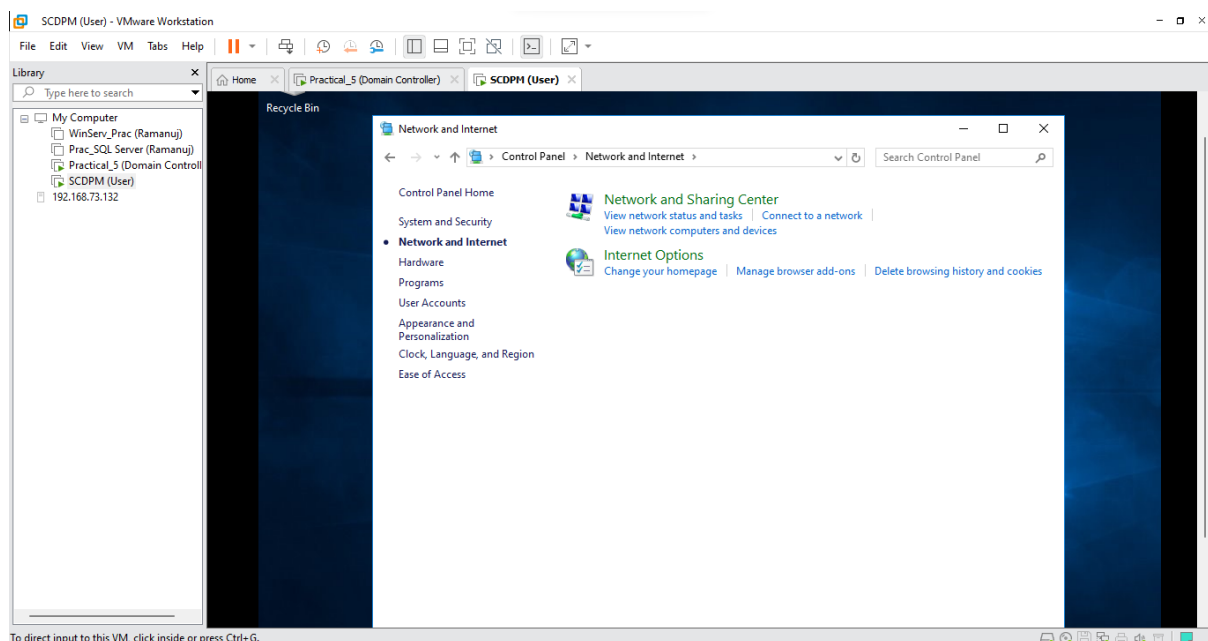


Step 3: Establishing a connection with the User VM and Domain Controller by connecting the User VM to the Domain Controller's Workgroup

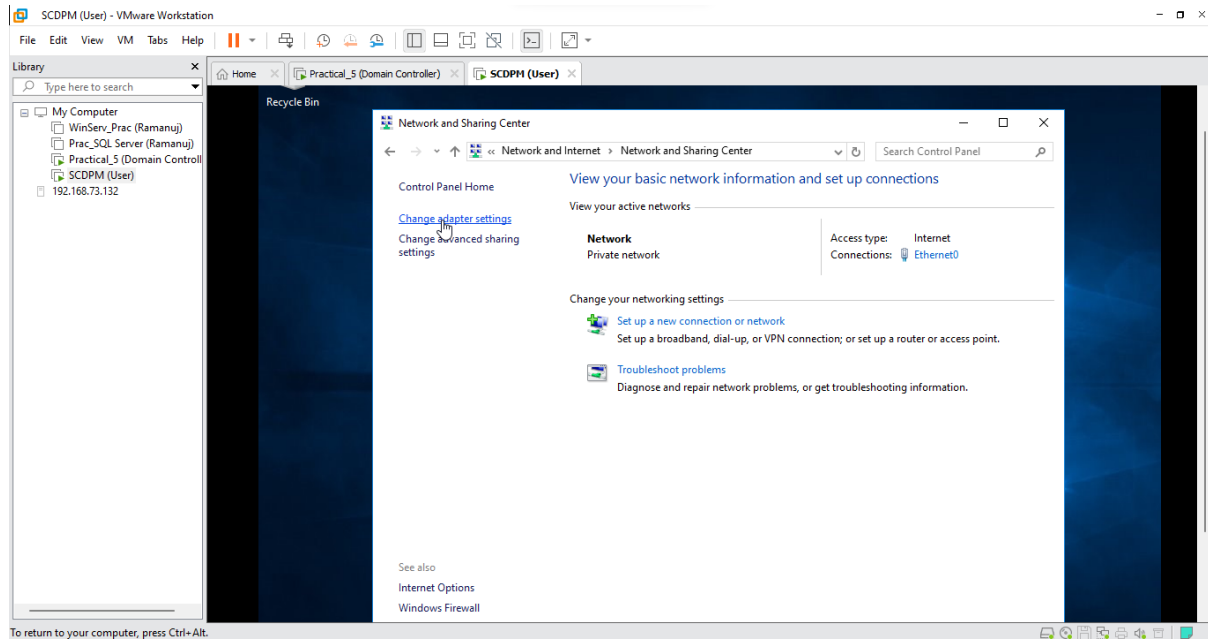
- Open **Control Panel** and Click on **Network and Internet**



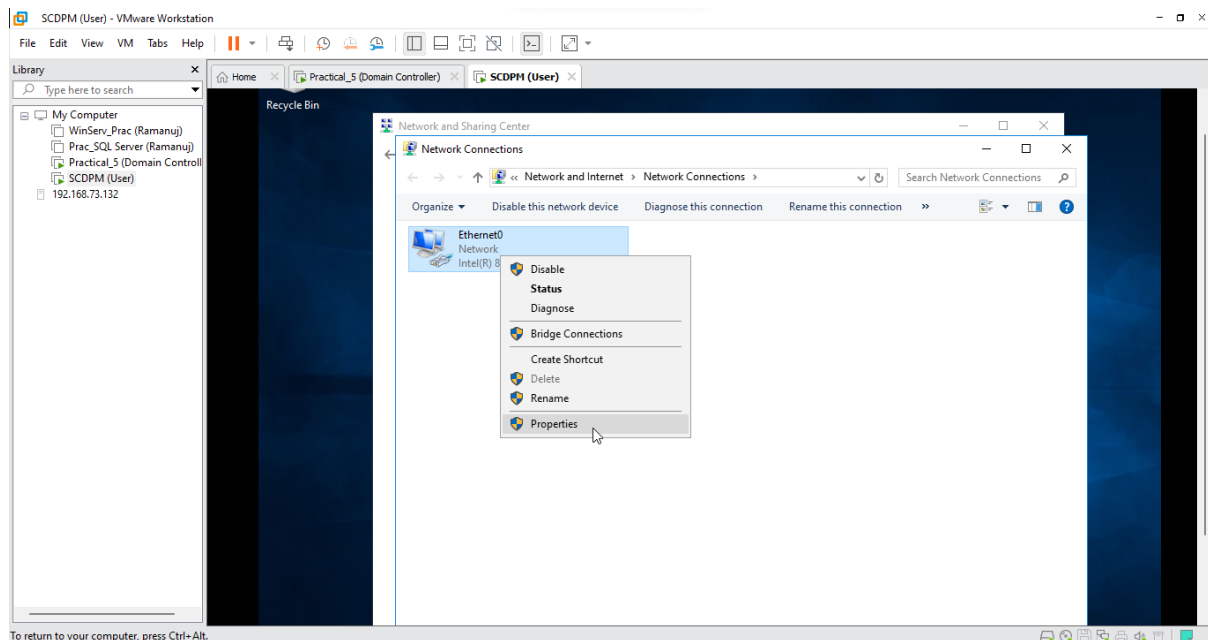
- Select **Network and Sharing Center**



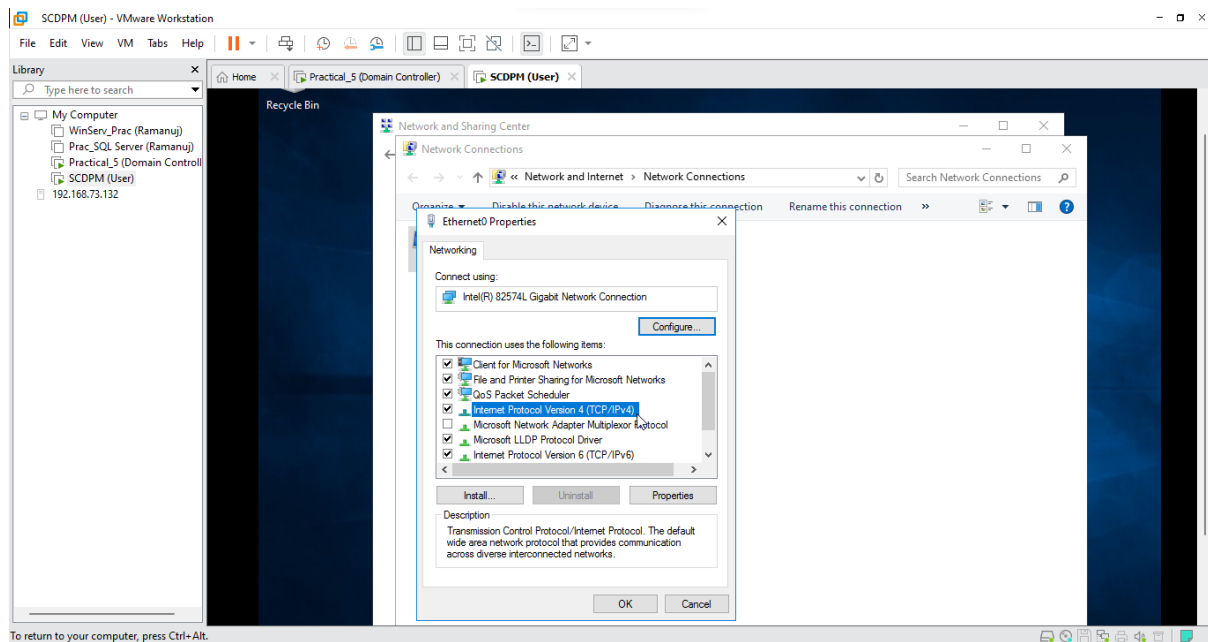
- Click on **Change adapter settings**



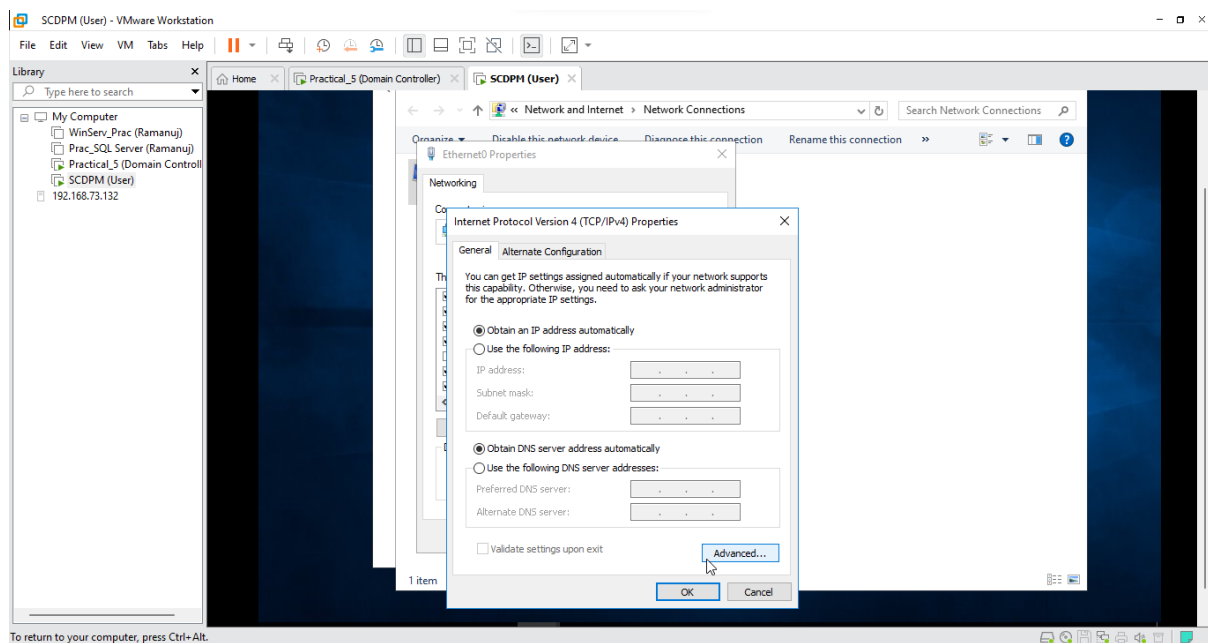
- Right Click on **Ethernet0** and Select **Properties**



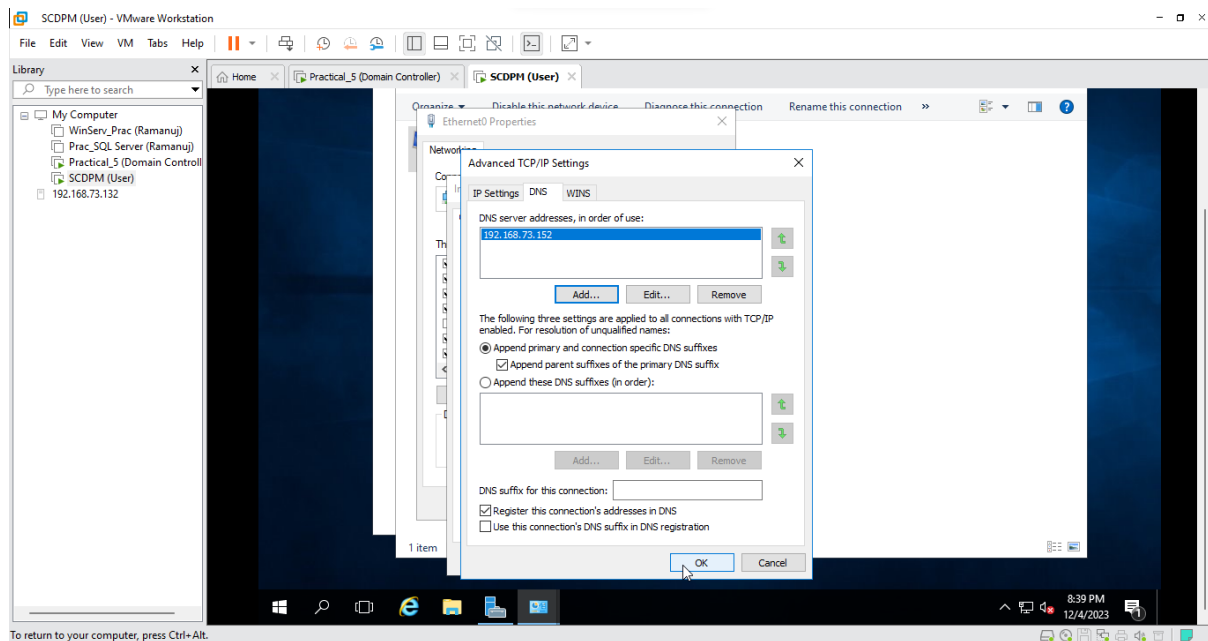
- Click on **Internet Protocol Version 4**



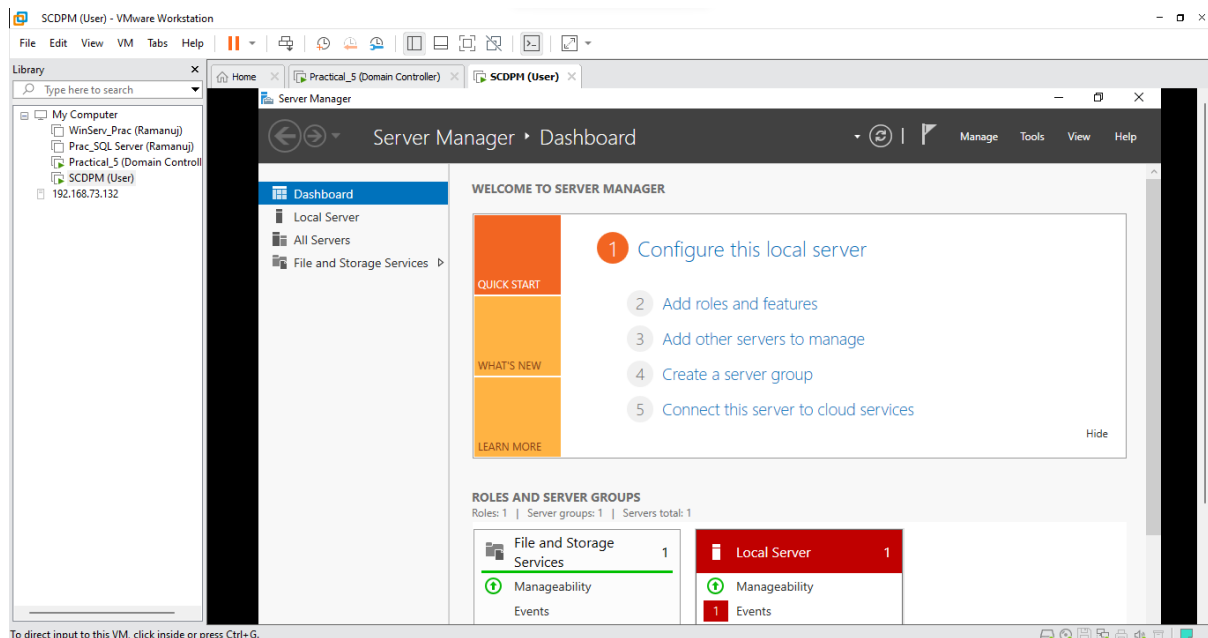
- Within Internet Protocol Version 4 Click on **Advanced**



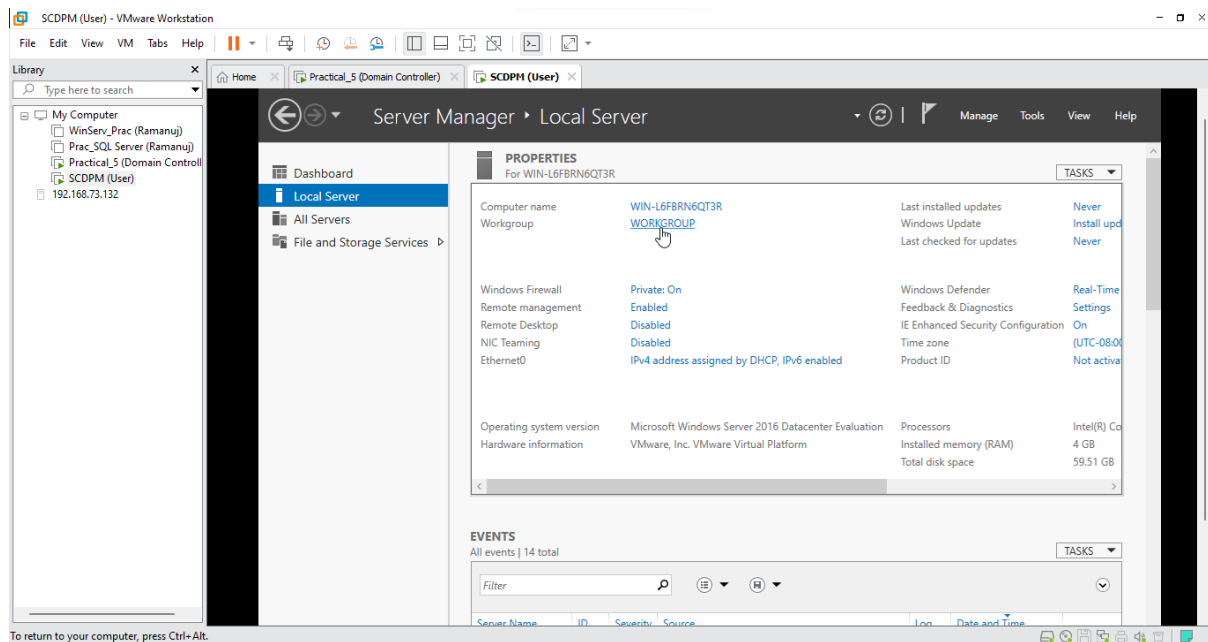
- Select **DNS** and Click on **Add** with the IP Address of the Domain Controller and Click **Ok**



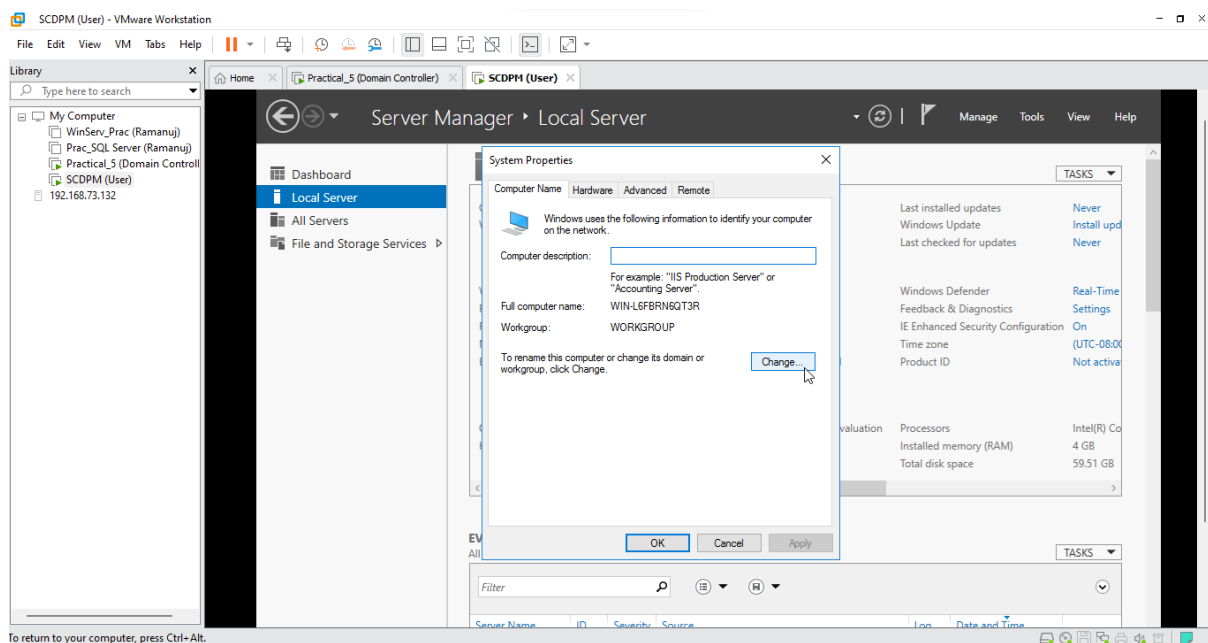
- Now Exit out of network settings and Click on **Server Manager**



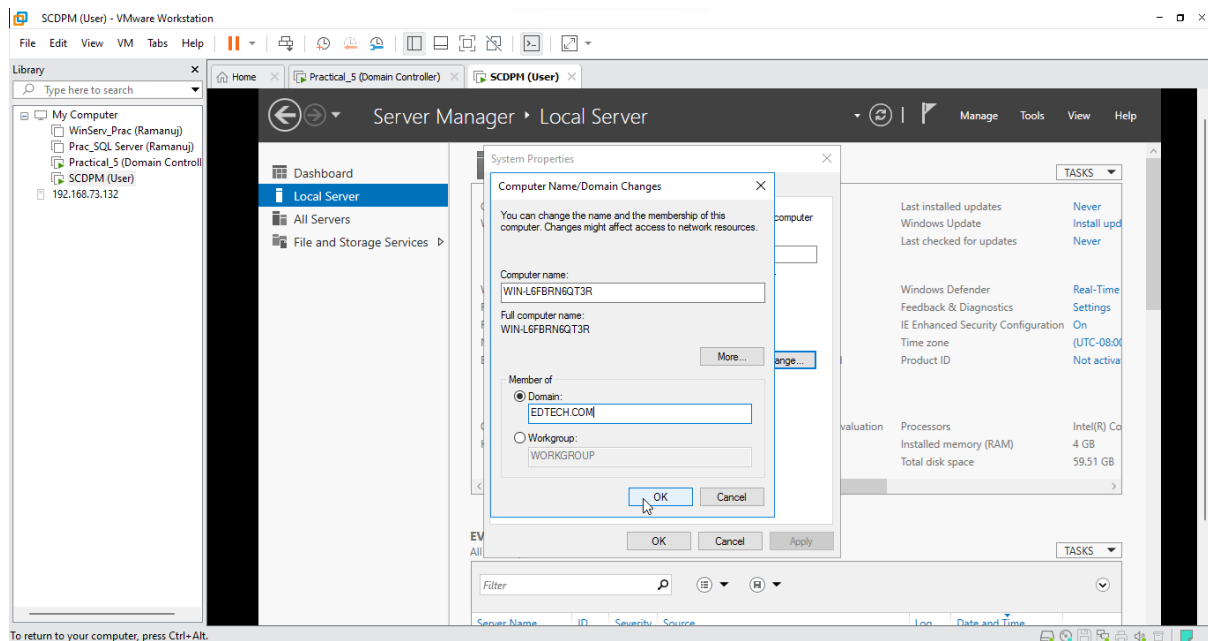
- Within Server Manager Click on **Local Server** and Select **Workgroup**



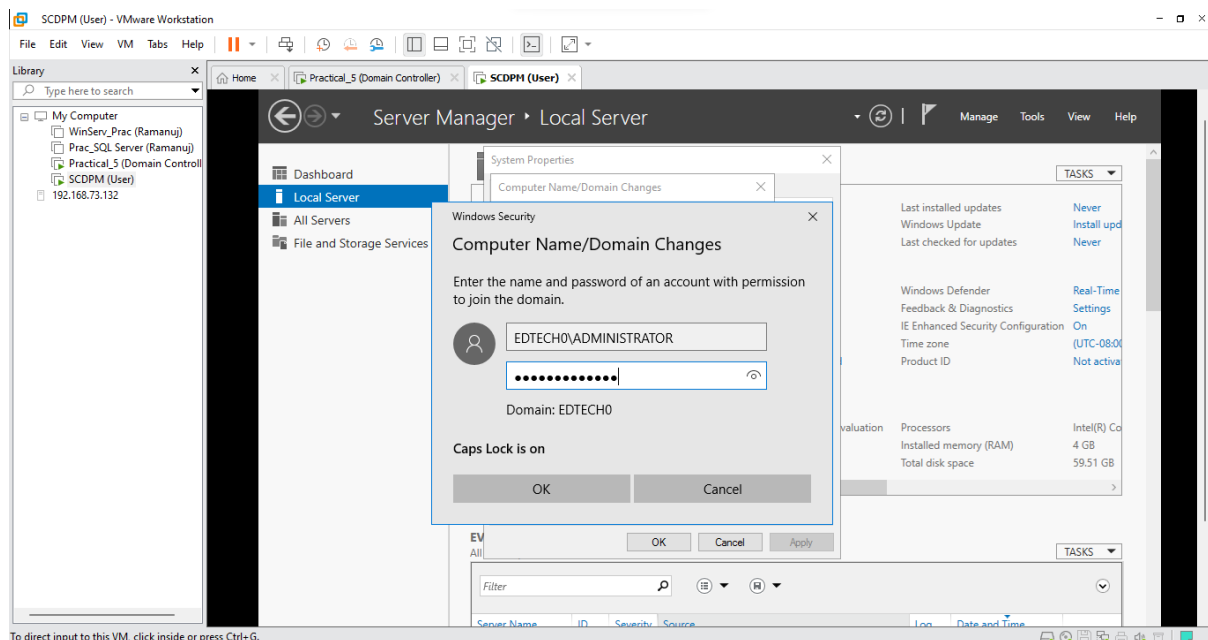
- Within Workgroup Click on **Change**



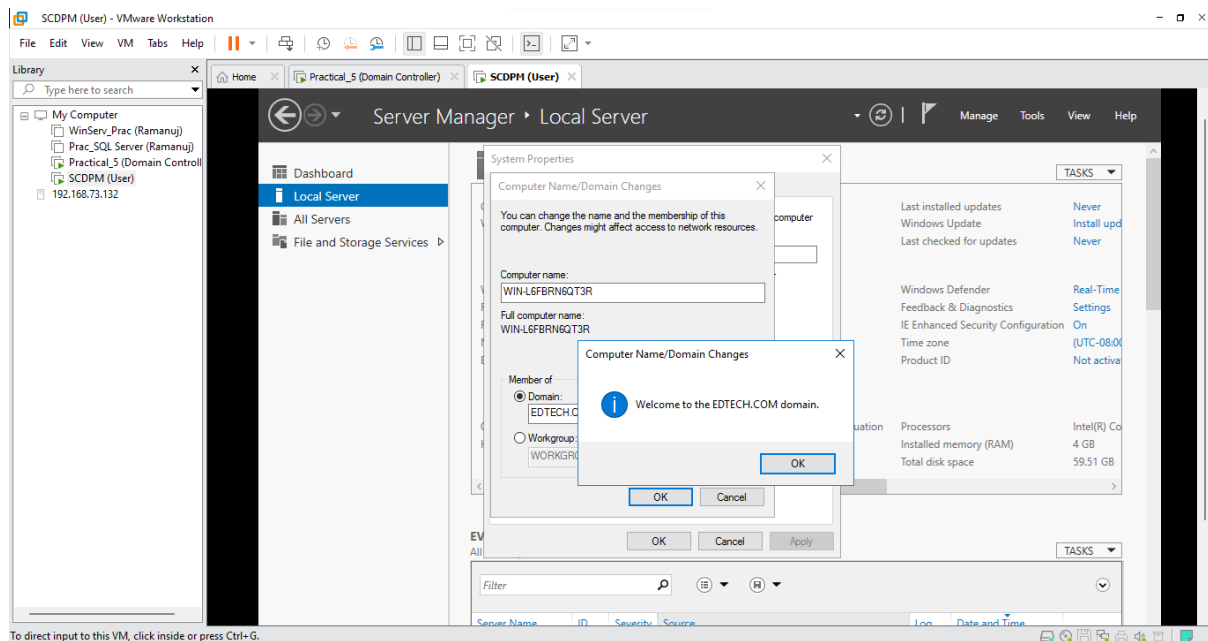
- In Member of Select **Domain** and type the Domain name (here it is EDTECH.COM) and Click **Ok**



- Enter the credentials of the Domain Controller and Click on **Ok**

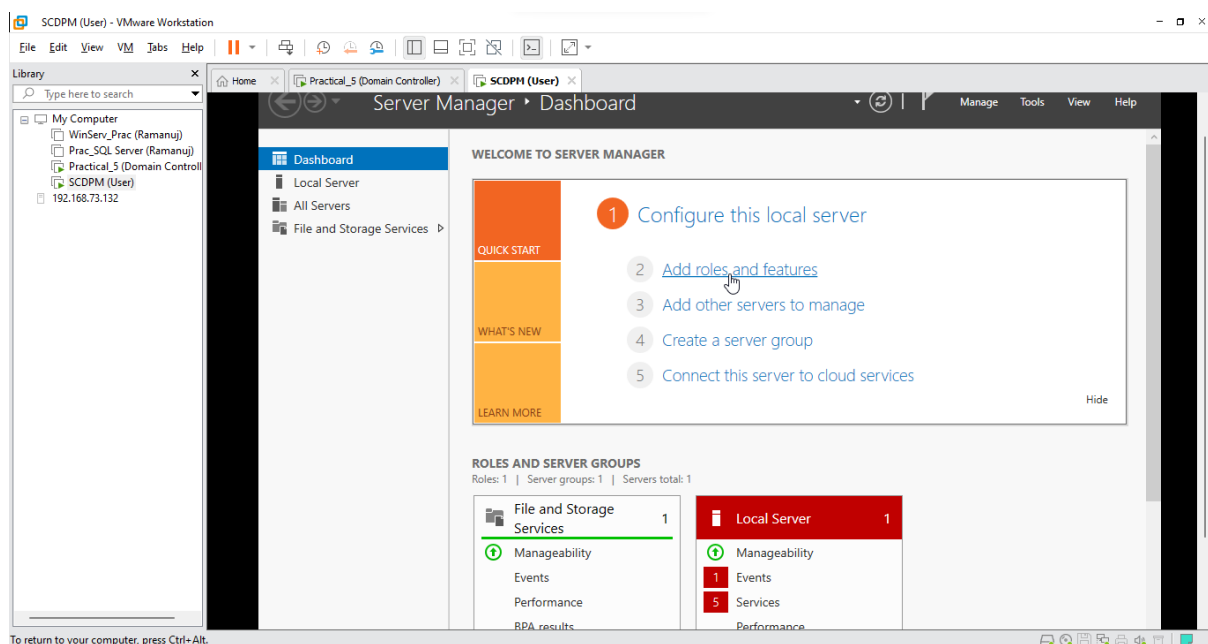


- The User VM is now in the same Domain as the Domain Controller and the system will require a restart for the changes to take effect

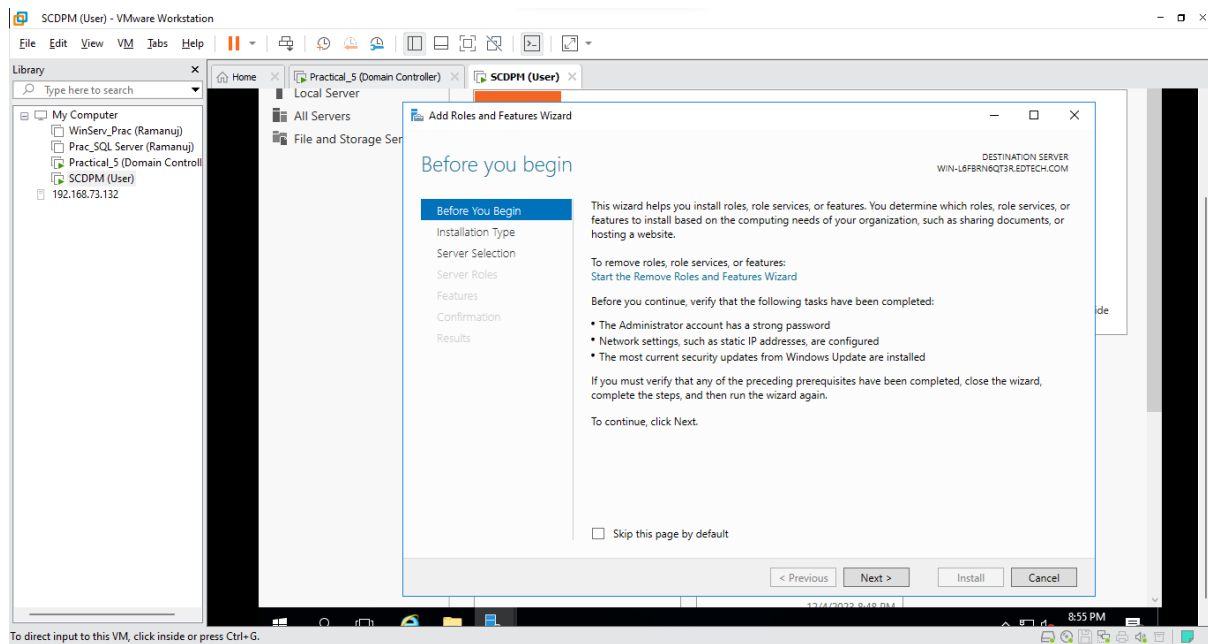


Step 4: Creation of User (SCDPM)

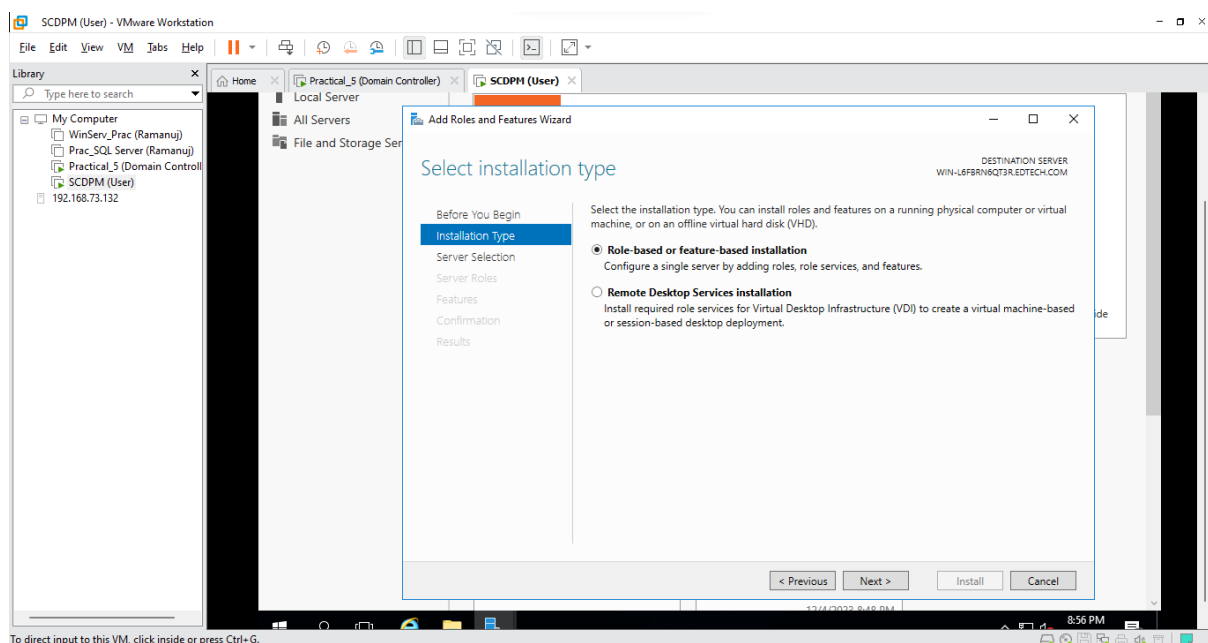
- To create a User first we require the **Active Directory Domain Service** which can be added through the Server Manager
- Open **Server Manager** and Click on **Add Roles and Features**



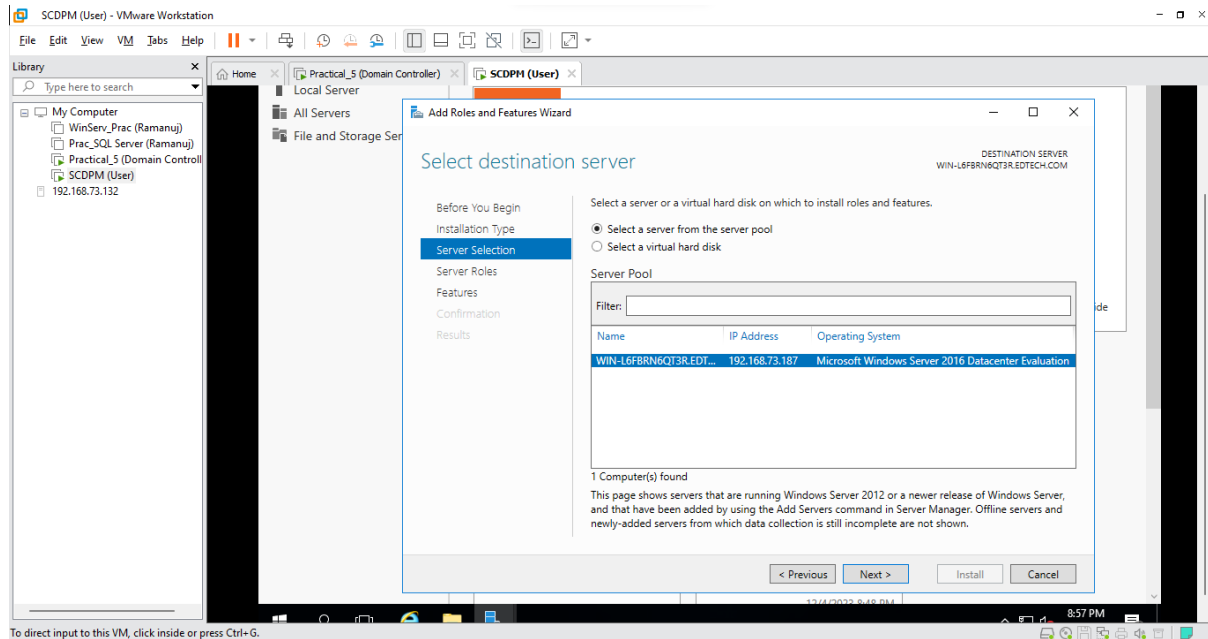
- Click Next



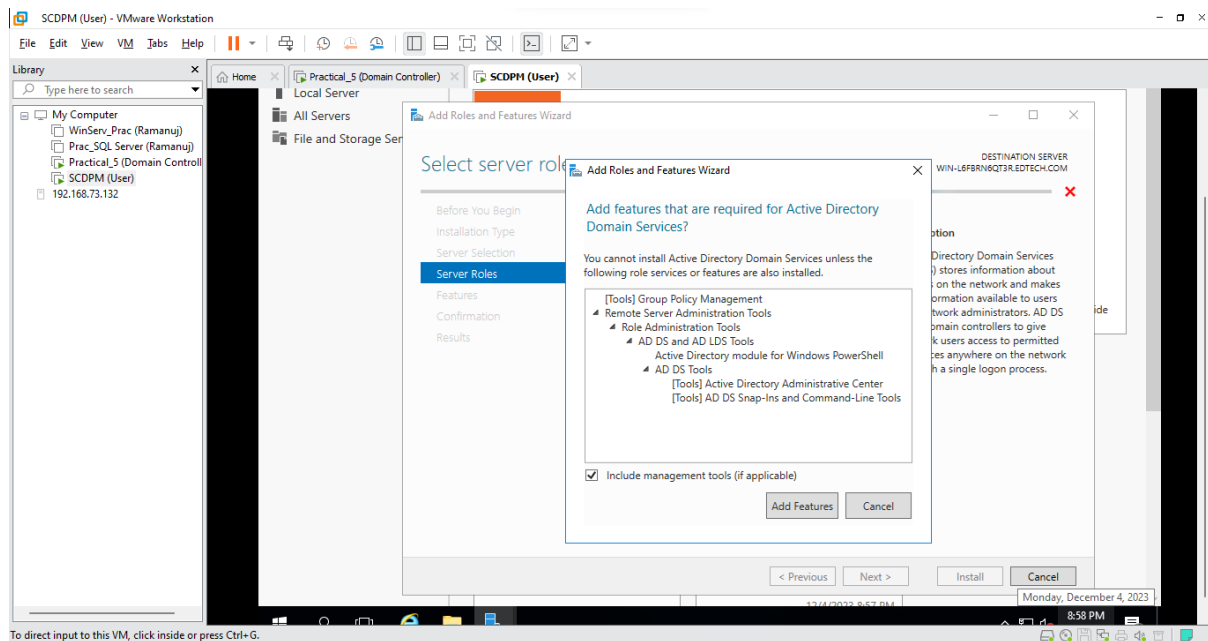
- Select **Role-based or Feature-based installation** and Click Next



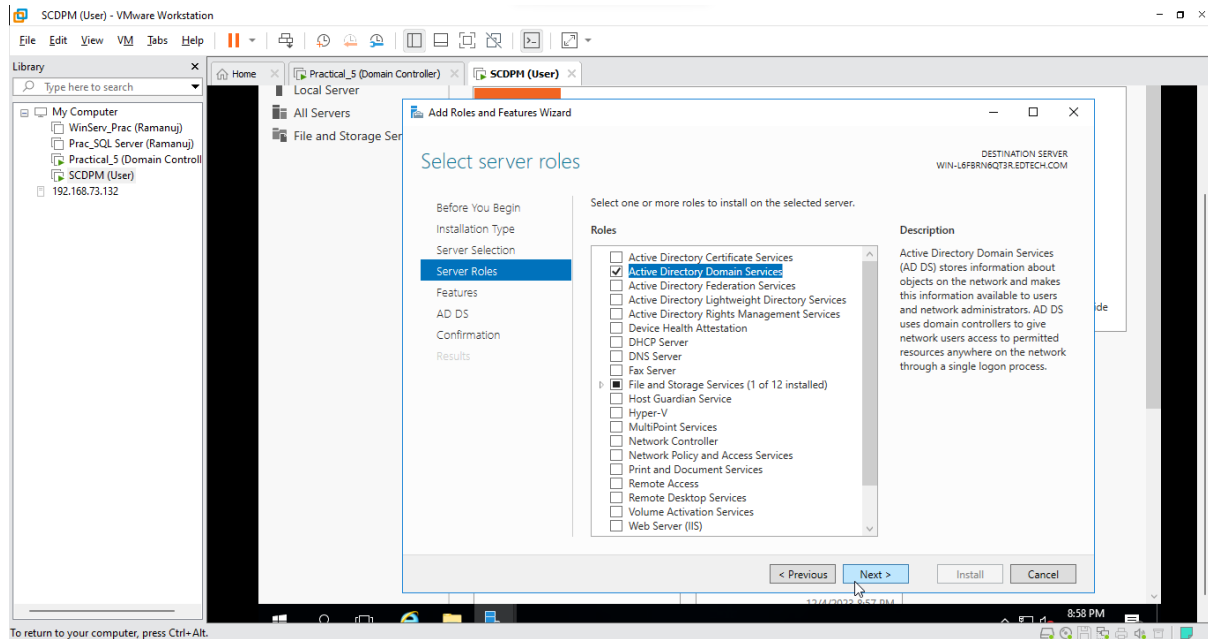
- Keep default values and Click **Next**



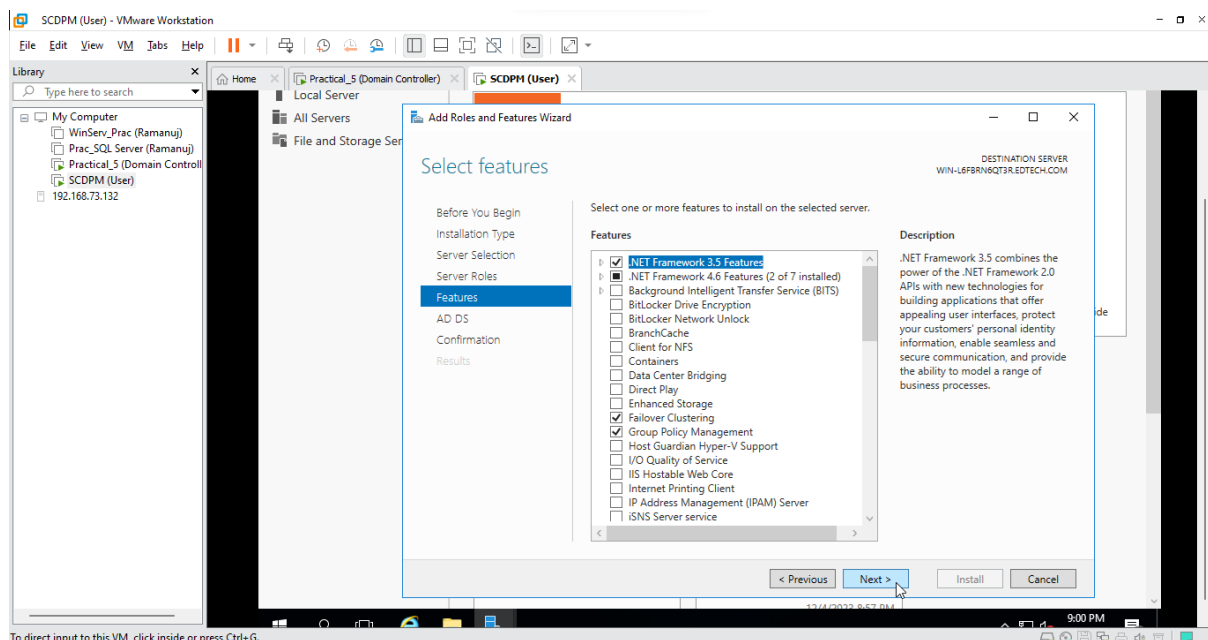
- Select **Active Directory Domain Service** and Click on **Add Feature**



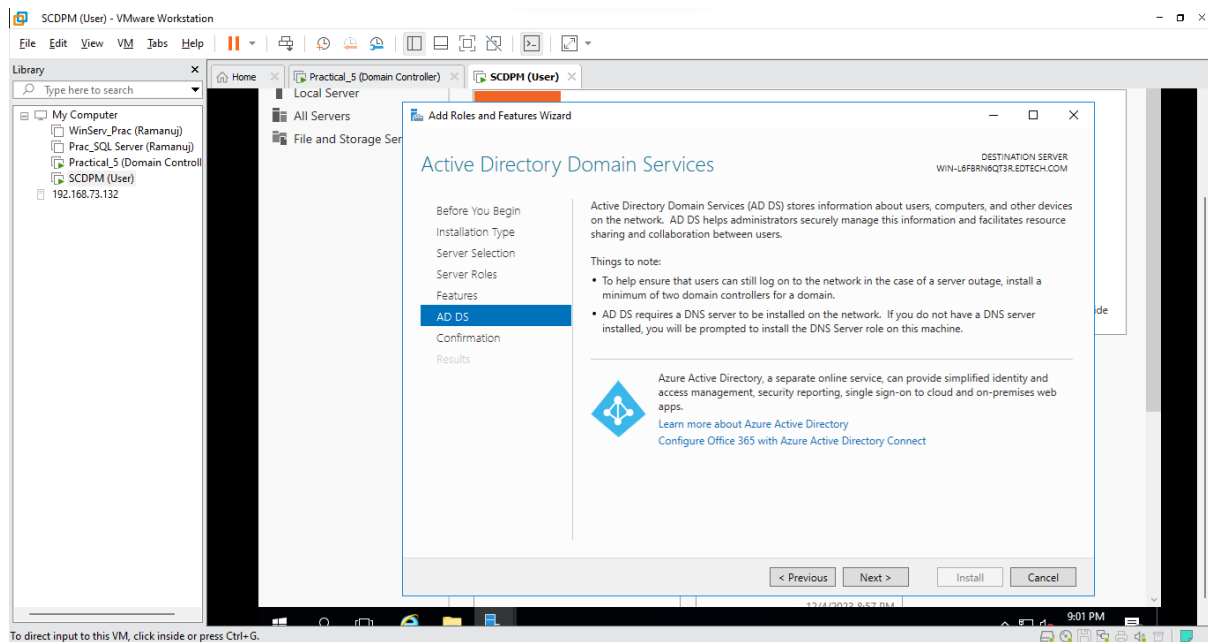
- Click Next



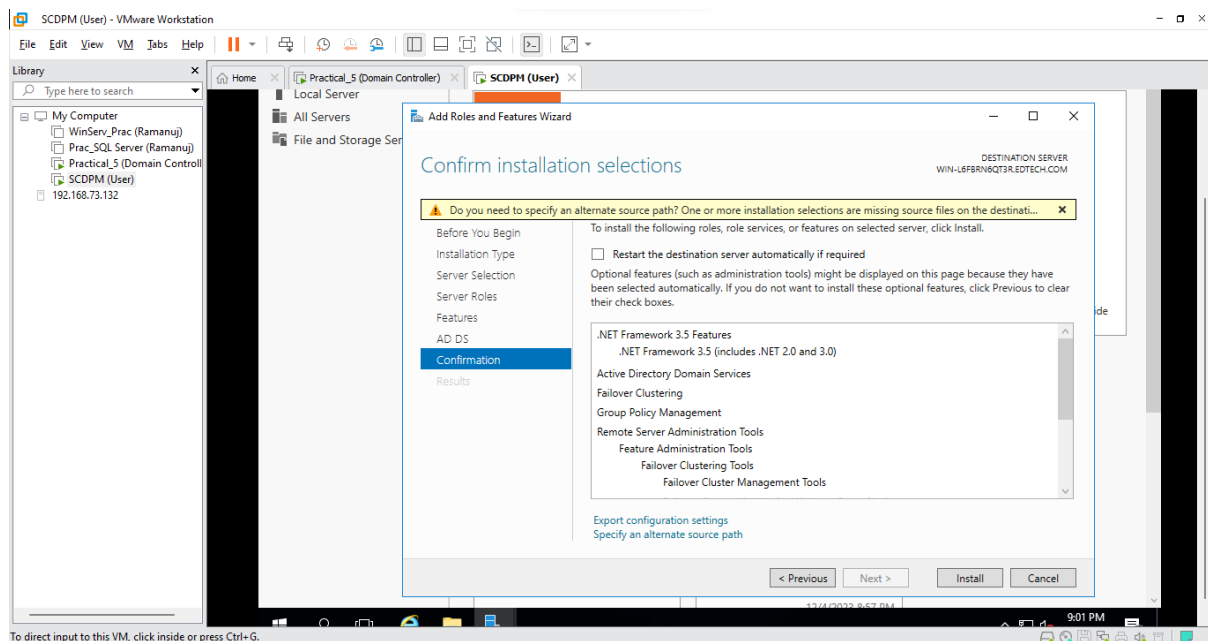
- Select .NET 3.5 Features and Failover Clustering and Click Next



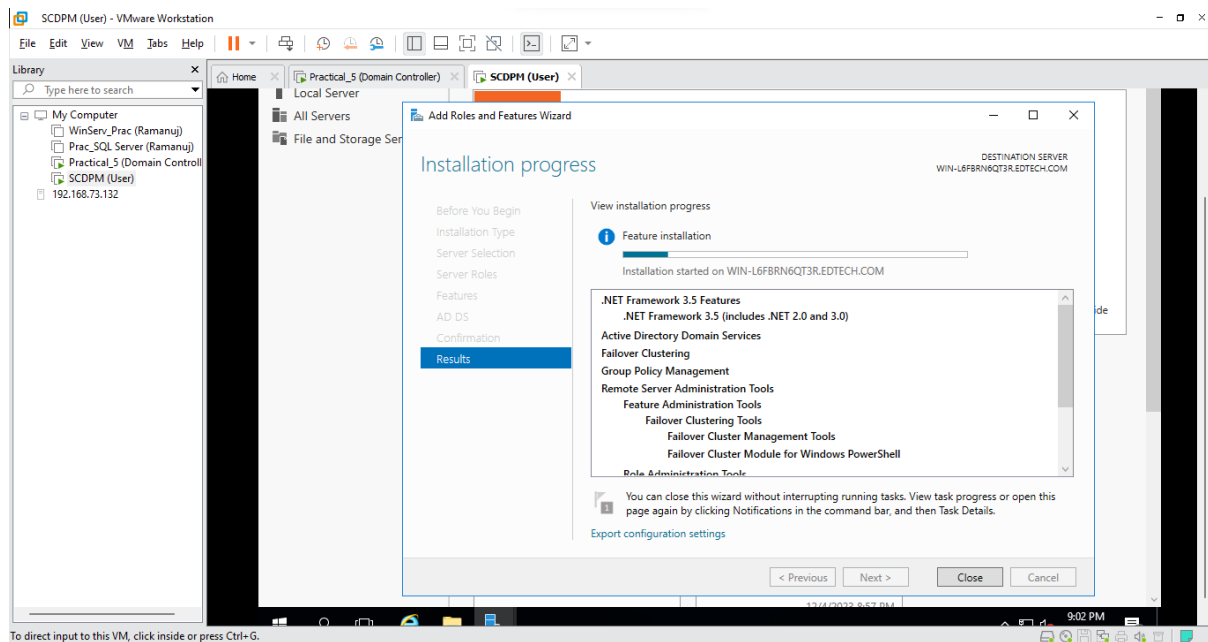
- Click **Next**



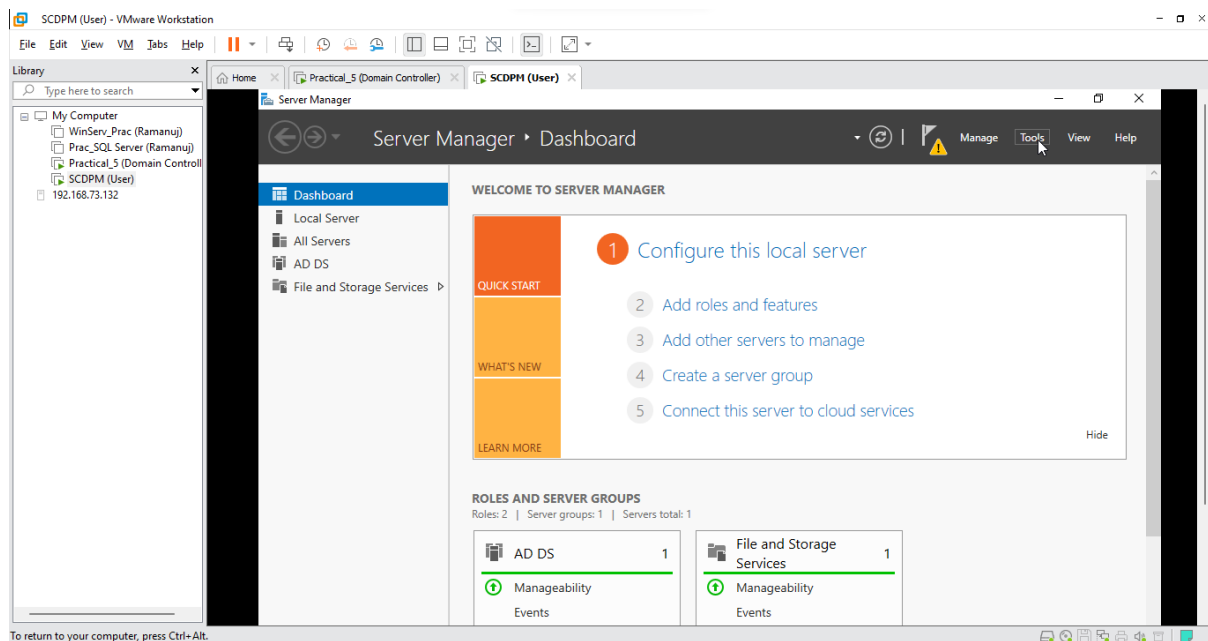
- Click **Install**



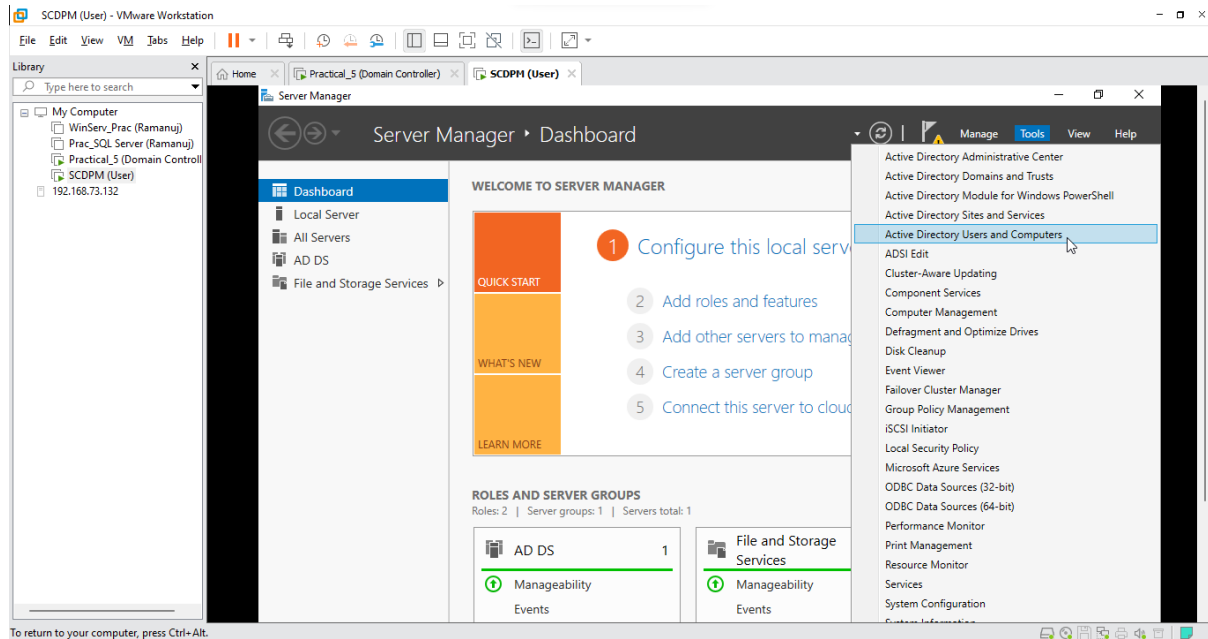
- Features will start installing in the system



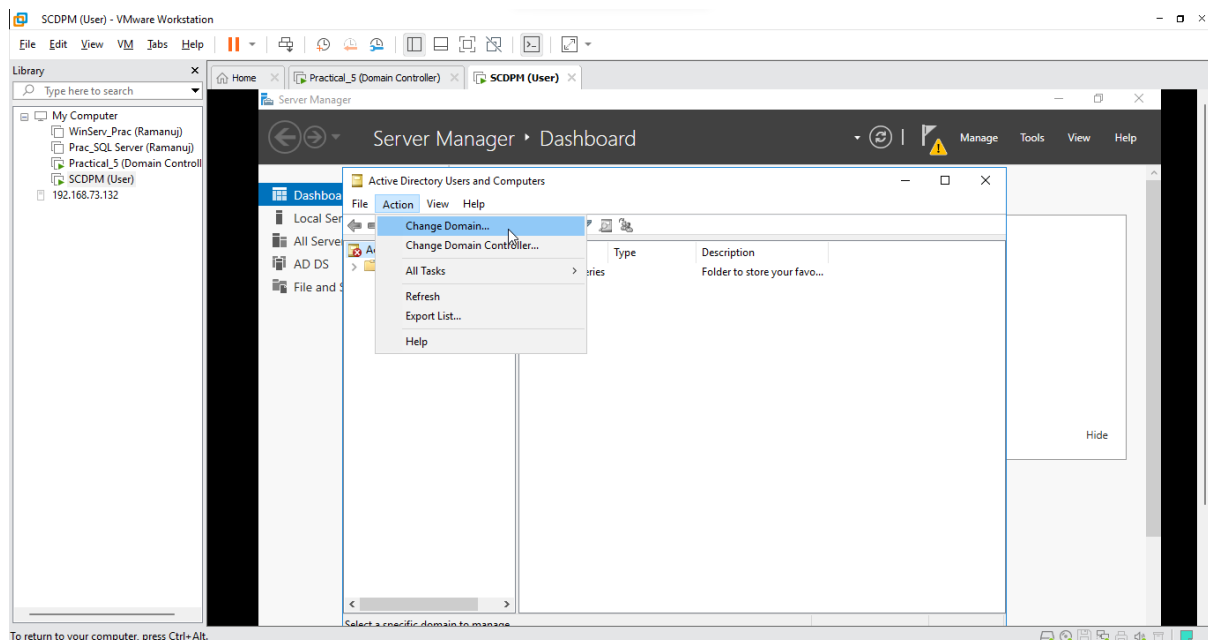
- After installation, within Server Manager Click on **Tools**



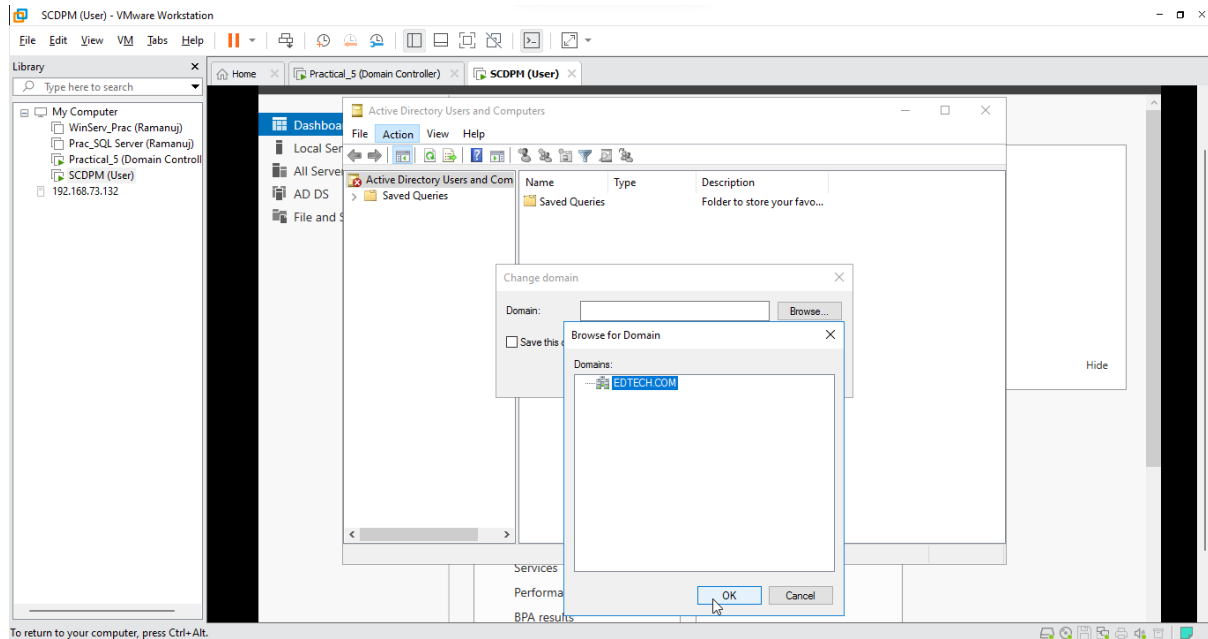
- Within Tools Click on **Active Directory Users and Computers**



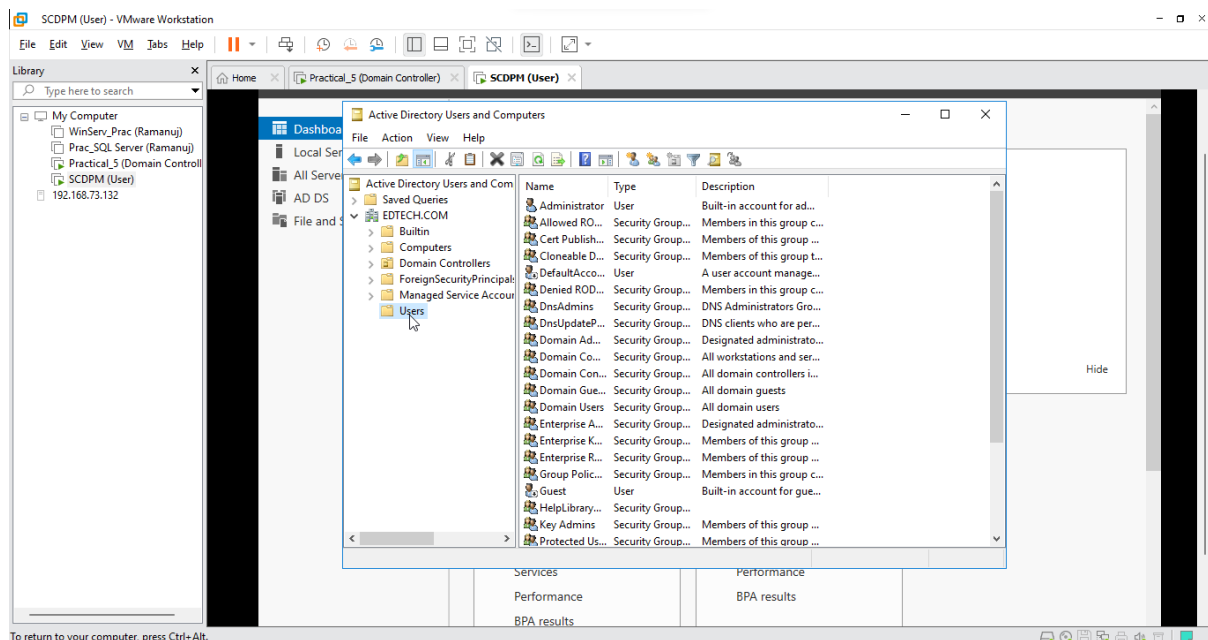
- Within Active Directory Users and Computers, Click on **Action** and Select **Change Domain**



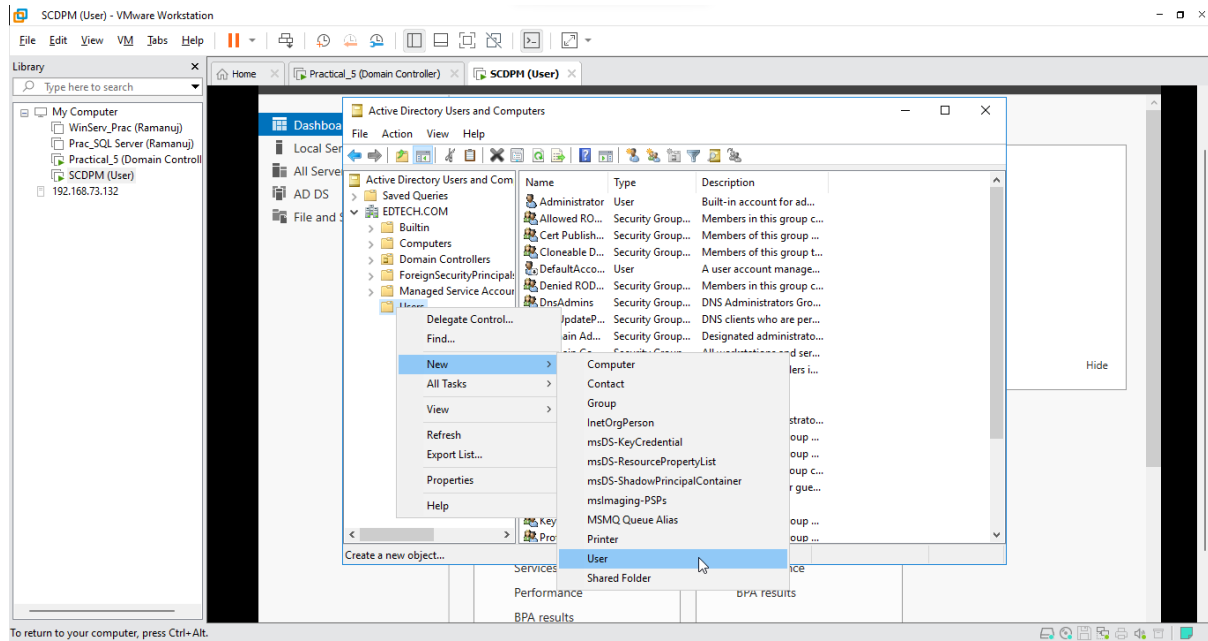
- Click on **Browse** and Select the **Domain Controller** and Click **Ok**



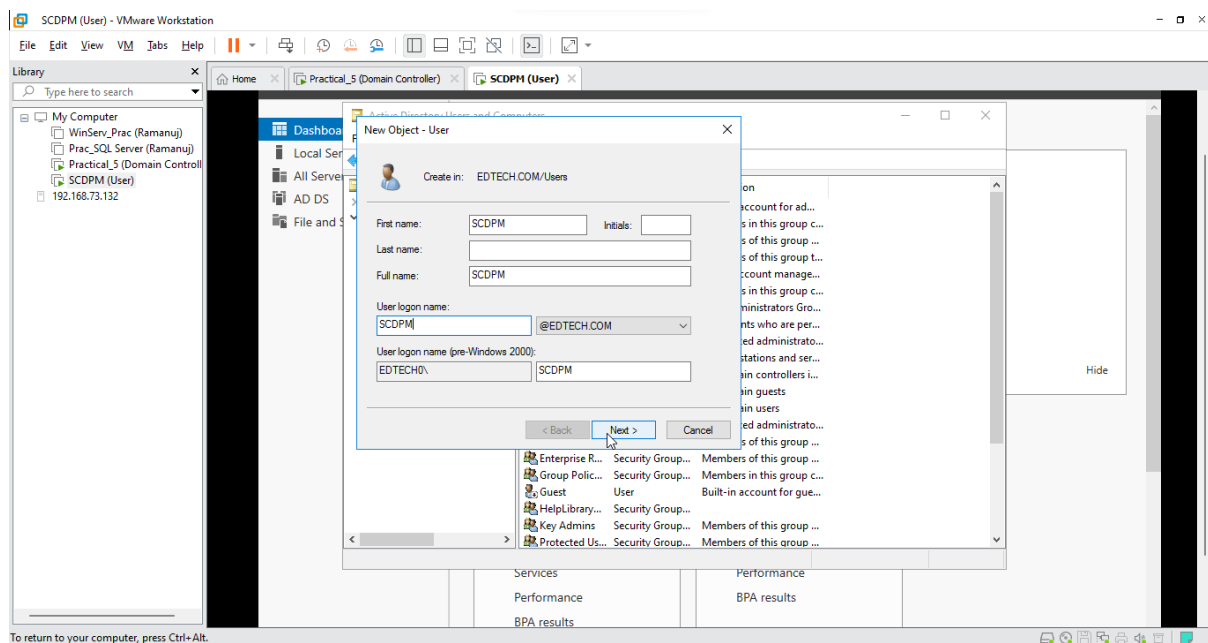
- Now the Domain Controller will be added, Click on the **Domain Controller** and Click on **Users**



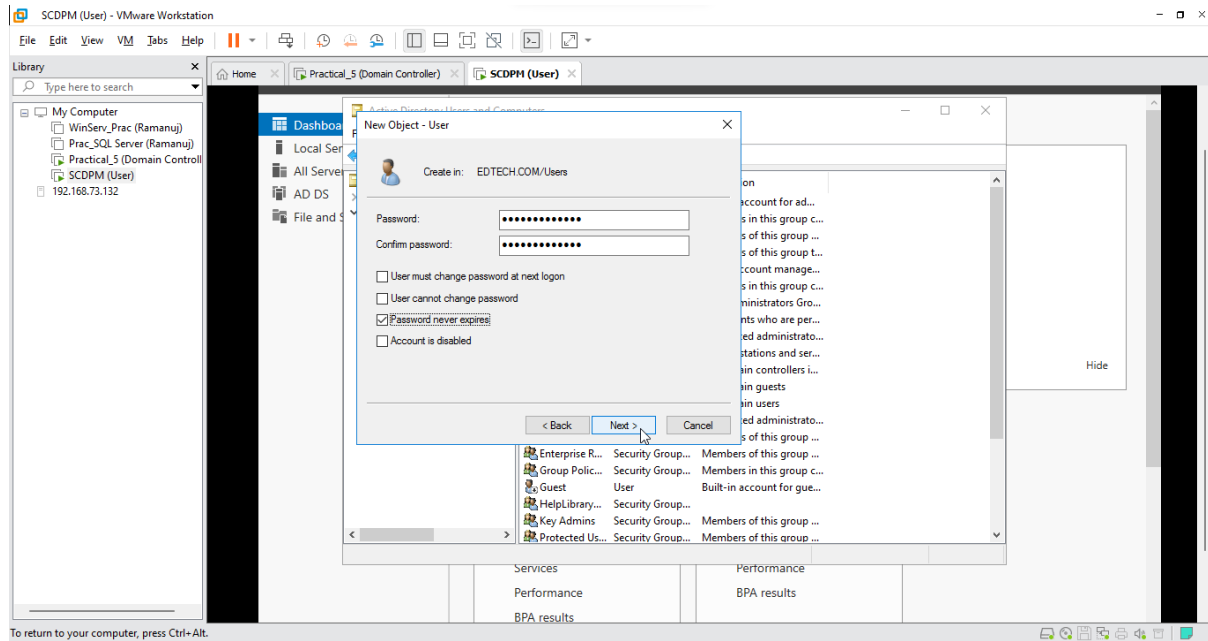
- Right Click on **Users** and Click **New** and **User**



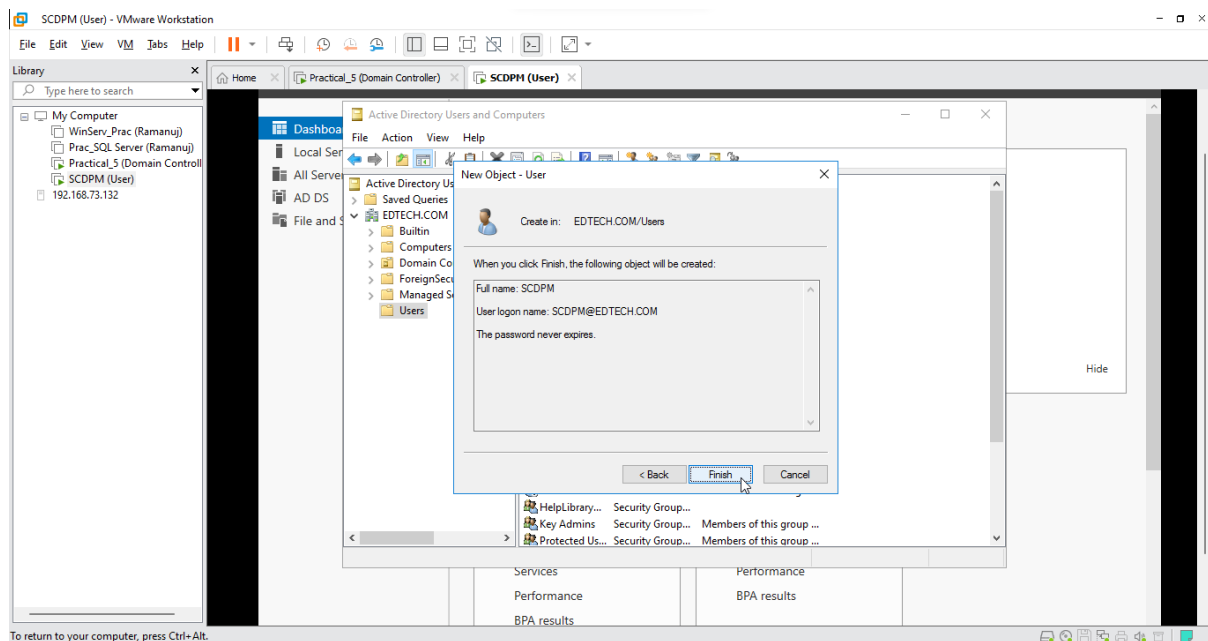
- Type in a new User name (here it is SCDPM) and Click **Next**



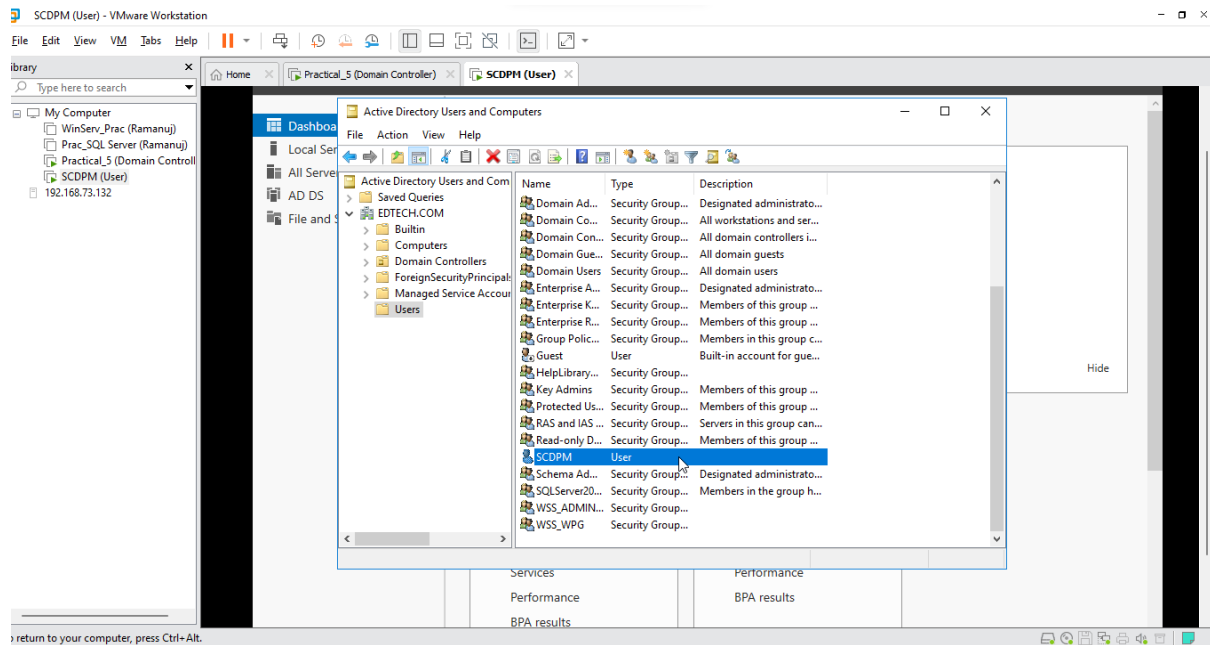
- Type in a password and Select **Password never expires** and Click **Next**



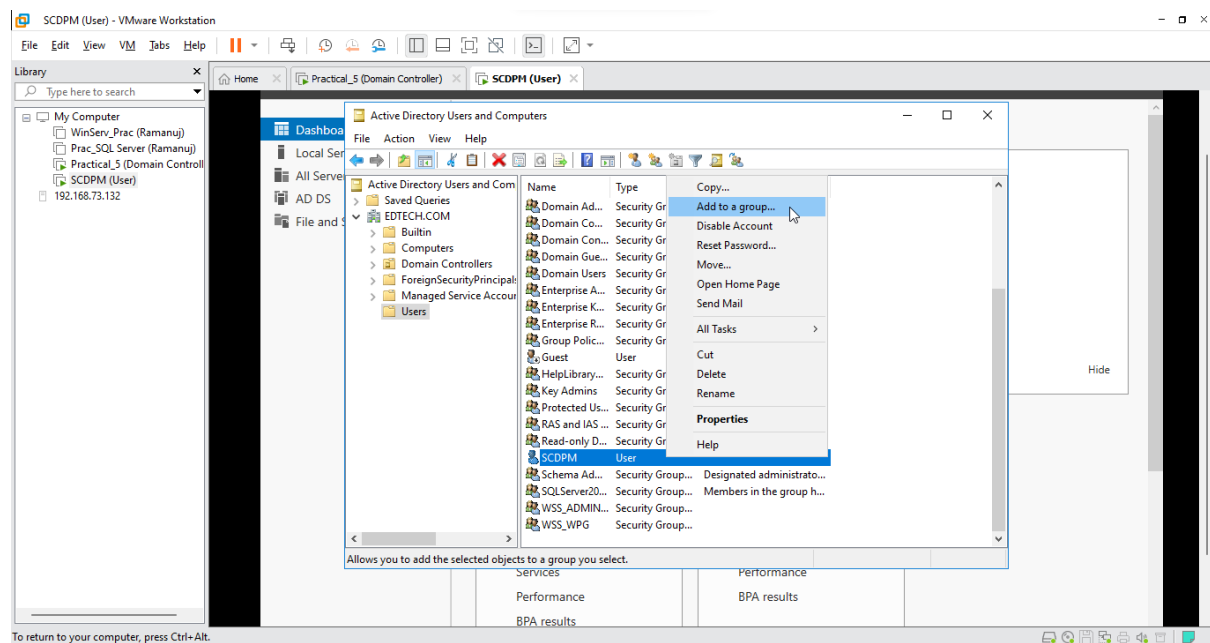
- Click on **Finish**



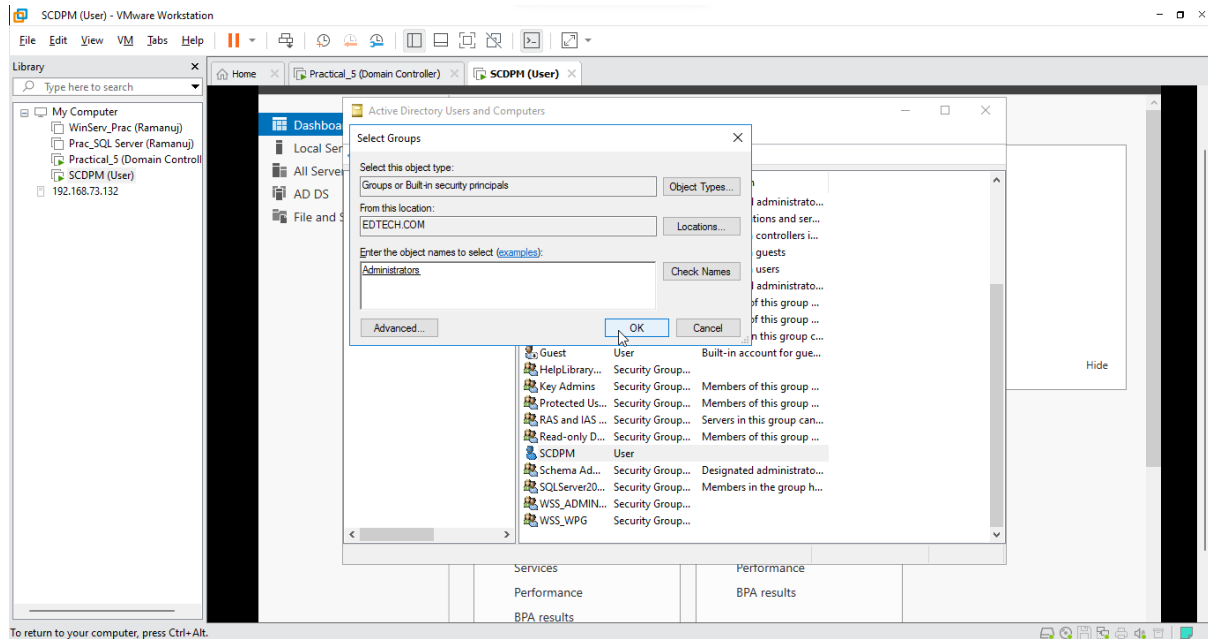
- The SCDPM User has been created



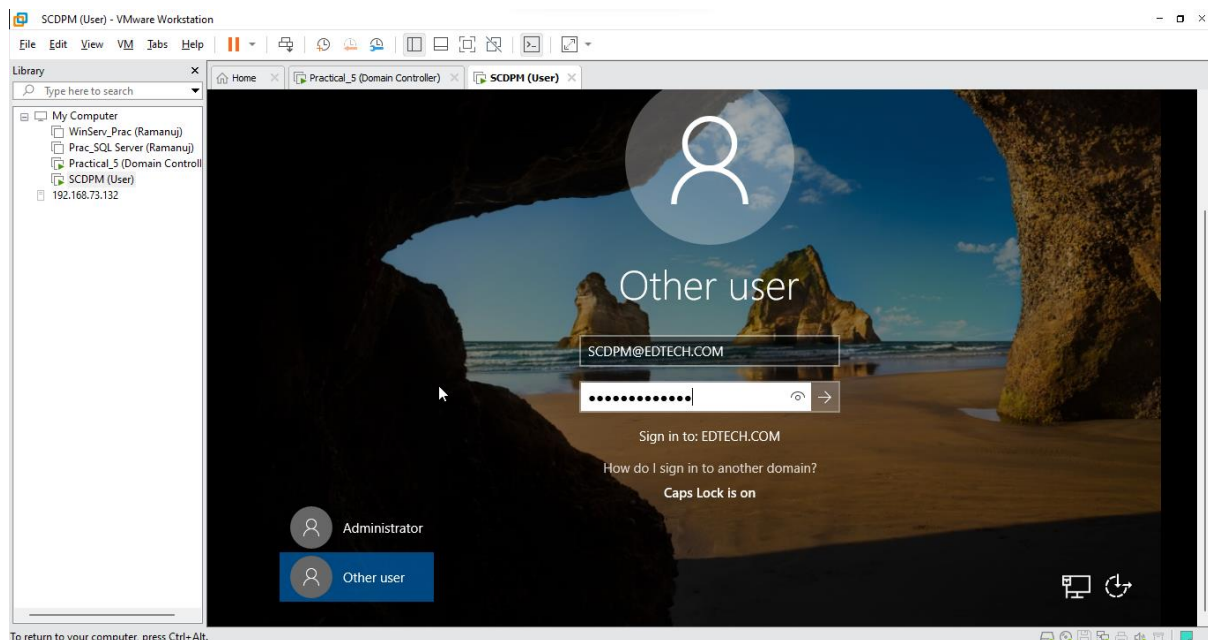
- Add the User to the **Domain Group**, Right Click on SCDPM and Select **Add to a group**



- Type in **Administrator** and Click on **Check Names** and Click on **Ok**



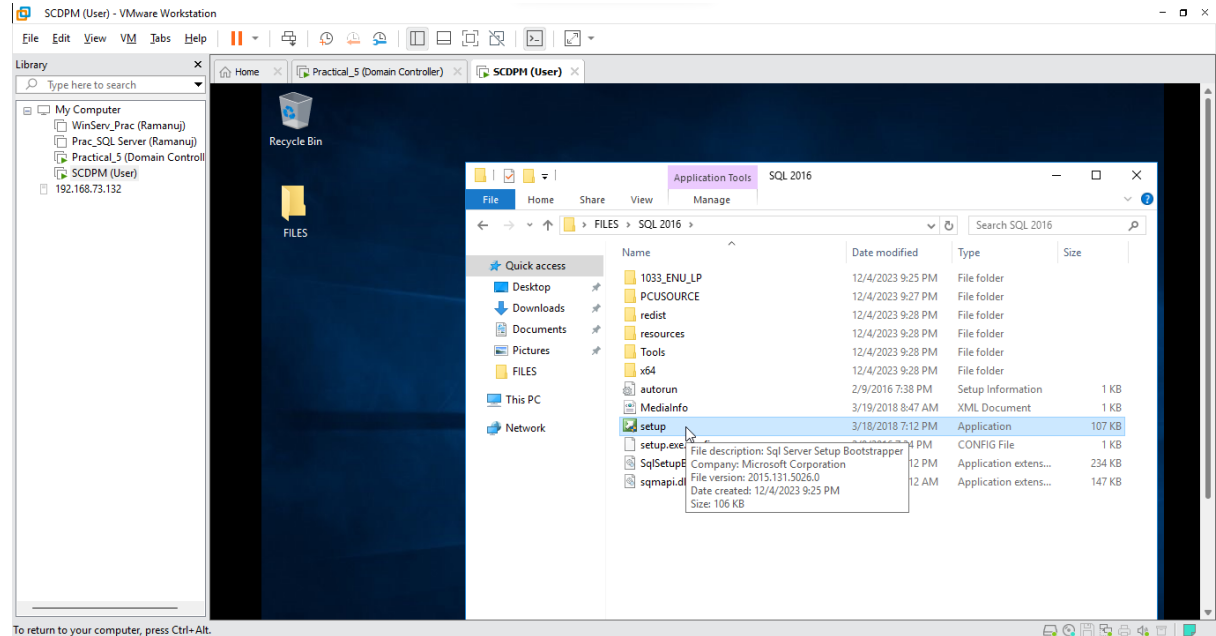
- Now Login as SCDPM



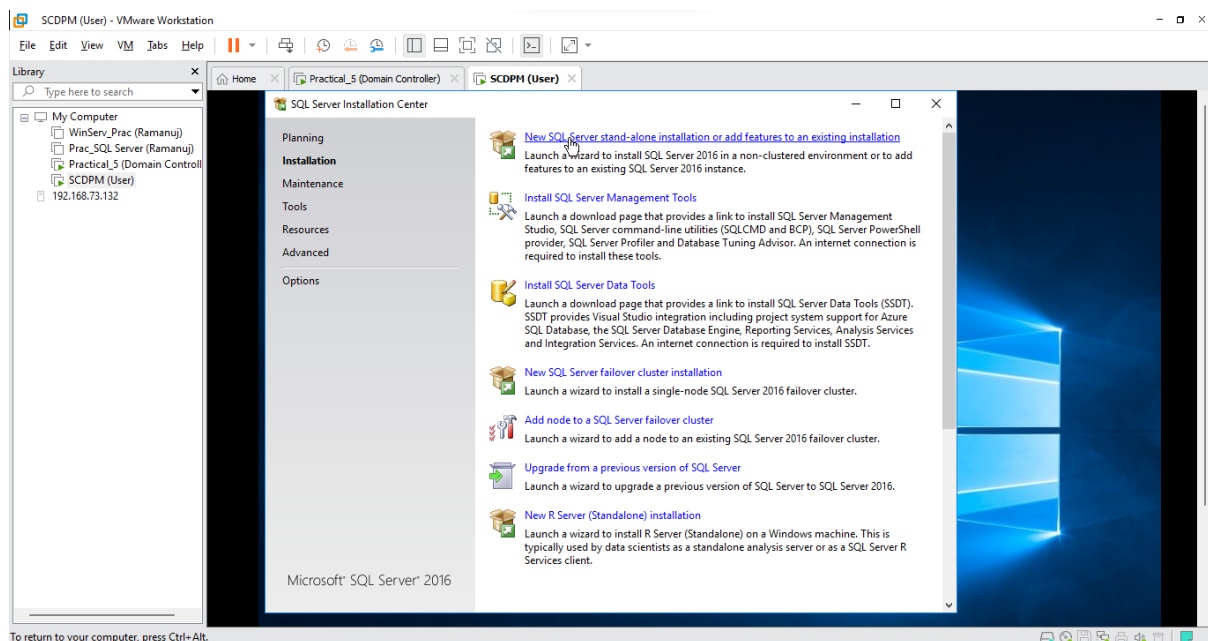
Step 5: Installing SQL Server

- Now Open the SQL 2016 setup file from your folder

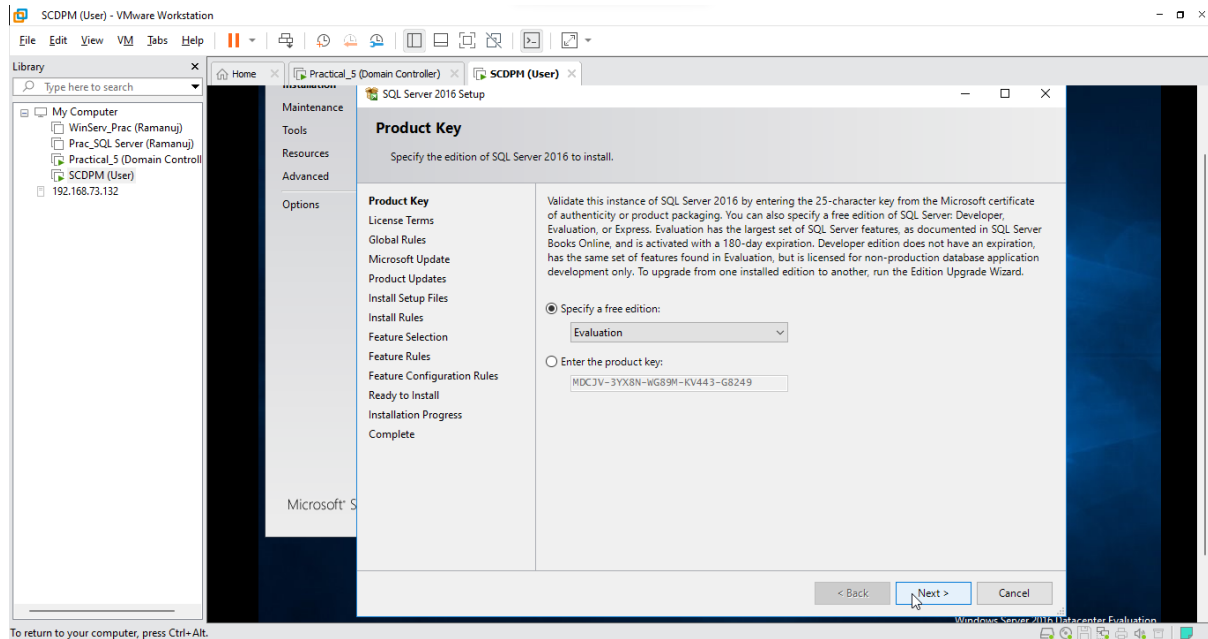
Note: This has to take place within SCDPM User and not Administrator



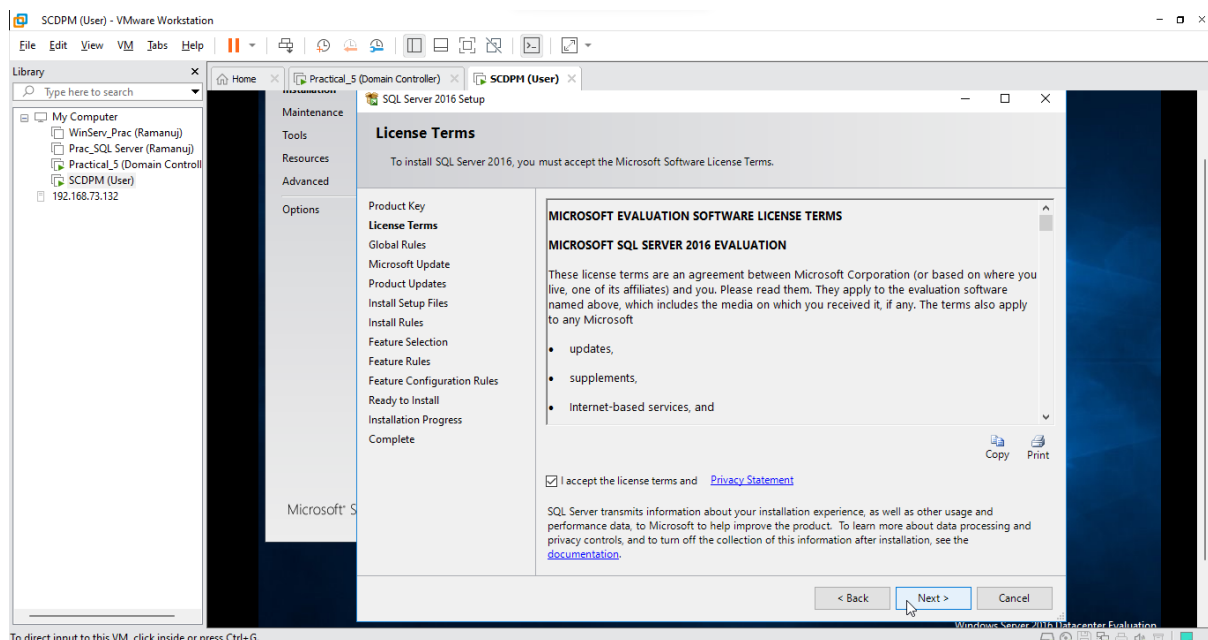
- Select **Installation** and Click on **New SQL Server Stand-alone installation**



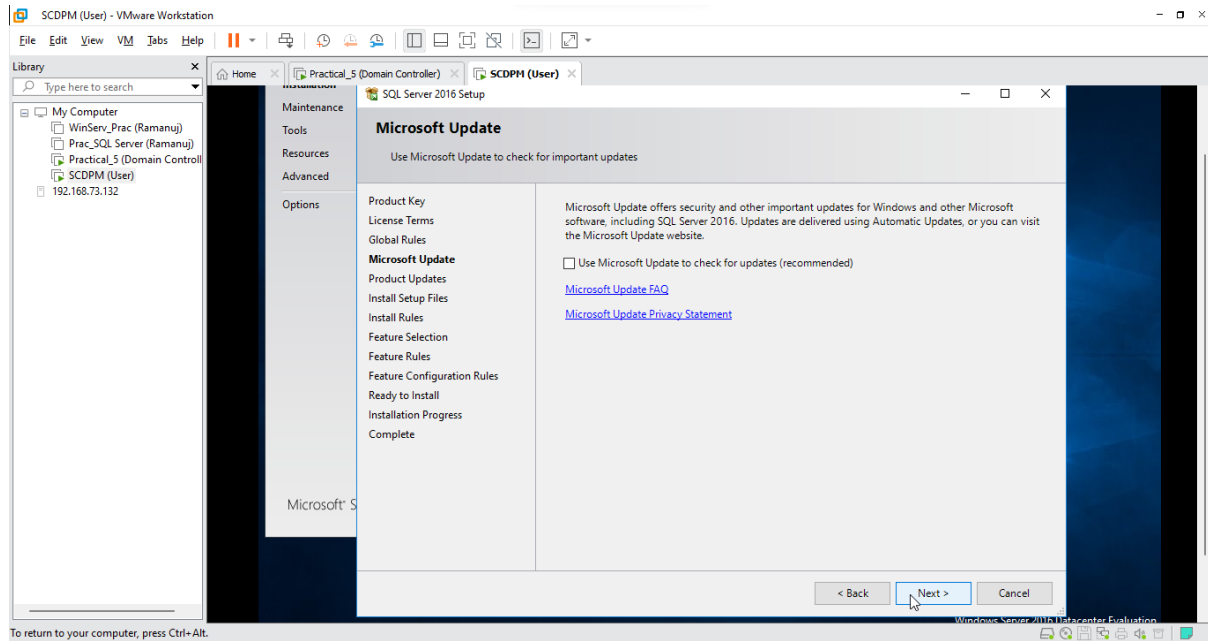
- Select **Specify a free edition** and Click on **Evaluation** and Click **Next**



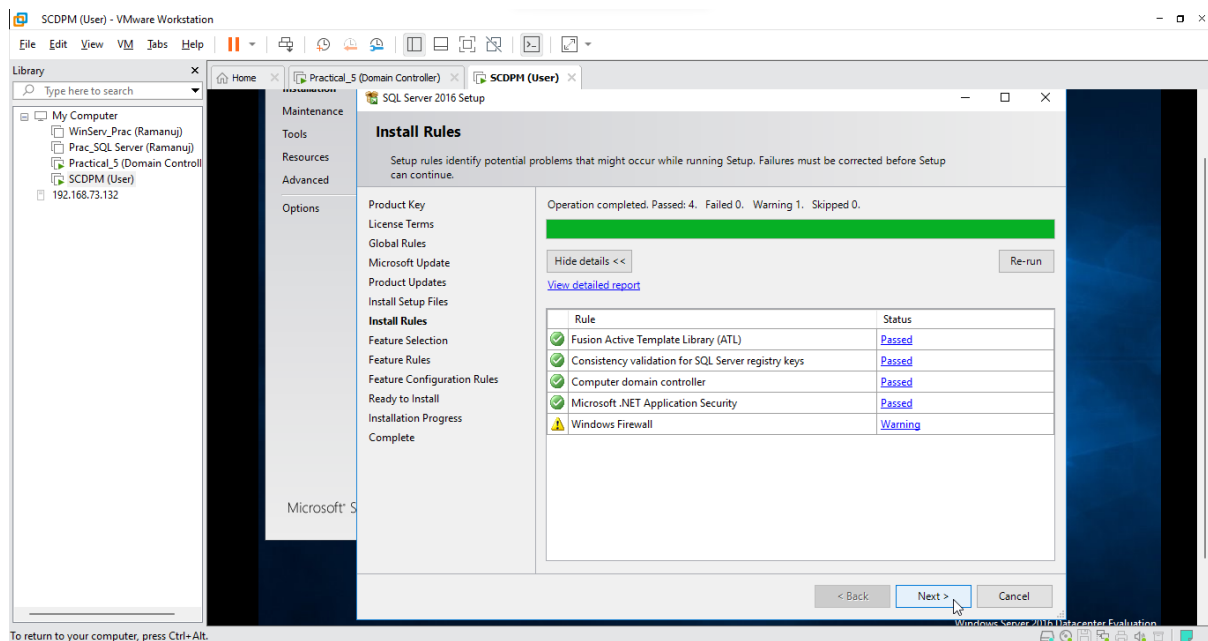
- Select **I accept these license terms** and Click **Next**



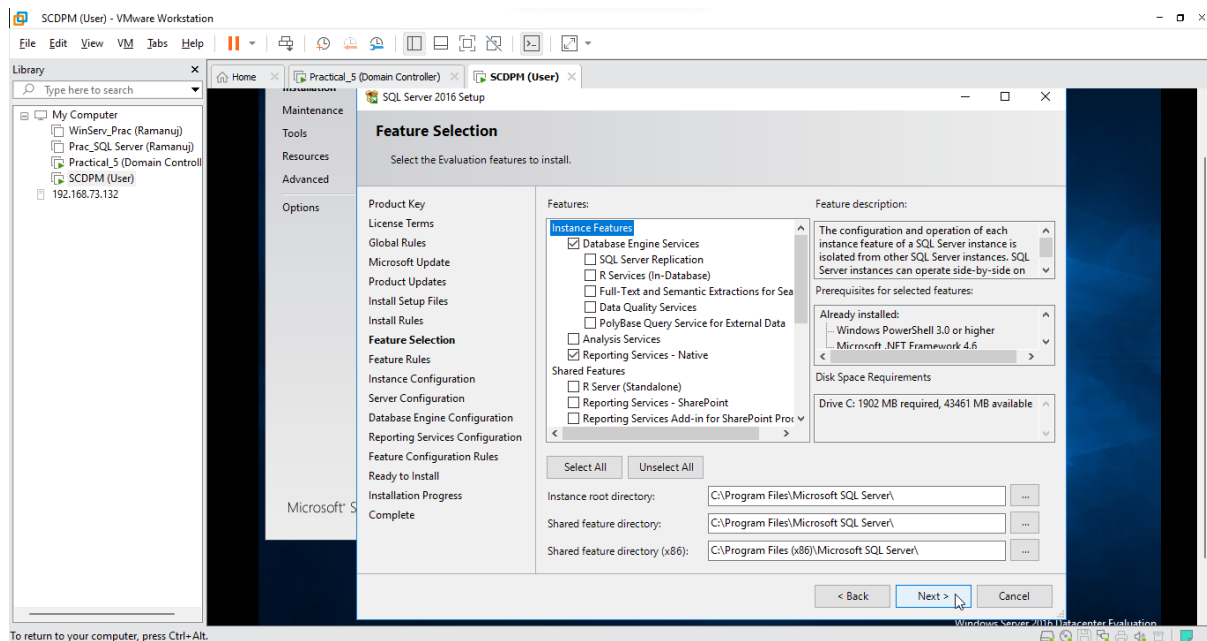
- Keep default values and Click **Next**



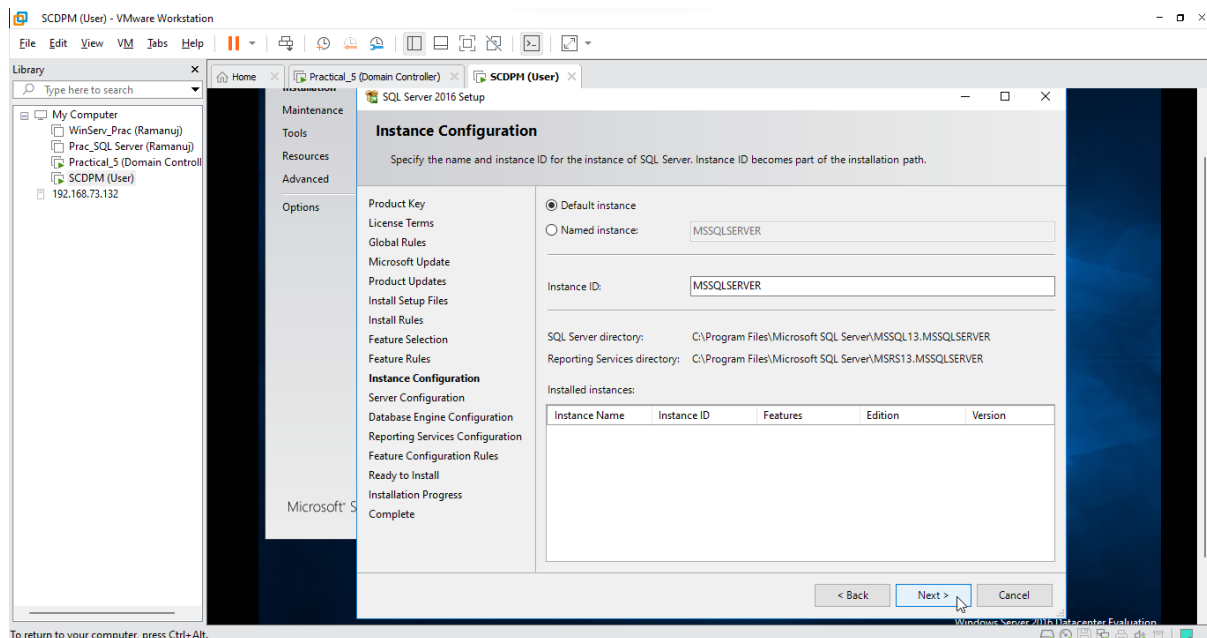
- Warnings are acceptable, Click **Next**



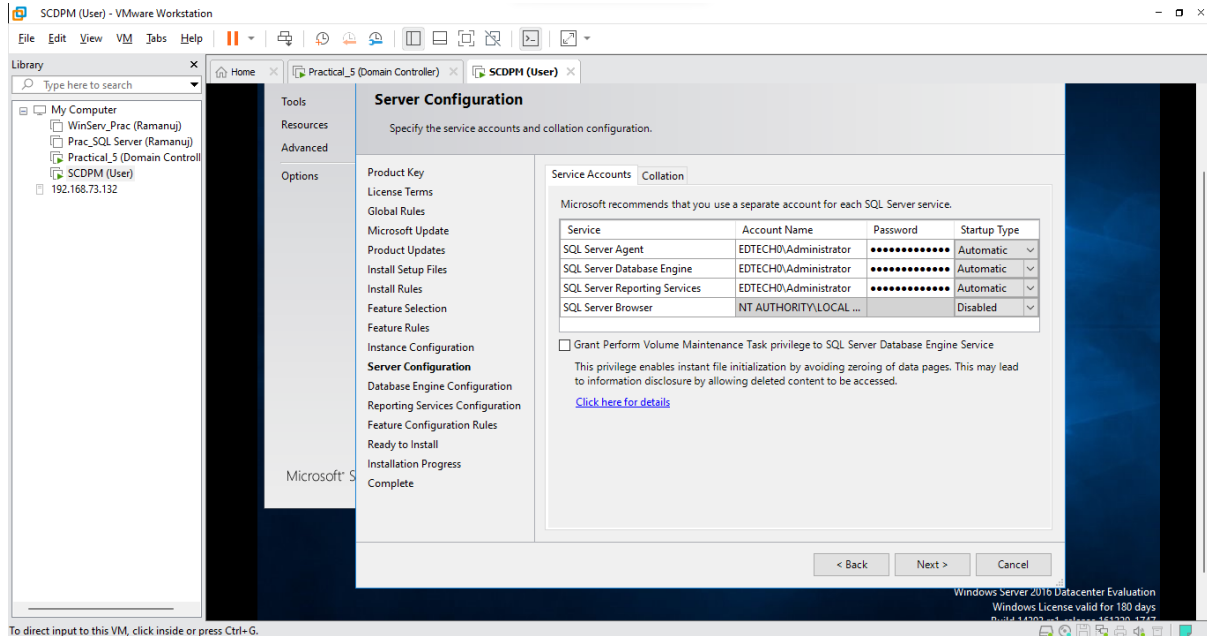
- Select two features: **Database Engine Services** and **Reporting Service-Native** and Click **Next**



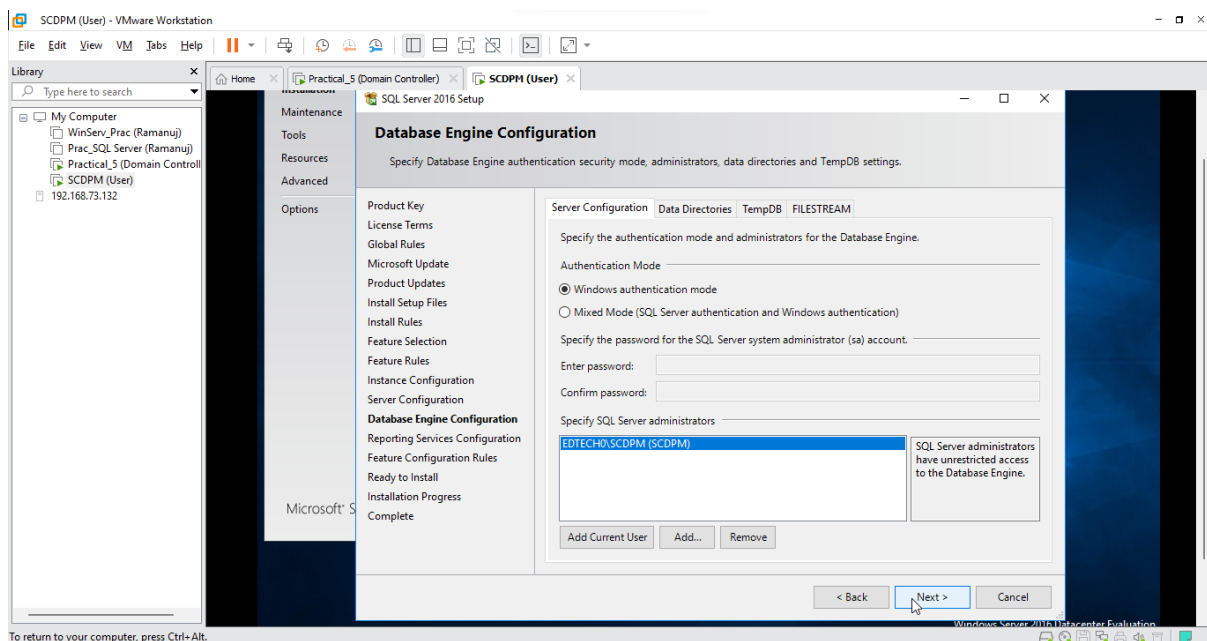
- Keep default values and Click **Next**



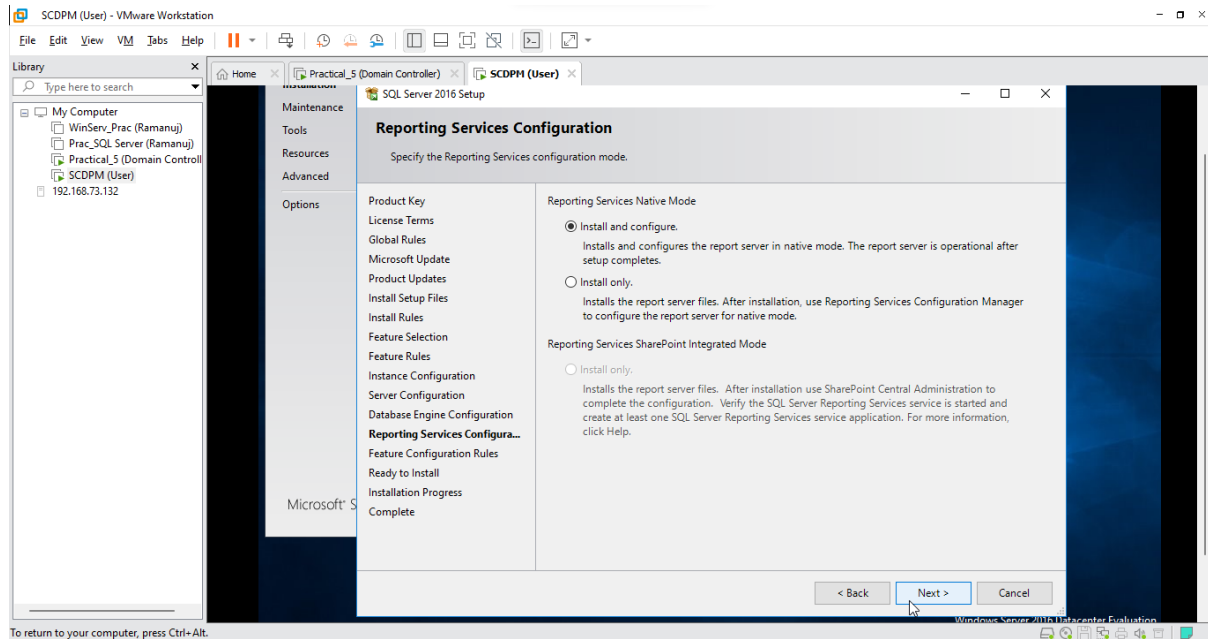
- Change the Account Name for **SQL Server Agent, Database Engine and Reporting Services** to the **Users Account** and Ensure the Start-up type is **Automatic** and Click **Next**



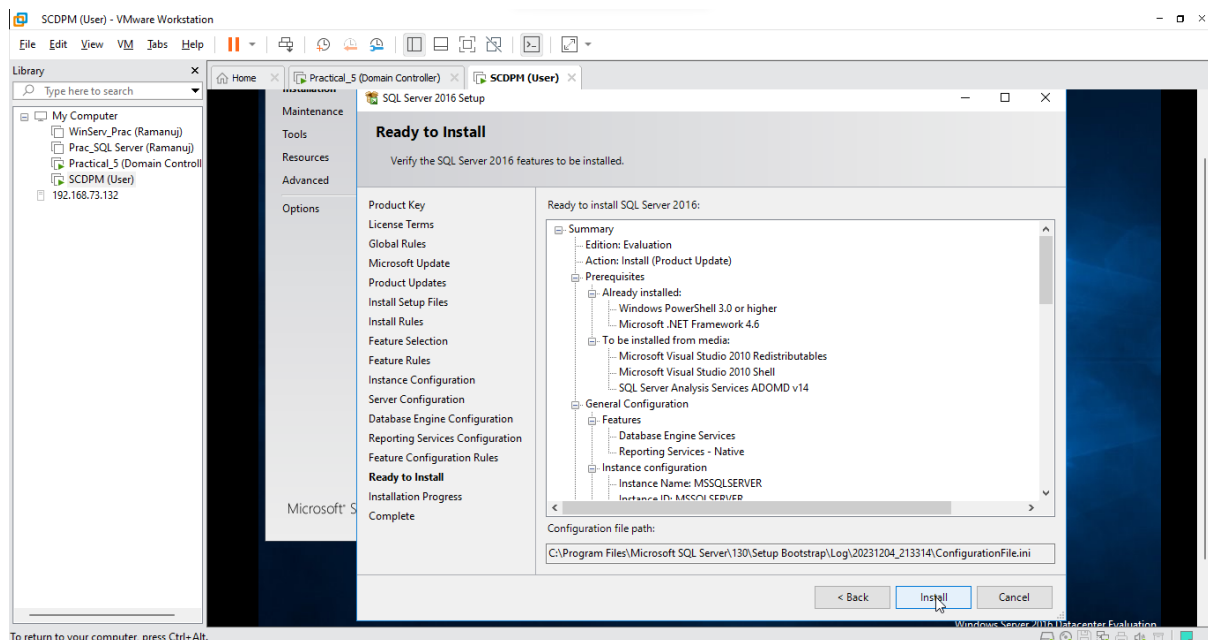
- Click on **Add Current User** (Ensure it is SCDPM and not the Administrator) and Click **Next**



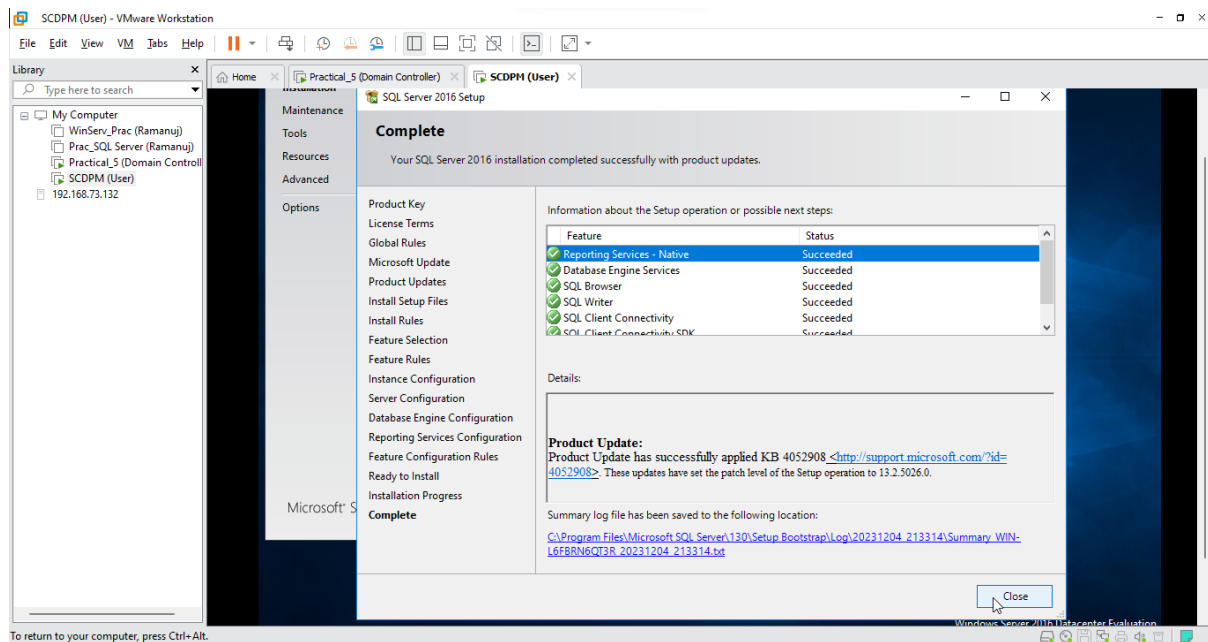
- Keep default values and Click **Next**



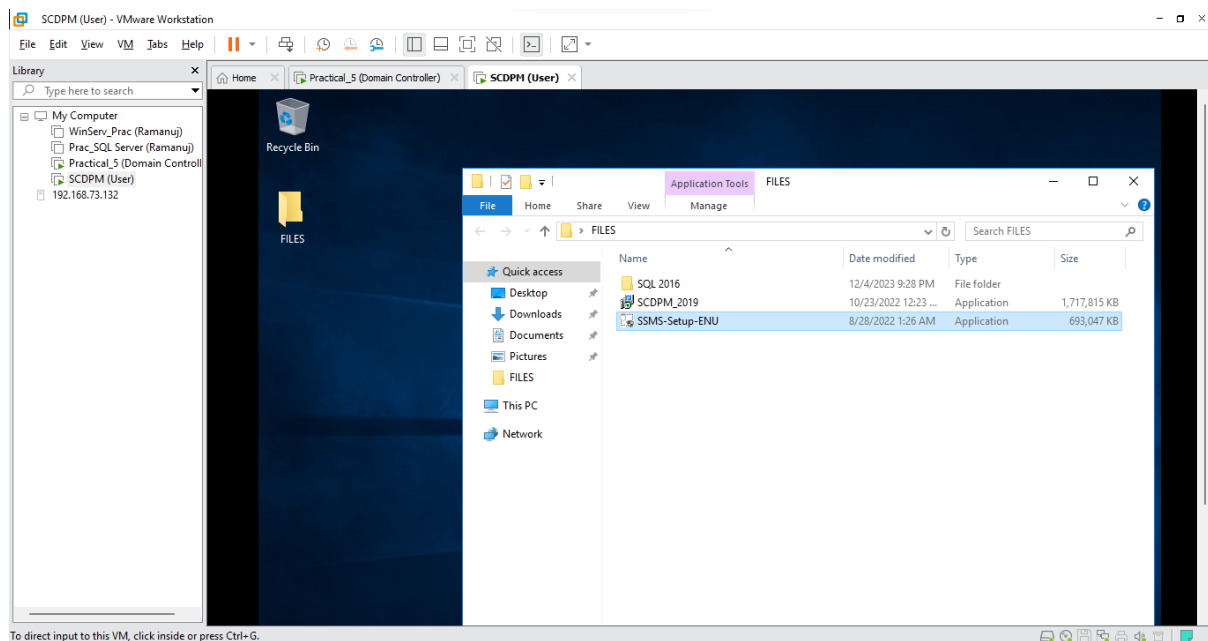
- Review the summary of the SQL Server and Click on **Install**



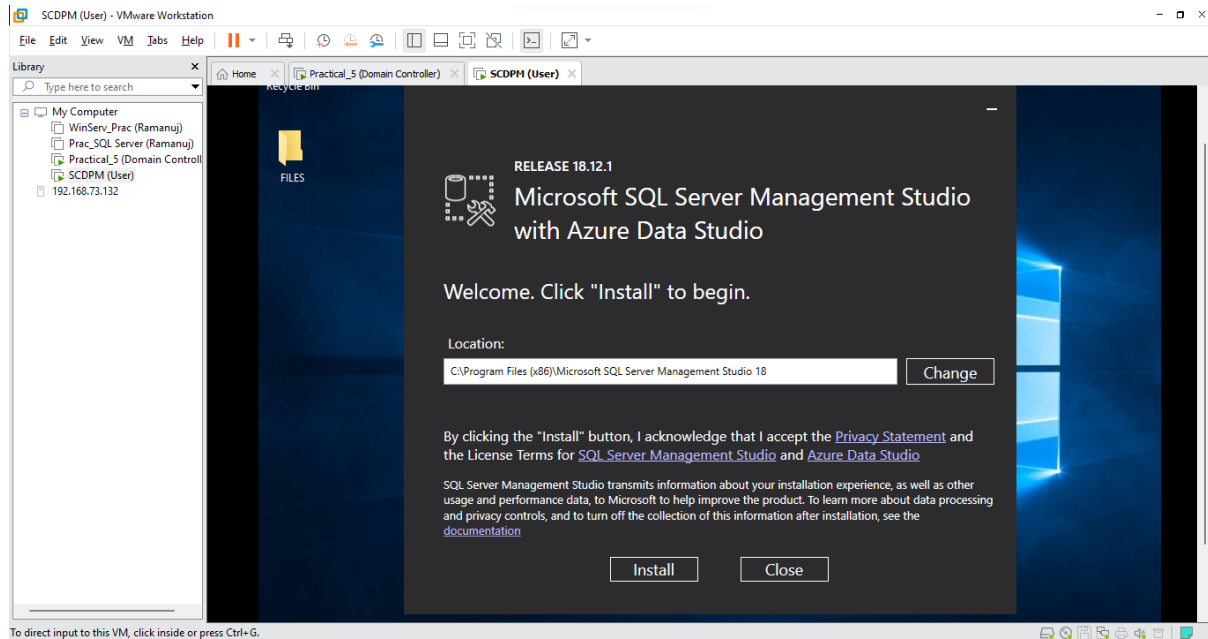
- After installation Click **Close**



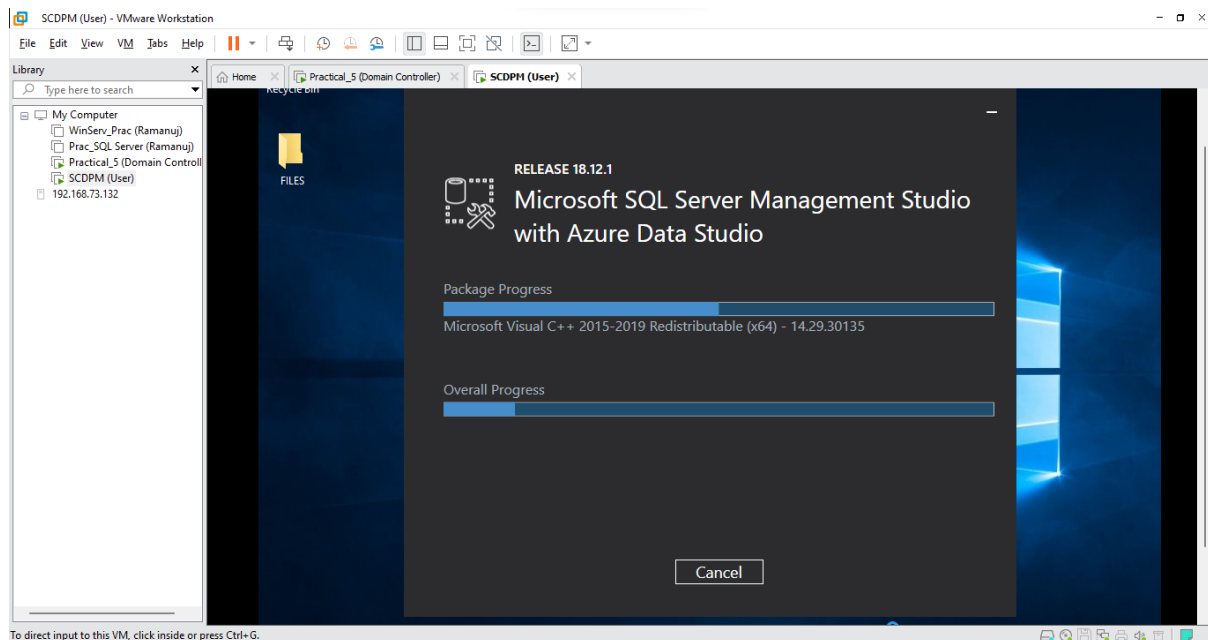
- Now Open **SSMS** Setup from your folder



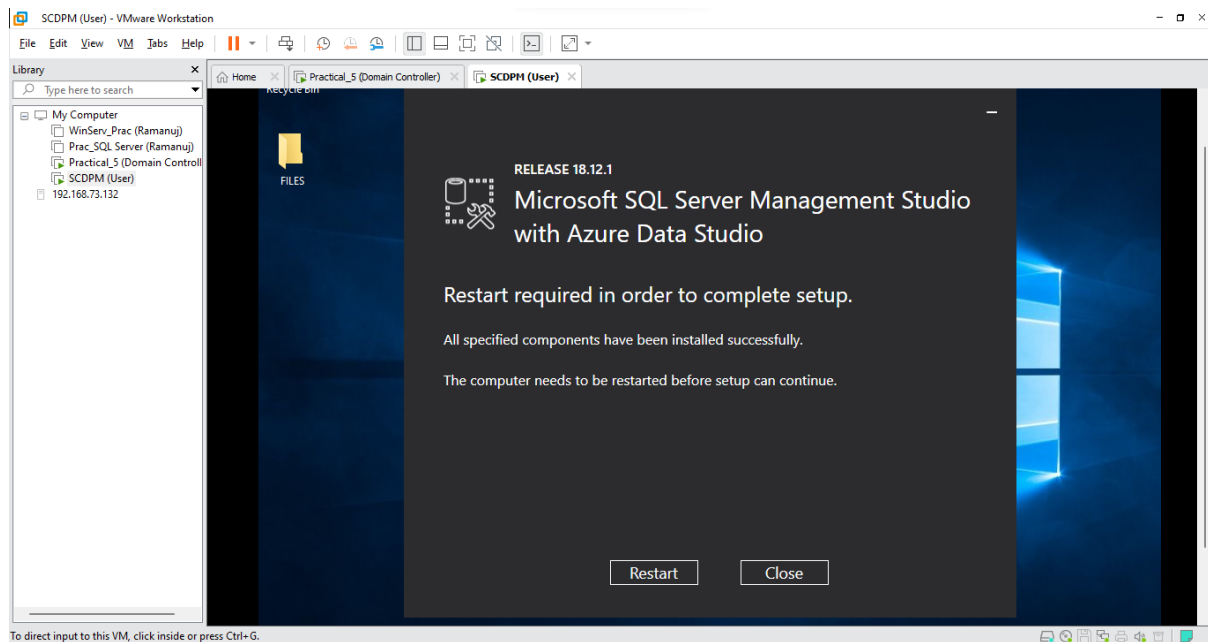
- Click on **Install**



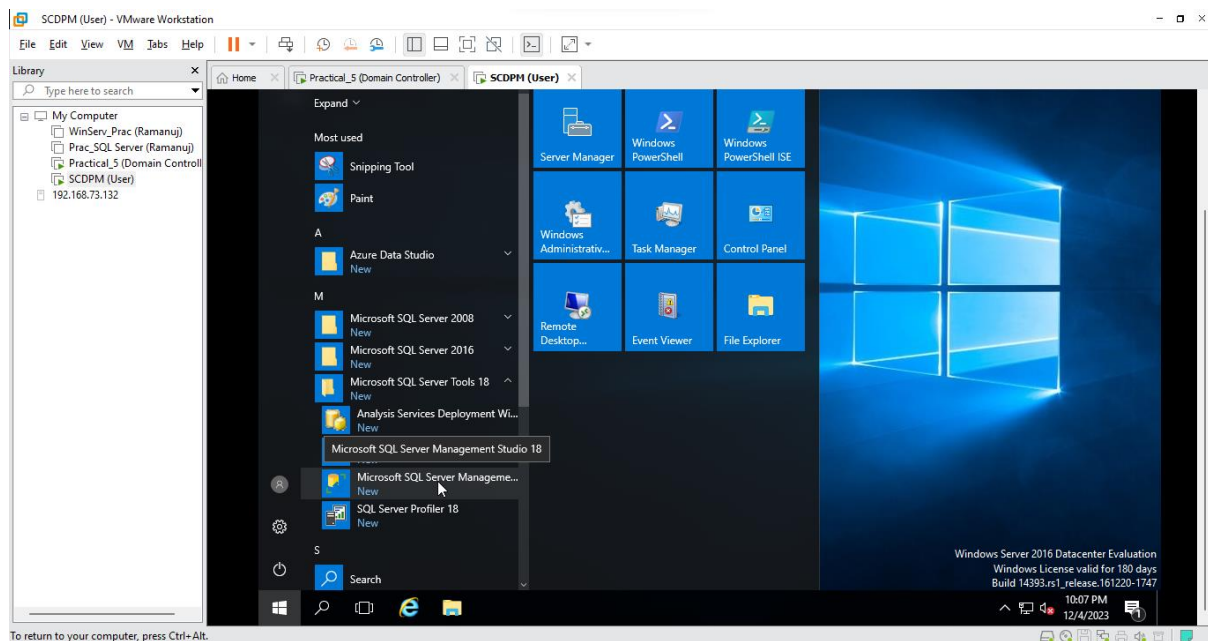
- SSMS will start installing onto your system



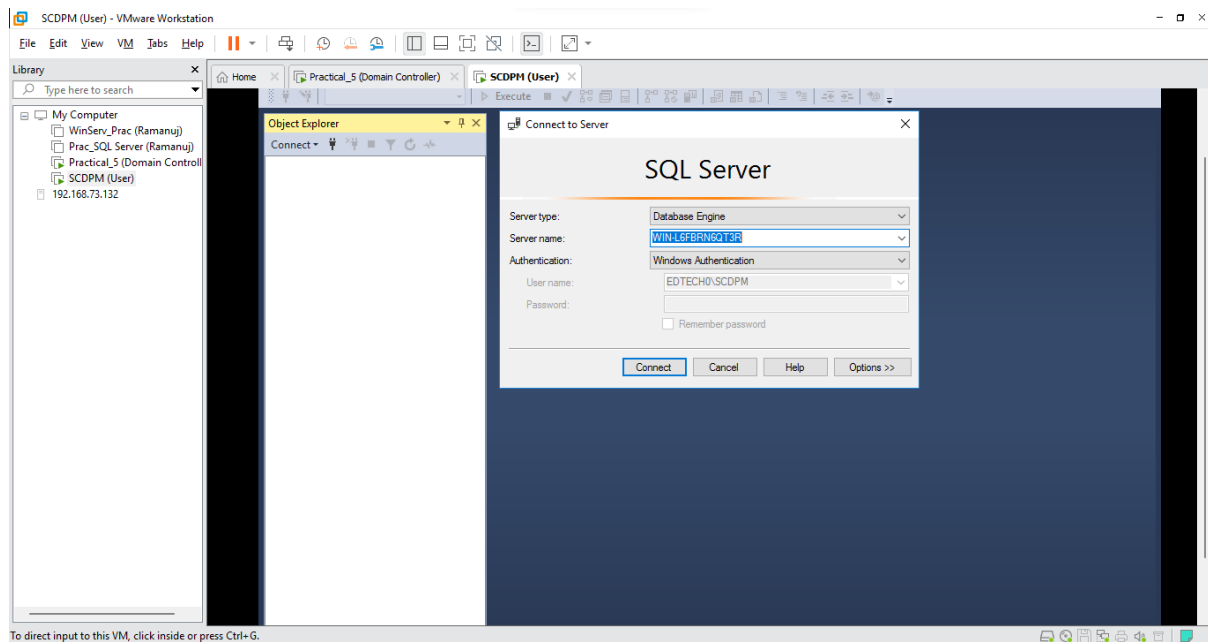
- After installation a restart is required



- Open SQL Server Management Studio 18

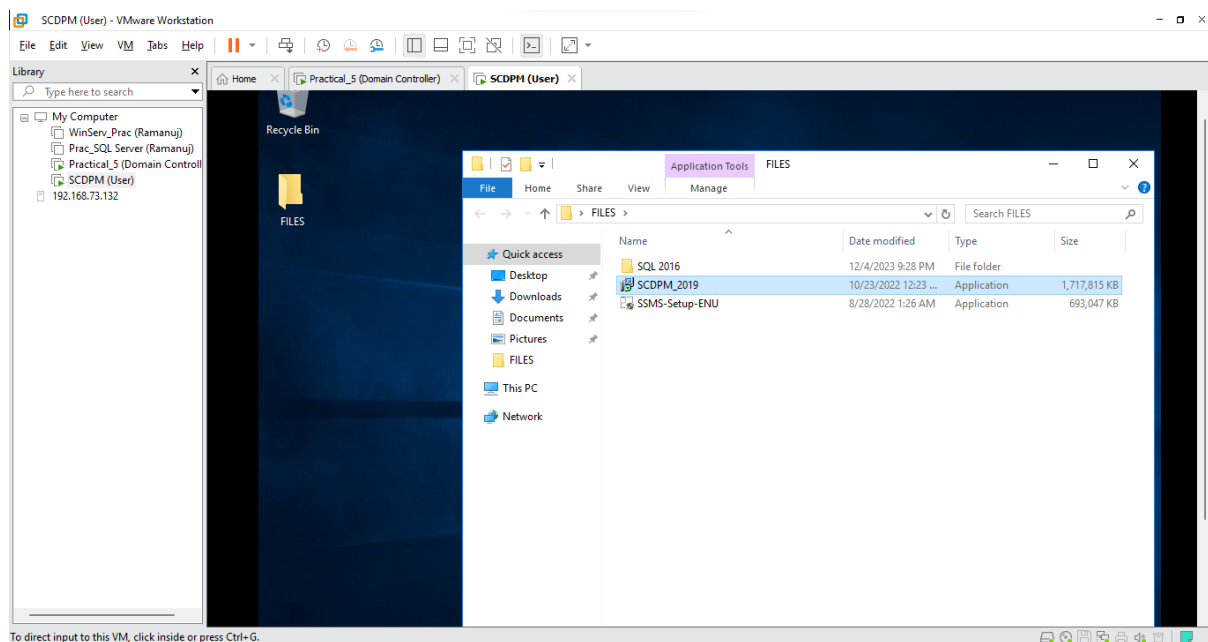


- Keep default values and Click on **Connect**

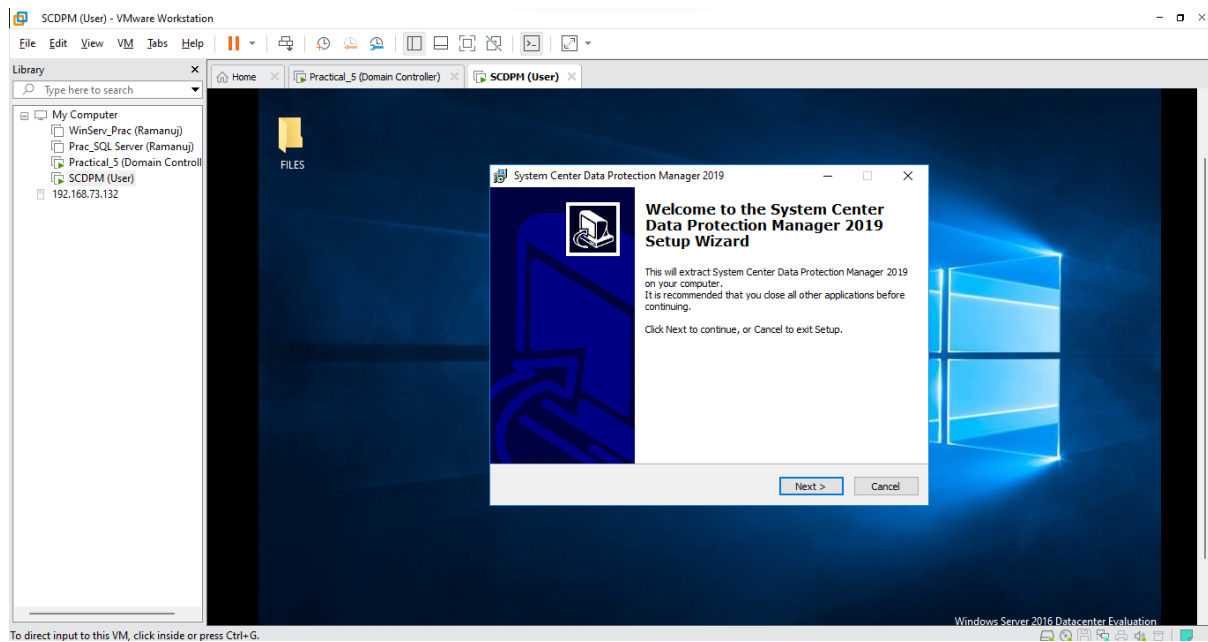


Step 6: Installing Data Protection Manager

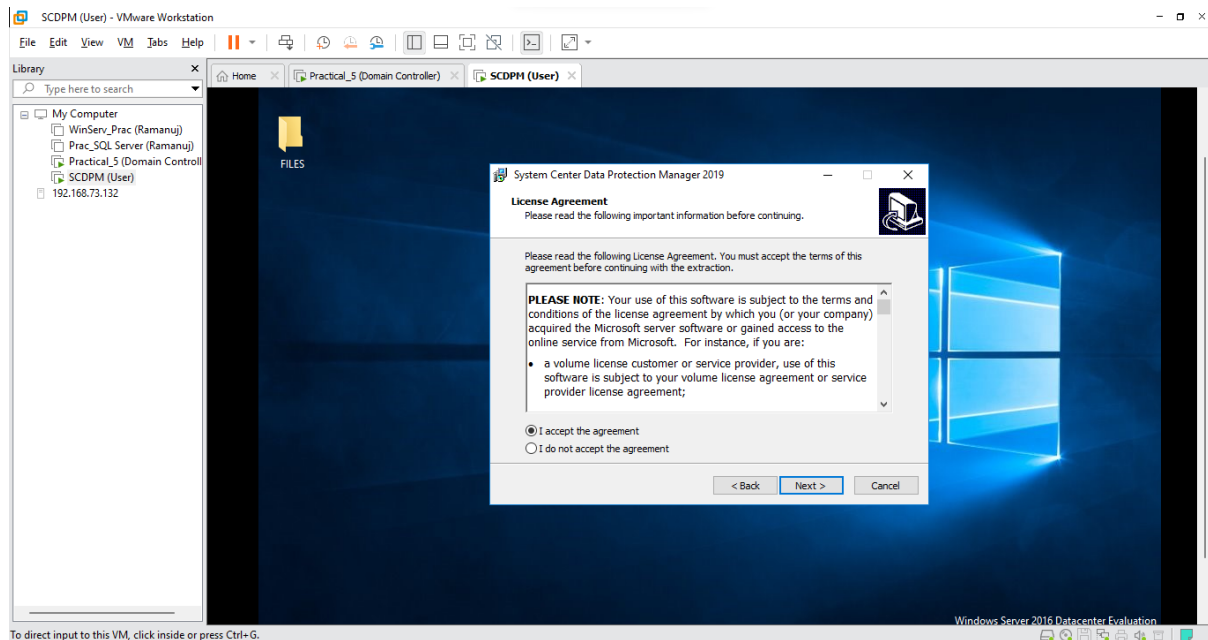
- Open **SCDPM_2019** from your folder



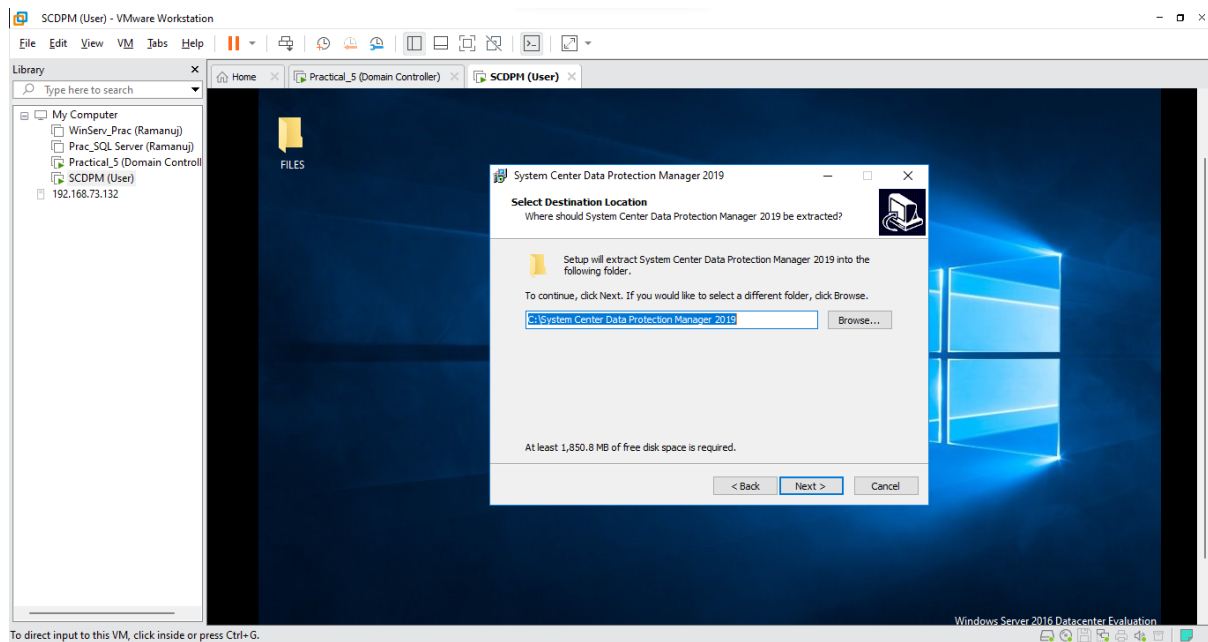
- Click Next



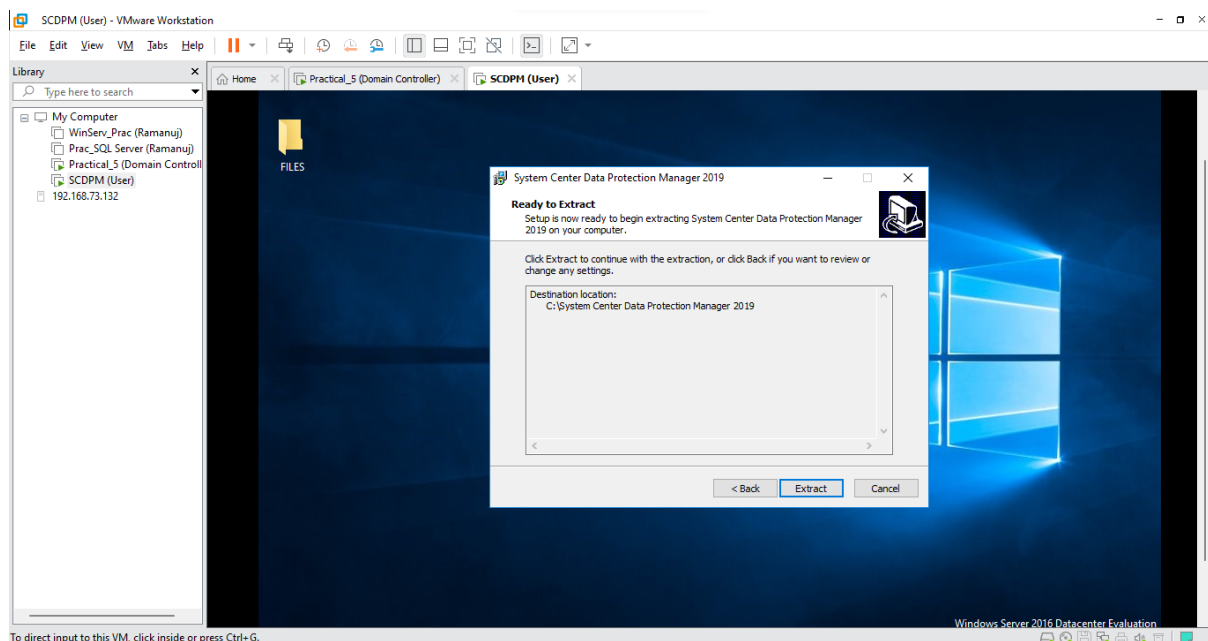
- Select **I accept the agreement** and Click Next



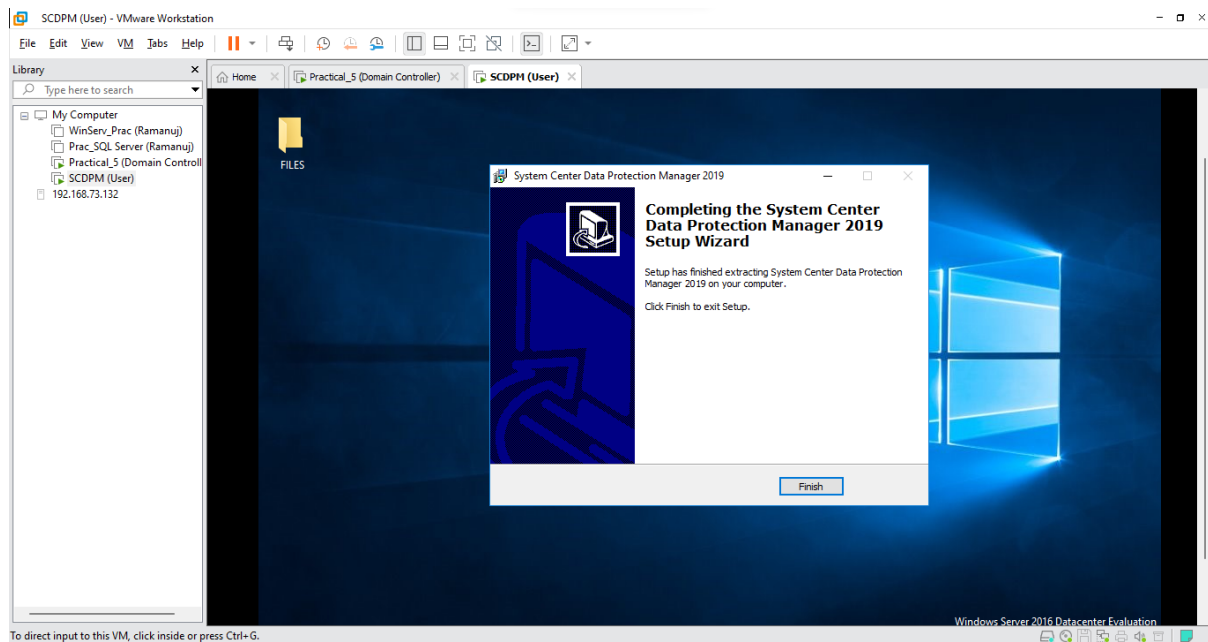
- Keep default values and Click **Next**



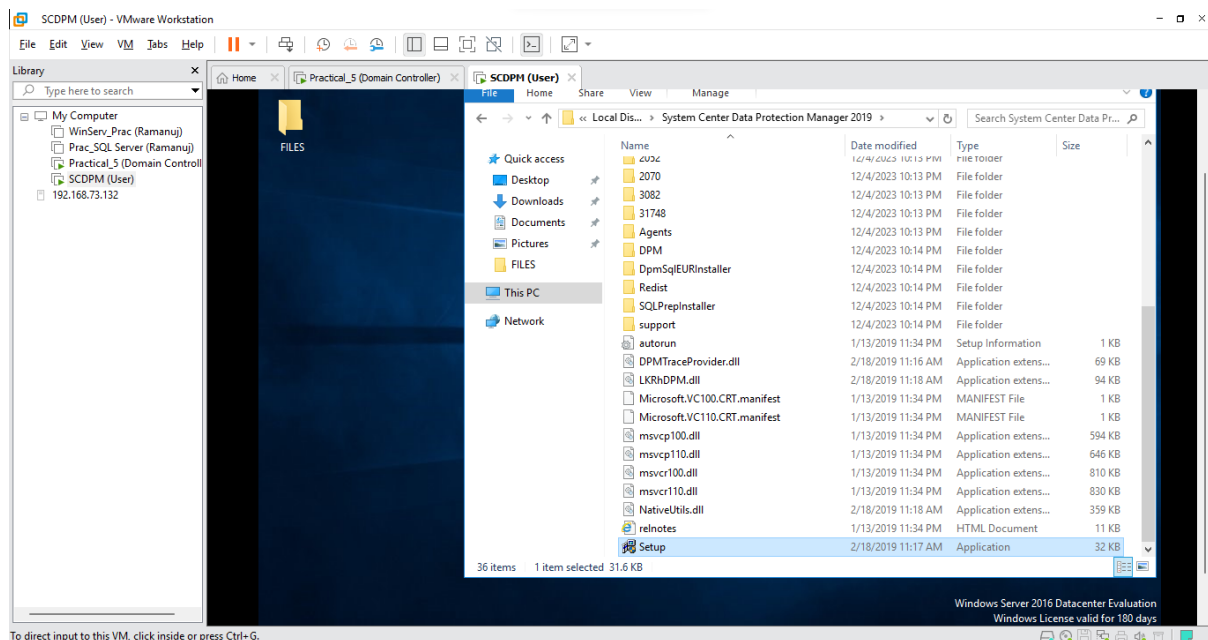
- Click on **Extract**



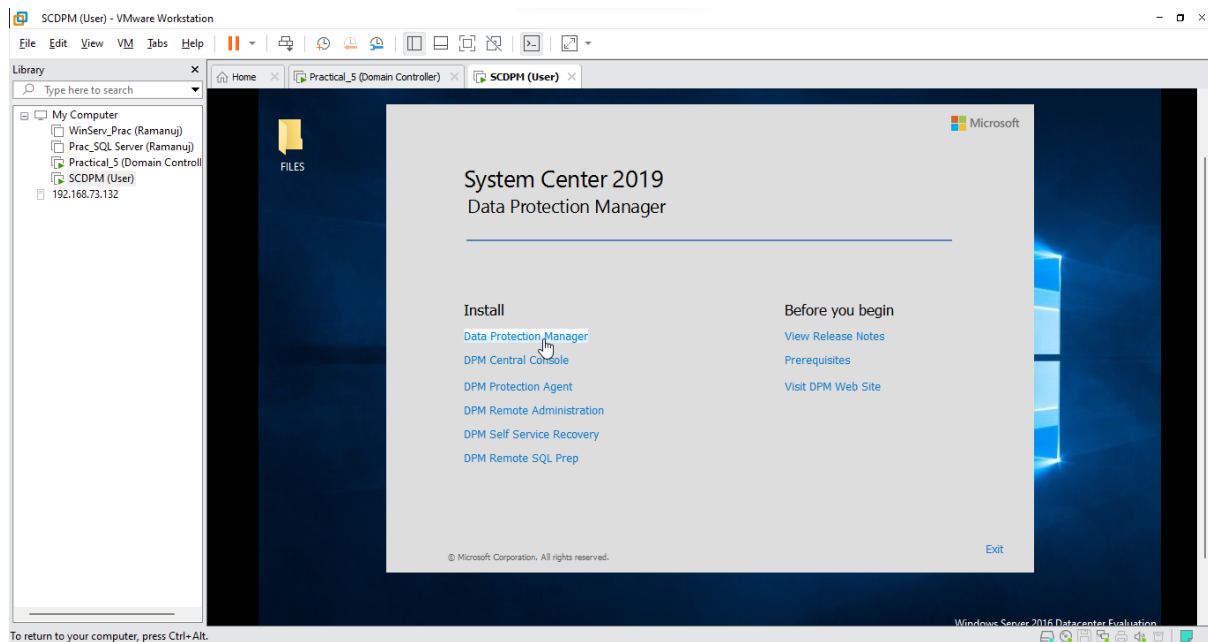
- Click on **Finish**



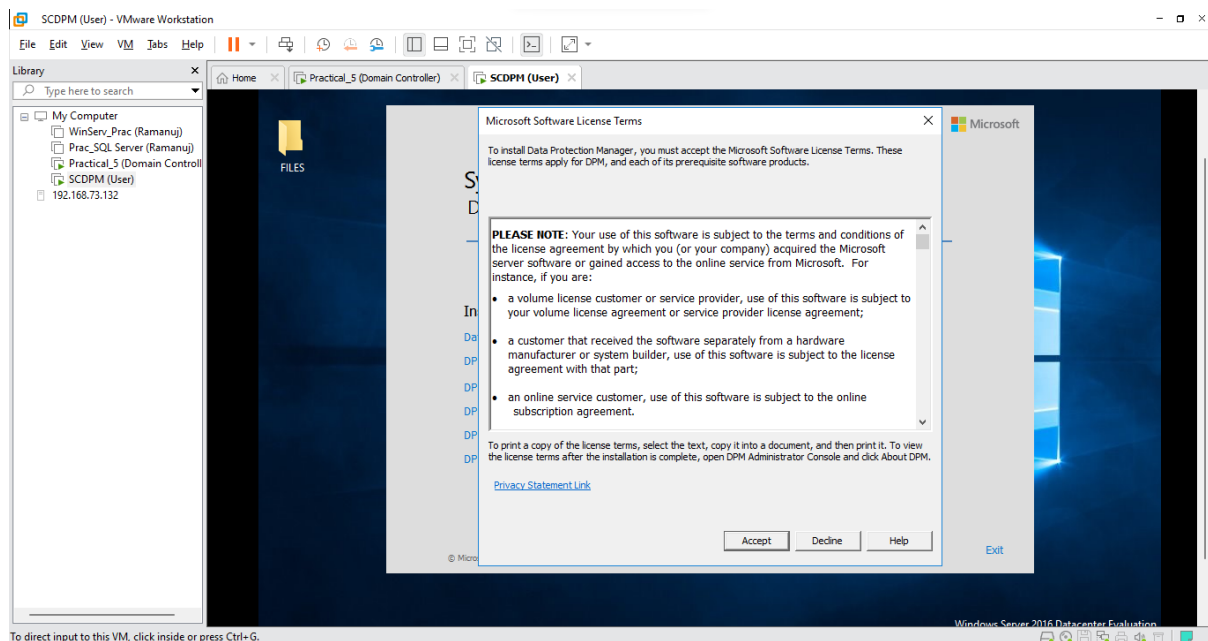
- Now In you C: Directory Click on the **System Center Data Protection Manager** folder and Click on **Setup**



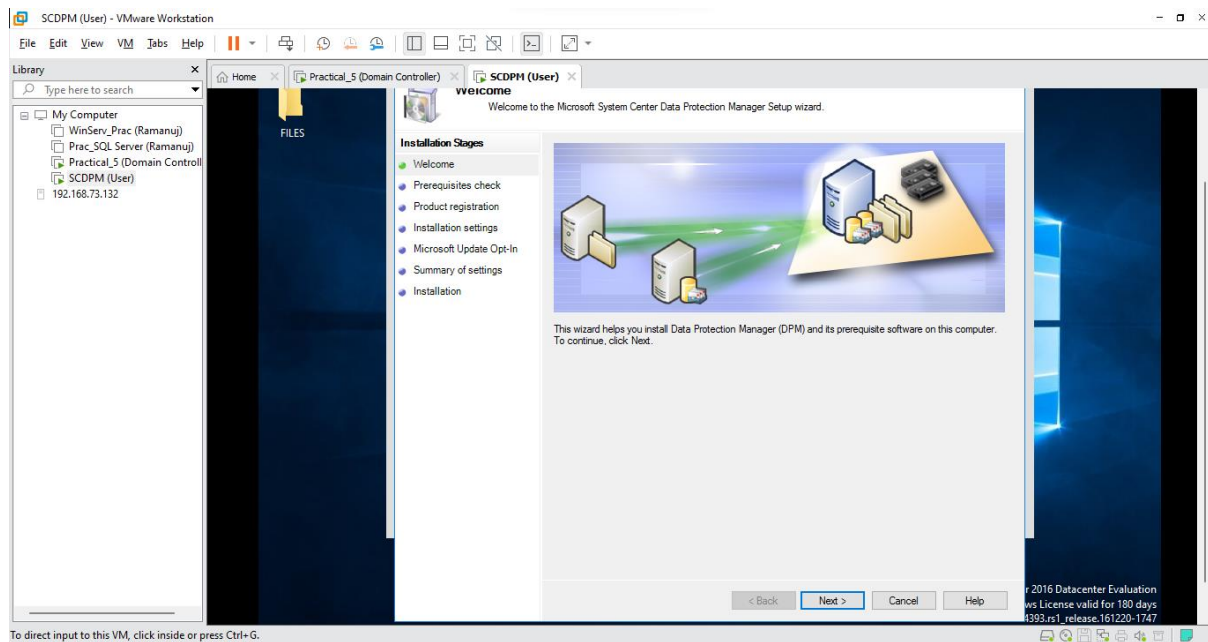
- Click on **Data Protection Manager**



- Click on **Accept**

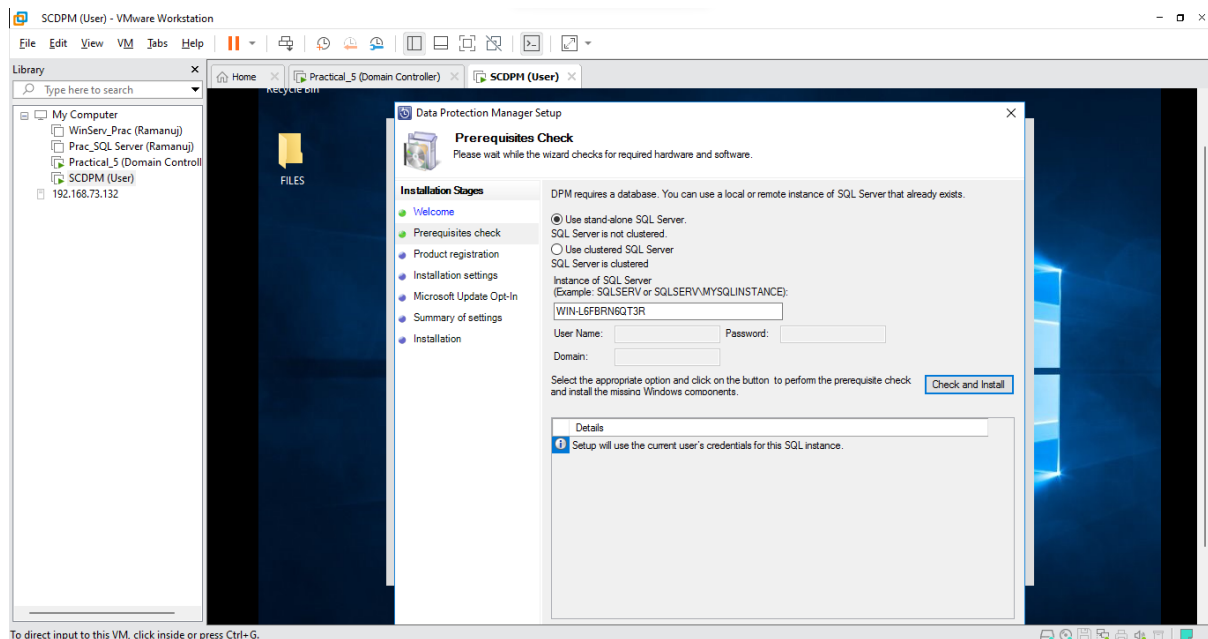


- Click **Next**

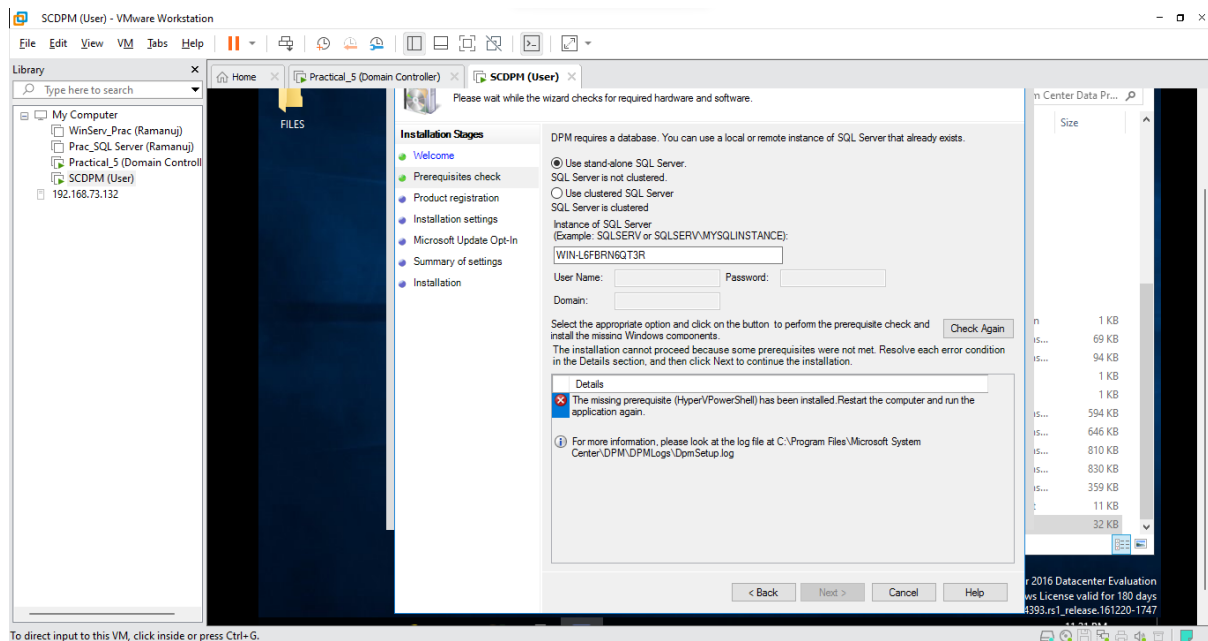


- Select **Use Stand-alone SQL Server** and Enter the Server Name and Click on **Check and Install**

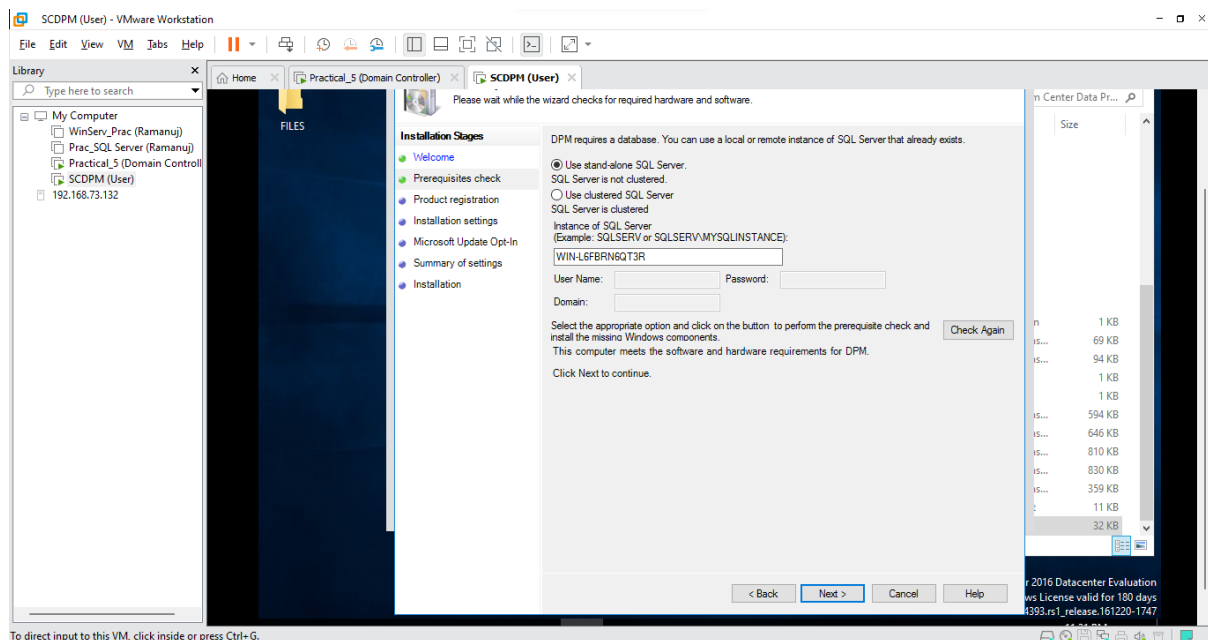
Note: If there is an Error try to elevate the control of your User



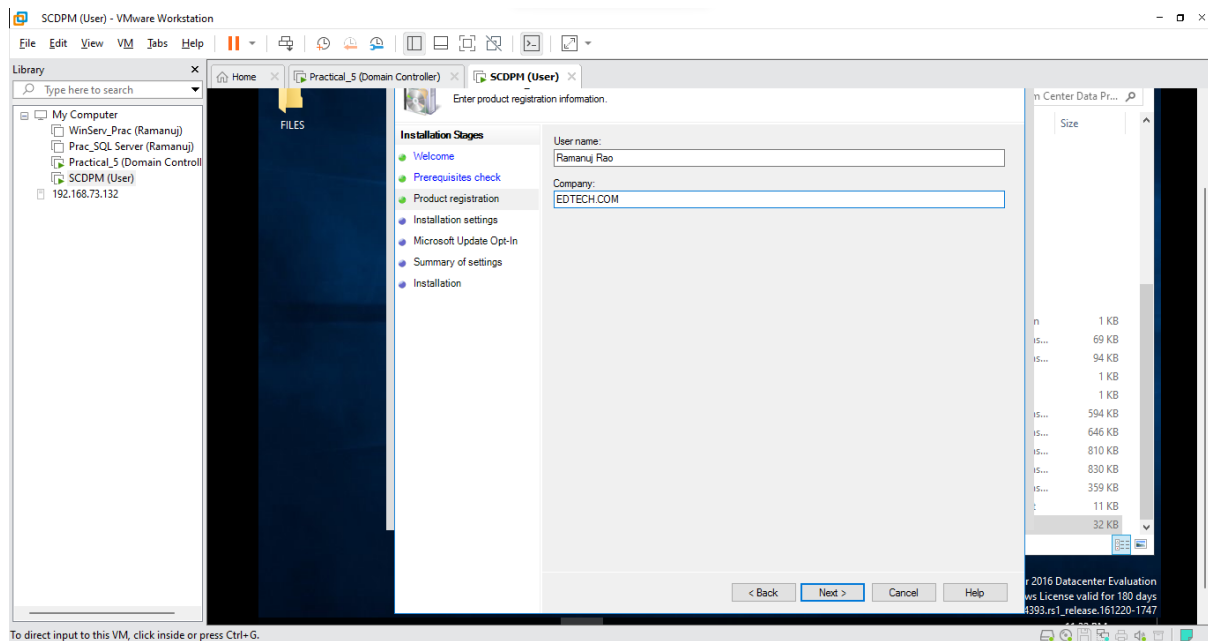
- While checking this error will pop-up which requires a restart of the system



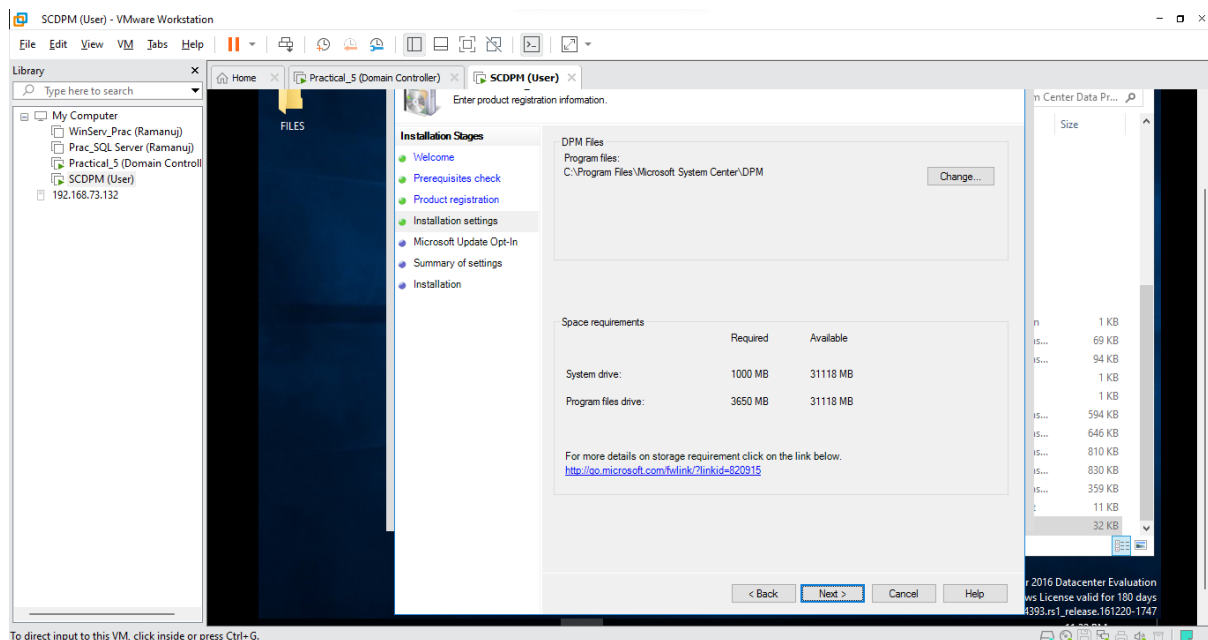
- After the checks are completed Click **Next**



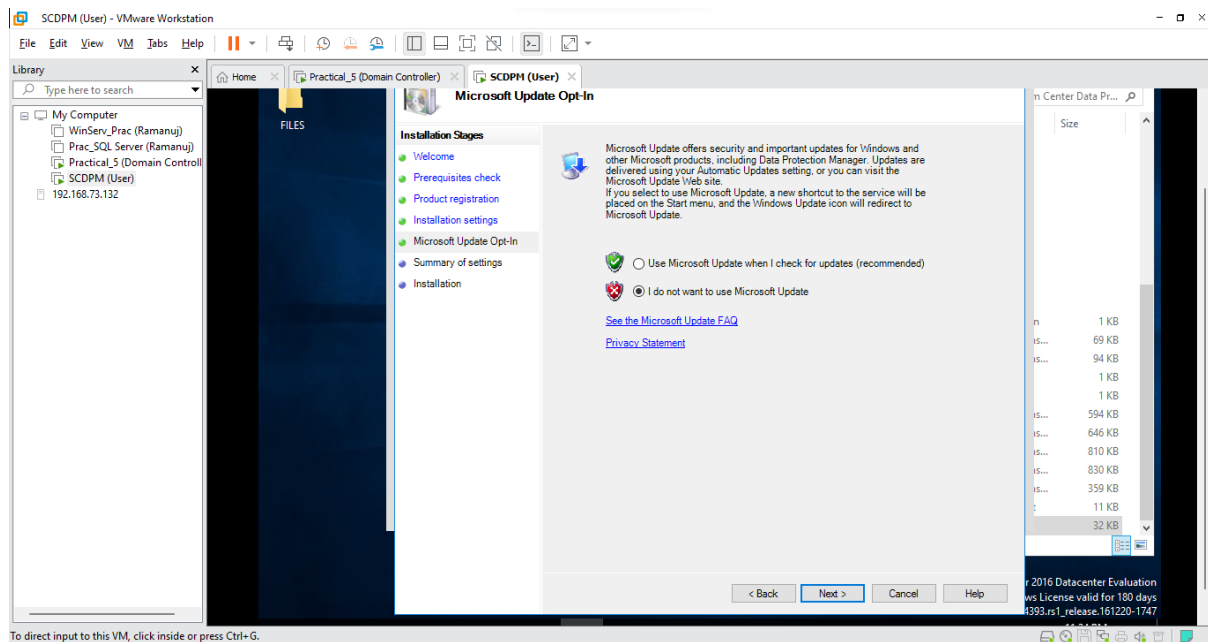
- Enter a User name and Company Name (Here it is Ramanuj Rao & EDTECH.COM) and Click **Next**



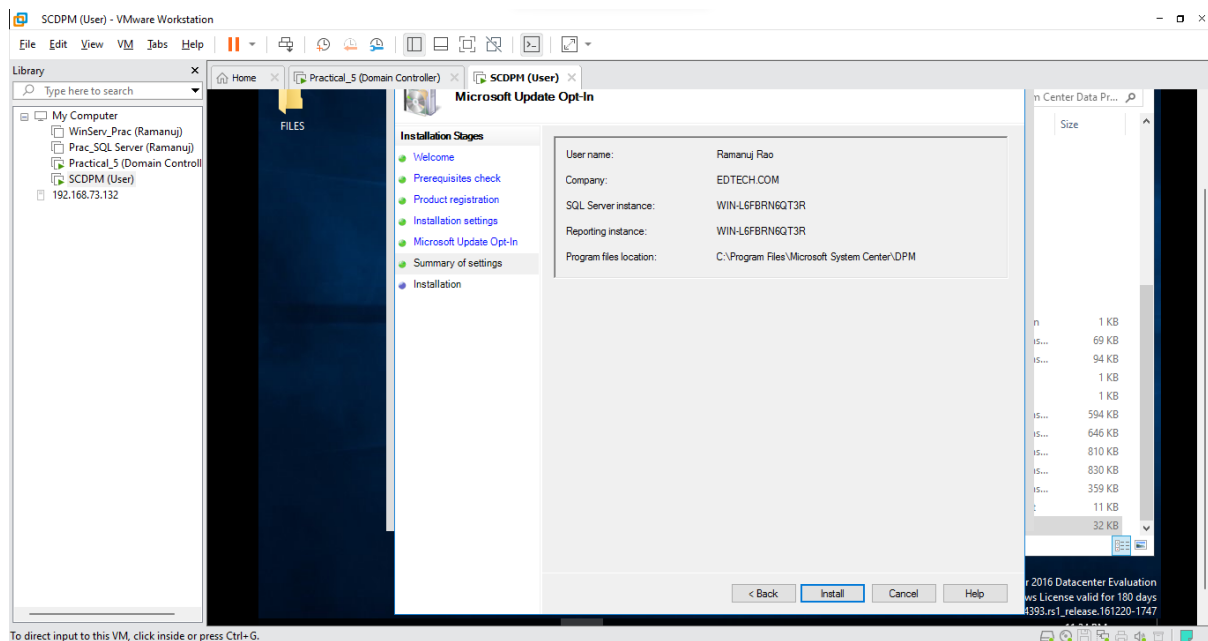
- Click **Next**



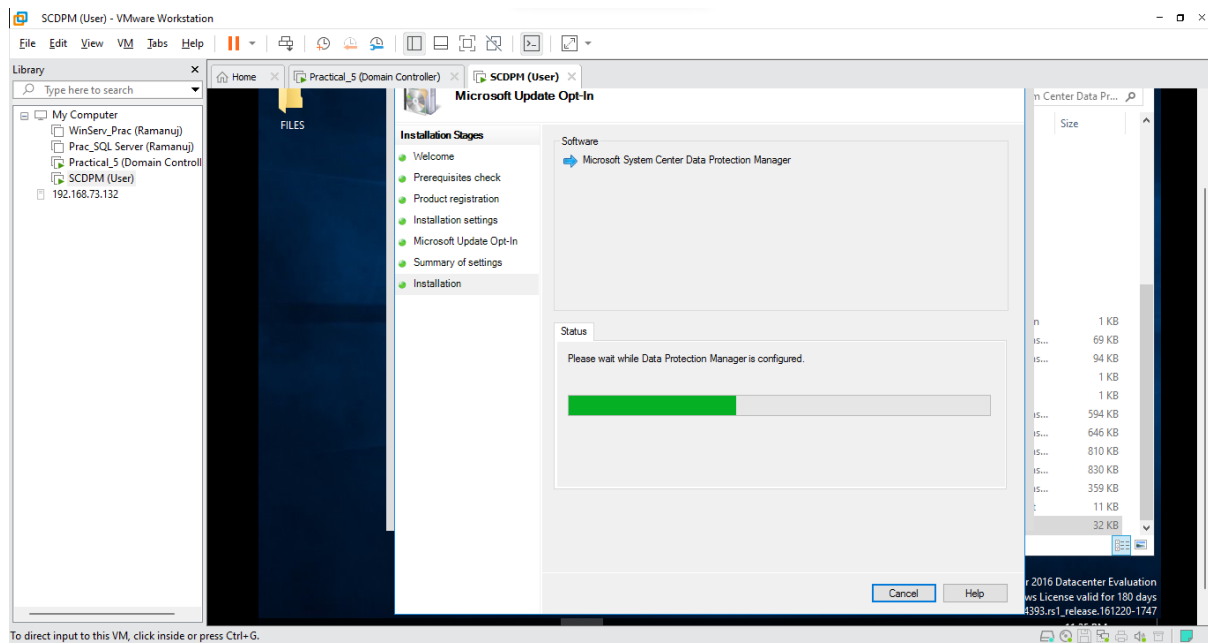
- Select **I do not want Microsoft Updates** and Click Next



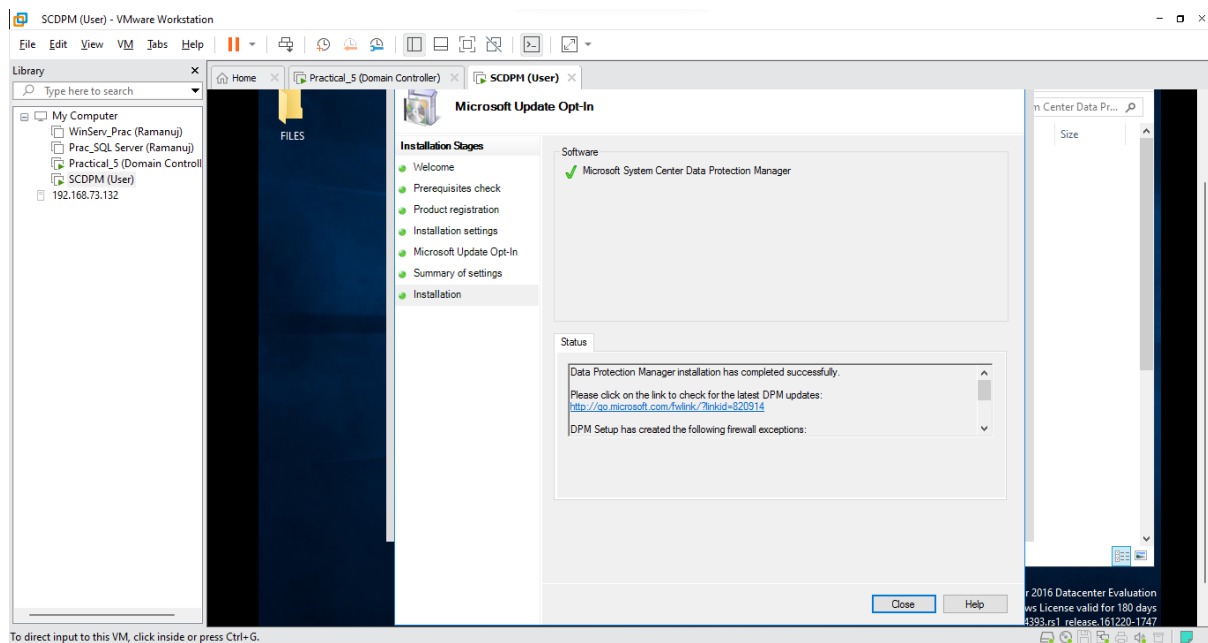
- Click on **Install**



- DPM will start installing



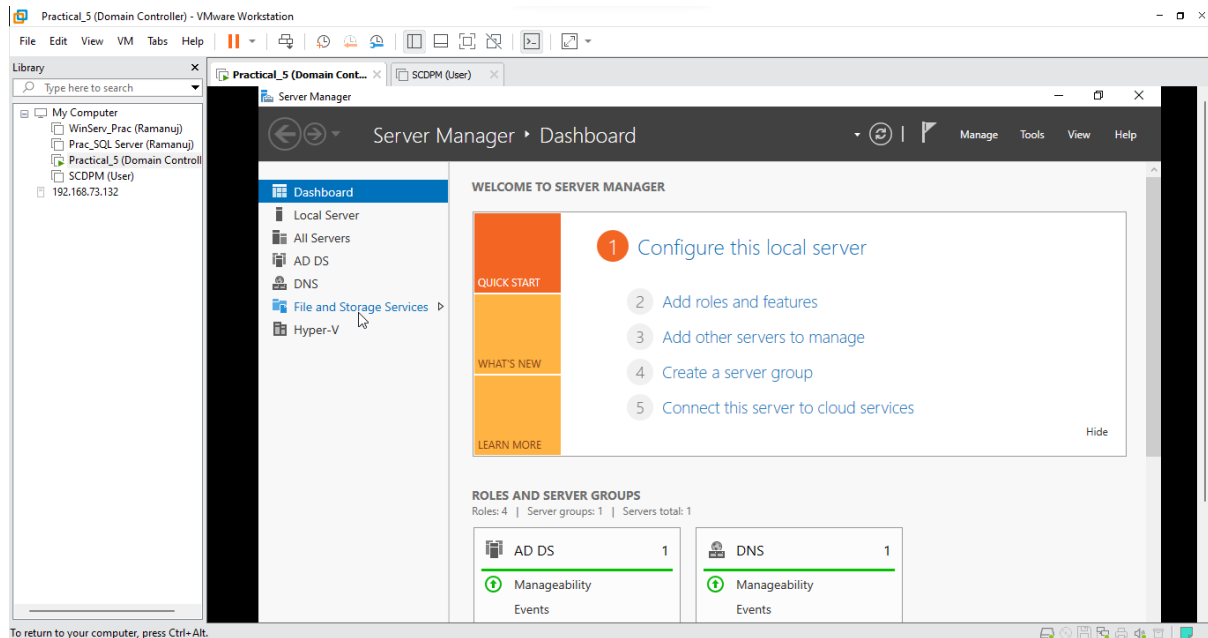
- Once DPM is installed Click **Close**



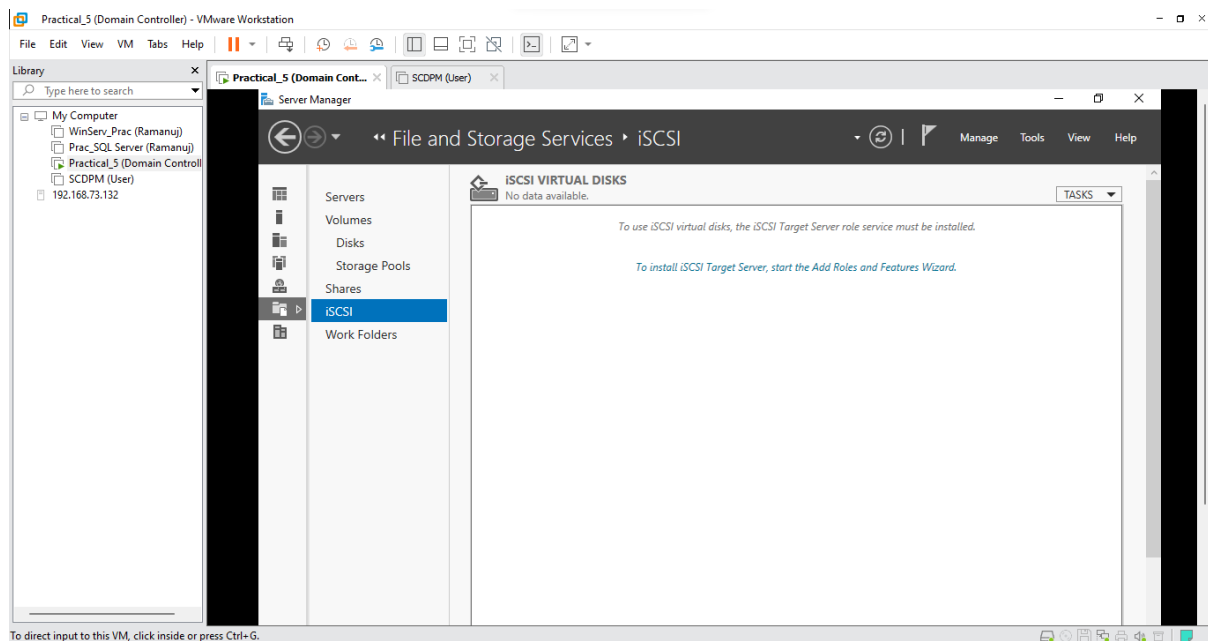
Step 7: Creating and initializing iSCSI Disk Storage

The following steps have to be executed in the Domain Controller:

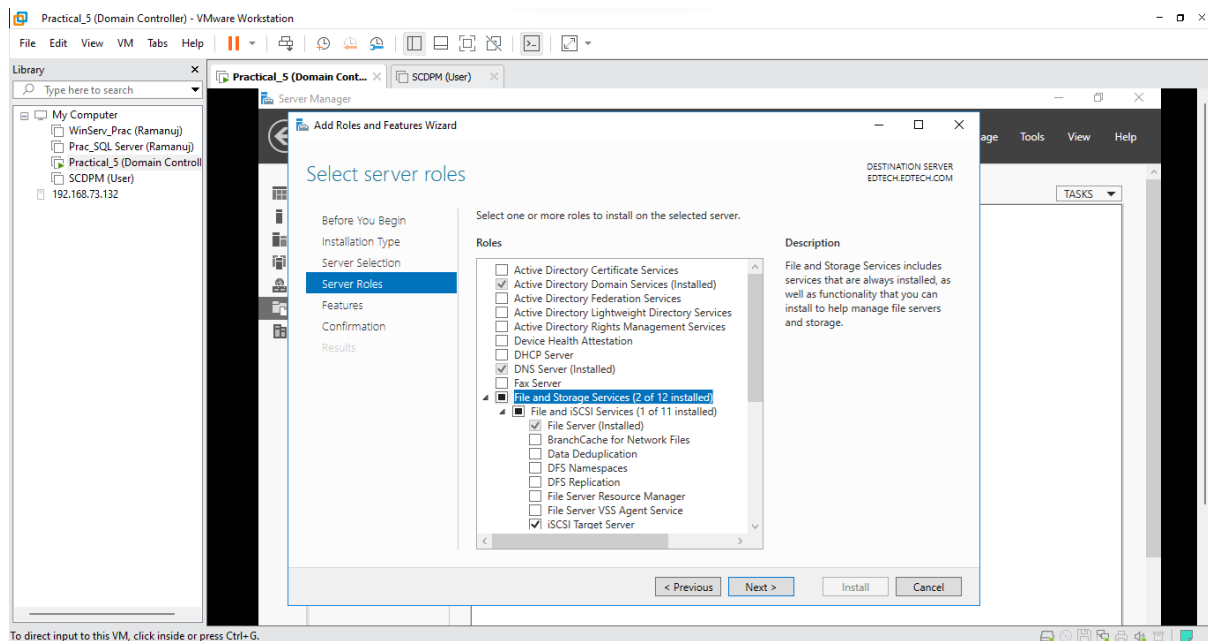
- Open **Server Manager** and Click on **File and Storage Services**



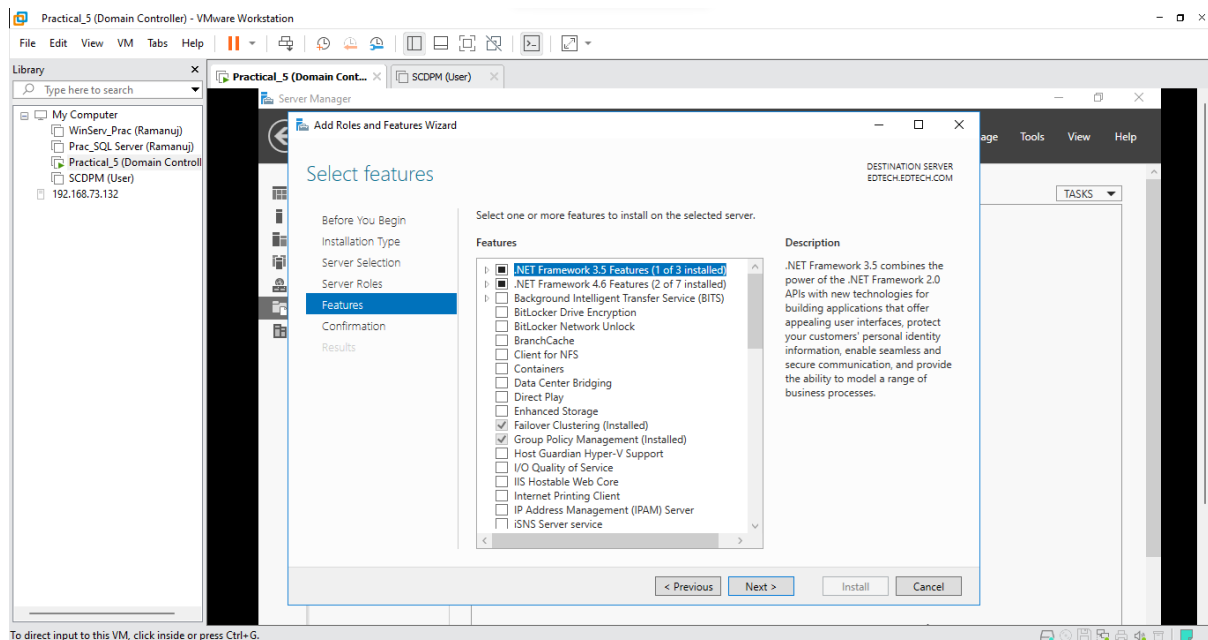
- Within File and Storage Services Click on **iSCSI** and Click on **To install iSCSI Target Server, Start the Add Roles and Features Wizard**



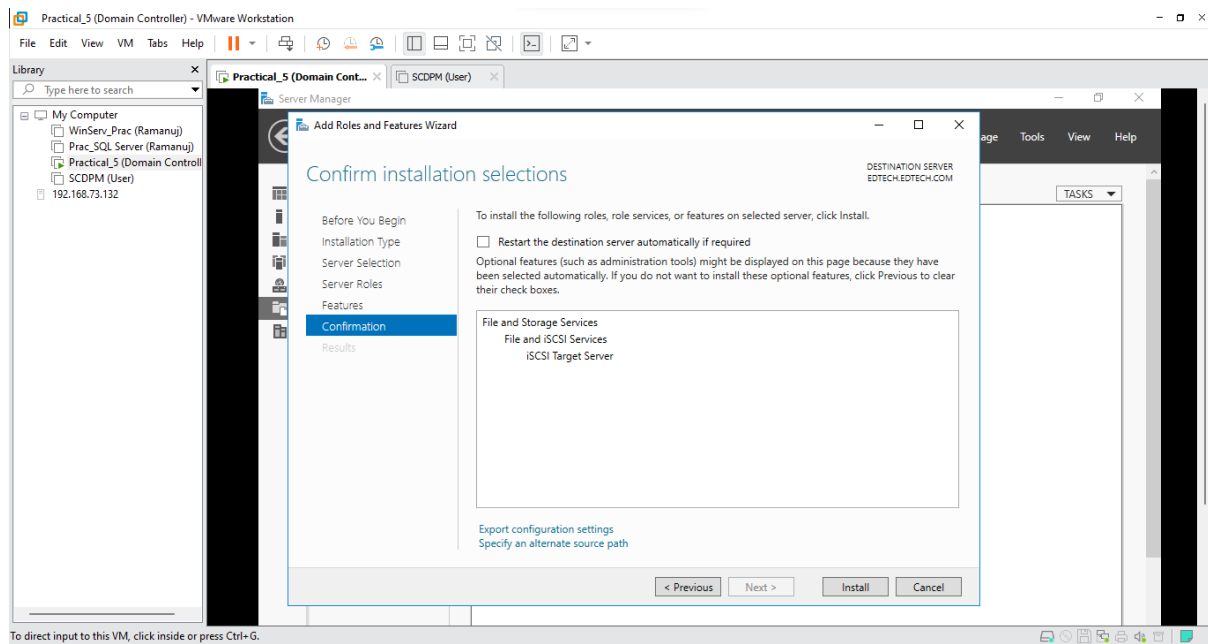
- After Click it will open the **Add Roles and Features Wizard**, Keep the default values and Click **Next**



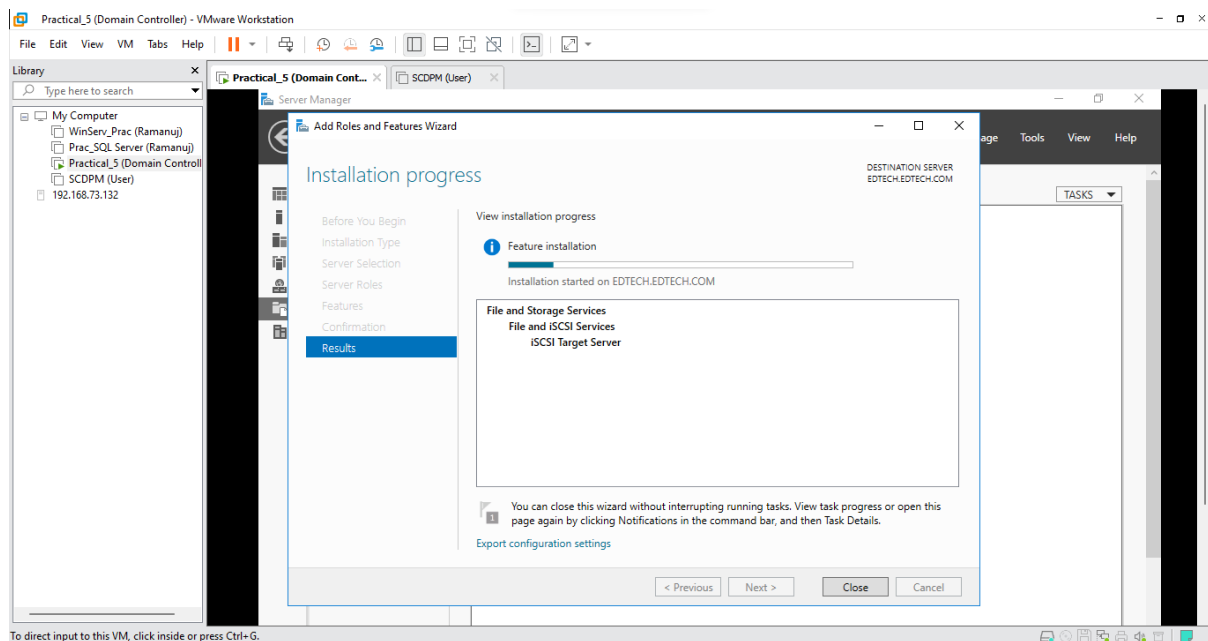
- Keep default values and Click **Next**



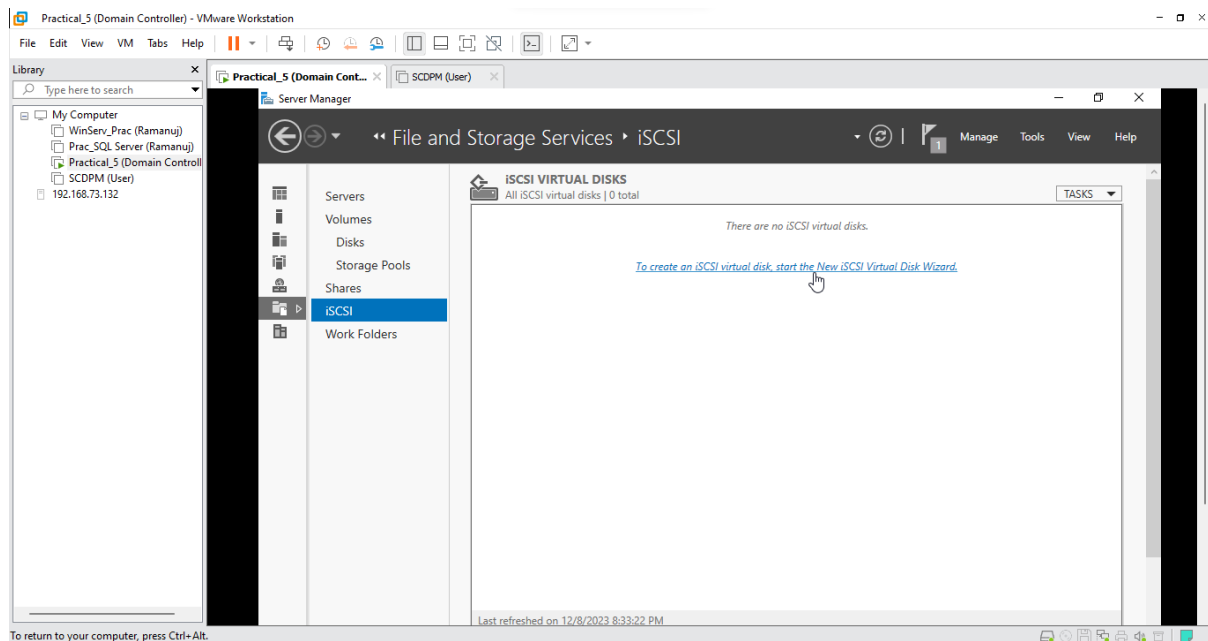
- Click on **Install**



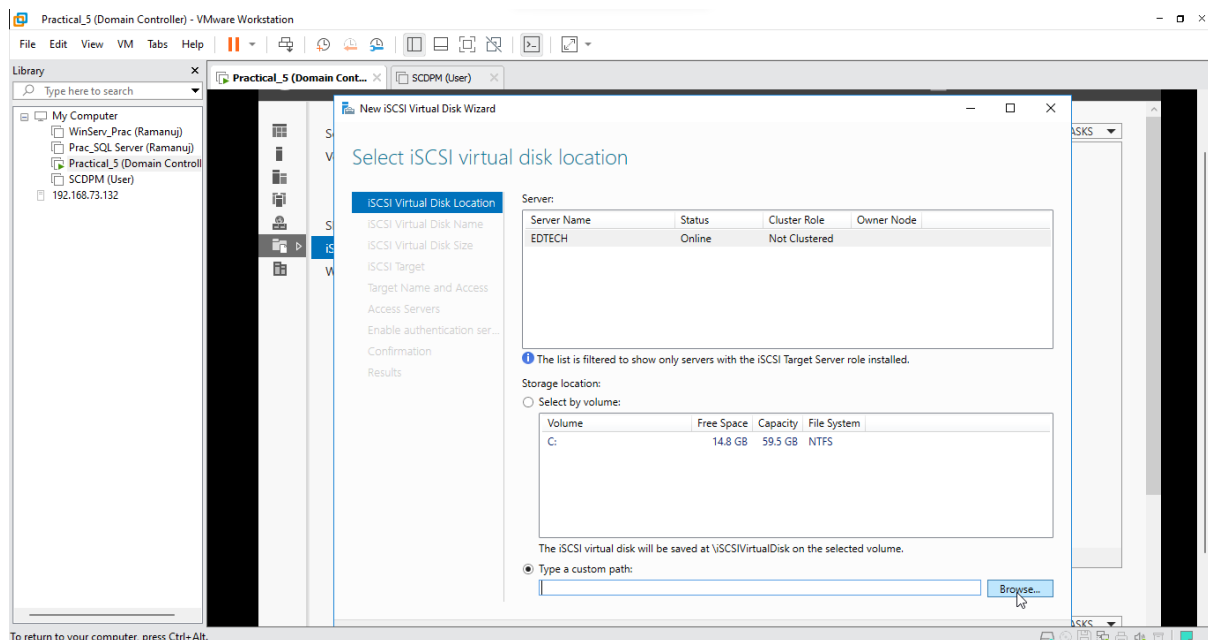
- Features will start installing



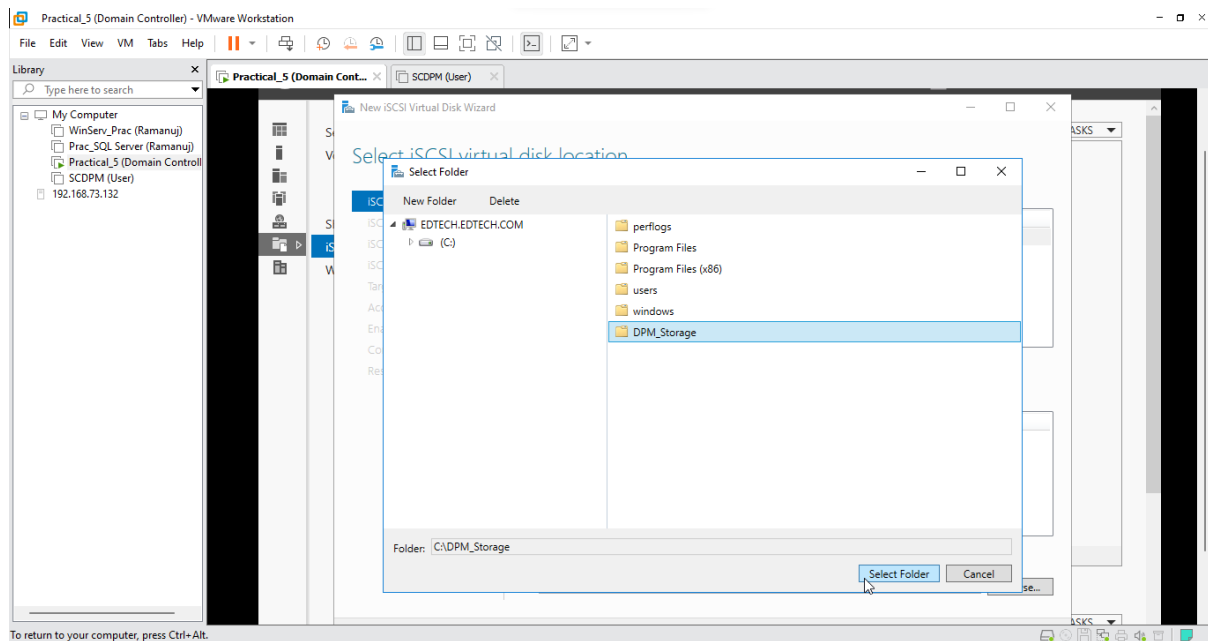
- After the installation is completed Click on **To create an iSCSI virtual disk, start the New iSCSI Virtual Disk Wizard**



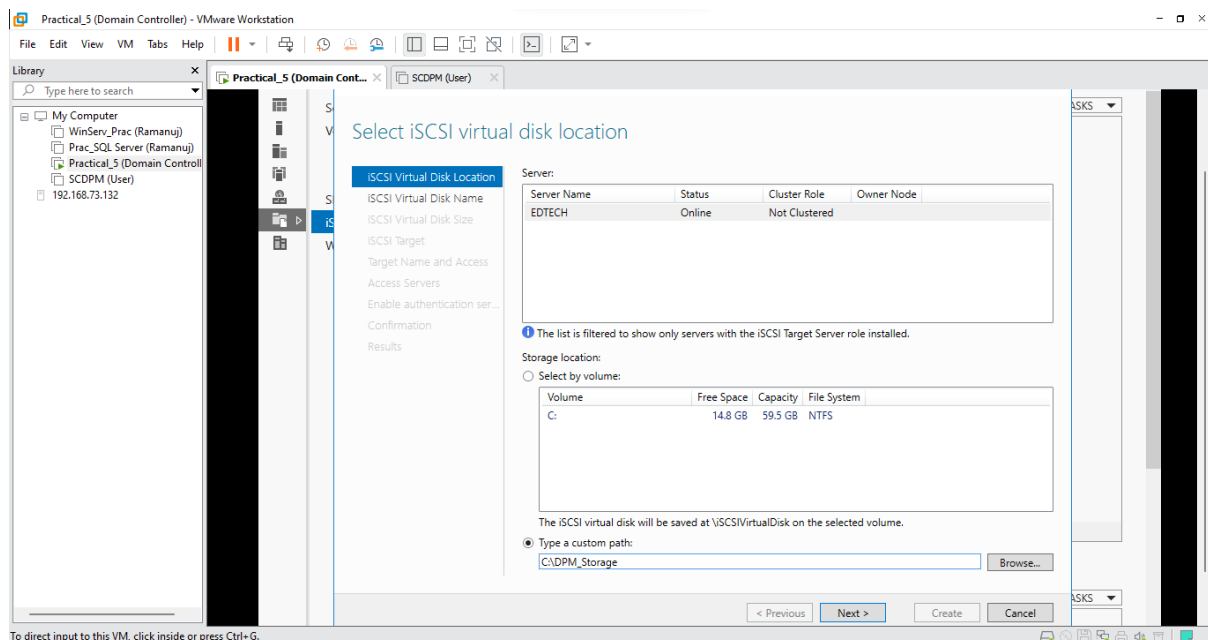
- Select **Type a custom path** and Click on **Browse**



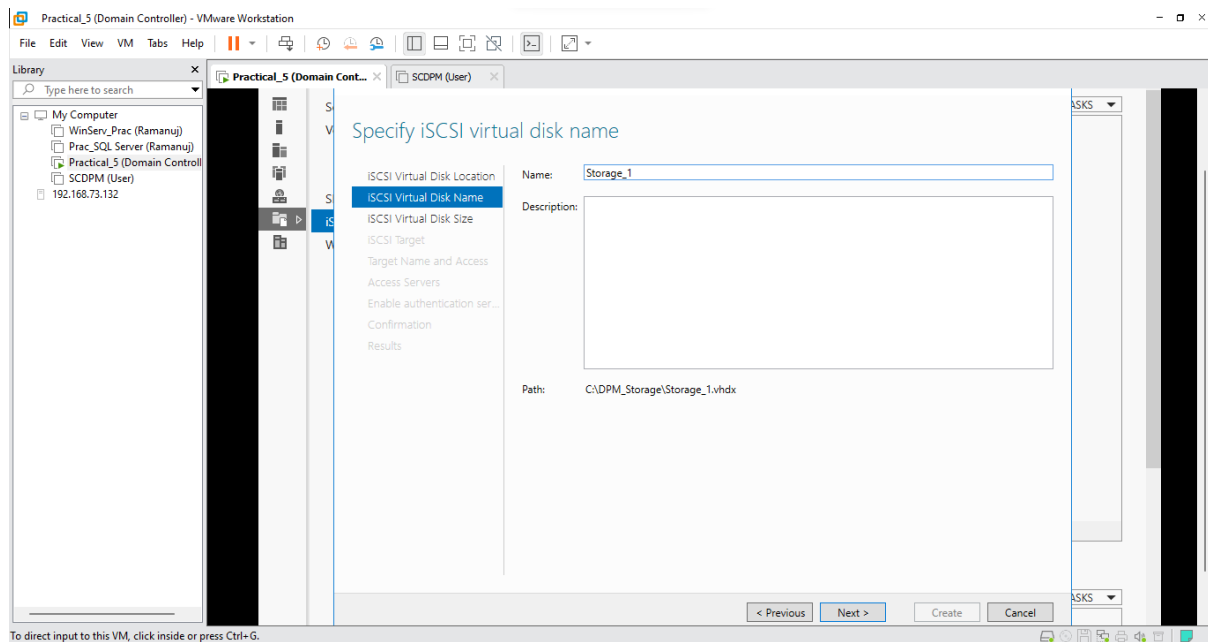
- Within Browse create a new folder and give it a name (Here it is DPM_Storage) and Click on **Select Folder**



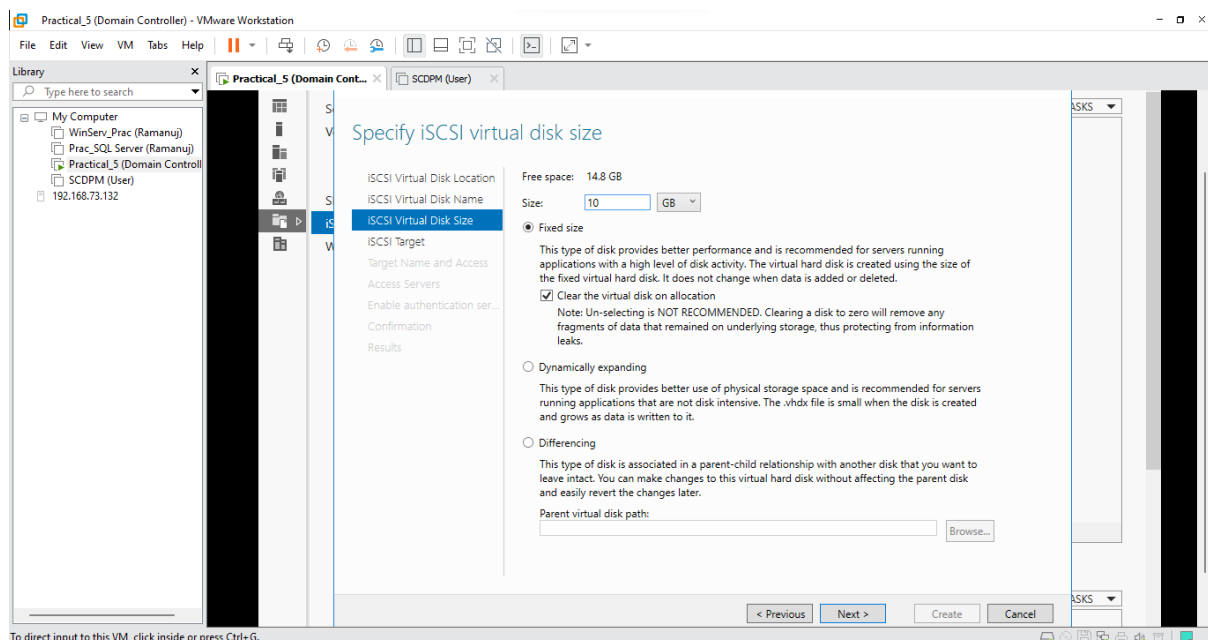
- Click Next



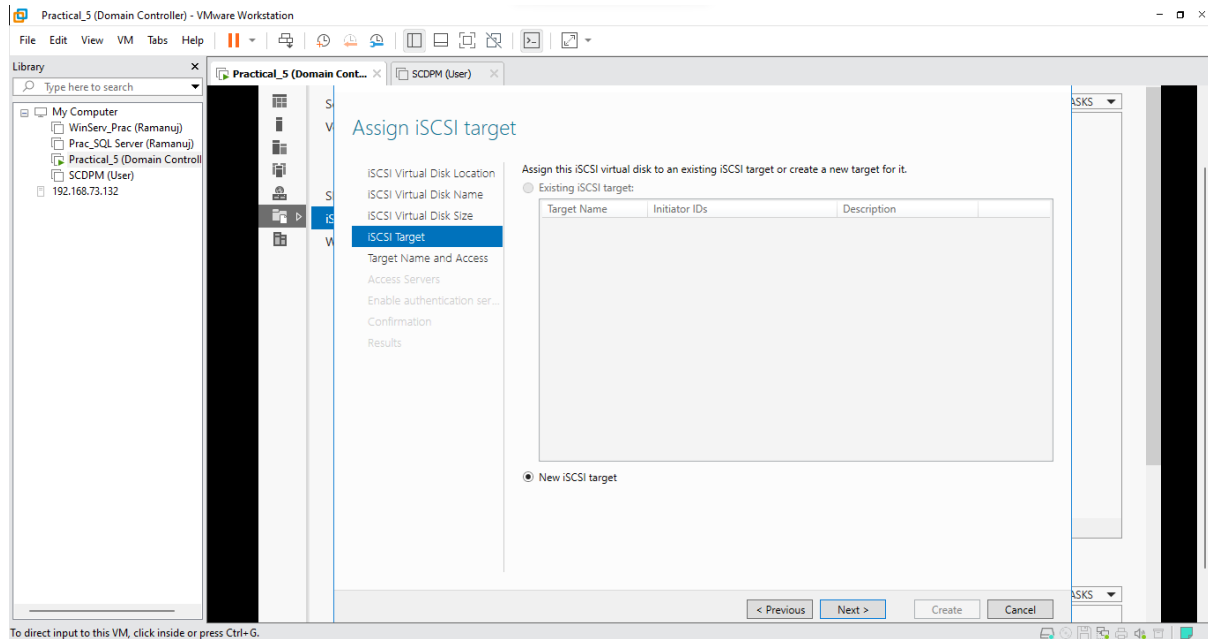
- Give your virtual disk a name (here it is Storage_1) and Click **Next**



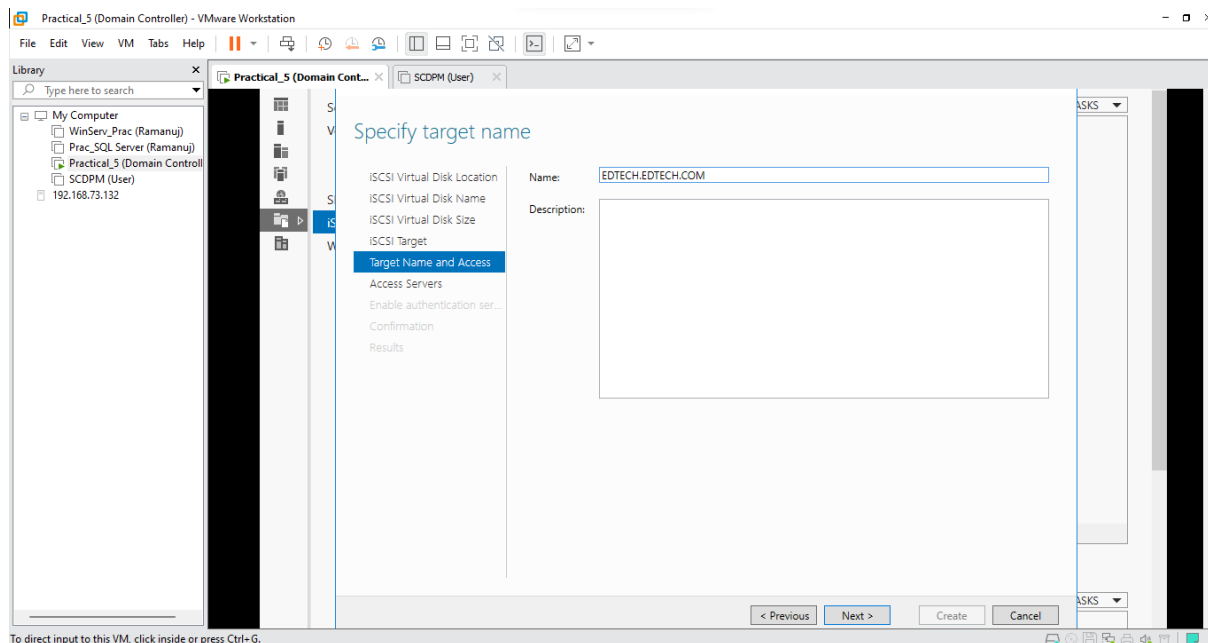
- Type in **10 GB** for Size and Select **Fixed Size** and Click **Next**



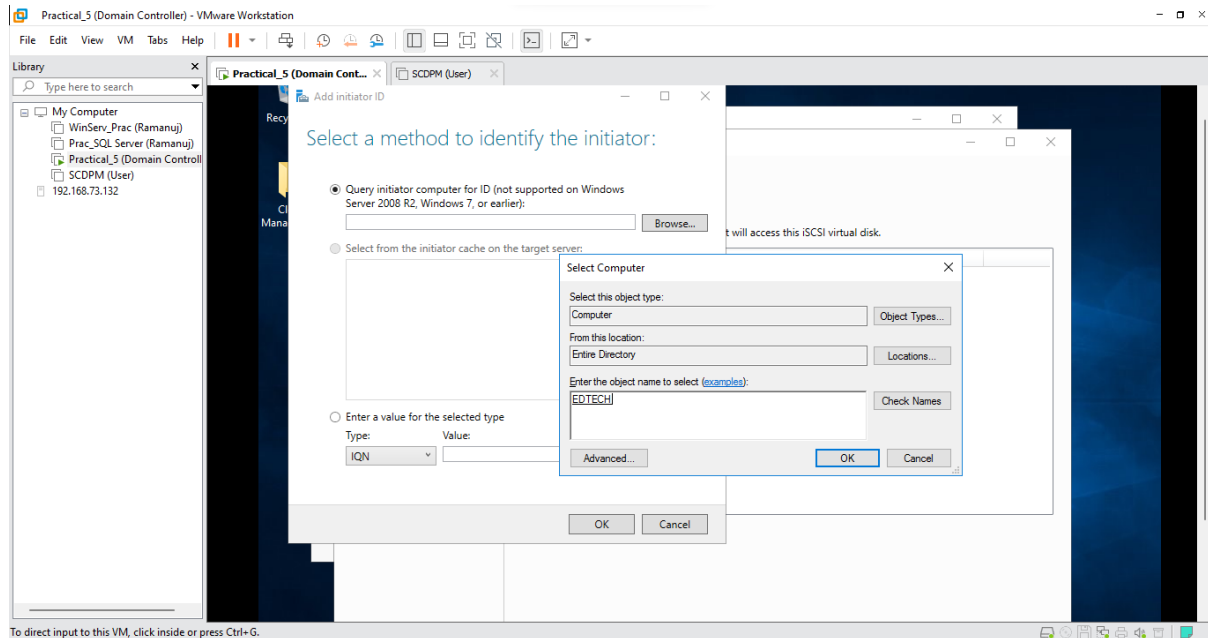
- **Select New iSCSI Target and Click Next**



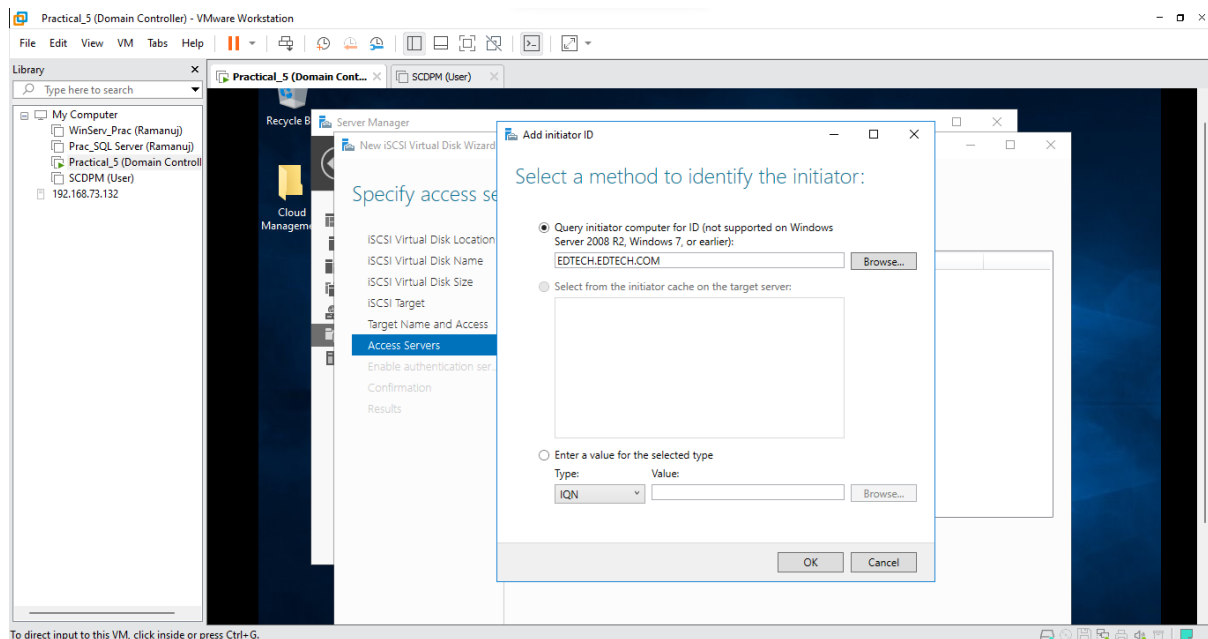
- In Target Name type in your Domain Name (here it is EDTECH.EDTECH.COM) and Click Next



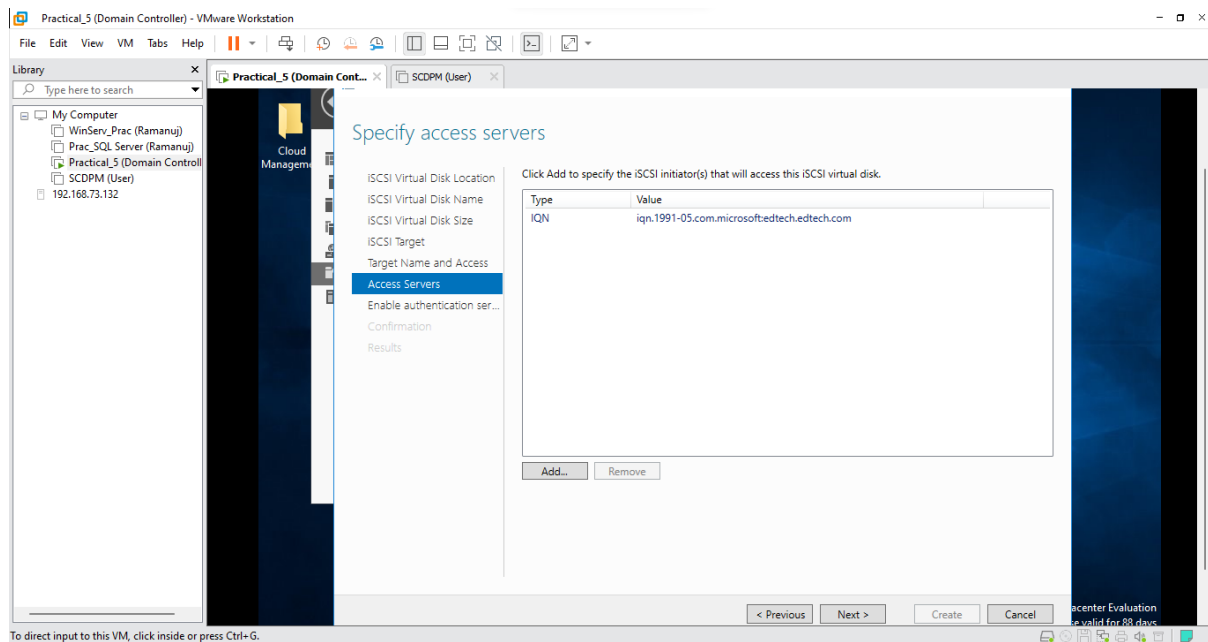
- Click on **Add** and Click on **Browse**, Type in the Name of your Domain Controller (Here it is EDTECH) and Click on **Check Names**, then Click **Ok**



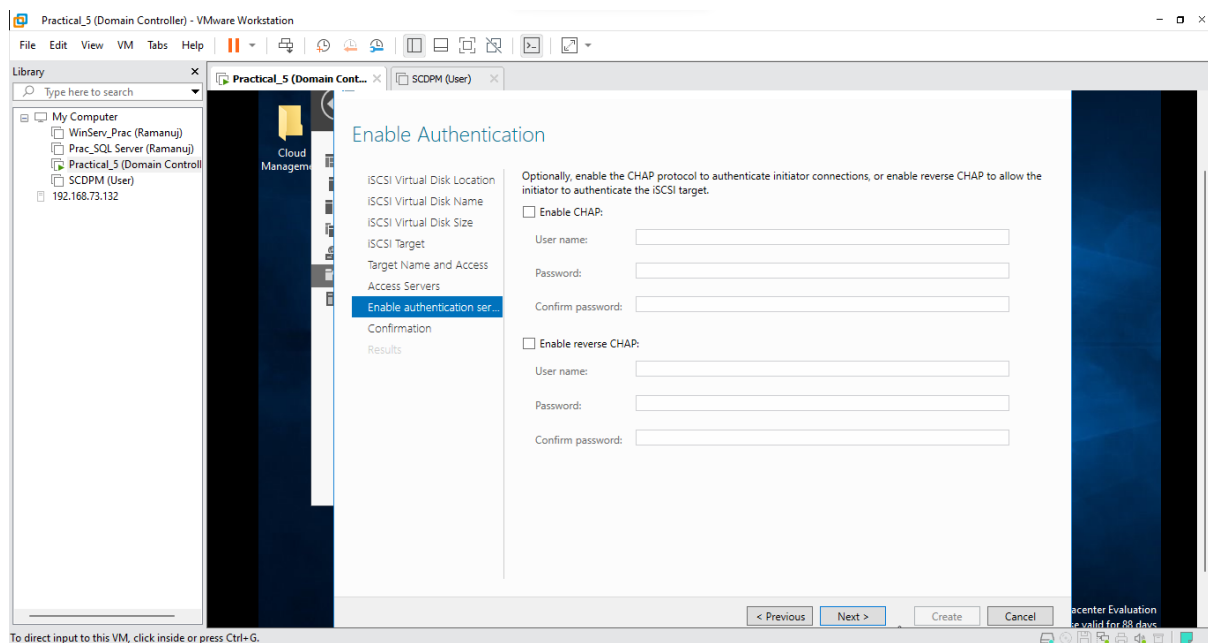
- Click **Ok**



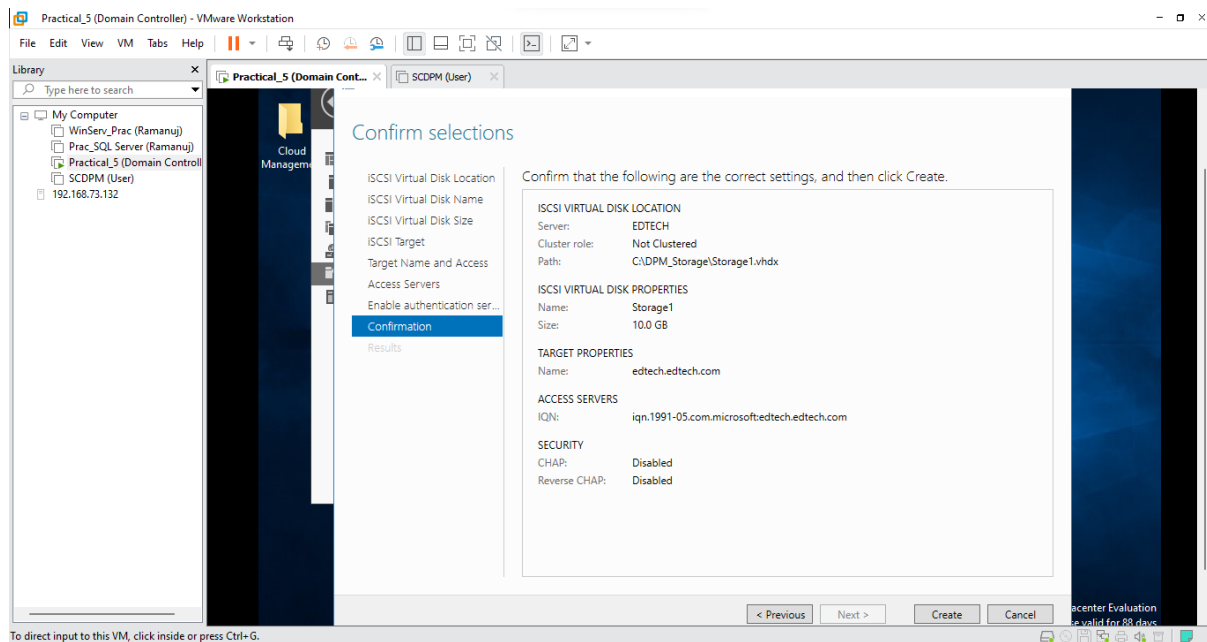
- Click Next



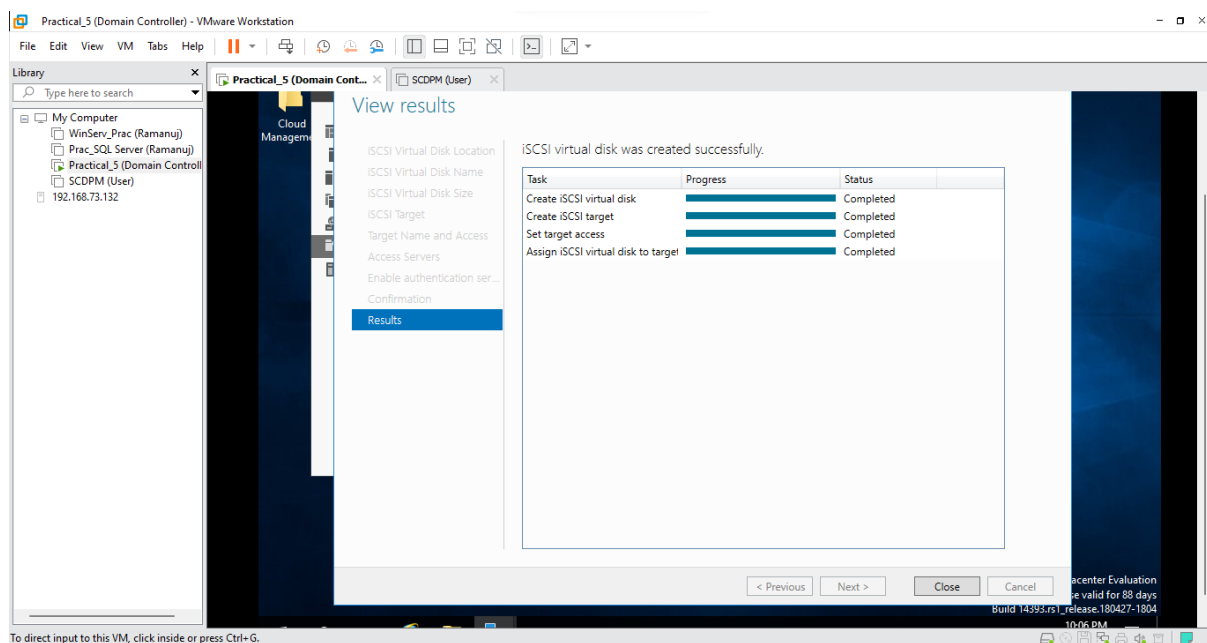
- Keep default values and Click Next



- Review the summary of your iSCSI instance and Click **Create**

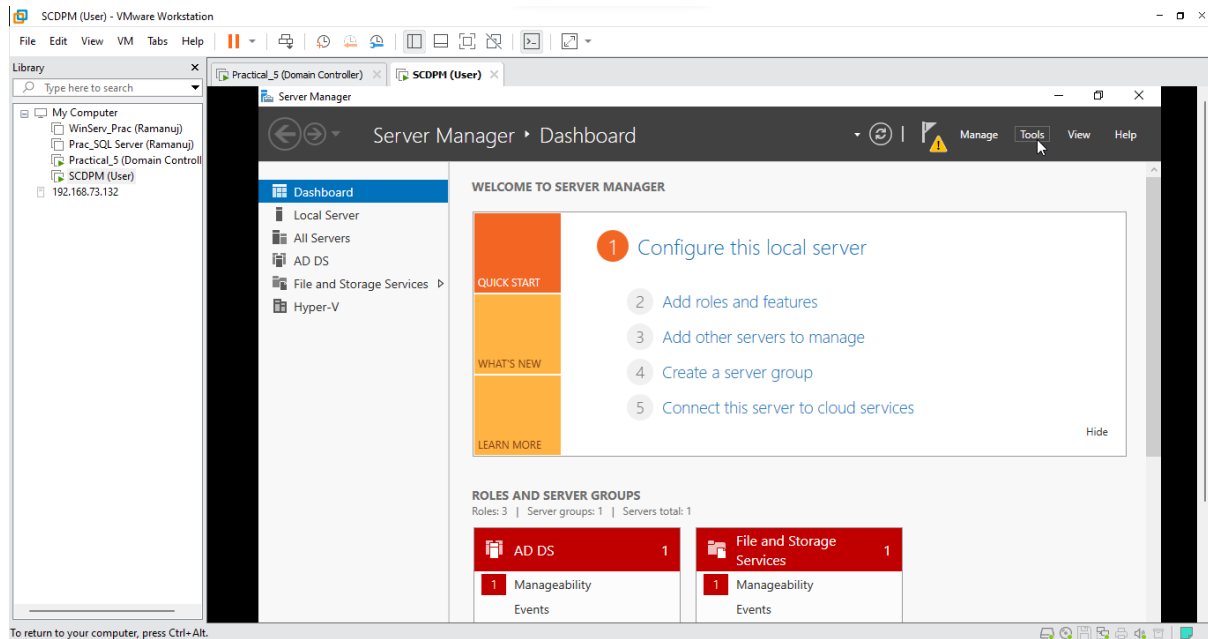


- Your iSCSI instance has been created successfully

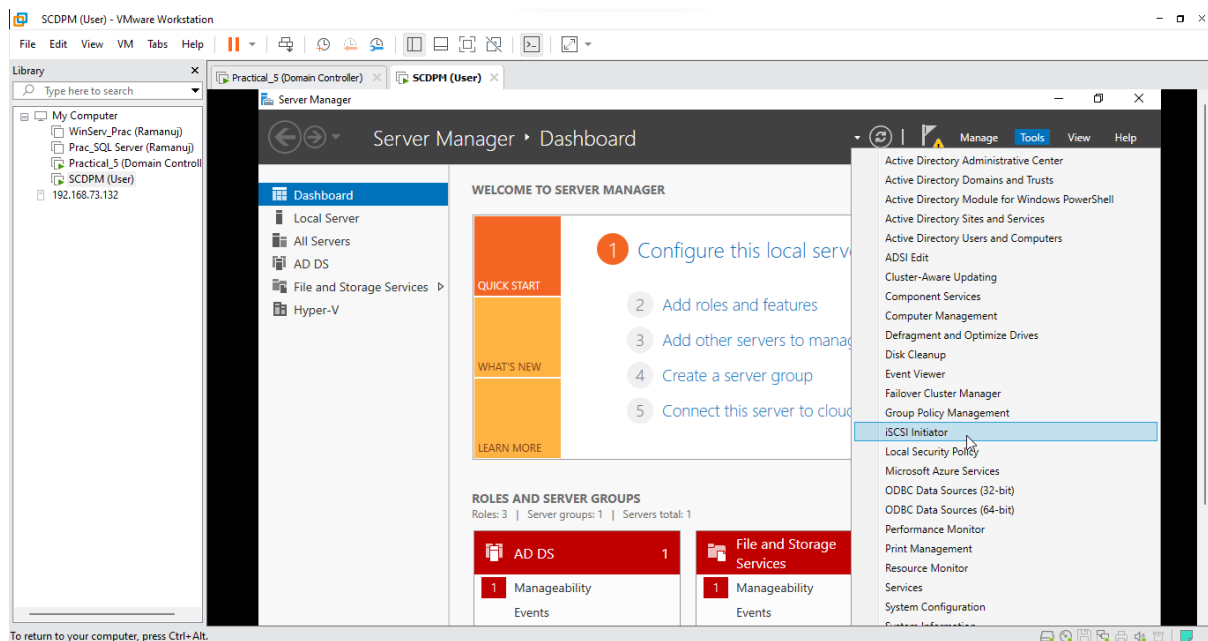


Note: The following steps are for the **SCDPM VM** instance, where we will initialize the iSCSI instance created in the domain controller. Ensure that both VM are running

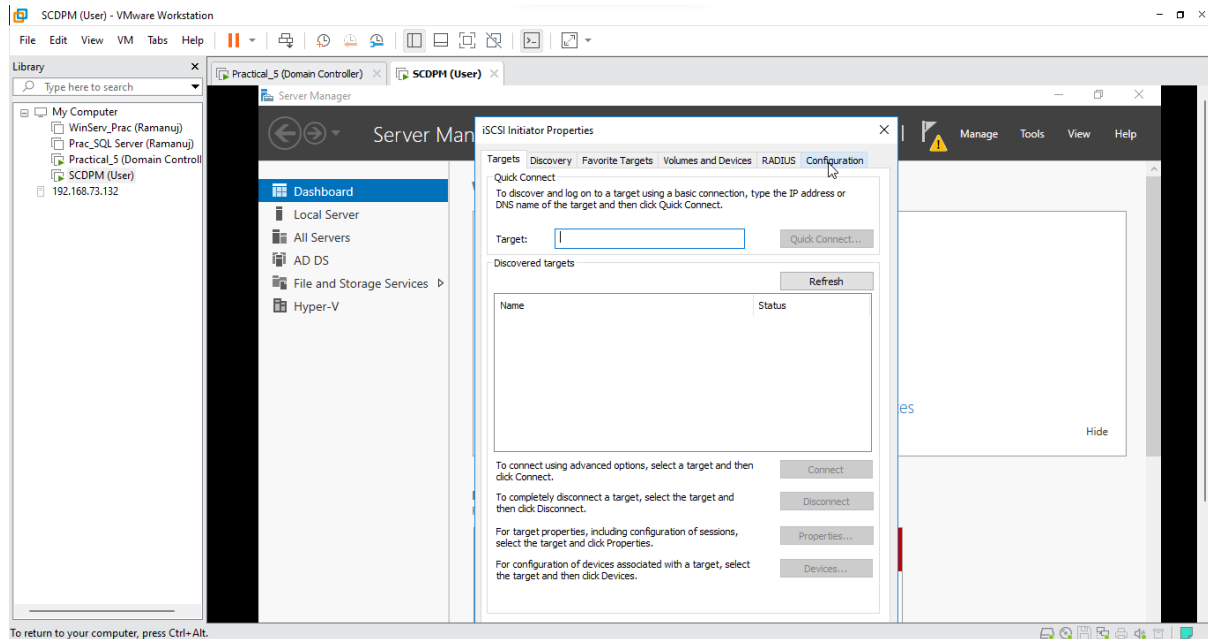
- Open **Server Manager** and Click on **Tools**



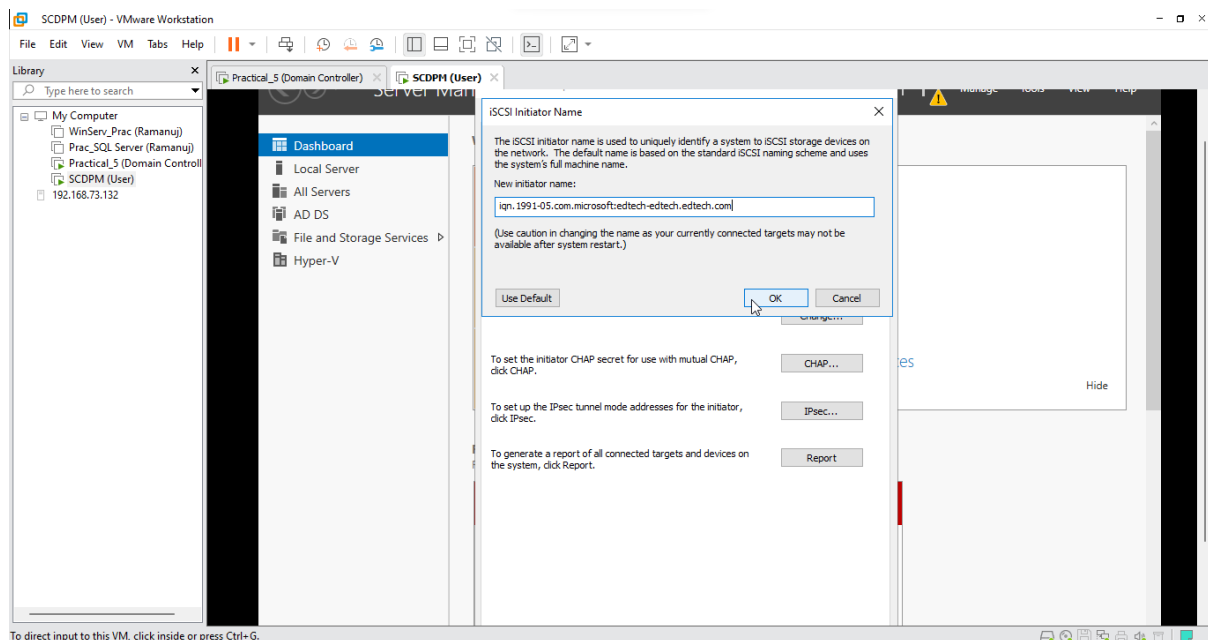
- Within Tools Click on **iSCSI Initiator**



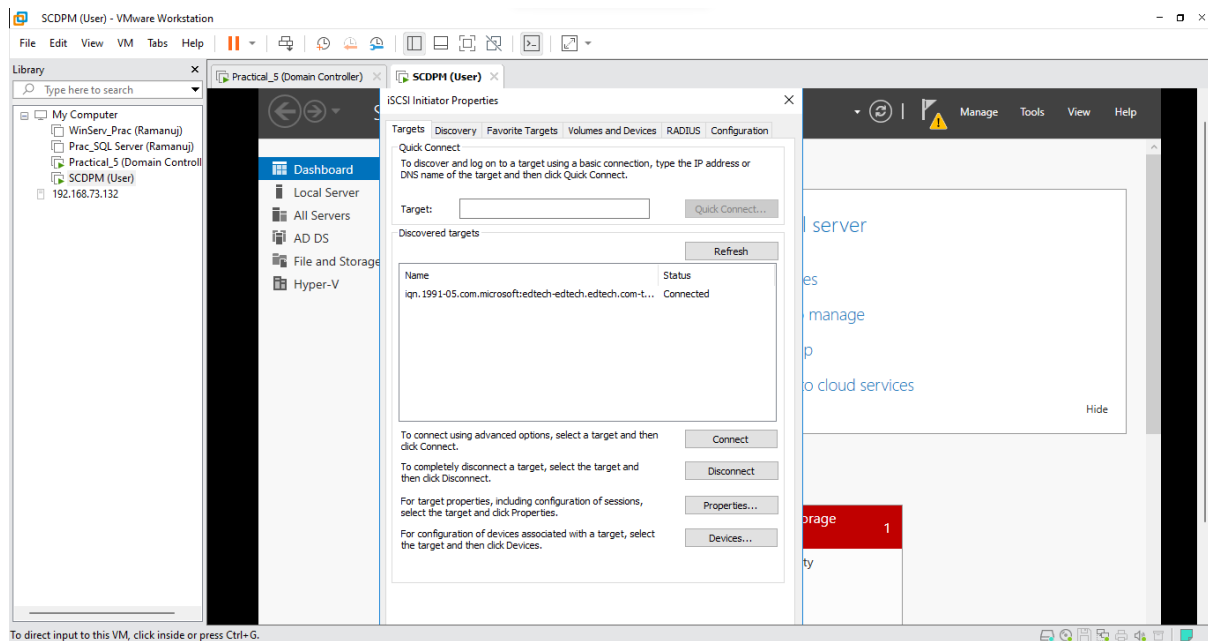
- Within iSCSI Initiator Click on **Configuration**



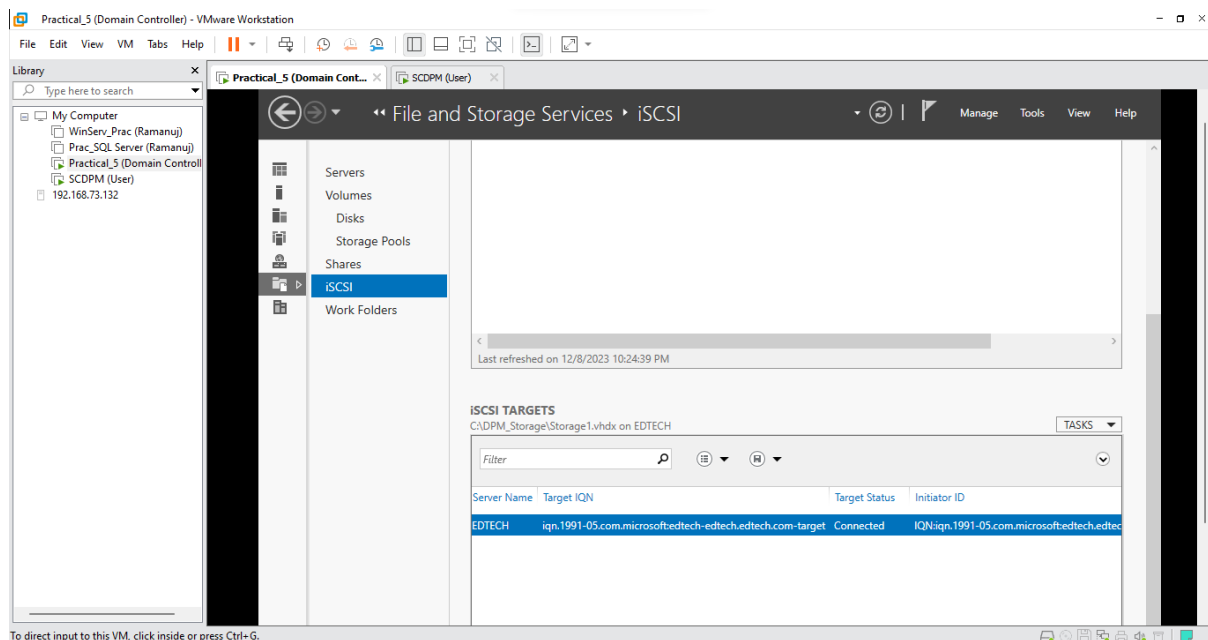
- Click on **Change** and paste in the IQN value from the Domain Controller and Click **Ok**



- Now Click on **Targets** and Enter the IP of the Domain Controller and Click on **Quick Connect**

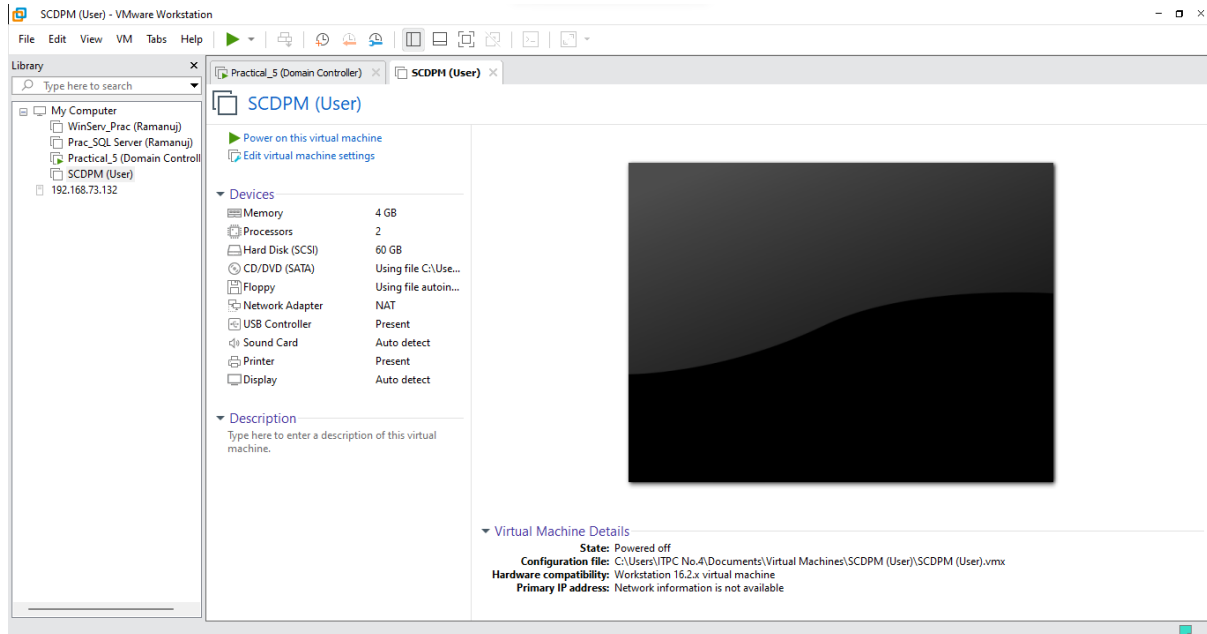


- Confirm the connection on the iSCSI Server on the Domain Controller

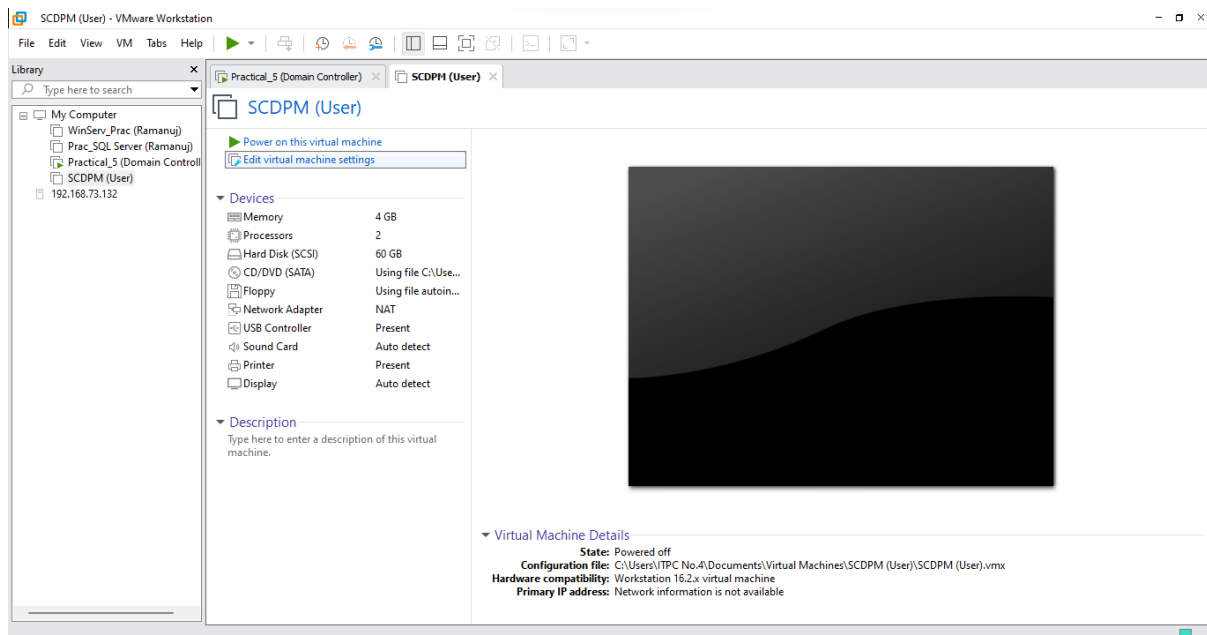


Step 8: Creating Disk Storage in SCDPM

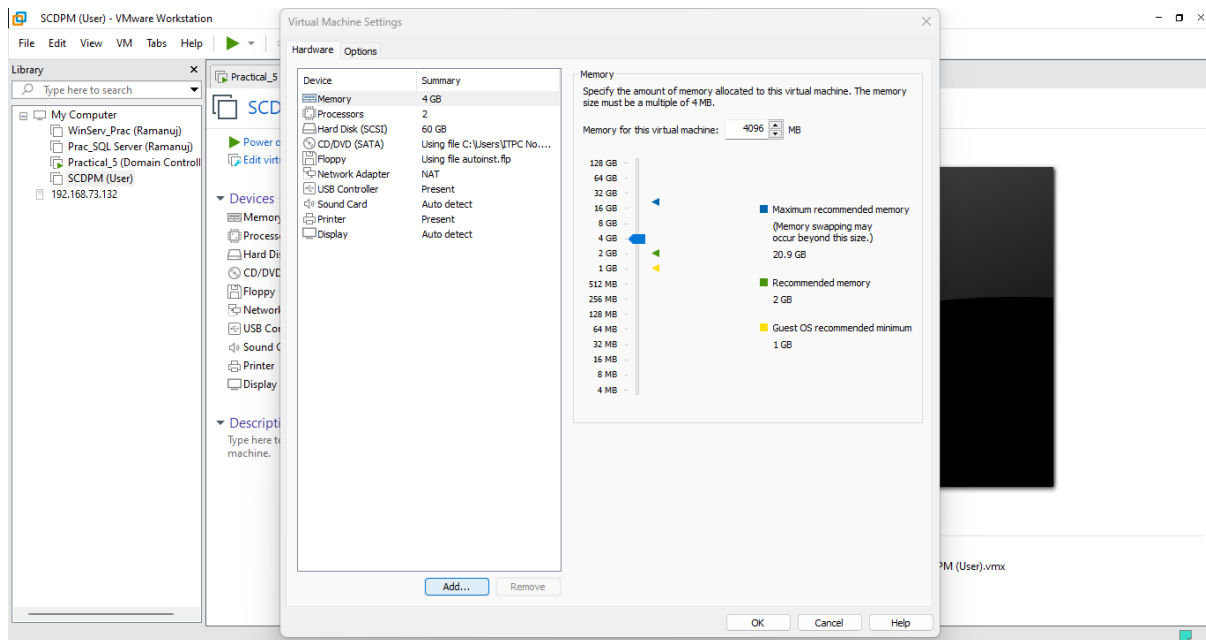
- There are a few modification that need to be made to the SCDPM VM
- Shutdown the VM



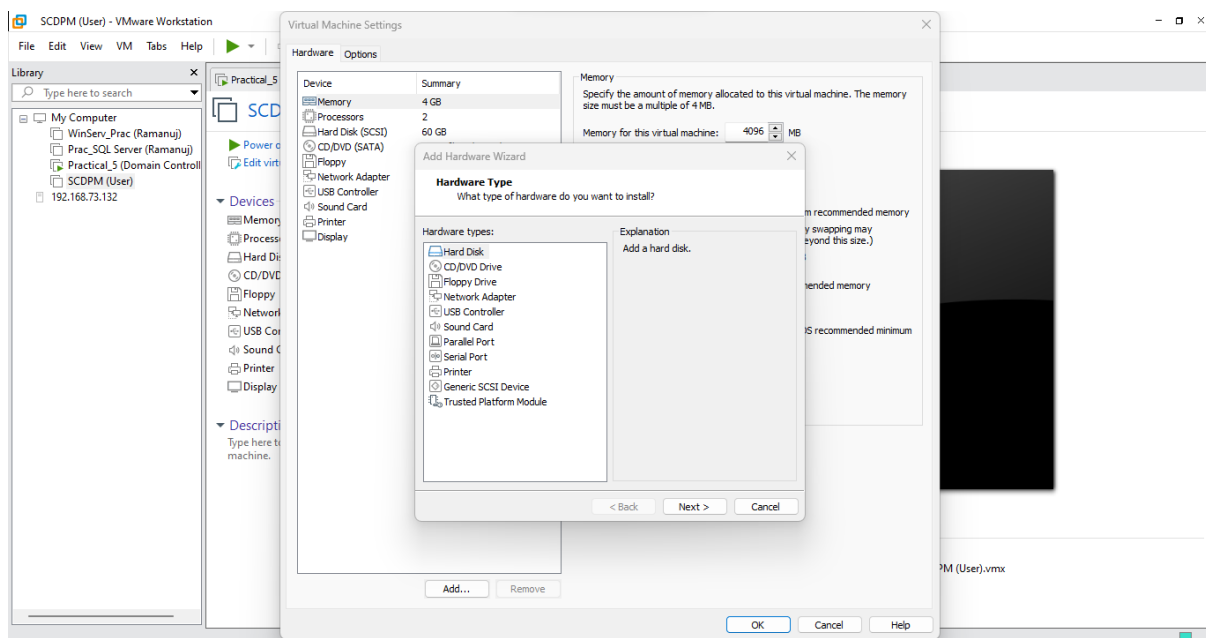
- Click on **Edit virtual machine settings**



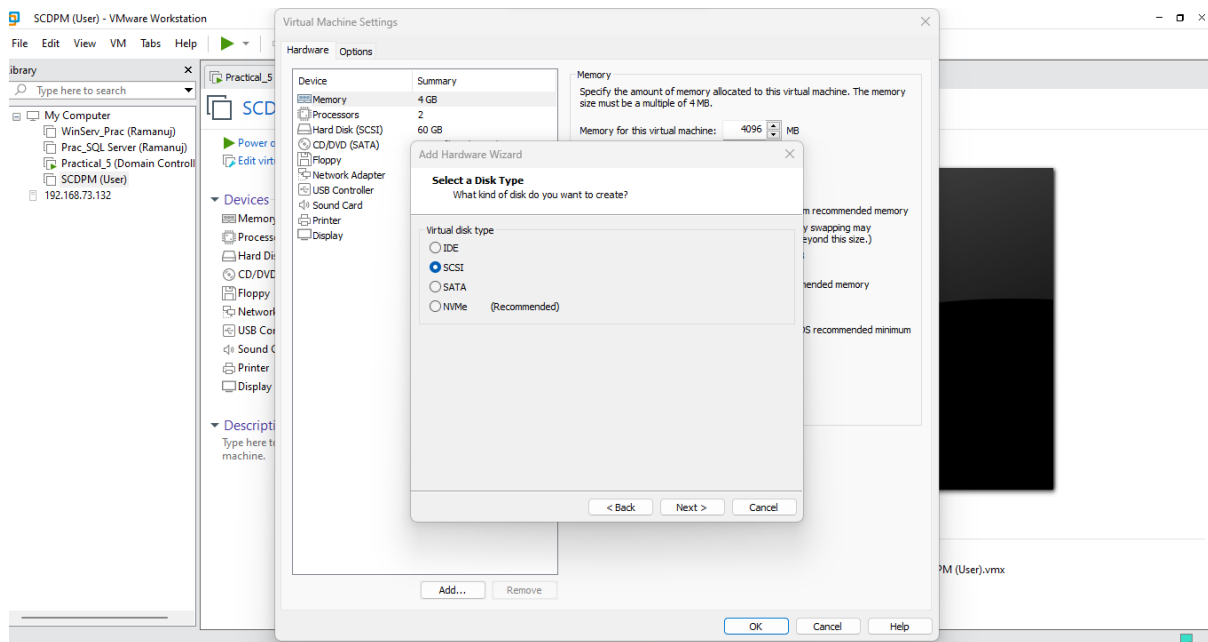
- Now within Edit virtual machine settings Click on **Add**



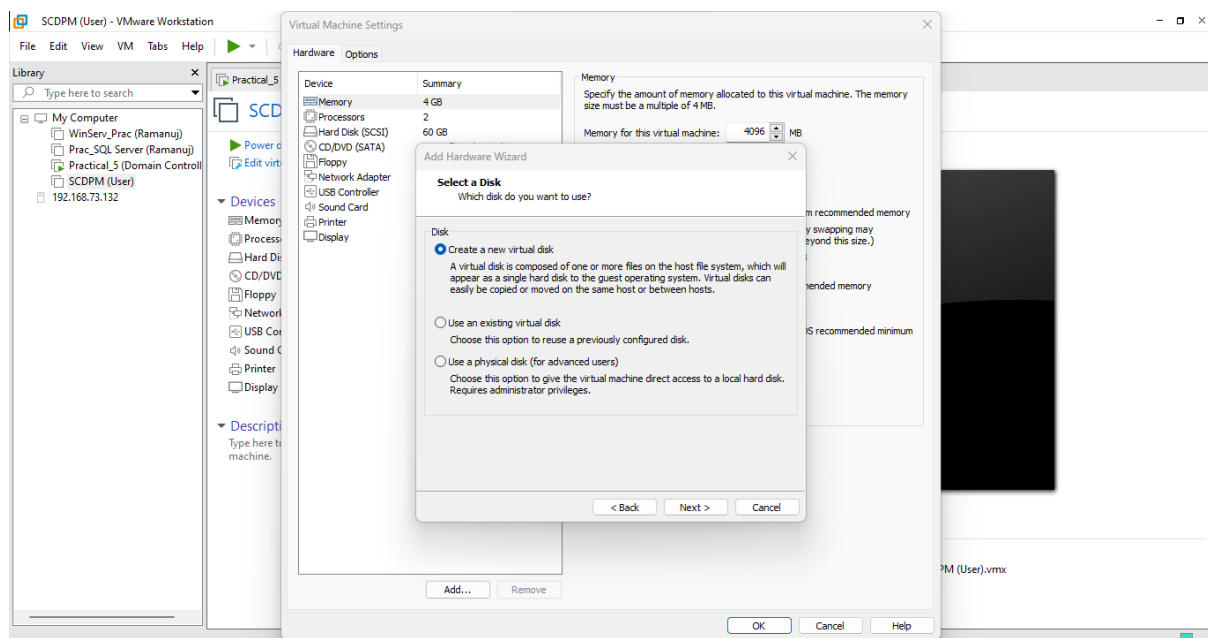
- Select **Hard Disk** and Click **Next**



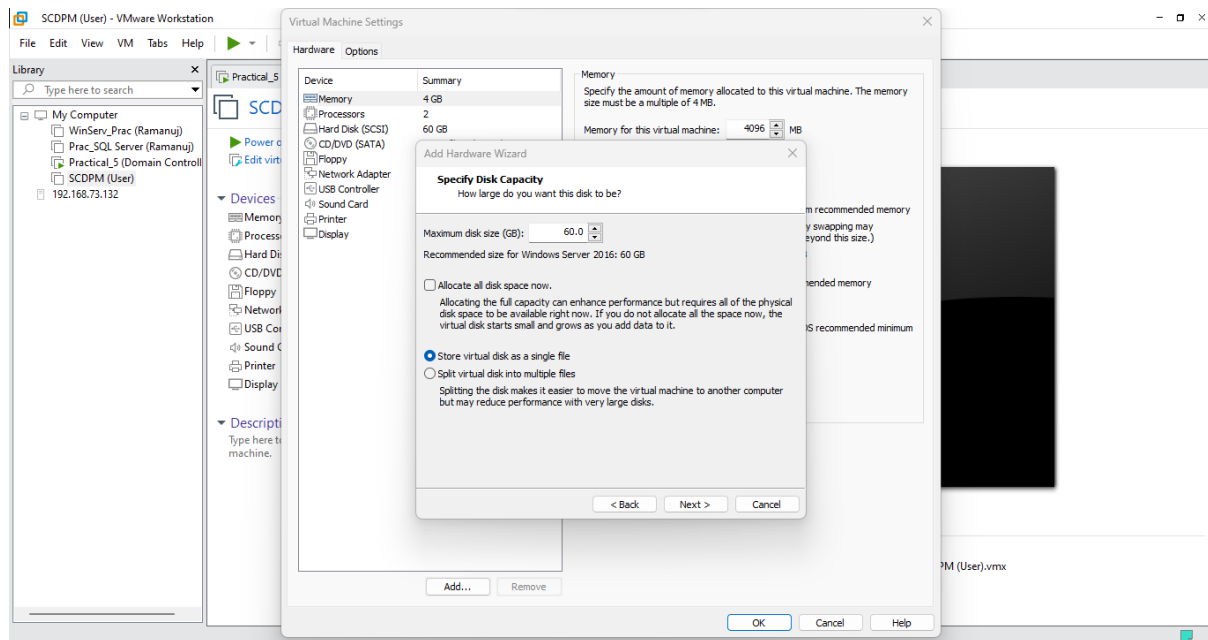
- Select **SCSI** and Click **Next**



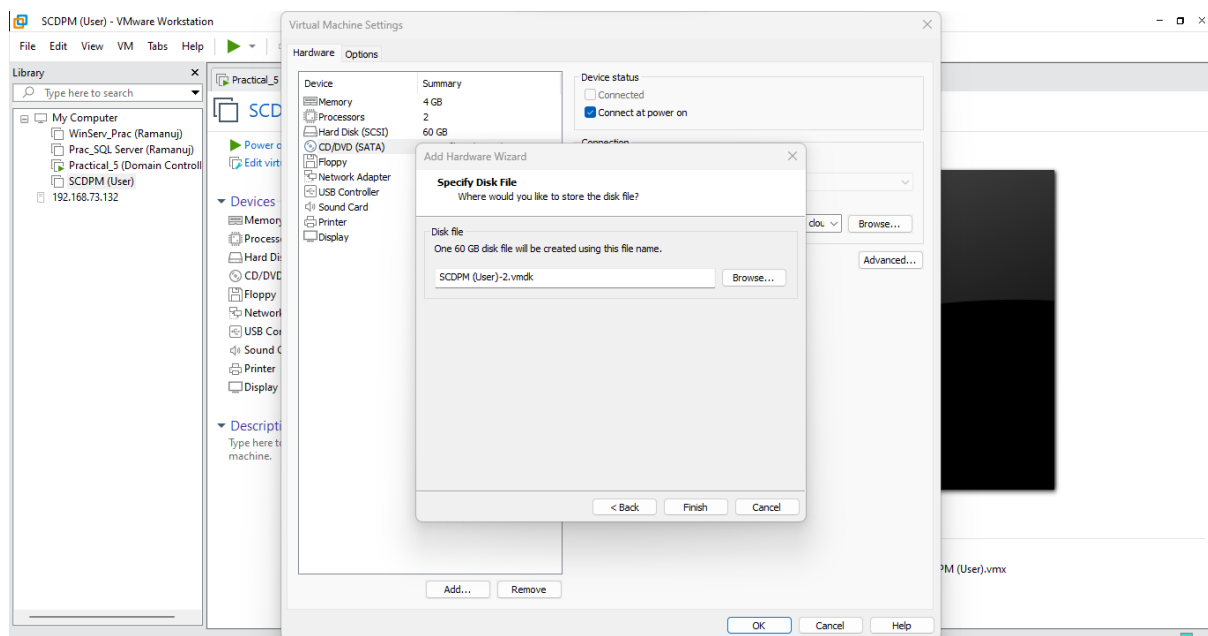
- Keep default values and Click **Next**



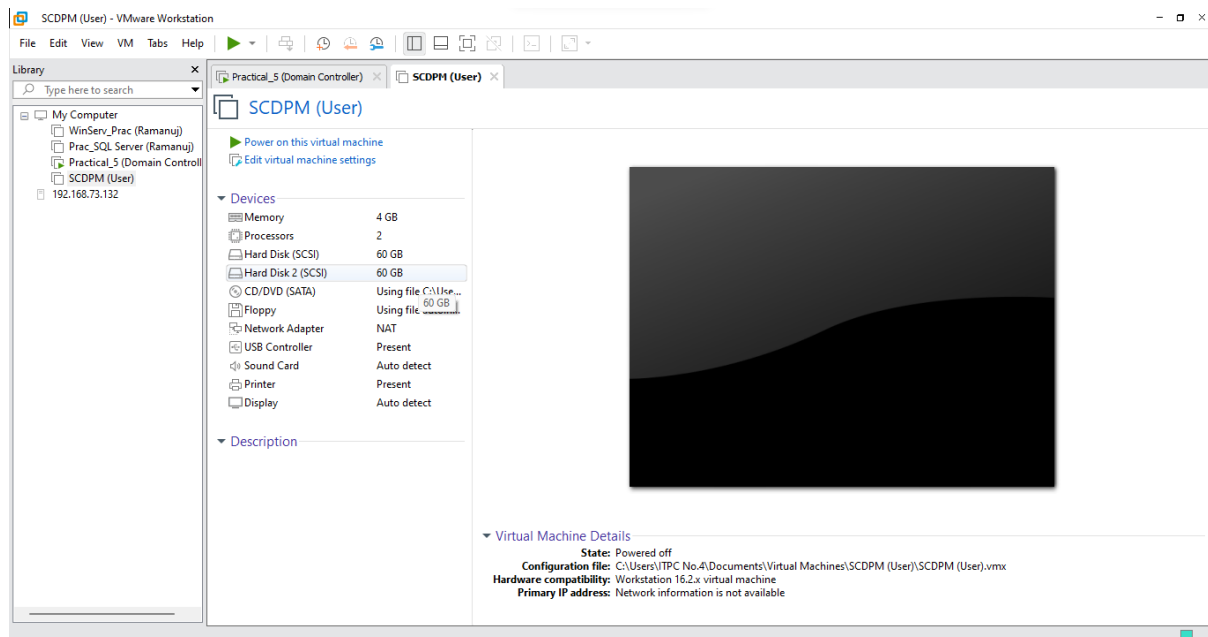
- Select **Store virtual disk as a single file** and Click **Next**



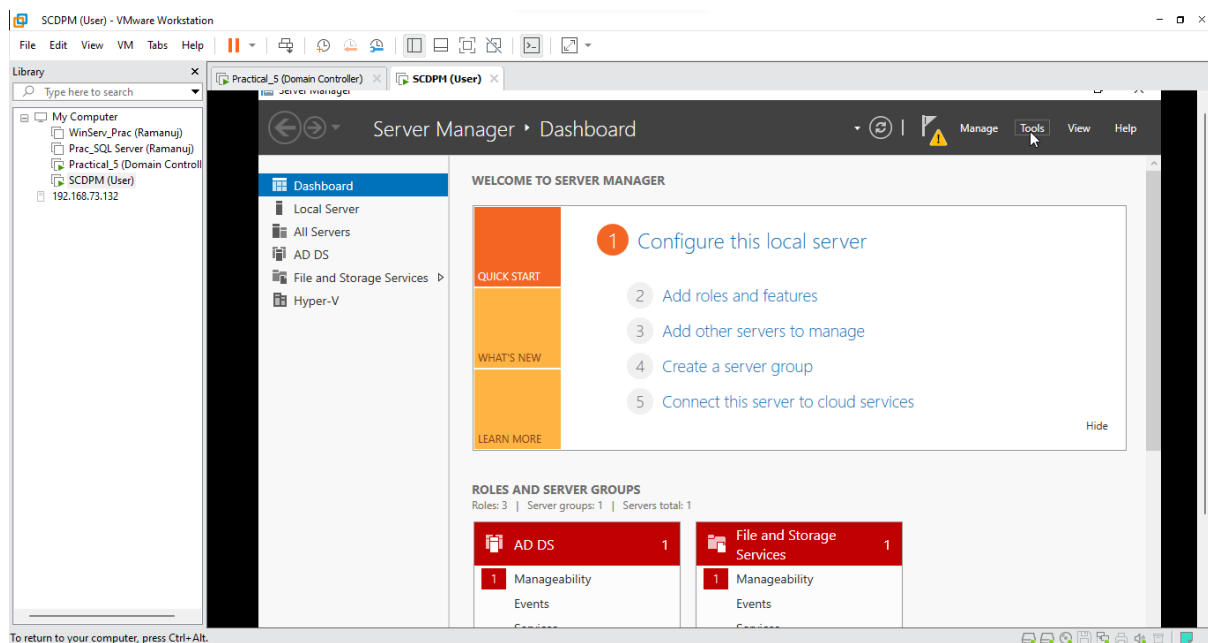
- Click on **Finish**



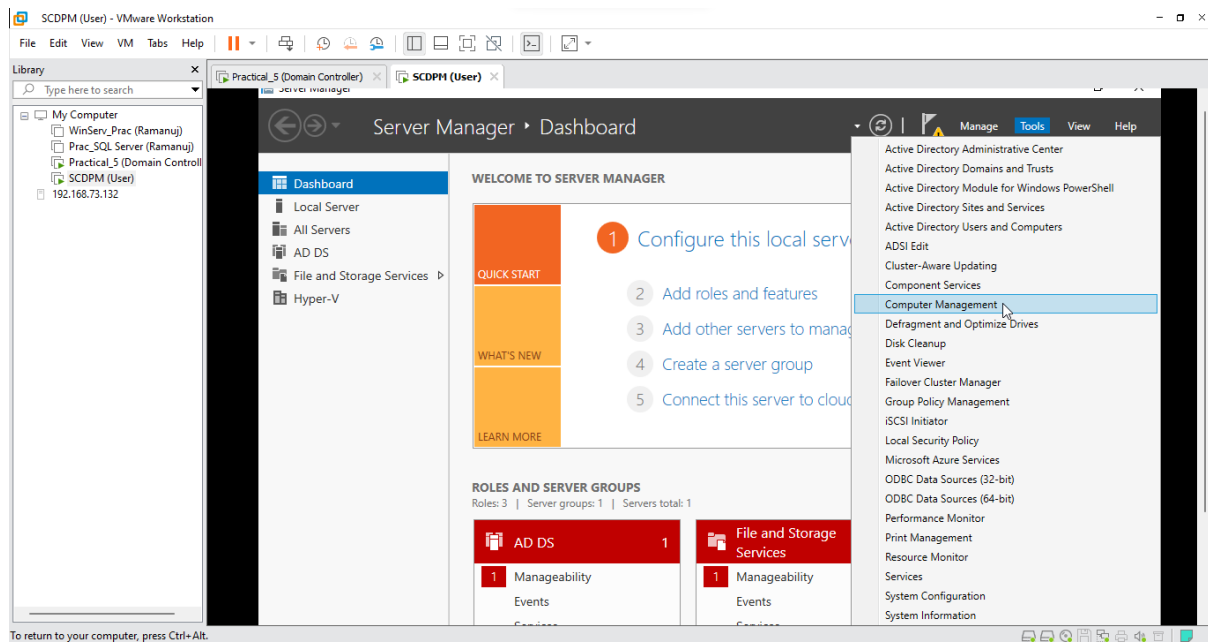
- Click on **Ok** and Check if the hard is detected by the VM



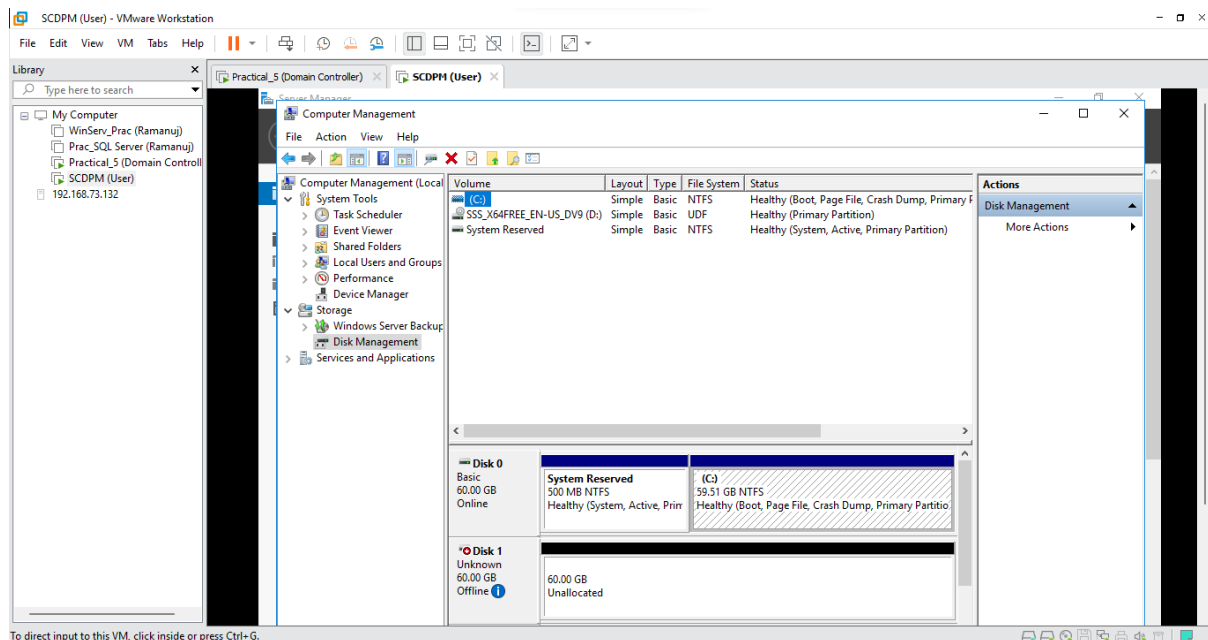
- Now Turn On the VM and Open **Server Manager** Click on **Tools**
- Note: This is to be done in the Administrator and not as SCDPM



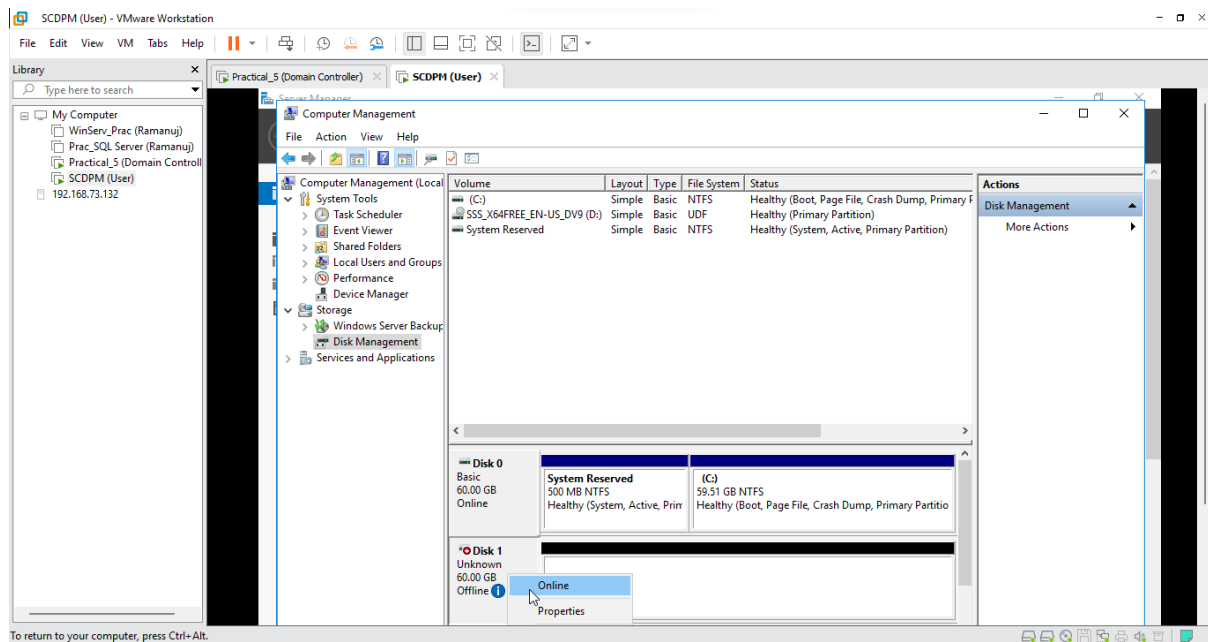
- In Tools Click on **Computer Management**



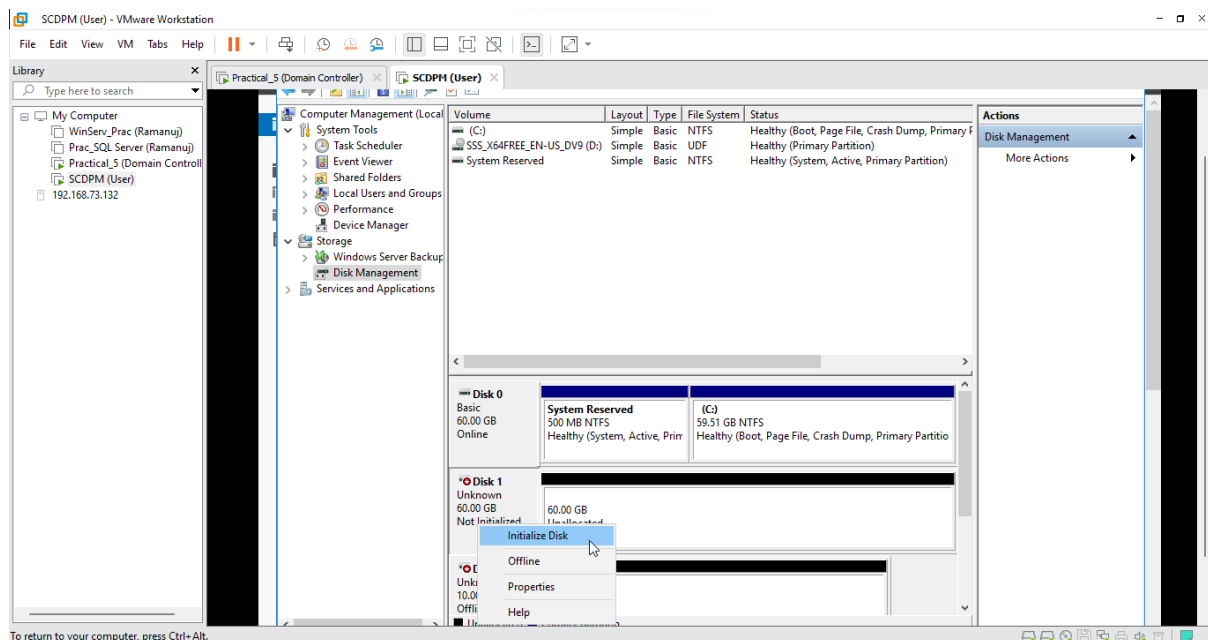
- Click on **Disk Management**



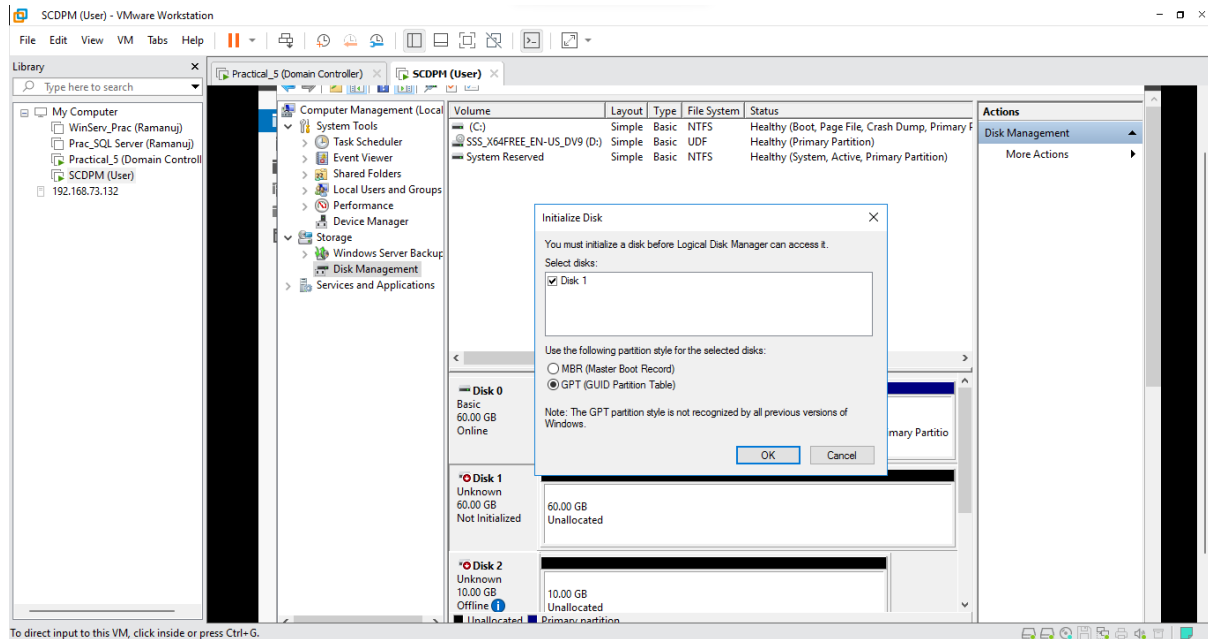
- Right Click on **Disk 1** and Select **Online**



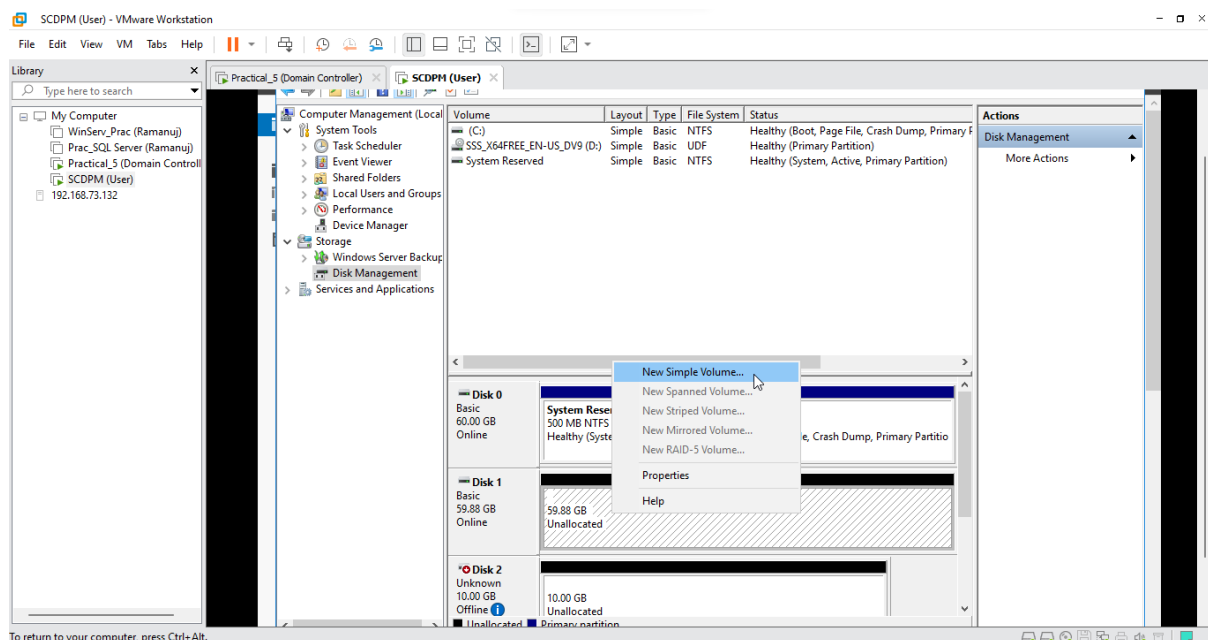
- Again Right Click on **Disk 1** and Select **Initialize Disk**



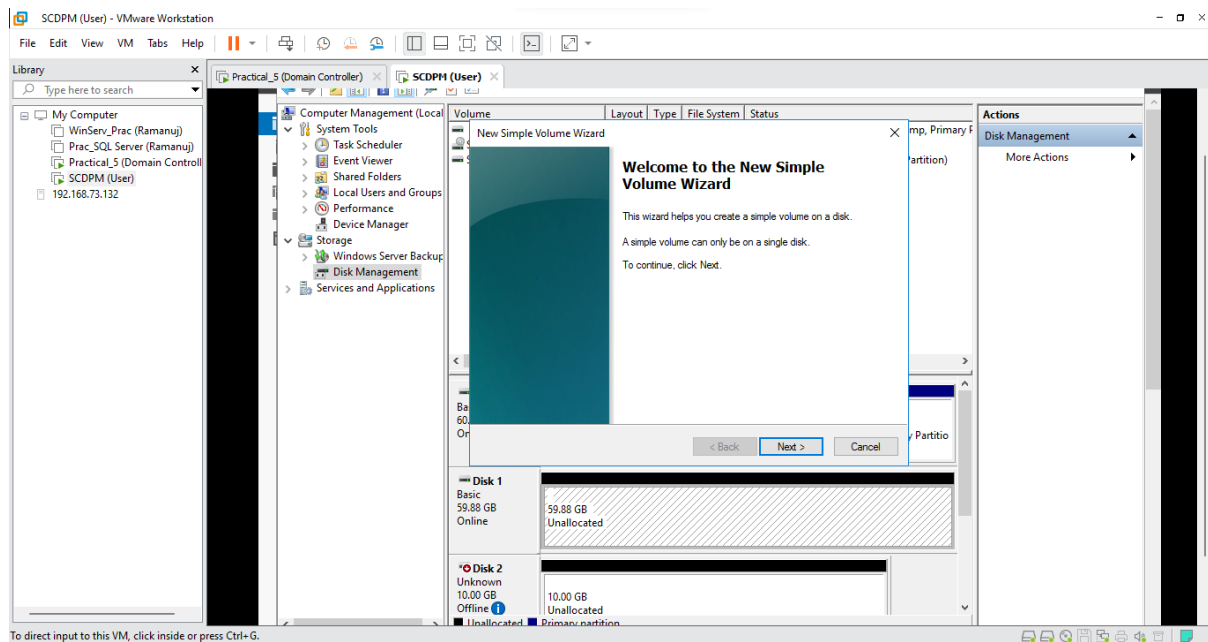
- Select **GPT** and Click **Ok**



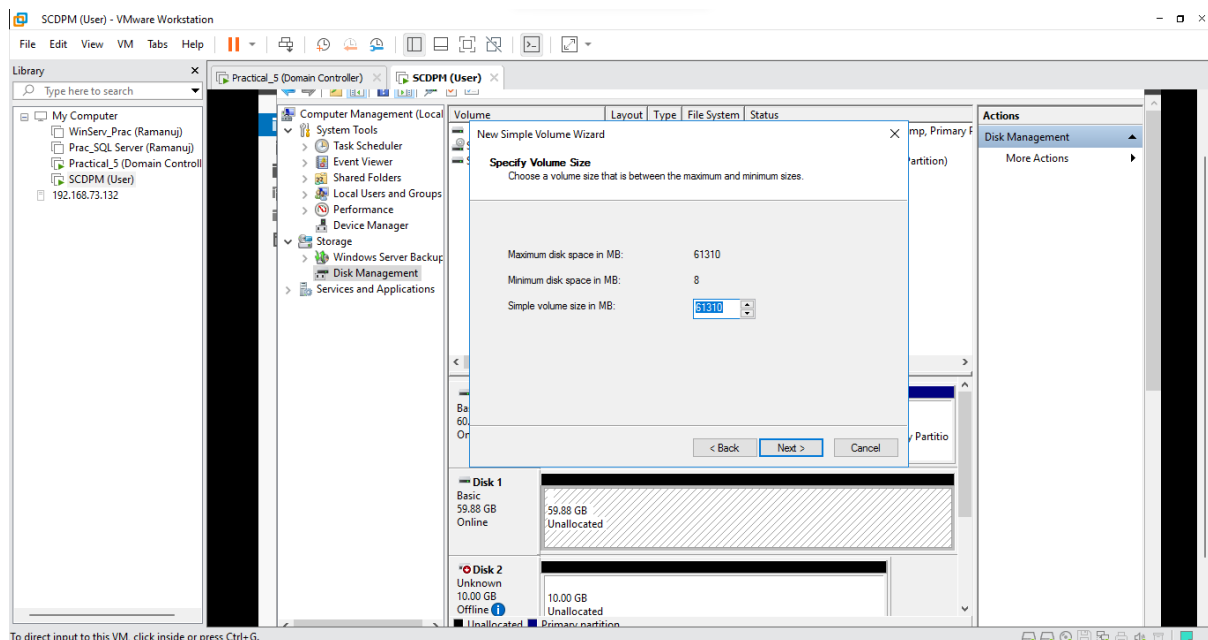
- Right-Click on the storage of **Disk 1** and Click on **Create new simple volume**



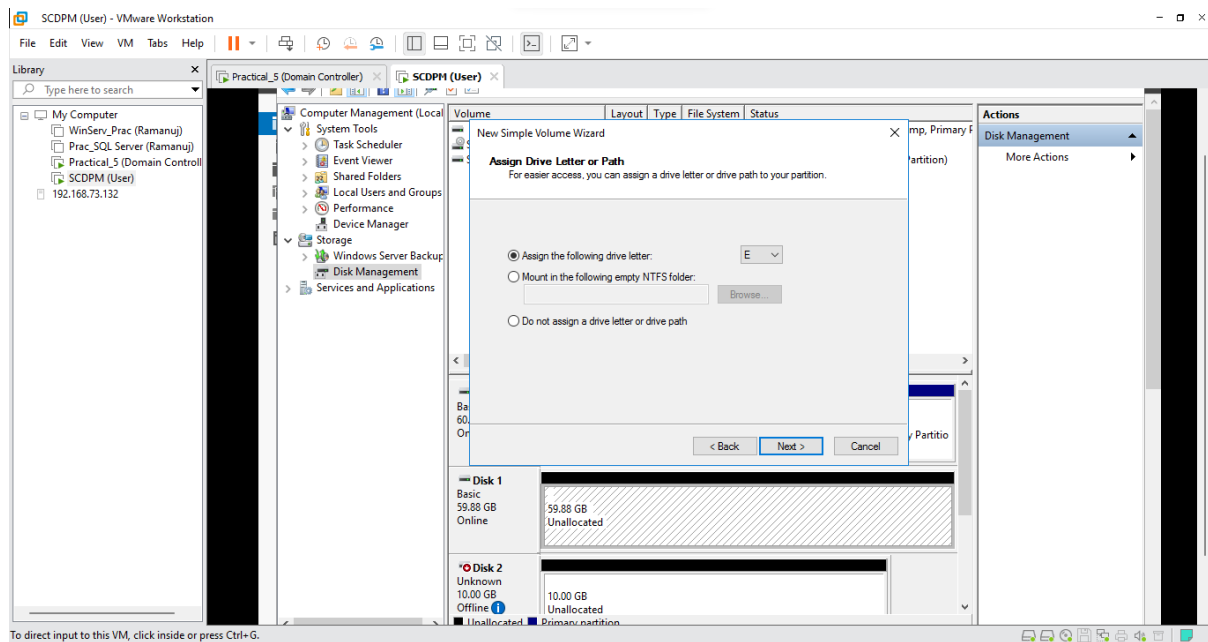
- Click Next



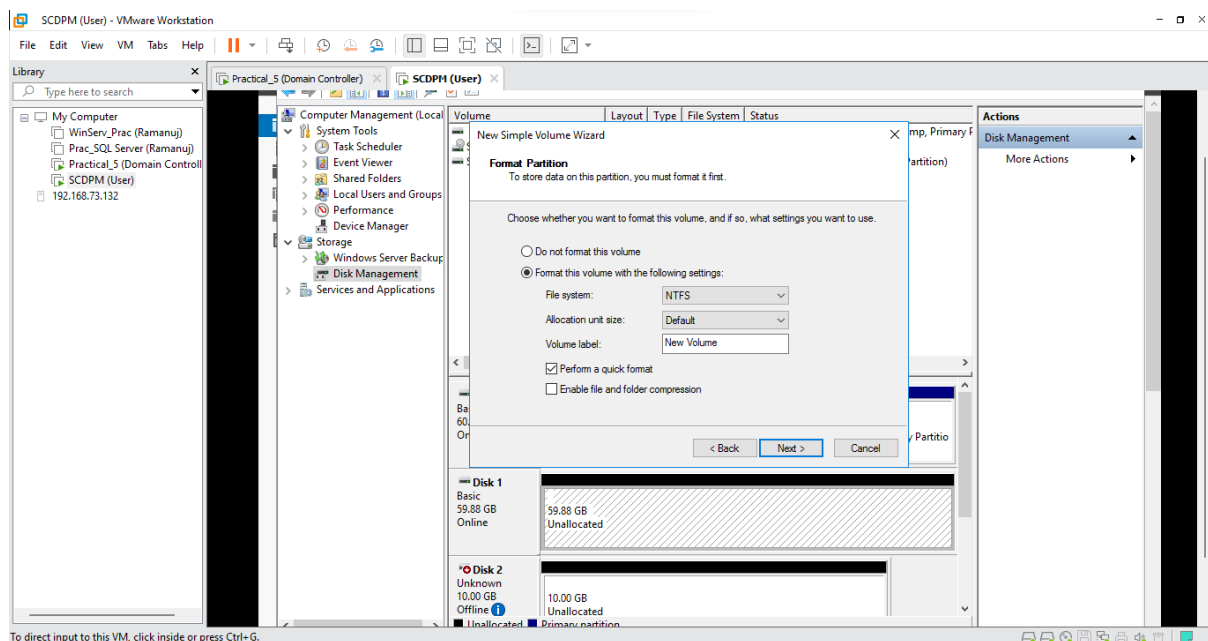
- Keep default values and Click Next



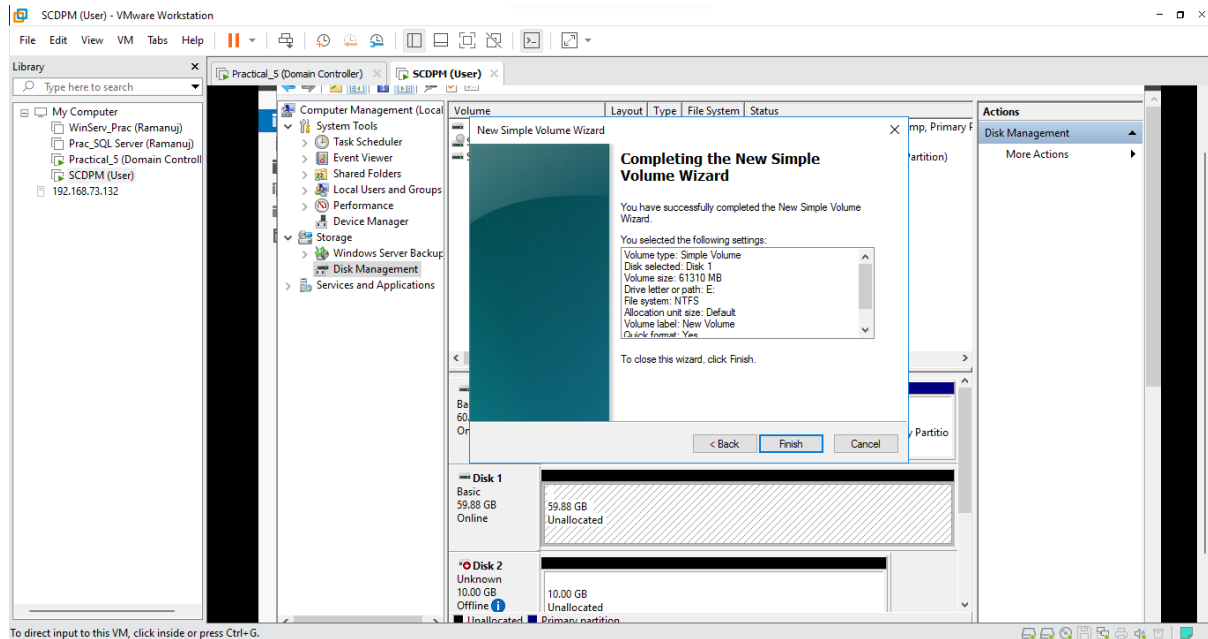
- Keep default values and Click **Next**



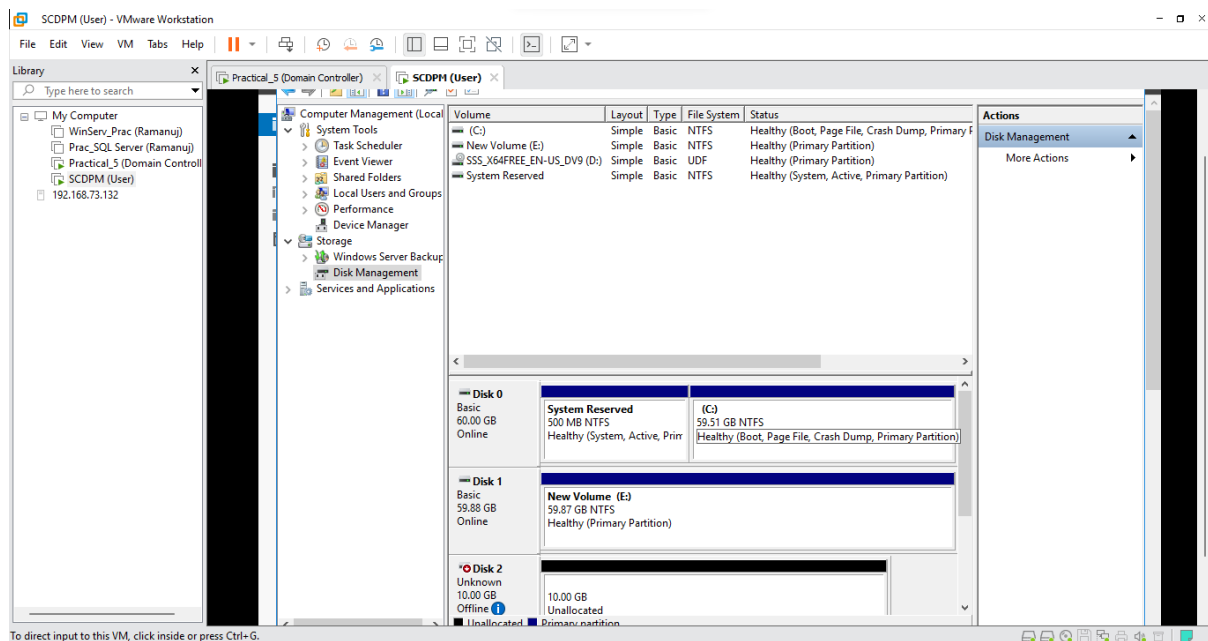
- Keep default values and Click **Next**



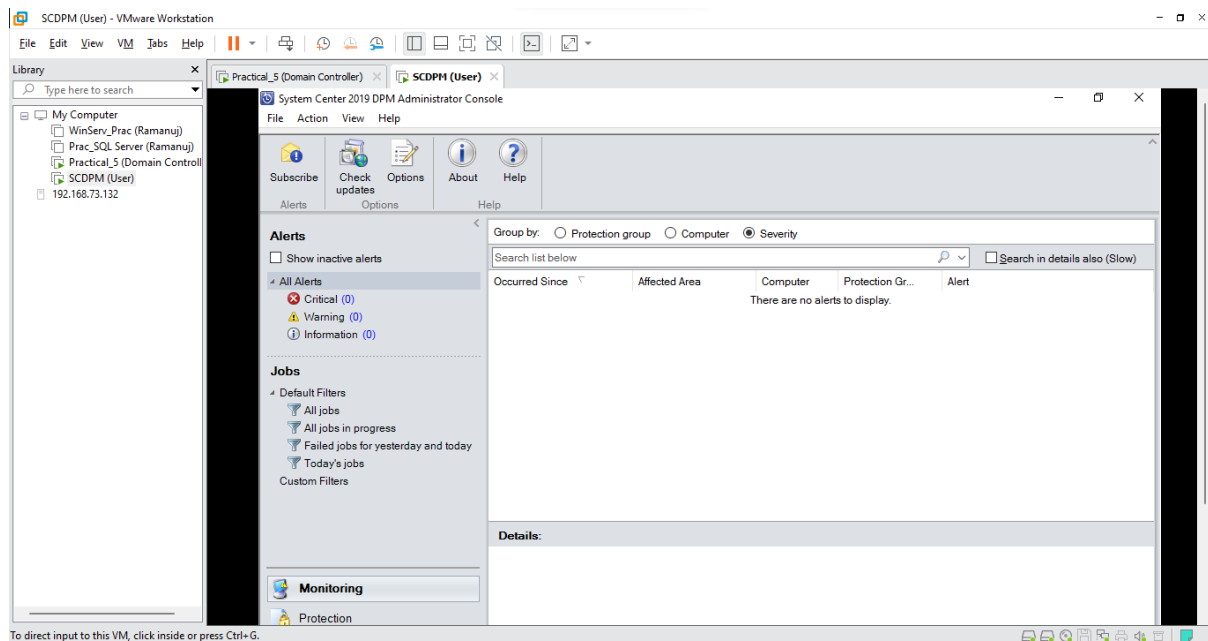
- Click on **Finish**



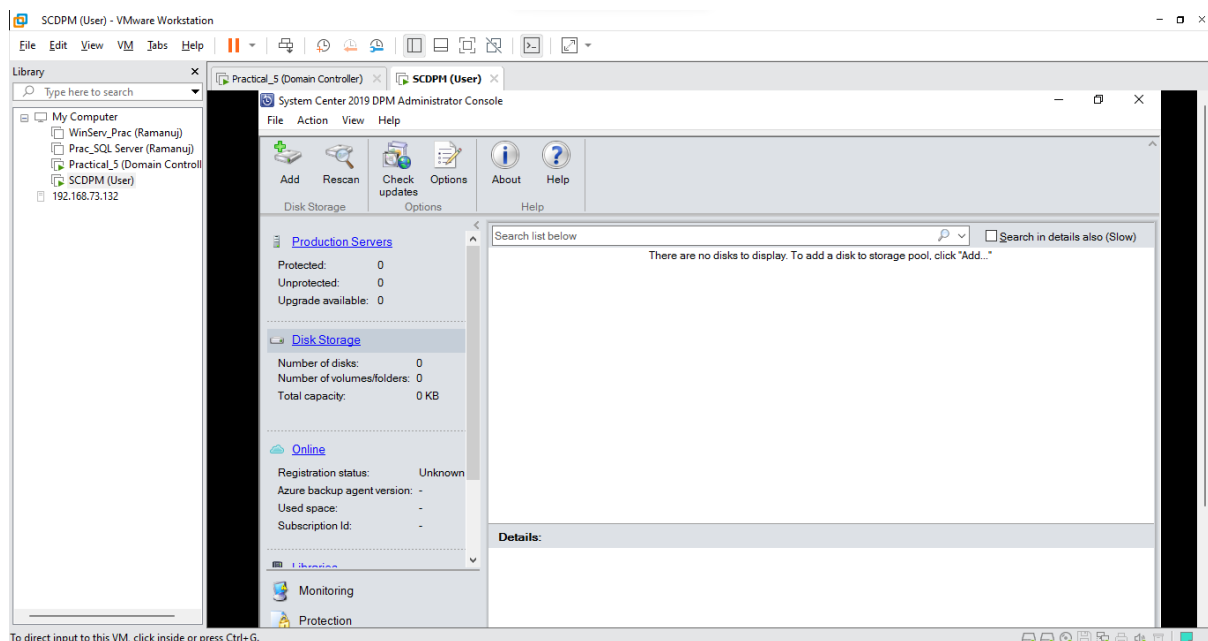
- The new volume has been created



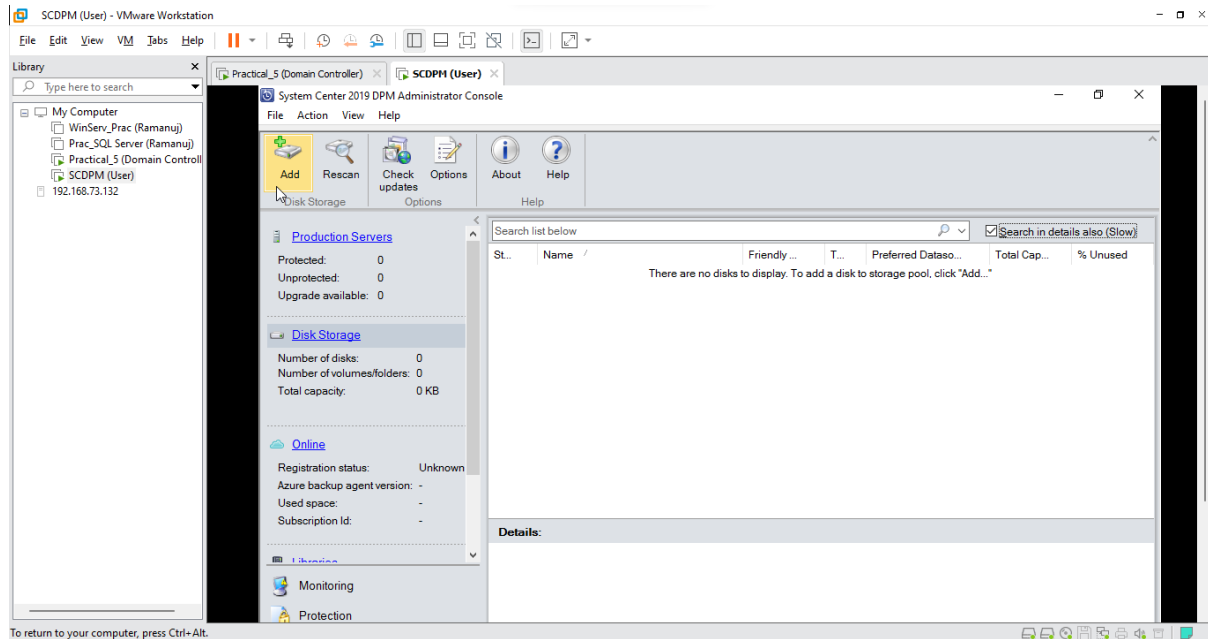
- After the volume is created change the user back to SCDPM and Run SCDPM



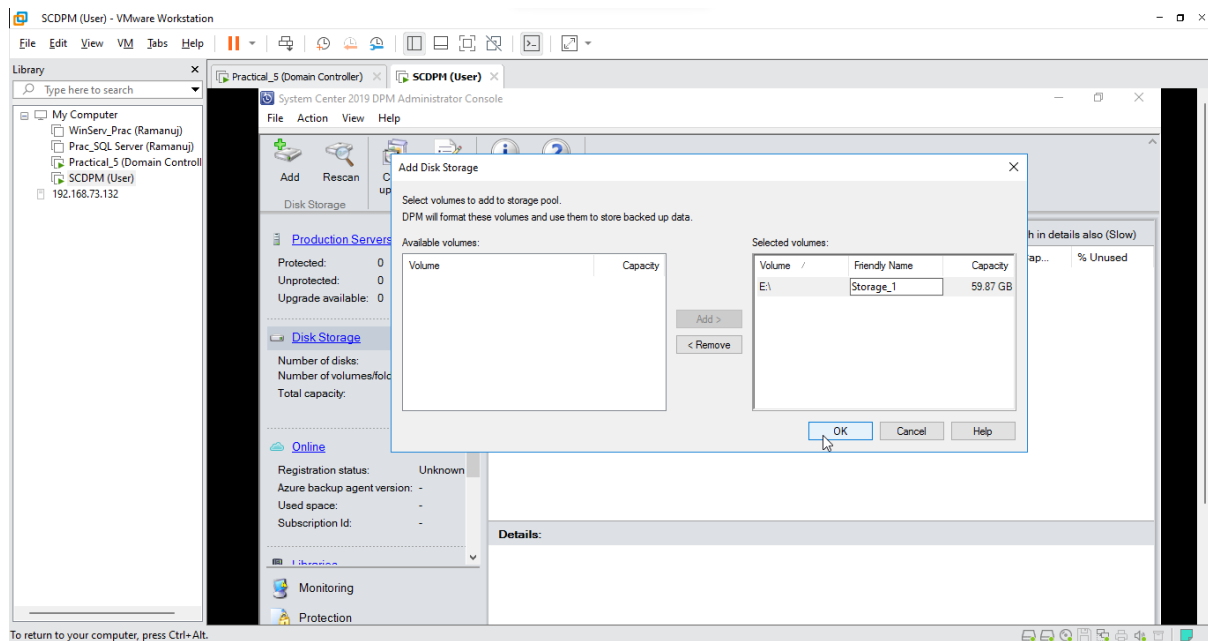
- Click on **Management** and Select **Disk Storage**



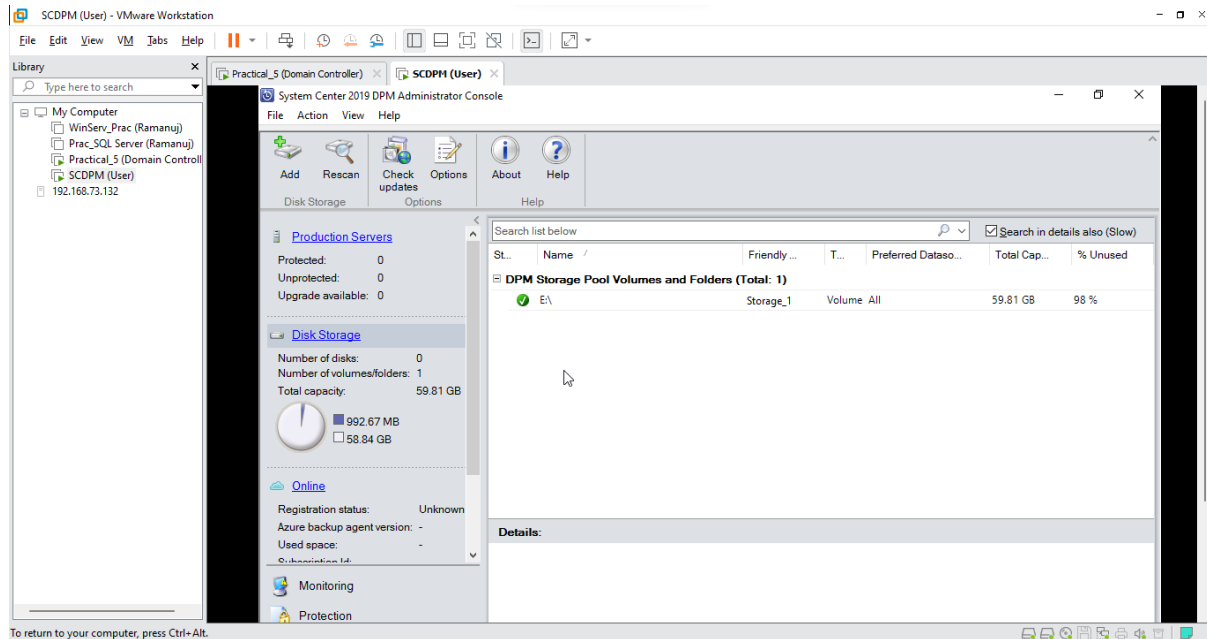
- Click **Add**



- Select volume E and Click **Add**, then Name the Storage and Click **Ok**



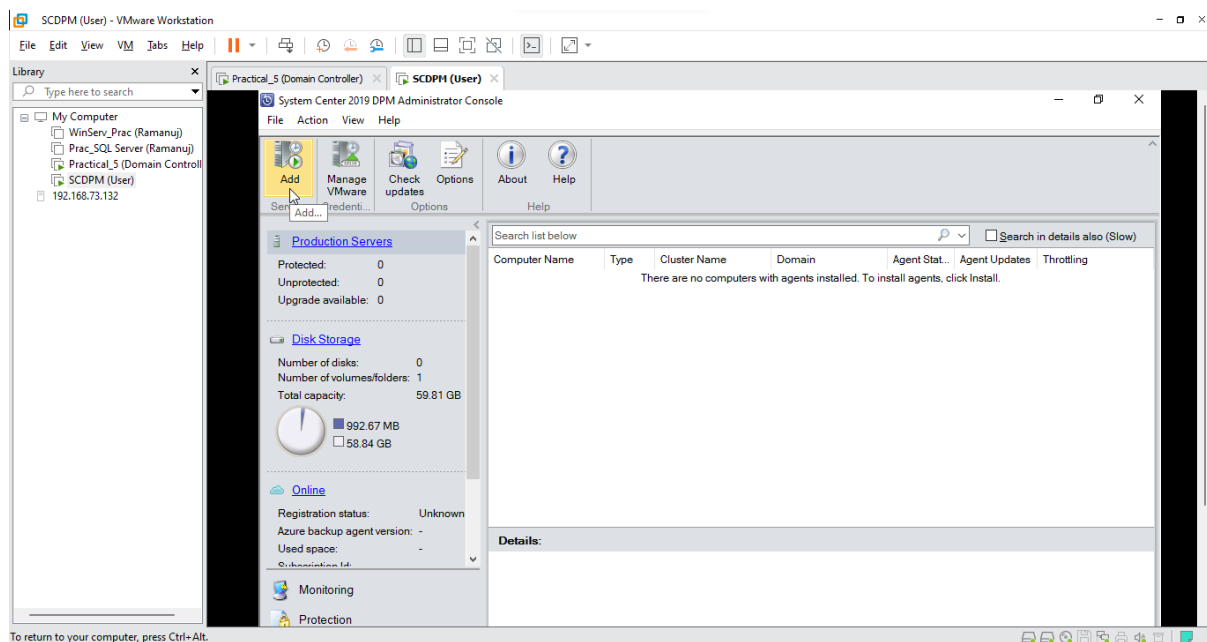
- The Storage has been created



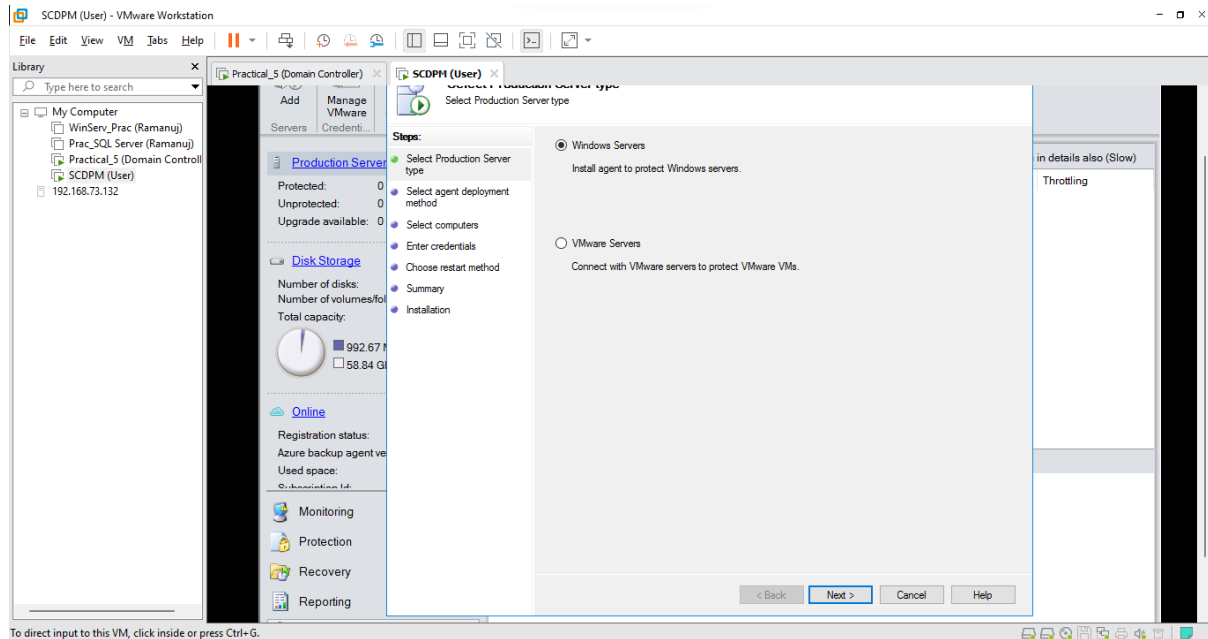
5-B: Deploy Protection Agent

Step 9: Creating a Production Server

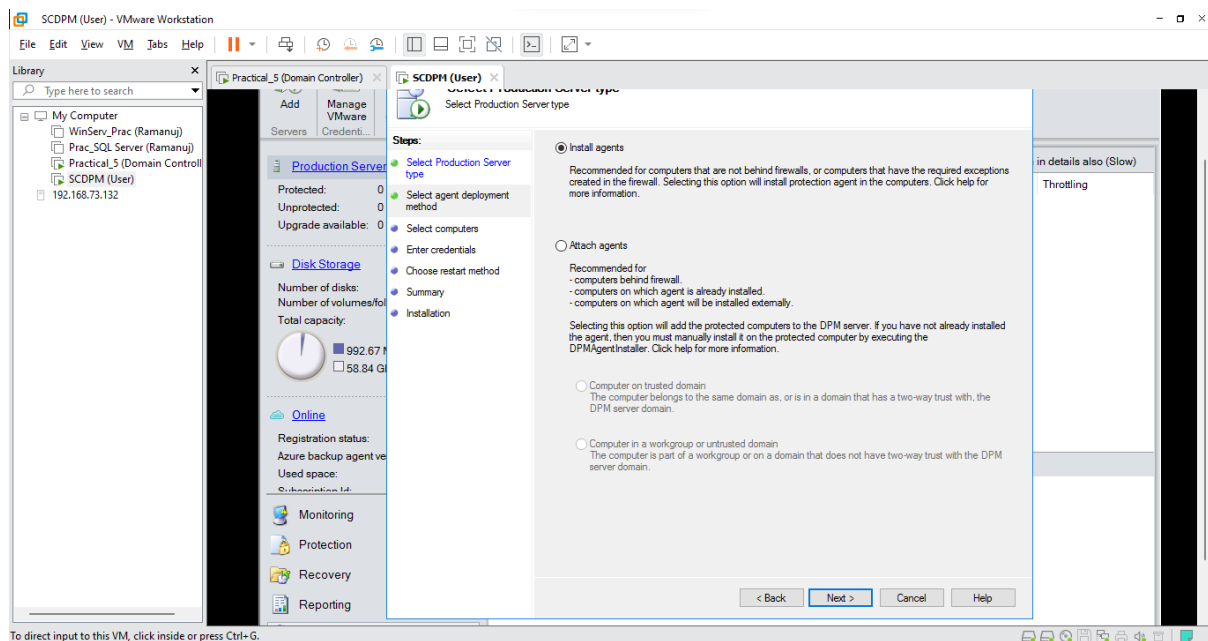
- Within Management Click on **Production Servers** and Click on **Add**



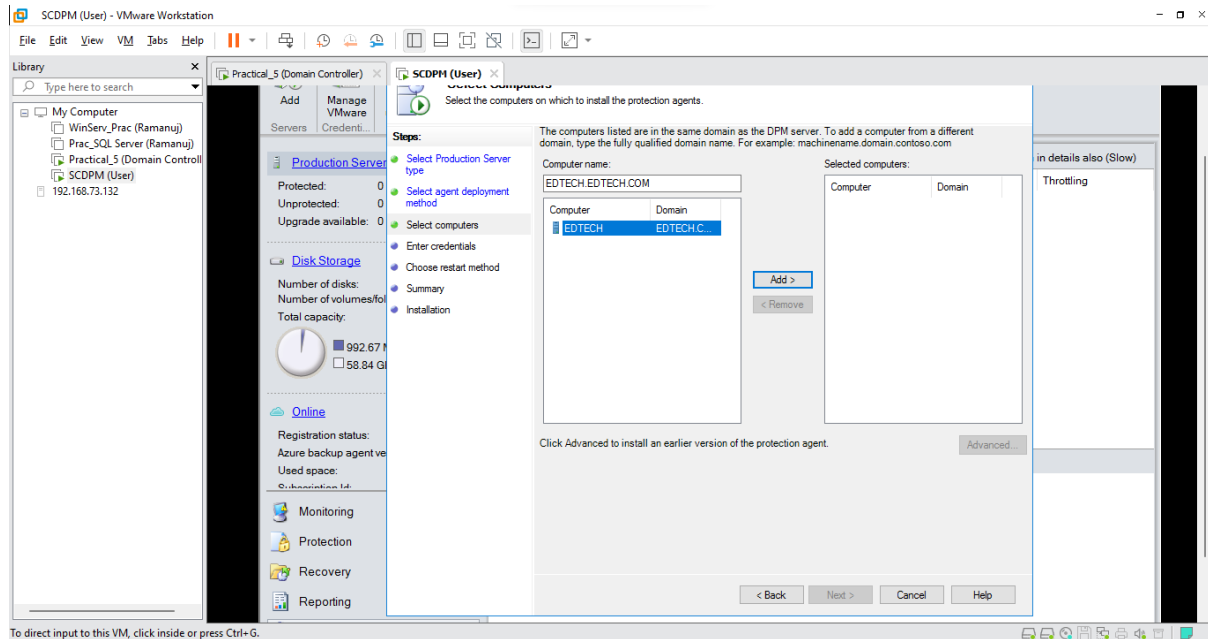
- Select **Windows Server** and Click **Next**



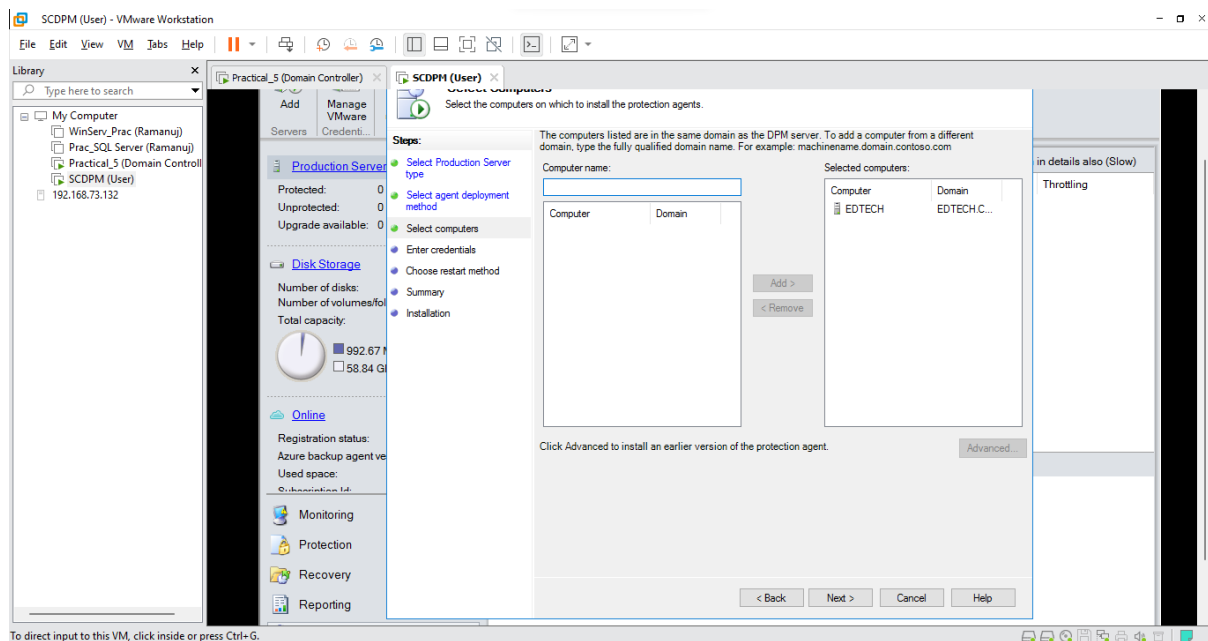
- Select **Install Agents** and Click **Next**



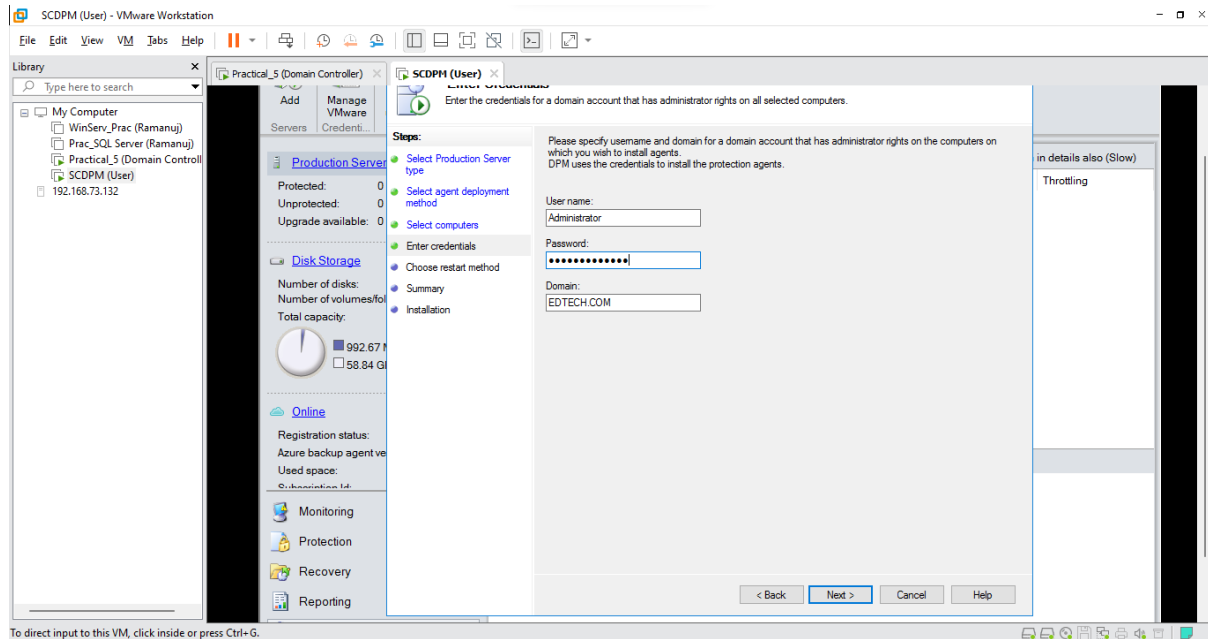
- Select **EDTECH** and Click **Add**



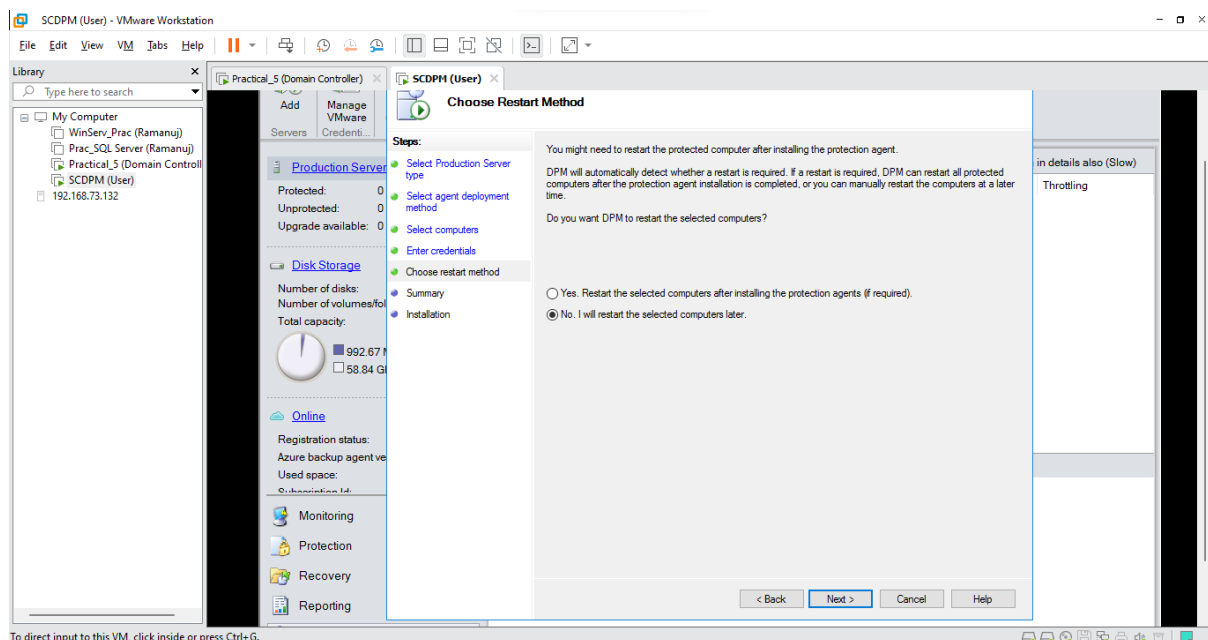
- Click **Next**



- Type in the credentials and Click on **Next**

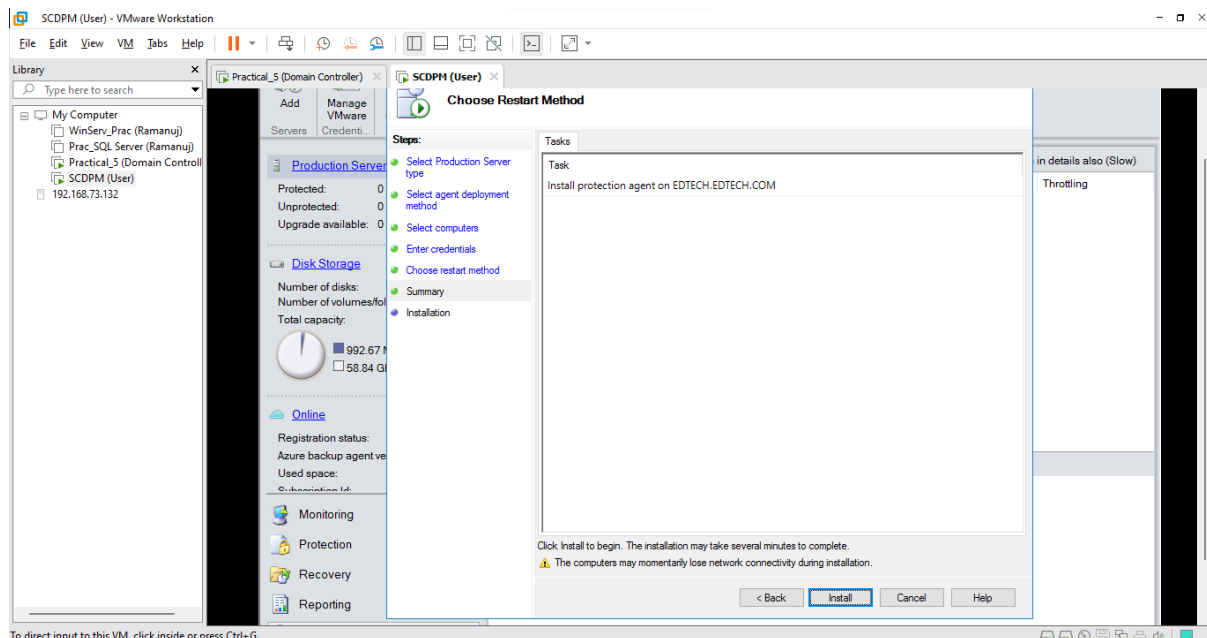


- Select **No, I will restart the selected computers later** and Click **Next**



- Click on **Install**

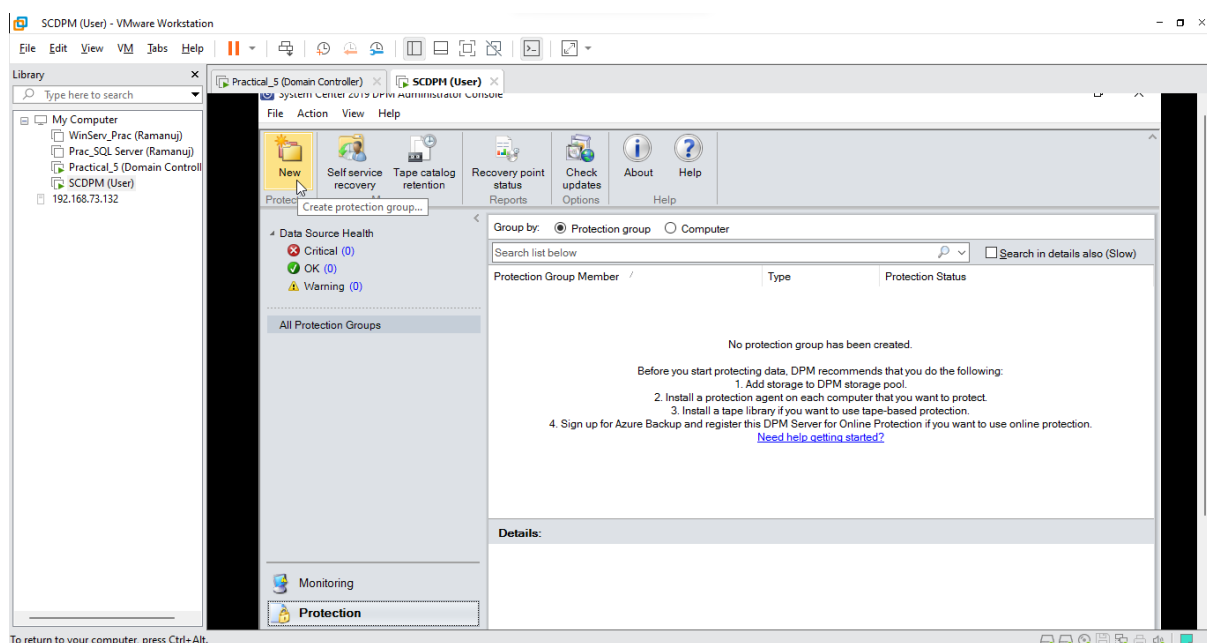
Note: If there is an error 313 or 319 then **Disable** the firewalls of both the Domain Controller and SCDPM. You can also refer to **Step 10** for the cmd commands for allowing particular files in the firewall



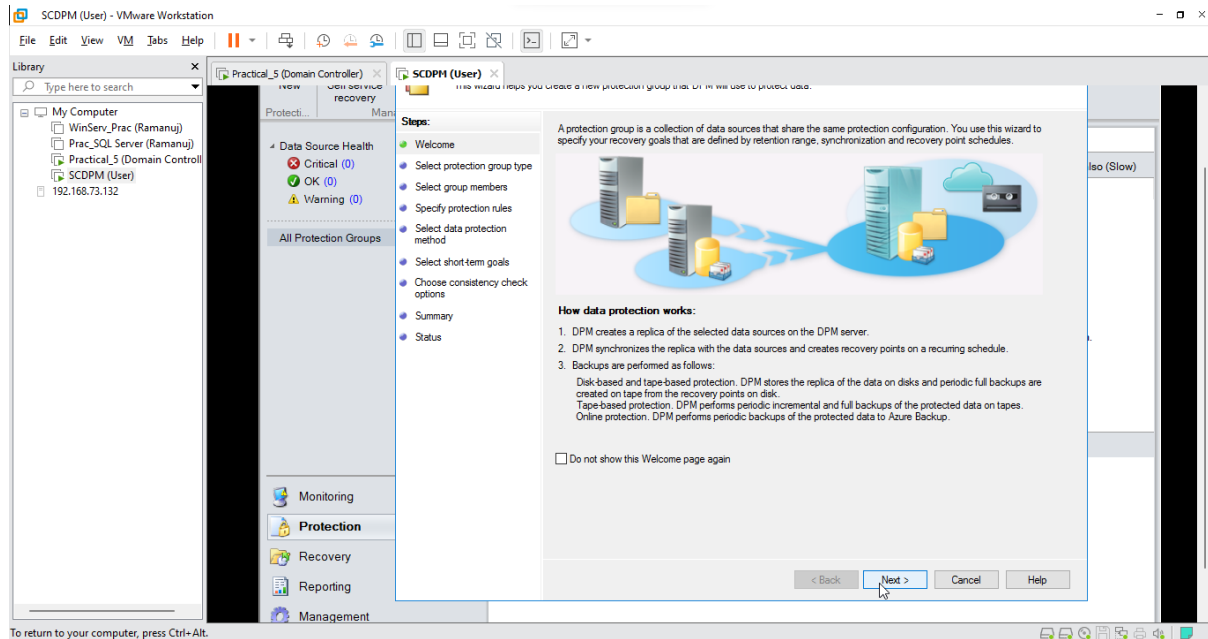
5-C: Deploy Protection Groups

Step 9: Creating a Protection Group in SCDPM

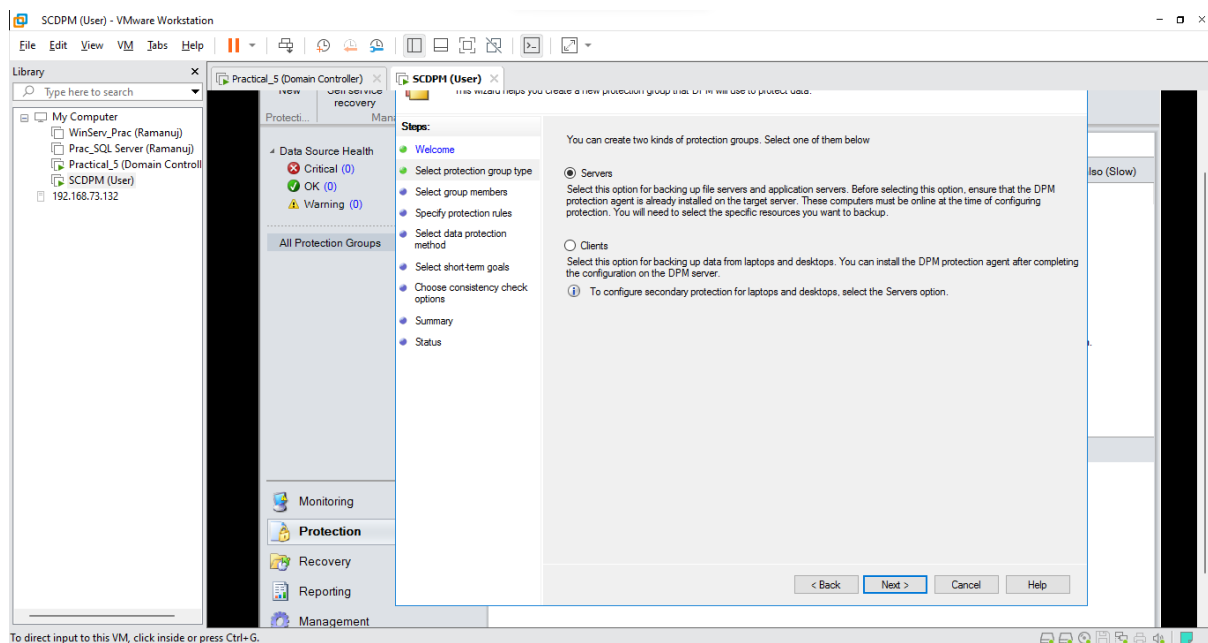
- Click on **Protection** and Click on **New**



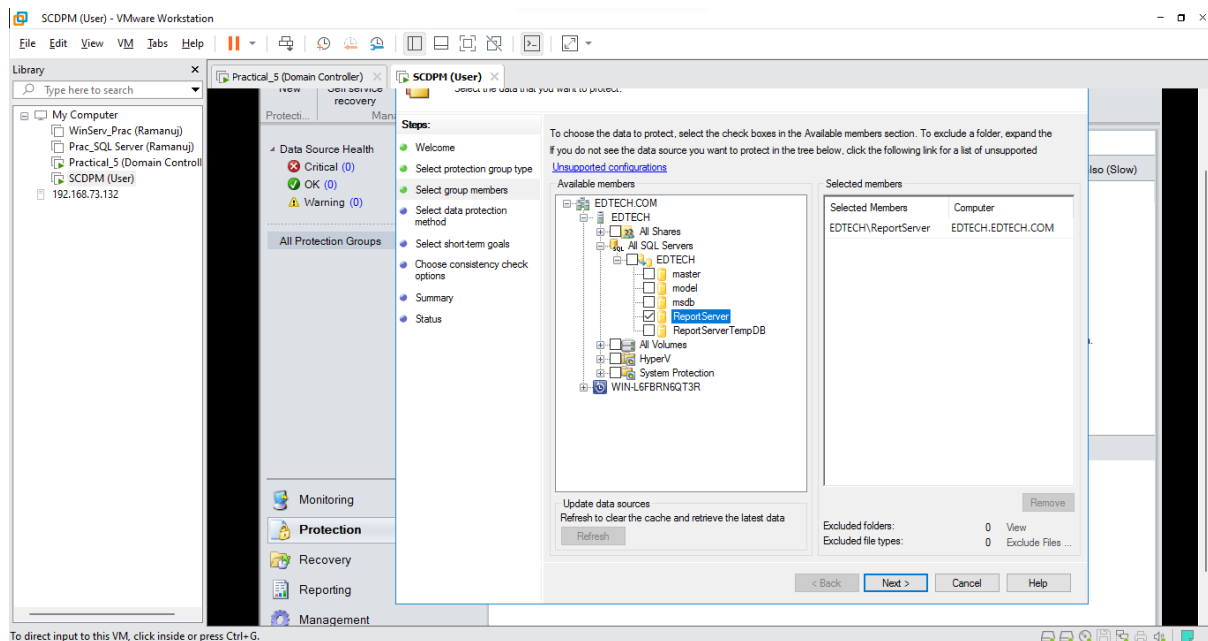
- Click Next



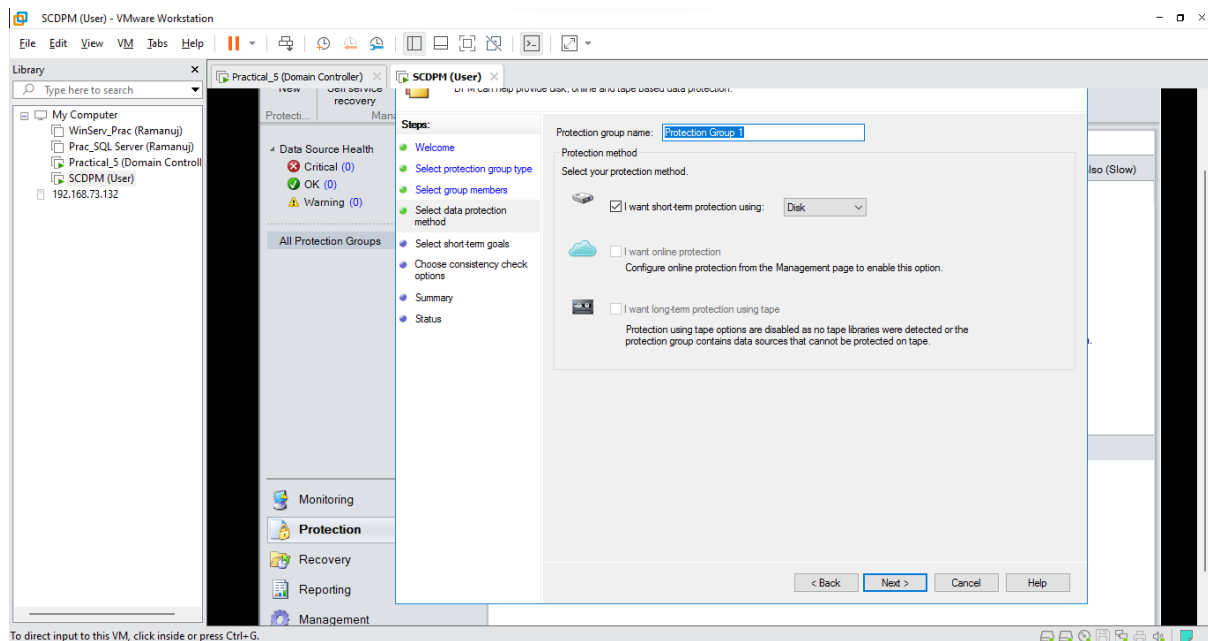
- Select Server and Click Next



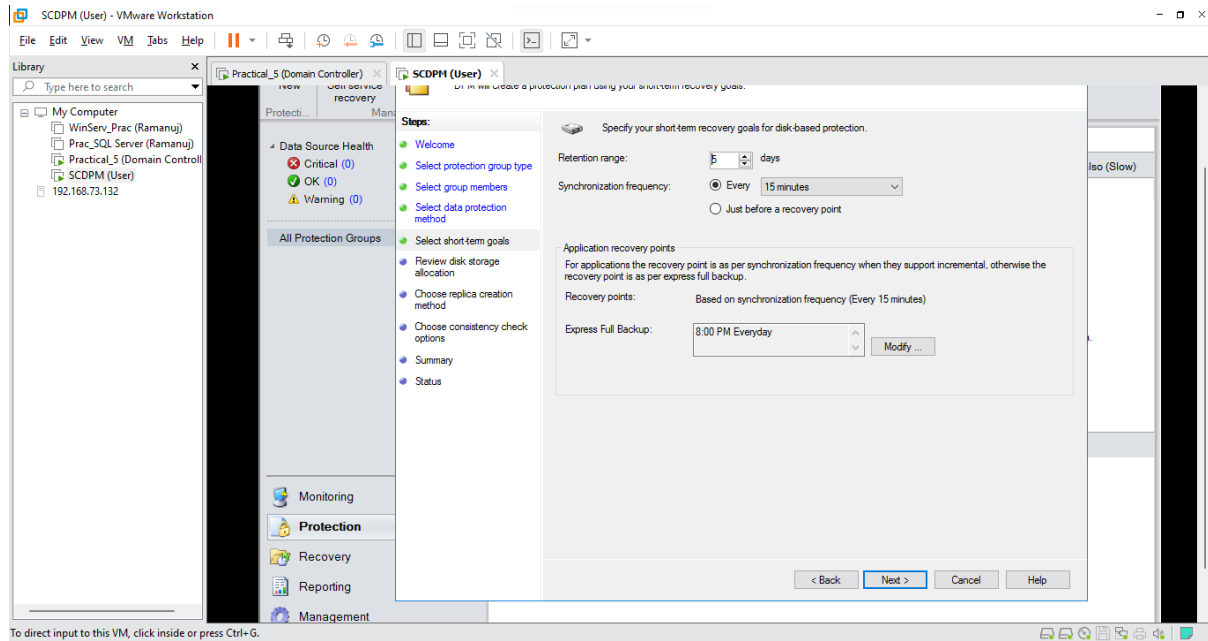
- Select a Folder from your SQL Server (Here it is Report Server) and Click **Next**



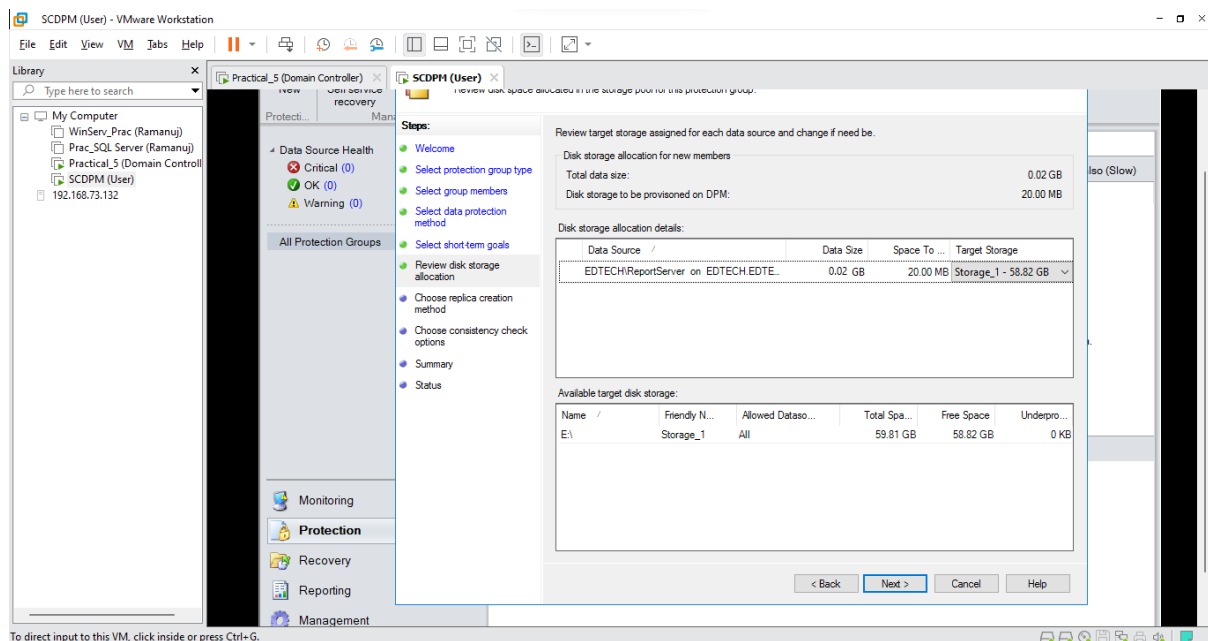
- Name the Protection Group or Keep the default value and Click **Next**



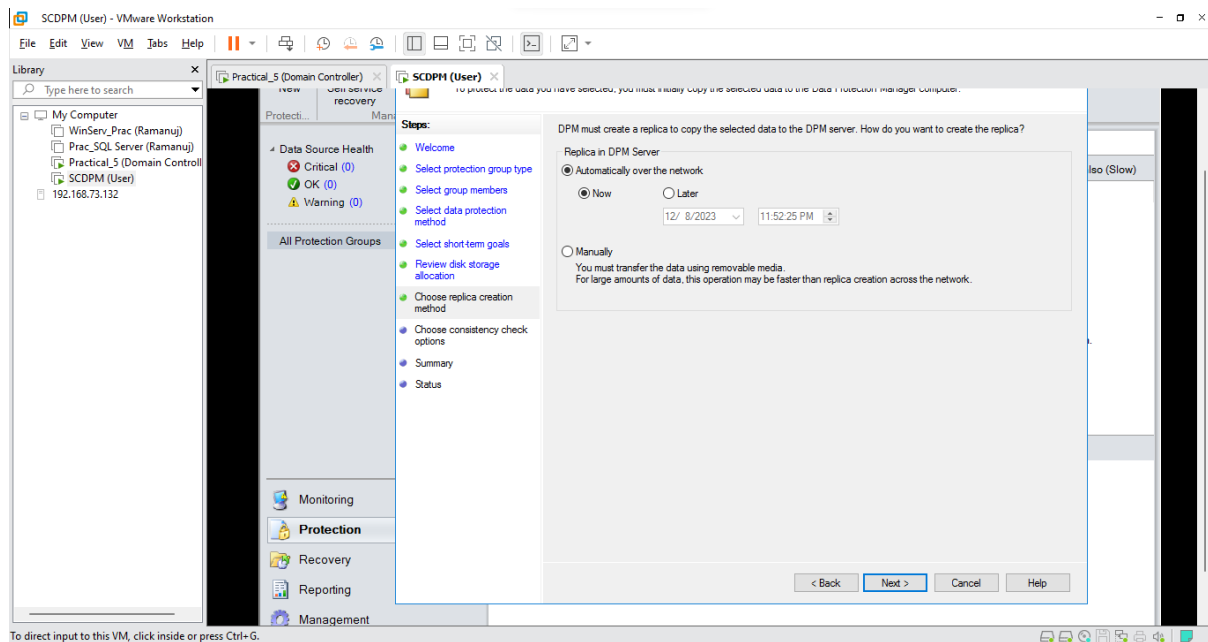
- Keep default values and Click **Next**



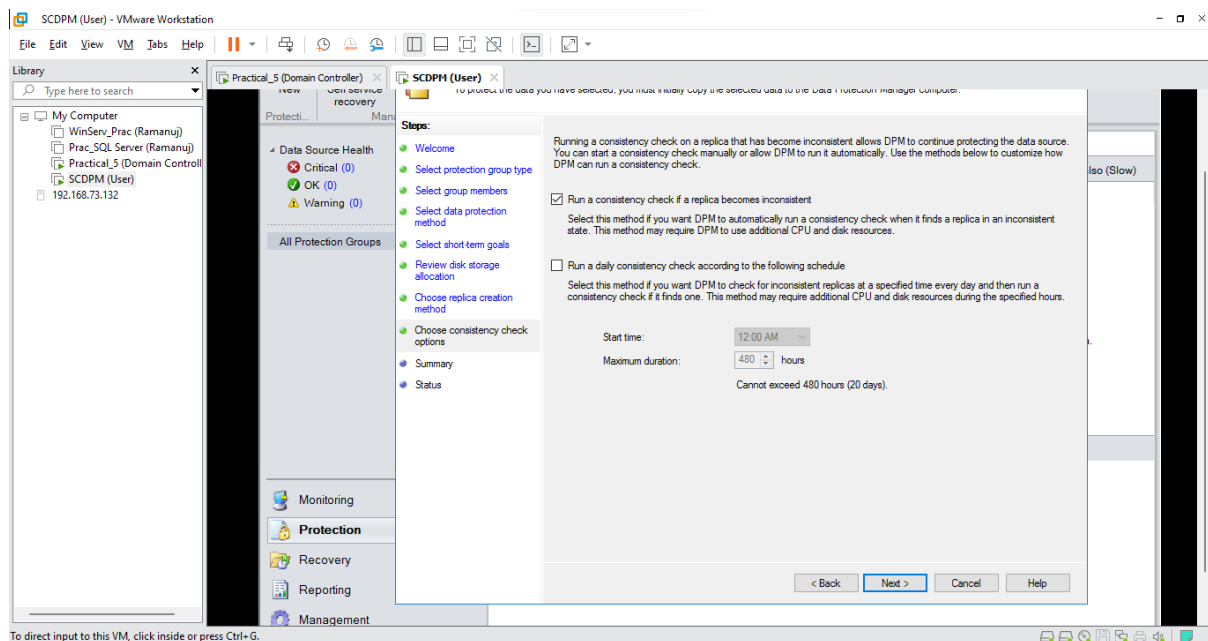
- Click **Next**



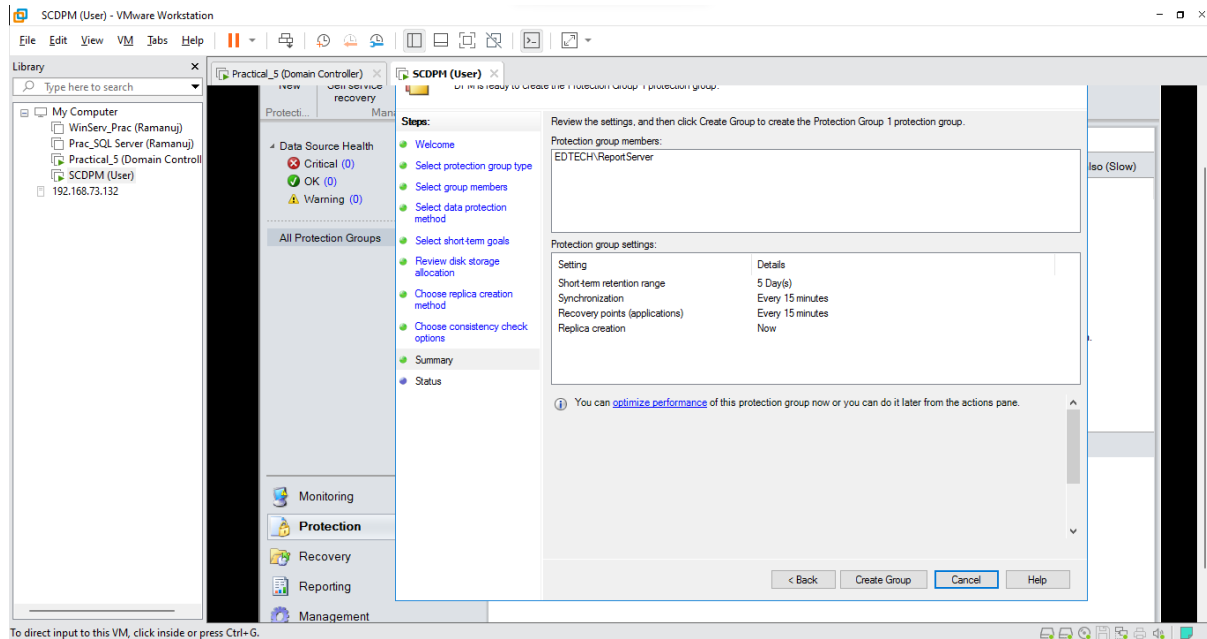
- Select **Automatic** and Click **Next**



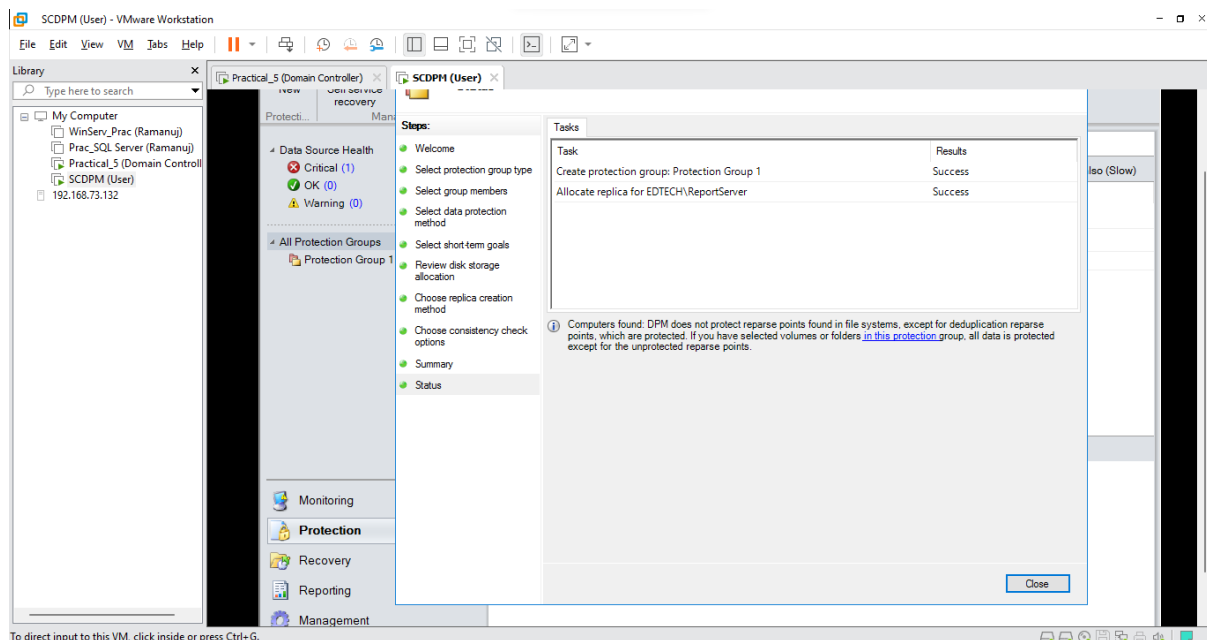
- Keep default values and Click **Next**



- View the summary of your protection group and Click **Create Group**



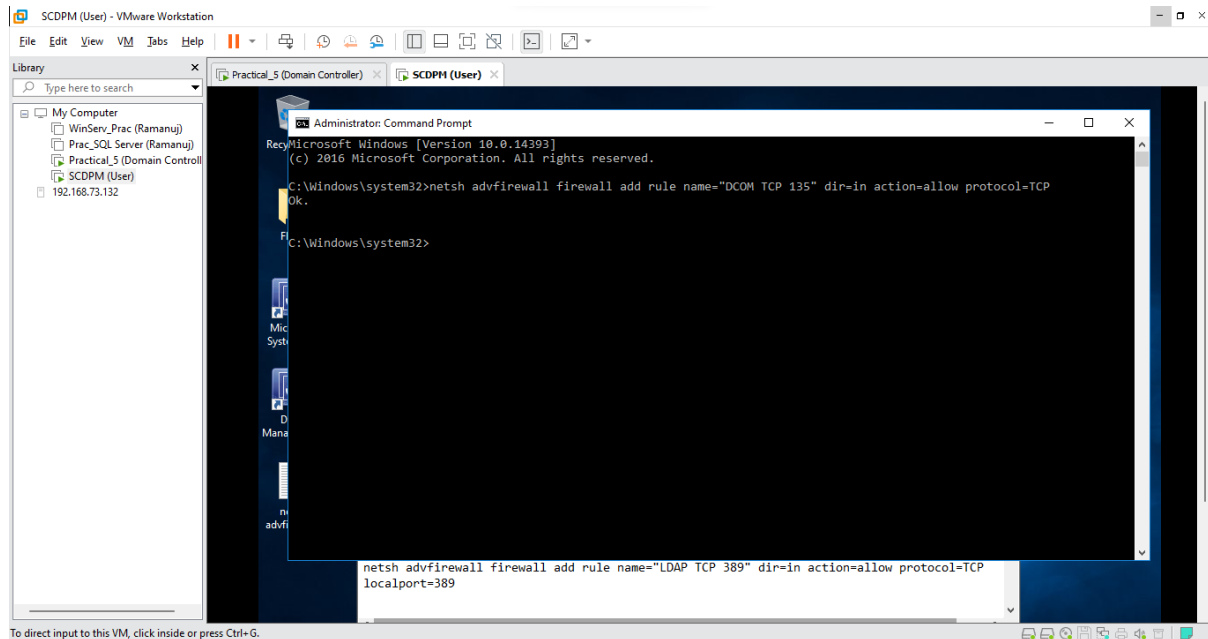
- Protection Group has been successfully created



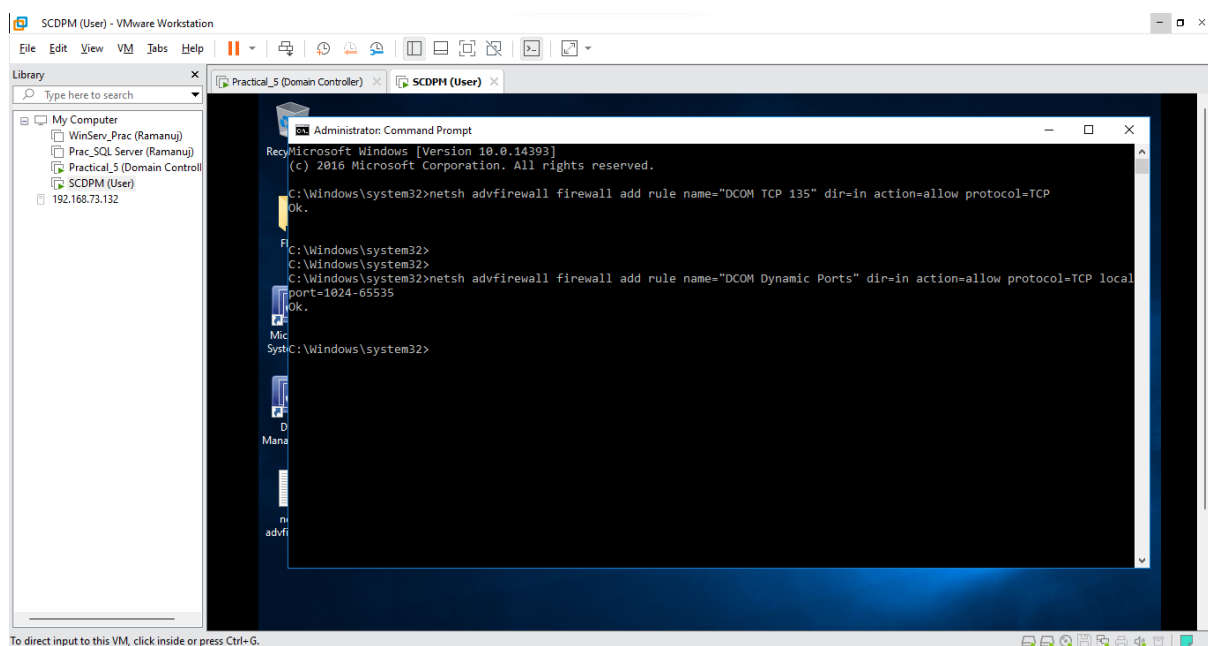
5-D: Configure Firewall Settings

Step 10: Run firewall commands in command prompt

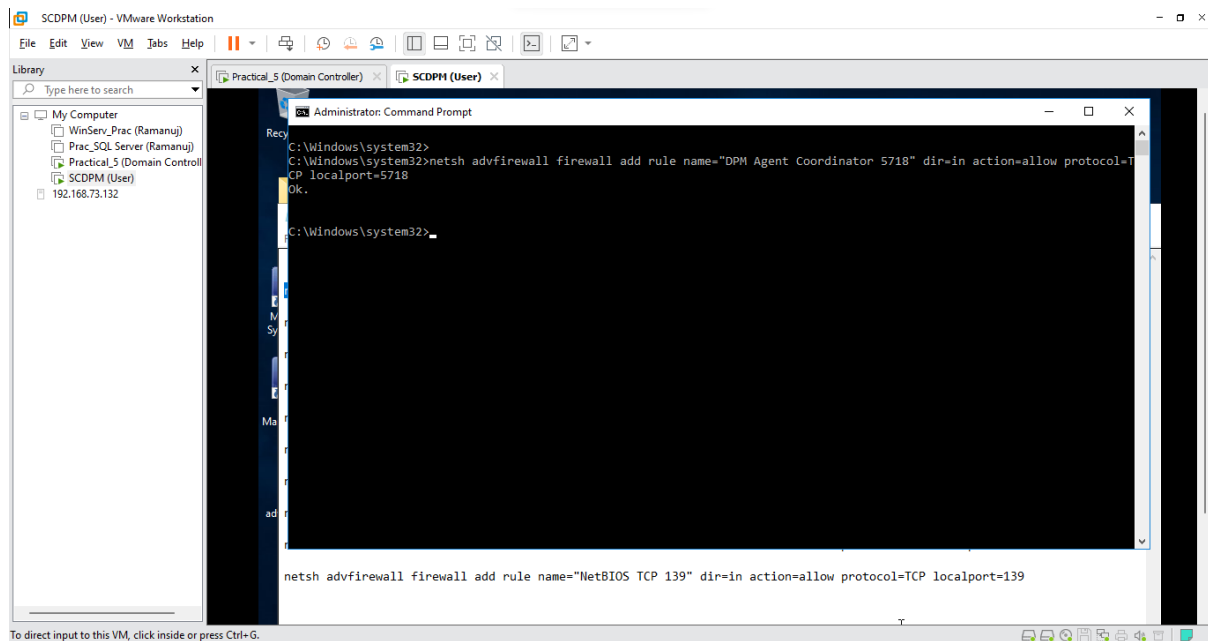
- **netsh advfirewall firewall add rule name="DCOM TCP 135" dir=in action=allow protocol=TCP localport=135**



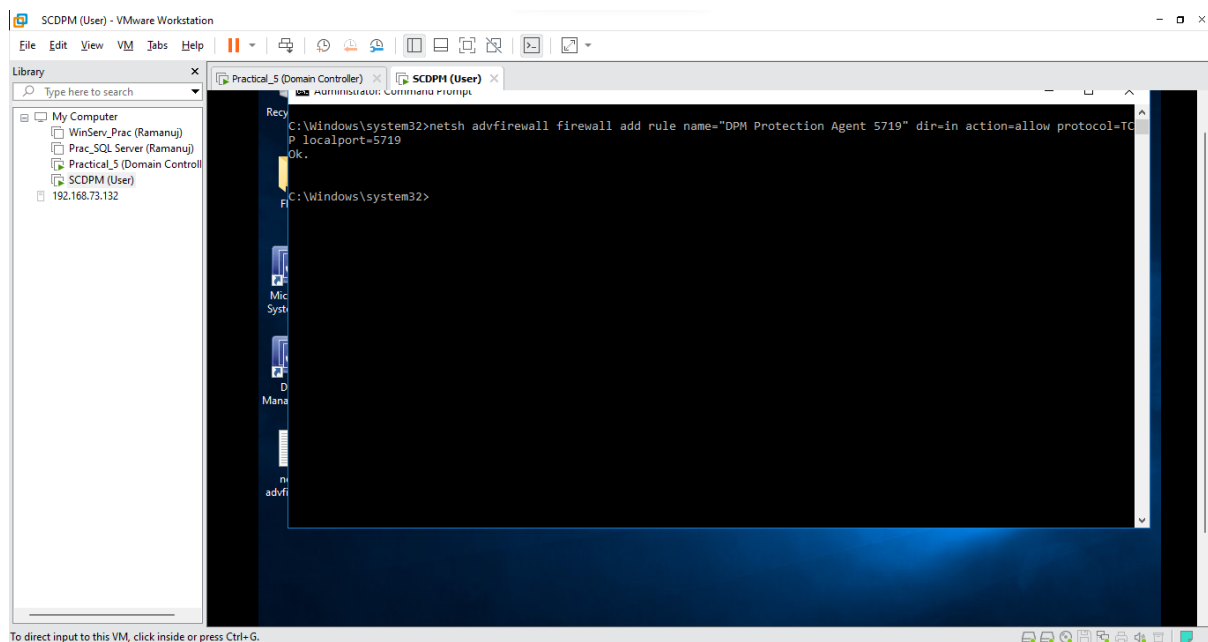
- **netsh advfirewall firewall add rule name="DCOM Dynamic Ports" dir=in action=allow protocol=TCP localport=1024-65535**



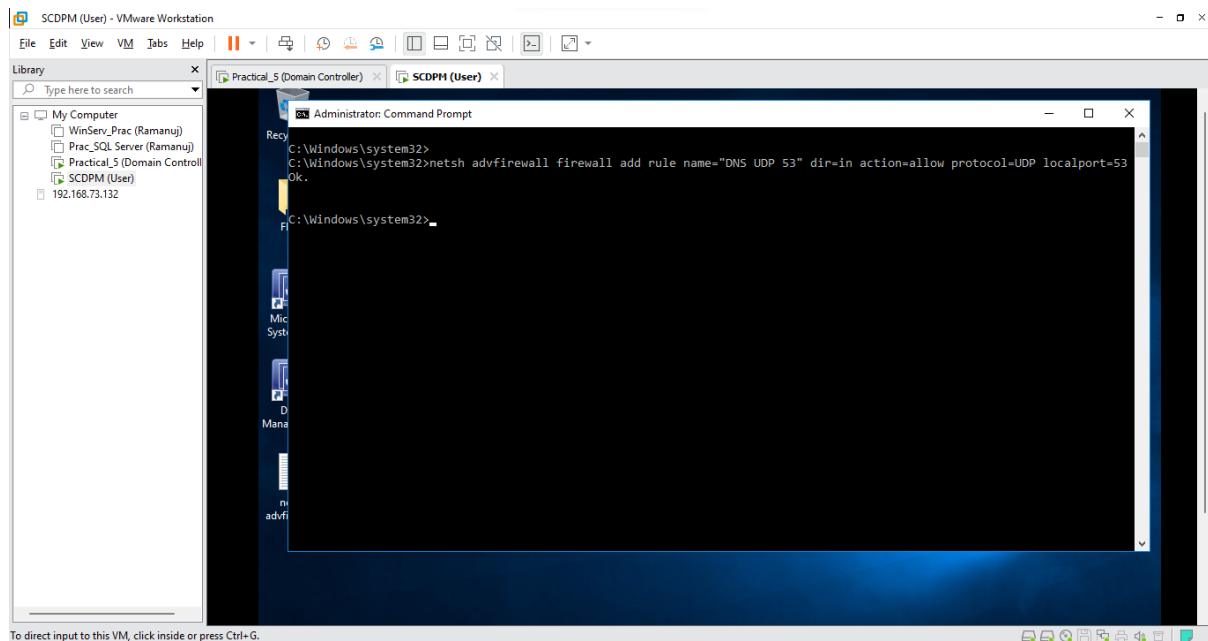
- **netsh advfirewall firewall add rule name="DPM Agent Coordinator 5718" dir=in action=allow protocol=TCP localport=5718**



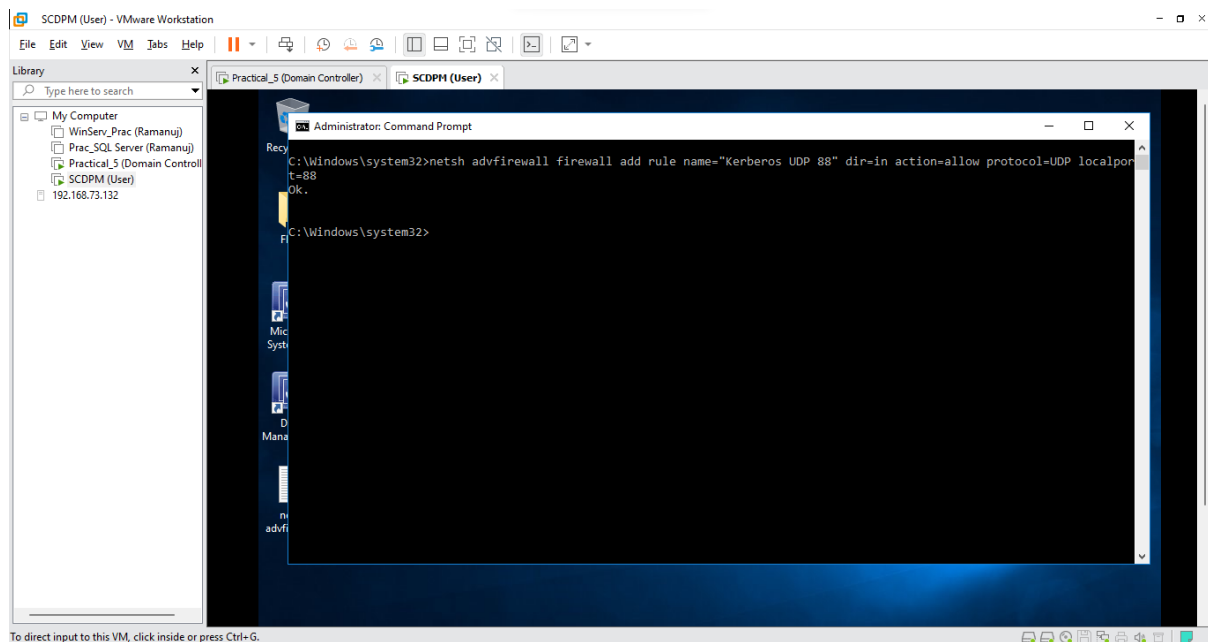
- **netsh advfirewall firewall add rule name="DPM Protection Agent 5719" dir=in action=allow protocol=TCP localport=5719**



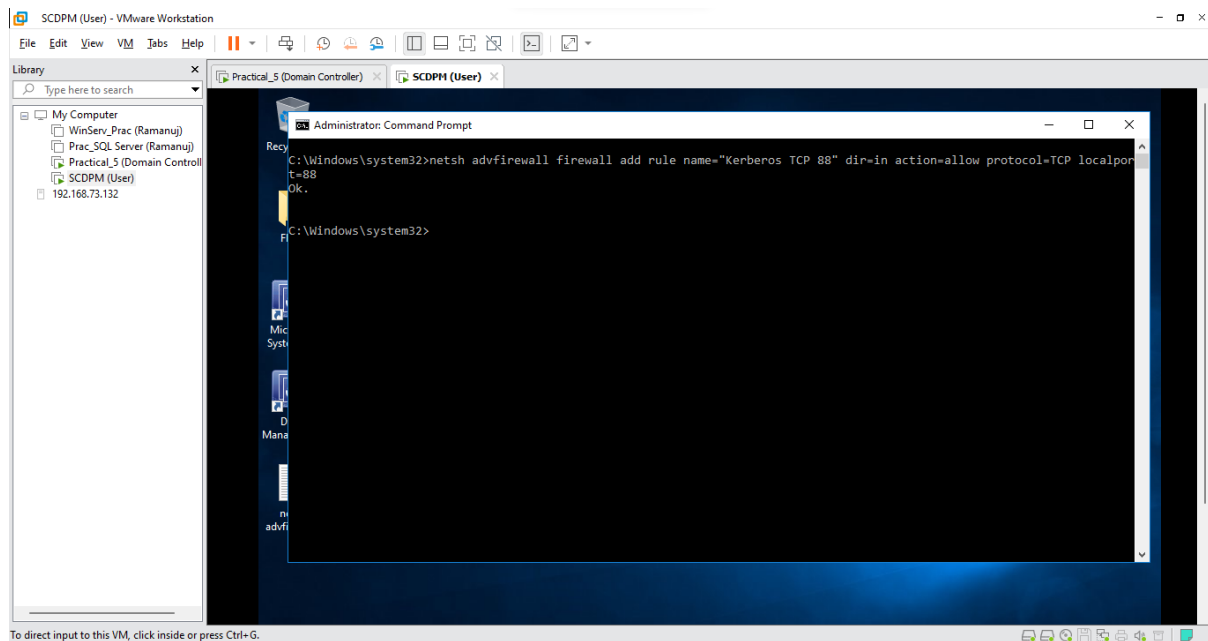
- **netsh advfirewall firewall add rule name="DNS UDP 53" dir=in action=allow protocol=UDP localport=53**



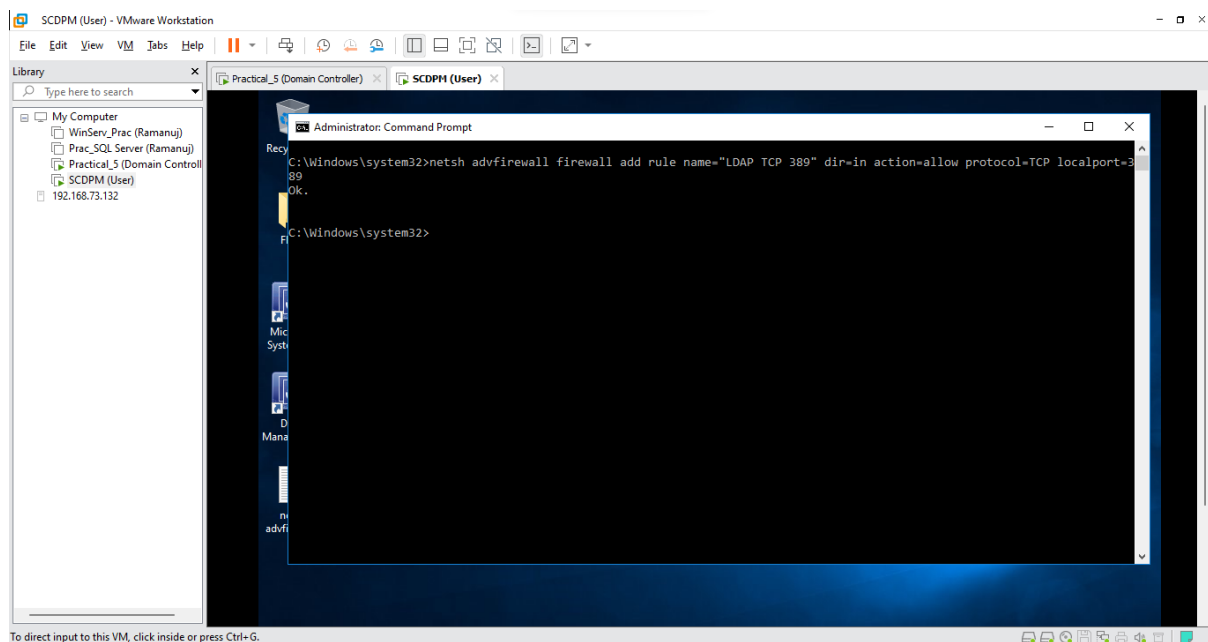
- **netsh advfirewall firewall add rule name="Kerberos UDP 88" dir=in action=allow protocol=UDP localport=88**



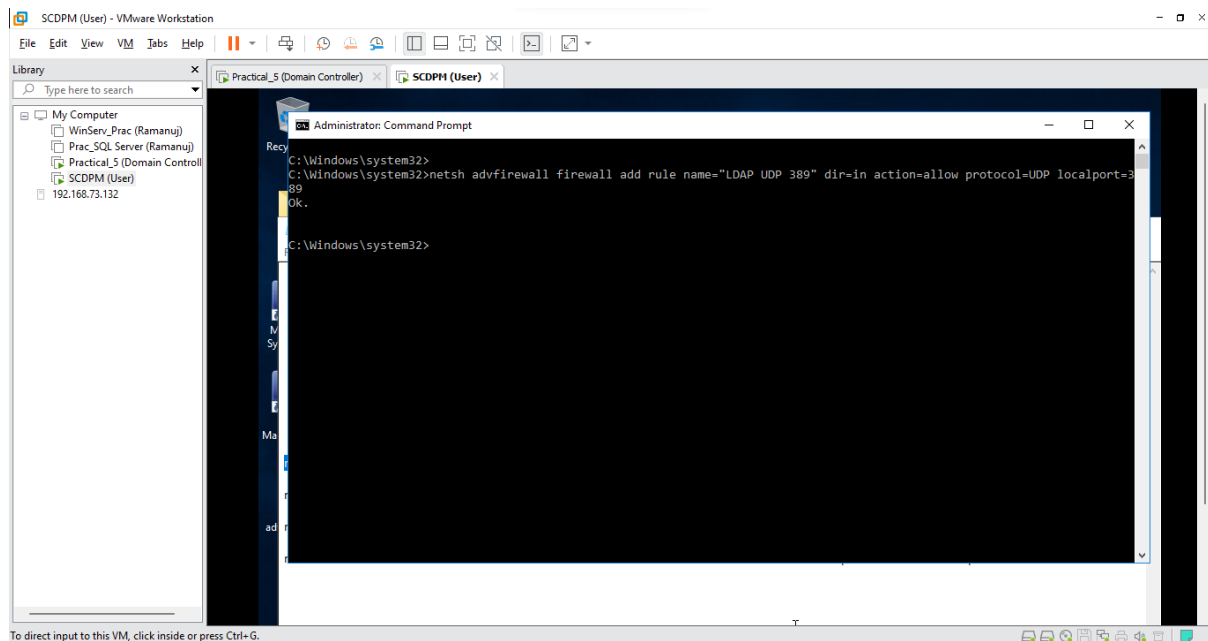
- **netsh advfirewall firewall add rule name="Kerberos TCP 88" dir=in action=allow protocol=TCP localport=88**



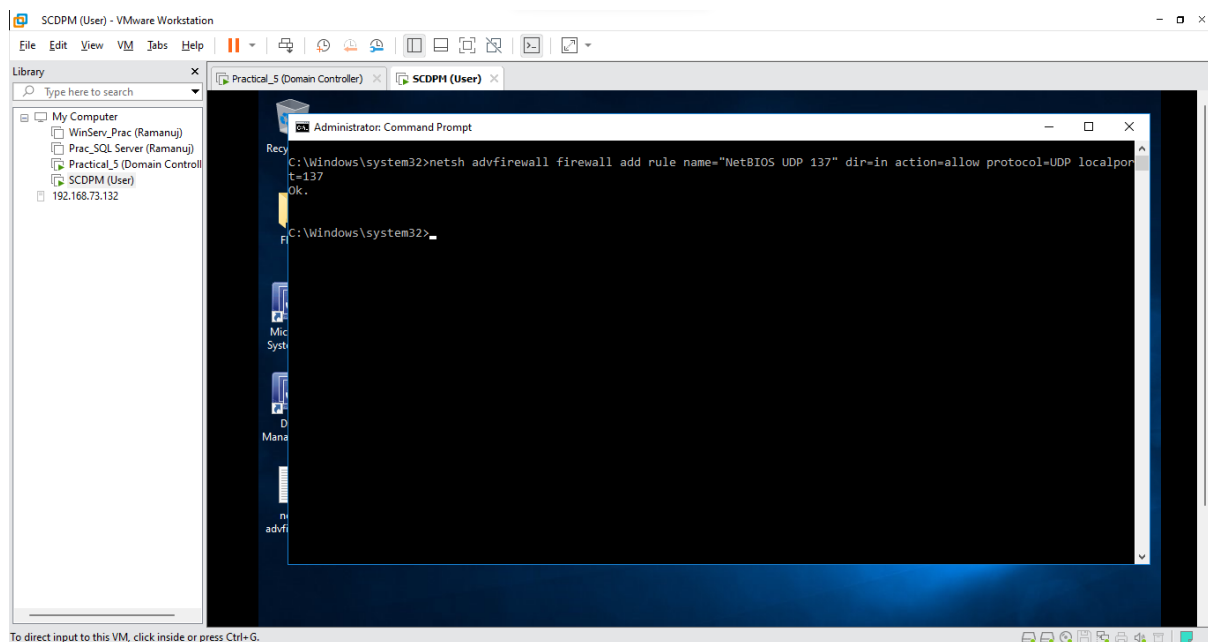
- **netsh advfirewall firewall add rule name="LDAP TCP 389" dir=in action=allow protocol=TCP localport=389**



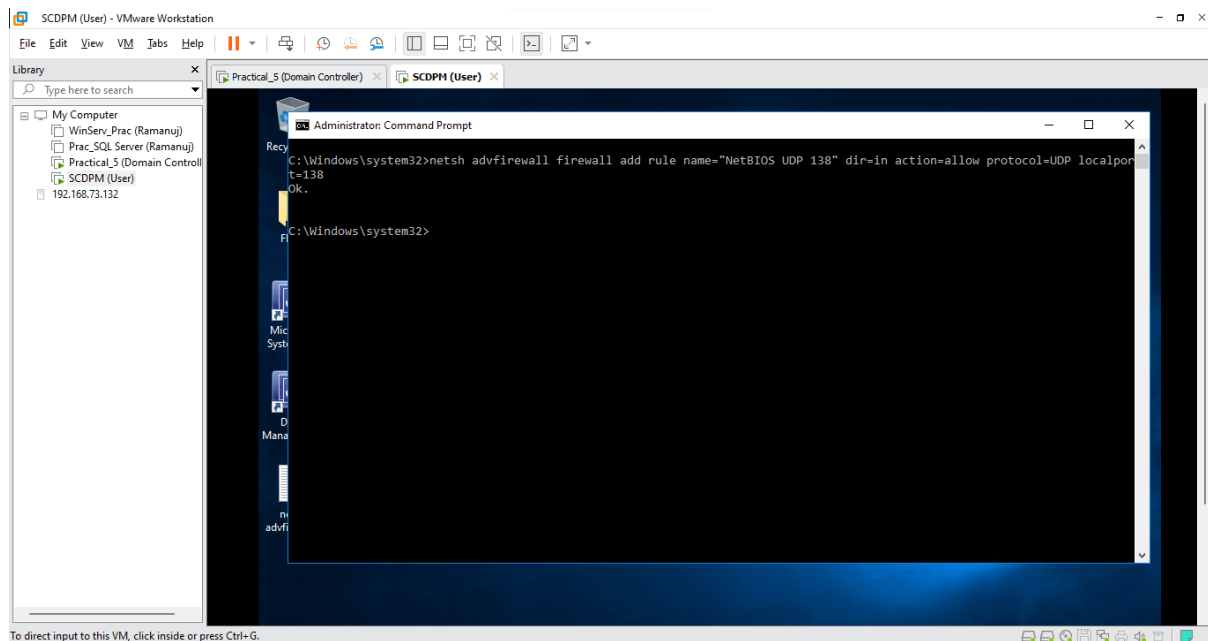
- **netsh advfirewall firewall add rule name="LDAP UDP 389" dir=in action=allow protocol=UDP localport=389**



- **netsh advfirewall firewall add rule name="NetBIOS UDP 137" dir=in action=allow protocol=UDP localport=137**



- **netsh advfirewall firewall add rule name="NetBIOS UDP 138" dir=in action=allow protocol=UDP localport=138**



- **netsh advfirewall firewall add rule name="NetBIOS TCP 139" dir=in action=allow protocol=TCP localport=139**

